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The Syntax of Clausal Prefixes in Tigrinya

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Abstract

This dissertation investigates the syntax of clausal prefixes in Tigrinya, an Ethio-Semitic language which, despite its head-finality, exhibits clause-initial complementizers that attach as prefixes to embedded verbs (*clausal prefixes*). It challenges the view that prefixes in head-final languages have an intrinsic morphological property, proposing instead that their surface position results from specific syntactic constraints. Four main clausal prefixes are analyzed. $Z(i)-$, found in relative clauses, noun clausal complements, and comparative clauses, is argued to be a reflex of successive-cyclic $\bar{A}$ -movement triggered by Spec-Head agreement in a phasal aspectual projection. $K\epsilon mz(i)-$, a bimorphemic element consisting of a preposition plus $z(i)-$, introduces complement clauses that are argued to be underlying relative clauses involving the $\bar{A}$ -movement of a null nominal. $K(i)-$, a subjunctive marker, is claimed to occur in reduced clauses whose temporal interpretation depends on the matrix clause and to appear either as a low MoodP head or as a reflex of temporal operator movement. $?aj\text{---}(i)n$, the circumfix used to mark sentential negation, reflects split negation, with the prefix merged in low NegP and the suffix in high PolP. These findings support a more local view of $\bar{A}$ -movement than standard phase theory and point to a refined notion of cyclicity, offering a novel perspective on the syntax of prefixes in head-final languages.

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List Of Glossing Abbreviations

ACC	Accusative
ADJ	Adjective
AUX	Auxiliary
BEN	Benefactive
CL	Clitic
CMPR	Comparative
COMP	Complementizer
COORD	Coordination
COP	Copula
DAT	Dative
DEF.ART	Definite Article
DEM	Demonstrative
DIST	Distal Demonstrative
DIR	Directional
DOM	Differential Object Marking
FUT	Future
GER	Gerund
IMP	Imperative
IND	Indicative
INF	Infinitive
IPFV	Imperfective
LOC	Locative
NEG	Negation
NMZ	Nominalizer
OBL	Oblique
OM	Object Marking
PART	Participle
PASS	Passive
PAST	Past
PFV	Perfective
POSS	Possessive
PREP	Preposition
PRES	Present
PROS	Prospective
PROX	Proximal Demonstrative
PURP	Purposive
REL	Relativizer
SBJ	Subjunctive
SM	Subject Agreement

Transliteration System

	ε	u	i	a	e	i/ə/-	o	
h	ሀ	ሁ	ሂ	ሃ	ሄ	ህ	ሆ	
l	ለ	ሉ	ሊ	ላ	ሌ	ል	ሎ	
ḥ	ሐ	ሑ	ሒ	ሓ	ሔ	ሐ	ሑ	
m	መ	ሙ	ሚ	ማ	ሜ	ም	ሞ	
(s	ሠ	ሡ	ሢ	ሣ	ሤ	ሥ	ሦ	old style)
r	ረ	ሩ	ሪ	ራ	ራ	ር	ሮ	
s	ሰ	ሱ	ሲ	ሳ	ሴ	ሰ	ሱ	
ʃ	ሸ	ሹ	ሺ	ሻ	ሼ	ሸ	ሹ	
q	ቀ	ቁ	ቂ	ቃ	ቄ	ቅ	ቆ	
q ^w	ቈ		ቊ	ቋ	ቌ	ቍ		
ḳ'	ቐ	ቑ	ቒ	ቓ	ቔ	ቕ	ቆ	
b	በ	ቡ	ቢ	ባ	ቤ	ብ	ቦ	
v	ቨ	ቩ	ቪ	ቫ	ቬ	ቨ	ቩ	
t	ተ	ቱ	ቲ	ታ	ቲ	ት	ቶ	
ʈ	ቸ	ቹ	ቺ	ቻ	ቼ	ቸ	ቹ	
ɸ	ጥ	ጥ	ጥ	ጥ	ጥ	ጥ	ጥ	
h	ኀ	ኁ	ኂ	ኃ	ኄ	ኅ	ኆ	
n	ነ	ኑ	ኒ	ና	ኔ	ን	ኖ	
ɲ	ኘ	ኙ	ኚ	ኛ	ኜ	ኝ	ኞ	
ʔ	አ	ኡ	ኢ	ኣ	ኤ	አ	ኡ	
k	ከ	ኩ	ኪ	ካ	ኤ	ከ	ኩ	
x	ኸ	ኹ	ኺ	ኻ	ኼ	ኸ	ኹ	
w	ወ	ዉ	ዊ	ዋ	ዌ	ው	ዐ	
ʕ	ዐ	ዑ	ዒ	ዓ	ዔ	ዐ	ዑ	
z	ዘ	ዙ	ዚ	ዛ	ዞ	ዘ	ዙ	
ʒ	ዝ	ዞ	ዟ	ዠ	ዡ	ዝ	ዞ	
j	የ	ዩ	ደ	ደ	ደ	ደ	ደ	
d	ደ	ደ	ደ	ደ	ደ	ደ	ደ	
ɖ	ጀ	ጀ	ጀ	ጀ	ጀ	ጀ	ጀ	
g	ገ	ገ	ገ	ገ	ገ	ገ	ገ	
t'	ጠ	ጡ	ጢ	ጣ	ጤ	ጠ	ጡ	
ɣ	ጨ	ጨ	ጨ	ጨ	ጨ	ጨ	ጨ	
p ^c	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ	
ts'	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ	
(ts ^c	ፀ	ፀ	ፀ	ፀ	ፀ	ፀ	ፀ	old style)
f	ፈ	ፉ	ፊ	ፋ	ፈ	ፍ	ፎ	
p	ፐ	ፑ	ፒ	ፓ	ፔ	ፕ	ፖ	

Chapter 1

Introduction

1.1 Setting the Scene

At the beginning of the 1960s, the American linguist Joseph Greenberg examined a sample of thirty languages and proposed a well-known series of forty-five linguistic universals spanning typological, morphological and syntactic phenomena. The fourth of these universals states the following:

“With overwhelmingly greater than chance frequency, languages with normal SOV order are postpositional.” (Greenberg 1963)

After reading this postulate, one might expect not to encounter a prepositional head-final language easily.¹ However, Greenberg himself adds in a footnote that he is aware of the existence of at least four prepositional SOV languages: Iraqw (Cushitic), Khamti (Thai), standard Persian and Amharic (Greenberg 1963:footnote 8). One may therefore ask: are these really the only four languages exhibiting this “typological anomaly” (Baker & Kramer 2014:142)? In fact, no. This dissertation focuses on another language that behaves similarly in an exceptional way from the perspective of Greenberg’s Universal 4: Tigrinya. This Ethio-semitic language — a sister of Amharic — is an SOV language with a typical head-final word-order in both its nominal and verbal systems. It is thus surprising that such a language should have prepositions as those in (1) shown in bold.

¹I use the term *easily* because Greenberg’s universals are not absolute rules, but statistical tendencies.

- (1) a. **mis** dim-ay ji-ts'awet
 with cat.POSS.1S SM.1S-play.IPFV
 'I play with my cat.'
- b. **nab** gezi-ʔa ti-xɛjid
 to home-POSS.3FS SM.3FS-go.IPFV
 'She goes home.'
- c. **kab** ʔertra mɛts' i-u
 from Eritrea come.GER-SM.3MS
 'He comes from Eritrea.'

(adapted from [Cacchioli 2023:235](#), (2))

In the Tigrinya example above, *mis* 'with', *nab* 'to' and *kab* 'from' are prepositions that modify, respectively, the nominals *dimay* 'my cat', *geziʔa* 'her home' and *ʔertra* 'Eritrea' preceding them. This is a classic exception that proves the rule! [Baker & Kramer 2014](#) — and [Bach 1970](#) and [Tremblay & Kabbaj 1990](#) before them — discuss the existence of prepositions in Amharic and resolve this typological exception by claiming that these elements are actually case markers inserted post-syntactically. As [Gebregziabher 2013:38](#), fn. 16 notes, it would be interesting to determine whether the same analysis could be proposed for Tigrinya.

The expectation that head-final languages are postpositional is not merely an empirical generalization, but can also be framed in more theory-driven terms. In head-final languages, heads *follow* their complements, whereas in head-initial languages, heads precede their complements.² Postpositions are relational elements that *follow* their complements, while prepositions precede them. Accordingly, it is theoretically plausible to expect that adpositions in head-final languages are postpositional.

Now, let us extend this reasoning to affixation. Greenberg takes into consideration affixes in his Universal 27.

“If a language is exclusively suffixing, it is postpositional; if it is exclusively prefixing, it is prepositional.” ([Greenberg 1963](#))

In this postulate, Greenberg asserts that the presence of suffixes predicts postpositional order (and thus head-finality), while the presence of prefixes predicts prepositional order

²For a more detailed definition of 'head' in morphology, cf. [Hoeksema 1992](#).

(and thus head-initiality). A natural question arises: can this implication be reversed? That is, can one claim that if a language is postpositional (and thus head-final), it is necessarily suffixing, and if a language is prepositional (and thus head-initial), it is necessarily prefixing?

If we treat affixes as (phonological realizations of) functional heads that attach to lexical heads (which can be considered their complements), then in a head-final language, where heads *follow* their complements, affixes would *follow* the lexical head (the complement) and thus manifest as suffixes. In contrast, in a head-initial language, where heads *precede* their complements, affixes would *precede* the lexical head (the complement) and thus manifest as prefixes. In other words, from a purely theoretical perspective, affix ordering is expected to correlate with head-directionality. Empirically, this prediction is borne out: for example, the head-initial language Swahili (Bantu, SVO) exhibits prefixes, whereas the head-final language Turkish (SOV) exhibits suffixes. Now, what about Tigrinya?

Once again, this language diverges from this prediction and displays a range of nominal *prefixes*, many of which are cognate with Standard Semitic prefixes. In (2) below, we see the use of the prefix *b(i)-* (like *bi-* in Standard Arabic) which is equivalent to the *by*-phrase in English passive constructions, and the prefix *n(i)-* that differentially marks the direct object.³

- (2) a. ʔit-a finistra **b-it-om** sirɛf'ti ti-sɛyr-a
 DIST-FS window.FS by-DIST-MP thieves PASS-break.PFV-SM.3FS
 ‘The window was broken by the thieves.’
- b. Tɛsfay **n-it-i** sɛbʔaj qɛtil-w-o
 Tɛsfay.MS DOM-DIST-MS man kill.GER-SM.3MS-OM.3MS
 ‘Tɛsfay killed the man.’

(from [Cacchioli 2023:234](#), (1))

Now, let us extend this reasoning one step even further and consider complementizers.⁴ Complementizers are functional heads that introduce and type (embedded) clauses. The-

³In these prefixes [i] is an epenthetic vowel.

⁴I use the term *complementizer* here in an atheoretical sense.

oretically, in head-*initial* languages, heads precede their complements, and consequently complementizers are clause-*initial*; in head-*final* languages, heads follow their complements, and complementizers are clause-*final*. Empirically, this generalization is borne out, for example, by the head-initial language Italian (Romance, SVO), which exhibits clause-initial complementizers, and by the head-final language Japanese (SOV), which exhibits clause-final complementizers. Now, is this pattern also attested in Tigrinya?

No. Yet again, this language diverges: it displays a range of clause-initial complementizers that occur as *prefixes* attached to the embedded verb. Consider the Tigrinya examples in (3): the subordinate clauses appear to the left of the matrix verbs, as befitting a head-final language (Bayer 1999) and the complementizers *z(i)-*, *kɛmz(i)-* and *k(i)-* occur attached to the embedded verbs. In the examples in (3), *z(i)-* is prefixed to the embedded verb in a relative clause, *kɛmz(i)-* introduces a complement clause and *k(i)-* is found in a control construction with the verbal element /dlj/ ‘want’ (many other uses of these prefixes will be presented later).

- (3) a. n-it-a n-iz-a mɛts’ħaf **z**i-ts’ɛħaf-ɛt sɛbɛjiti
DOM-DIST-FS DOM-PROX-FS book.FS REL-write.PFV-SM.3FS woman
rɛkib-ɛ-ja
meet.GER-SM.1S-OM.3FS
‘I met the woman who wrote this book.’
- b. tufah **kɛmzi**-beli?-u ji-fellit’
apple COMP-eat.PFV-SM.3MP SM.1S-know.IPFV
‘I know that they ate an apple.’
- c. jay **k-i**-seti dɛlɛj-ɛ
tea SBJ-SM.1S-drink.IPFV want.GER-SM.1S
‘I wanted to drink tea.’

From this point onward, to describe the prefixes in example (3), I do not use the term “subordinators” (Haumann 1997) or “complementizers” (Bresnan 1972:6-7). I avoid the first because it is too restrictive and I do not use the second because it suggests the C° head in the clausal spine, and, as I show throughout the chapters, none of these elements appear in that position. As in the title of this thesis, I use the atheoretical term “clausal prefixes” to describe them and not “verbal prefixes” to avoid confusion with the

prefix-suffix issue in Afroasiatic (the investigation of the phonological, morphological, and syntactic properties of verbal agreement affixes). Note that usually, when one speaks of *nominal* prefixes, these prefixes are attached to nouns, when one speaks of *verbal* prefixes, they are attached to verbs; and so on. However, the *clausal* prefixes under investigation in this thesis are not attached to clauses. By “clausal prefixes” I emphasize that these prefixes function at the level of the clause.

Lastly, to complete the list of Tigrinya properties that seem to “glitch” Greenberg’s matrix, consider negation. Sentential negation in Tigrinya is partly prefixal: it is expressed with the circumfix *ʔaj...-(i)n* attached to the verb, as illustrated in (4).

- (4) mis dim-ay **ʔaj-Ø-ts’aweti-n**
 with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG
 ‘I do not play with my cat.’

So, what is going on in (3) and (4)? How is it possible that Tigrinya, a language that is verb- and auxiliary-final has clausal *prefixes* (Biberauer et al. 2017:22)? Let us also consider the broader picture. We are dealing with a *Semitic* language that (as will be discussed in more detail in section 1.5.2) shares all the standard Semitic properties - phonetics, phonology, non-concatenative morphology, verb system, and so forth - found in Arabic and Hebrew, but exhibits a completely different word order. Arabic and Hebrew are VSO and SVO head-initial languages, while Tigrinya is an SOV head-final language. How can one account for these shared Semitic features alongside these different syntactic properties? From a morphological perspective this question might not appear relevant. Traditional accounts assume that a prefix is *intrinsically* a prefix, a suffix is *intrinsically* a suffix. In other words, such elements are assumed to have the property (or feature) that allows them to attach to other constituents, with word order emerging simply from morphemes attaching where they must. End of story. However, for a syntactician the question is not only interesting but also relevant. Again, how can Semitic languages with very similar morphology be both head-initial (like Arabic and Hebrew) and head-final (like Tigrinya and Amharic)? And, even more importantly, why do head-final languages have (clausal) prefixes?

In this dissertation, I address the second question, leaving the first to other adventurous researchers. I investigate clausal prefixes in Tigrinya and challenge the view that these elements are intrinsically prefixes. I propose instead that clausal prefixes are subject to specific syntactic constraints that result in their appearing in front of verbs.⁵ Therefore, the goal of this dissertation is to answer the question: *What is the underlying architecture of sentences that allows elements in a head-final language to surface as prefixes?* I show not only that Tigrinya is not a typological anomaly but also that its syntactic profile can be accounted for if morphology is considered part of syntax.

Tentative and preliminary ideas have already been proposed along these lines by Roberts 2017a and Girma 2003. In the first work, it is suggested in passing that languages with OV order and prefixing should be analyzed as involving “a combination of V-movement and ‘roll-up’ movement of the remnant VP and so on” (Roberts 2017a:341), while admitting that the details remain to be worked out. The second work develops a more detailed analysis, which will be mentioned and compared throughout the following chapters.

It is important to note that the idea of morphology as syntax stands in opposition to the principles of Distributed Morphology (Halle & Marantz 1993), which posits that affix placement is determined by post-syntactic mechanisms. However, it aligns with literature that seeks to link more closely the operations involved in morphology and syntax. In fact, the notion of a parallelism between morphology and syntax was first articulated in the mid-1980s with Baker’s *Mirror Principle* (Baker 1985). This principle states that the linear order of morphemes reflects the hierarchical order of syntactic operations. In other words, if a syntactic operation α applies before an operation β , the corresponding morpheme is closer to the root than the morpheme introduced by β . A few years later, Baker 1988 extended these ideas, suggesting that being a prefix or being a suffix is an idiosyncratic property of the individual morpheme and that “each inflectional morpheme is a functional head which projects its own projection” (Speas 1991:184).⁶

⁵Another way to look at prefixes would be to consider them as suffixes on empty elements, an option not considered in this dissertation.

⁶In Cartographic terms (as explained in section 1.3) this would mean that each morpheme heads a different projection; this is also one of the tenets of Nanosyntax (Starke 2009, 2011, Caha 2009).

Even though, according to Baker, the order of attachment of affixes must reflect the syntactic derivation, he did not claim that affixes are attached syntactically. Such a claim emerged later in more contemporary approaches (Julien 2002, Kayne 2019, 2023, Shlonsky 2023), including the program of *Morphology as Syntax (MS)* proposed by Collins & Kayne 2020. In this work the authors advance the provocative claim that Universal Grammar contains no morphological component and that word formation and morphological generalizations are entirely derived from syntactic operations and principles. In this thesis, I do not adopt this strong position, nor do I endorse the MS program.

I argue, however, that the nature and distribution of clausal prefixes in Tigrinya can be understood as dependent on syntax rather than on morphology. In this sense, the thesis can be seen as a “proof of concept” showing that syntactic mechanisms can derive the same results that the Distributed Morphology framework attributes to postsyntactic processes. I will show that morpheme ordering — at least for clausal prefixes in Tigrinya — can be accounted for by syntactic operations such as $\bar{A}$ -movement, the operation AGREE, selection and head movement. This does not amount to a claim that postsyntactic mechanisms, the morphological component, or the Distributed Morphology framework lack motivation or empirical support; rather, the objective is to explore a theoretically coherent alternative within the broader landscape of generative grammar.

The goal of this dissertation is twofold. On the one hand, as already mentioned, it addresses the theoretical dilemma of head-final languages manifesting prefixes by providing a syntactic account of the underlying architecture of sentences and their surface word order. On the other hand, it contributes empirically to the field of linguistics by presenting a large body of novel data from an Afro-asiatic language whose morpho-syntax has received little attention and by opening new avenues for research. In fact, Tigrinya, and the Ethio-Semitic family to which it belongs, has been largely neglected and understudied. Existing grammars and descriptions (a.o. Conti Rossini 1940, Leslau 1941, Mason 1996, Kogan 1997, Bulakh 2019, 2023) are valuable but “leave unexpressed many of the basic regularities of the [spoken and informal] language” (Chomsky 1965). Moreover, they generally show what *can* be said in a language but not what *cannot* be said - a crucial point

in Linguistic Theory (Vaux et al. 2007). To advance the field of (not only theoretical) linguistics, it is vital to have descriptions and analyses of the widest possible variety of languages (Vaux et al. 2007:1). This dissertation is one such contribution.⁷

1.2 Thesis Overview

The plan of the thesis is as follows. Subsection 1.3 presents the theoretical framework. Subsection 1.4 explains the collection of the data, transcriptions and glosses. Subsection 1.5 provides background on Tigrinya, first placing it in its ethnographic context and then presenting several of its grammatical properties (covering the nominal and the verbal domains) to facilitate understanding of the subsequent sections. Finally, section 1.6 gives a syntactic account of declarative clauses in Tigrinya and claims that these are monoclausal when a periphrastic tense form (a verb with an auxiliary) expresses progressive aspect but with two aspectual layers.

Chapter 2 focuses on the clausal prefix $z(i)-$. I provide data on this prefix in different constructions - relative clauses, comparative clauses, noun clausal complements and *seem*-clauses. After a digression on the external syntax of relative clauses and on the syntax of DPs in Tigrinya (section 2.2.1) I turn to the analysis and propose that $z(i)-$ is a reflex of successive-cyclic movement and a marker of agreement arising from an $\bar{A}$ -dependency.

Chapter 3 deals with the clausal prefix $k\epsilon mz(i)-$. I show that this prefix introduces complement clauses with factive, cognitive non-factive, fiction, utterance and perception verbs, and compare it to a head-final complementizer in Tigrinya. After presenting data on $k\epsilon mz(i)-$ -clauses involving periphrastic tense forms, I argue that $k\epsilon mz(i)-$ is a bimorphemic element composed of the preposition $k\epsilon m$ and the clausal prefix $z(i)-$ studied in the first chapter. Adding to the literature on subordination as relativization, I suggest that $k\epsilon mz(i)-$ -clauses involve $\bar{A}$ -movement of non-phonologically overt nominals.

Chapter 4 focuses on the clausal prefix $k(i)-$. I claim from the outset that $k(i)-$ is a subjunctive marker and list its occurrence in various constructions with directive, desire,

⁷I would like to point out that there are preliminary versions (talks, proceeding papers, articles) of parts of what is presented in this dissertation. These are Cacchioli 2023, 2024, 2025, Cacchioli & Overfelt 2023, Cacchioli & Shlonsky 2023, Overfelt & Cacchioli submitted.

future-orienting, emotive, fiction, control, ECM and modal verbs. I also present data on $k(i)$ -clauses expressing purpose and future tense and show that sentential subjects also require this prefix. Finally, I discuss $k(i)$ -clauses expressing simultaneity, which results from $k(i)$ - appearing on both the verb and the auxiliary in periphrastic tense forms, similarly to $z(i)$ -. I provide syntactic derivations for all the aforementioned structures, claiming that $k(i)$ -clauses are generally control constructions.

Chapter 5 examines the circumfix $?aj\text{---}(i)n$. First, I discuss sentential negation in simple verb forms expressing present tense and show that, while the circumfix is needed to express negation in declarative clauses and some $k(i)$ -clauses, only its prefixal part is needed in $z(i)$ - and imperative clauses. After a digression on expletive negation, I offer a syntactic analysis claiming that the prefix $?aj$ - heads a NegP projection whereas the suffix $-(i)n$ heads a PolP projection. Second, I turn to the syntax of sentential negation in clauses expressing past tense and progressive aspect.

In the conclusion (Chapter 6) I summarize the phenomena investigated, the analyses proposed, and mention topics for further research both beyond clausal prefixes in Tigrinya and beyond Tigrinya itself.

1.3 The Theoretical Framework

This dissertation aims to adhere as closely as possible to the scientific method (Carnie 2003). In each chapter I present and carefully describe the data, make generalizations about the patterns found, and develop hypotheses and analyses to explain the data.

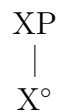
This work belongs to theoretical syntax, a subfield of linguistics that seeks to develop a general theory of the language module of human cognition by analyzing, among other things, the structure of specific languages. In simple terms, syntax formalizes speakers' intuitions (Cooper 1983) and is considered the “real engine of language” (Roberts 2017b).

The line of inquiry adopted here is situated within the conceptual framework developed by Noam Chomsky and his students and collaborators since the mid-1950s. Chomsky's Generative Grammar not only invites linguists to view grammatical sentences as generated by a system of formal rules, but also takes as its premise the innateness of linguistic prin-

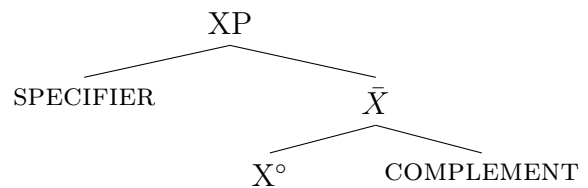
ciples and the consequent existence of linguistic universals. According to Chomsky, these universals form part of Universal Grammar, a grammar module in the human mind/brain that allows a child to learn and master a language despite limited input. Chomsky’s ideas began in *Syntactic Structures* (Chomsky 1957) and matured in *The Minimalist Program* (Chomsky 1995) and subsequent work.

The Minimalist Program elaborates the Principle and Parameters framework first developed in the 1980s (Chomsky 1993), which posits that a set of universal and innate principles common to every language are subject to parametric variation. Principles concern locality conditions, government, Case, theta-roles and so forth. Parameters are like switches on a machine: they can be set ‘on’ or ‘off’ depending on the language. One principle used throughout this dissertation, derives from X-bar Theory (Chomsky 1970, Jackendoff 1977). This theory was designed as a component of the computational system (or processing system of our brain) that projects structures from elements chosen from the Lexicon (Bianchi 1999). It suggests that all phrases (nouns, verbs, adjectives,...) share the same underlying structure and can thus be represented with an X-bar schema, such as the one in (6).

(5)



(6)



In X-bar Theory every phrase XP (“X phrase”) has a unique head X° (or simply X). Each X° head projects a phrase and determines the category of its maximal projection (the phrase itself), as shown in (5). Every node branches into two: the node in a sister relation with $\bar{X}$ is the Specifier position of XP (also Spec,XP), and the node in a sister relation with X° is the complement of XP, as in (6). Specifier positions are also called $\bar{A}$ -positions, contrasted with A-positions: theta-roles (semantic relations) can never be assigned in the former and can only be assigned in the latter (Müller 1995). Adjuncts - phrases that are not arguments - can optionally be attached to the $\bar{X}$ -schema by projecting an additional

maximal projection XP on top of the existent one.

X-bar Theory plays on the concept of endocentricity (every phrase has one head), binarity (a node can only have two daughters) and maximality (a head can only have one specifier projection) (Ogawa 2001). In the 1990s Chomsky moved away from strict X-bar Theory and advocated a looser phrase structure in which more than one specifier could in principle be licensed in a single maximal projection (Ogawa 2001). Debate continues today over the possibility of multiple specifiers.

An even larger debate at the center of current syntactic theory - and relevant throughout this dissertation - concerns the ordering of the daughters of $\bar{X}$. The head X° and its complement can, in principle, be ordered in two different ways: X° precedes the complement or X° follows it. This distinction has been used to explain the syntactic difference between head-initial and head-final languages: in the former X° precedes its complement and in the latter X° follows it. This is known as the *Head-directionality parameter* (a.o. Chomsky 1993).

That head-initial languages have a head-initial structure has always been uncontroversial; however, the idea that head-final languages are base-generated with a head-final structure has been called into question by several authors. It has been proposed that the Head-Directionality Parameter does not in fact exist and that all languages are underlyingly head-initial. Head-final languages would then result from the movement of complements to the left of heads. One of the most influential works arguing against a Head-Directionality Parameter is Kayne 1994. This seminal work on syntax proposes an antisymmetric approach in which “asymmetric c-command invariably maps into linear precedence” (Kayne 1994:3), as established by the *Linear Correspondance Axiom* (LCA). In other words, Kayne reformulated the standard concept of c-command (constituent command) proposed by Reinhart 1976, as shown in (7) and (8).

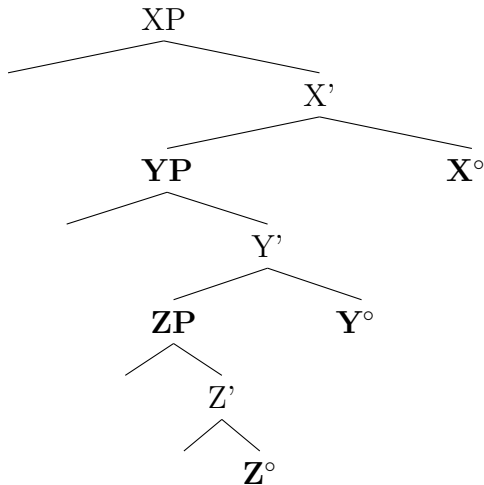
- (7) *A c-commands B if A does not dominate B and B does not dominate A and every X that dominates A also dominates B.*
(simplified version of the symmetric c-command proposed by Reinhart 1976)

- (8) *X asymmetrically c-commands Y iff X c-commands Y
and Y does not c-command X.*

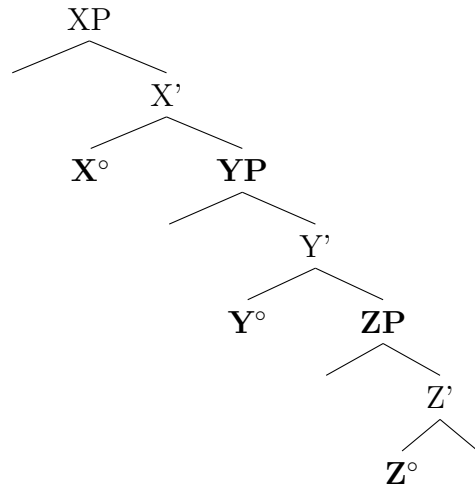
(the asymmetric c-command proposed by [Kayne 1994](#))

This means that all languages share an underlying SPECIFIER-HEAD-COMPLEMENT word order. In less abstract terms, if we take the head-final word order SOV (subject-object-verb), the object is assumed to be base-generated to the right of the verb (the head-initial order VO) and then to undergo leftward movement from its argument position to a higher functional projection, resulting in the surface head-final order OV ([Kayne 1994](#)). In the literature, the functional projection whose specifier hosts the moved object is not clearly defined; however, it is suggested to be higher than the verb, since movement to the specifier of VP would violate the Anti-Locality constraint ([Abels 2003](#)), according to which complements of phrases cannot move to their own specifiers. A more language-specific approach is, for example, [Haegeman 2000](#)'s derivation of West Germanic OV word order: according to the author, this is obtained by morphology-driven V-movement to an unlabelled functional head and remnant movement ([Thiersch 1985](#)) of the extended projection of V. In both [Kayne 1994](#) and [Haegeman 2000](#) (and other studies) the OV/VO alternation is thus derived rather than base-generated. The Head Parameter and Antisymmetry approaches are respectively illustrated in (9) and (10) below.

(9)



(10)



Theoretically, there are several ways in which a surface word order can be derived (Julien 2002). In this dissertation, which investigates an SOV language and thus head-final language, I adopt a head-parameter approach. This is a scientific choice, but it also reflects some skepticism about issues raised by Antisymmetry. First, this approach was “built almost exclusively on SVO data” and was therefore “tailored to SVO languages” (Haider 2013:x). In the past few years working on the syntax of Tigrinya, I noticed that an SOV grammar “does not fit neatly into a ‘universal’ model that has inadvertently been designed for SVO” (Haider 2013:x). Second, claiming that the OV word order is derived from the VO word order implies that the VO order is more basic than the OV one (Haider 2000); but, is it really? Does it mean that some languages are more basic than others? I do not wish to make that claim. Third, deriving OV word order from VO results in extensive leftward and semantically empty movement (Dékány 2018), which does not accord with the concept of grammatical economy (Chomsky 1989). Fourth, why should OV languages be derived from VO languages rather than the reverse? In other words, why can’t all languages be underlyingly head-final? Because of these four issues, I take a head-directionality parameter approach to head-finality in this dissertation.⁸

Lastly, a framework that I refer to throughout the dissertation, and which represents a more recent development within Generativism, is the cartographic approach developed in the late 1990s by Luigi Rizzi in *The Fine Structure of the Left Periphery* (Rizzi 1997) and by Guglielmo Cinque in *Adverbs and Functional Heads: A Cross-linguistic Perspective* (Cinque 1999). Cartography is a branch of theoretical syntax that examines the hierarchical structure and representation of functional projections while paying close attention to apparent or real variation and to the parameters along which languages can vary. Building on the extension of the nominal phrase by Steven Abney into a “split DP” (Abney 1987), the verb phrase by Richard K. Larson into a “VP shell” (Larson 1988), and Pollock’s discovery of two distinct landing sites for verbs in the clause (Pollock 1989), Cinque and Rizzi propose an even more radical dissection of CP and IP into an array of

⁸It is also important to note that in more recent years new approaches have proposed the existence of mixed-order languages in which “heads in the same extended projection may either precede or follow their complement” (Holmberg 2017a), meaning that some (functional) projections are head-initial and some head-final (Aboh 1999, Holmberg 2017b). The famously known *FOFC*.

universal functional projections. Syntactic structures are thus drawn as maps, as detailed as possible in their specific domain.

1.4 Notes on Data Collection, Glosses and Transcription

Unless otherwise noted, the data presented in this dissertation were collected through fieldwork carried out in Geneva between 2020 and 2025 with four native speakers of Tigrinya. Two of the language consultants are from the region of Debub, the other two from the region of Gash-Barka. They are fluent in French, have beginner to intermediate English and have been living in Geneva since the 2010s. Three of these consultants served as the main informants for this study, while the fourth was consulted only sporadically. I sometimes observed regional linguistic variation amongst my consultants and point this out in footnotes when relevant.

The data presented in the following chapters were gathered using different elicitation techniques ([Bowern 2008](#)): translation tasks (from French to Tigrinya), grammaticality and acceptability judgment tasks on manipulated data, and back-translating (from Tigrinya to French) using sentences constructed by me or provided by the consultants themselves. In some cases sentences were presented along with contexts constructed to show the intended meaning. The data presented in this thesis are also drawn from grammars and the literature. Transcriptions, glosses and translations of data from the literature is reproduced as faithfully as possible.

Transcription conventions in Tigrinya are subject to significant individual variation. For example, in the literature there is no uniformity in the realization of the “first order” vowel: for example, [Kogan 1997](#) uses /ä/, [Zellou 2010](#) uses /ɛ/ and [Gebregziabher 2013](#) uses /e/. Because the focus of this thesis falls outside the domain of phonology, I decided to use /ɛ/ to realize the first order vowel in Tigrinya after a brief e-mail exchange with Sharon Rose. Aware of variation in this phoneme and others, I have tried to account for it by using consistent glossing practices following the Leipzig Glossing Rules. My

two deviations are: using capital SM and capital OM respectively for subject- and object-marking morphology on verbal elements and hyphenating ϕ -features in front of verbs in the Imperfective.

I follow the rules proposed in [Meyer & Treis 2021](#) to cite Eritrean and Ethiopian authors, in order to do justice to their naming conventions. Accordingly, Eritrean and Ethiopian authors are cited by their first names, unless there is evidence in their work that they self-cite using their family name.

All inconsistencies, inaccuracies and mistakes are my own.

1.5 Preliminaries on Tigrinya

Because Tigrinya is an unfamiliar language to most readers, in this section I present some of its morphosyntactic properties that are relevant for understanding the data and the phenomena discussed in the main chapters.

1.5.1 Ethnographic Information

Tigrinya is an Ethio-semitic language of the Afro-Asiatic family. More specifically, it belongs to the Northern branch of Ethio-semitic, along with Ge'ez (Classical Ethiopic), Tigre and Dahalik ([Esayas 2003](#)).⁹ Tigrinya is closely related to Amharic and more distantly related to Arabic and Hebrew.

There are approximately 10 million speakers of Tigrinya worldwide. It is spoken predominantly in central Eritrea (2.5 million speakers) and in the northern region of Tigray in Ethiopia (4.3 million speakers) on the Horn of Africa ([Bulakh 2019:175](#)). Because of migration, Tigrinya is also spoken in several other countries, including Switzerland. [Figure 1](#) shows the geographical distribution of Tigrinya in Eritrea and Ethiopia.

⁹Note that [Bulakh & Kogan 2010](#) have argued against the genetic unity of a North Ethio-semitic. Further details on the subgrouping of Semitic languages can be found in [Rubin 2008](#).

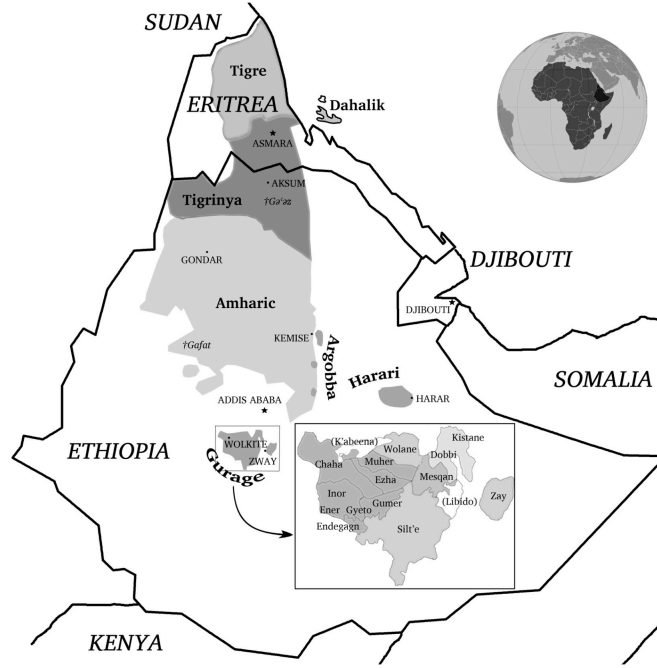


Figure 1: Distribution of Tigrinya speakers (adapted from The Semitic Languages (Weninger 2011); credit: Ronny Meyer)

Ethio-semitic developed from the speech of peoples who migrated from Yemen to Abyssinia (Dillmann 1974:3). Interestingly, the Semitic languages of Eritrea and Ethiopia were the only ones established in Africa before the Islamic conquest and the spread of Arabic (Goldenberg 2013:16).

As a Semitic language, Tigrinya exhibits non-concatenative root morphology, in which words are formed from a combination of (tri)consonantal roots - the semantic core of the words - and vowels templates that are transfixed into them. All Tigrinya roots recur in the other Semitic languages (Dillmann 1974). The Italian colonization of Eritrea in 1890 introduced many loanwords into the language (Negash 1999:5), such as *armadio* “cabinet”, *kamcha* (from *camicia*) “shirt” and *gazeta* (from *gazzetta*) “newspaper”. Despite the fact that Tigrinya is extremely understudied in linguistics, and the Ethio-Semitic branch as a whole has been largely neglected, its literary production has grown steadily, flourishing in particular since the 1990s (Goldenberg 2013:18).

Tigrinya is written from left to right with the *Gəŋəz* script, a syllabic writing system in which more than thirty consonants and seven vowels — called “orders” (Leslau 1941, Mason 1996) — are combined in a single glyph known as *fidəl* ‘syllograph’ (Nazareth

2011). The transliteration system used in this dissertation is presented immediately before the Introduction.

There are several “regional varieties” of Tigrinya between Ethiopia and Eritrea. At least a Northern (Eritrean) and a Southern (Ethiopian) dialect have been recognised in the literature (Goldenberg 2013:17). To my knowledge, there has been very little research conducted on dialectal variation in Tigrinya, with the exceptions of Tsehaye 2009 and Niguss 2021. The Northern dialect has been described as characterized by a substantial number of innovations, presumably arising under the influence of Amharic and the neighboring Oromo (Cushitic) (Bulakh 2019:175). As will also be shown in the following chapters, Eritrean Tigrinya has likewise been influenced by other languages, mainly Cushitic. As the reader will see by the end of this thesis, oral Tigrinya allows considerable variation (Shlomo 1980:240).

1.5.2 Grammatical Sketch

As already mentioned in the Introduction, Tigrinya’s clausal system is head-final, displaying an SOV(Aux) word order. In the following sections I present the morphosyntactic properties of Tigrinya, first in relation to S and O (the nominal domain) and then to V and Aux (the verbal domain).¹⁰

With regard to the phonetics and phonology of Tigrinya (both of which are outside the scope of this dissertation), I advise the reader to consult the Transliteration System provided immediately before the Introduction and to note that two phonological processes are quite common in Tigrinya: epenthesis — for example, when onsets and codas are complex, a [ɨ] is inserted — and spirantization — for example, [k] becomes [x] post-vocally (Godfrey 2011); and [b] becomes [β] post-vocally (Noam Faust, p.c.). Moreover, gemination also occurs in Tigrinya, but it is not phonemic.

¹⁰This section is partly taken from a short paper that I wrote in 2019 for my postgraduate Specialization Certificate in Linguistics entitled *Tigrinya Restrictive Relative Clauses*.

The Nominal Domain

In Tigrinya, nouns are inflected for number (singular or plural) and gender (masculine or feminine). Singular nouns are unmarked, whereas plurality is conveyed either by a suffix attached to the noun - most commonly *-tat* (Mason 1996:10) - or by a noun-internal phonological change (a form of ablaut) known as *Broken Plural* (Palmer 1955), which is also famously found in Arabic varieties. It is difficult to establish a rule that explains which nouns take a regular plural (with *-tat*) and which take an irregular plural (with a Broken Plural) form (Gebregziabher 2013). Example (11) shows a noun with a suffixal plural and a noun with broken plural.

- (11) a. *dimu*, *dimut**at***
 ‘cat, cats’
 b. *komadara*, *komid**ara***
 ‘tomato, tomatoes’

There are two genders in Tigrinya: masculine and feminine. Nouns are not normally inflected for gender (i.e., there is no clear morphophonological distinction between feminine and masculine nouns). Nouns that denote biological sex distinctions - and therefore animate entities such as people and animals - have semantic gender. However, nouns for inanimate objects (either material or abstract) do not have a stable grammatical gender: in fact, gender assignment appears to be very flexible (Mason 1996, Kogan 1997, Nazareth 2011), with a tendency to prefer masculine (Gebregziabher 2013).¹¹ For example, the reader will notice that in this dissertation the noun *meṣṣ’ḥaf* ‘book’ is sometimes glossed as masculine and sometimes glossed as feminine, depending on the choice of the informant to whom the sentence was first presented.

To serve the function of definite articles, Tigrinya uses distal demonstratives.¹² These demonstratives always appear before the noun they modify. The language also has proximal demonstratives, which are used when the speaker refers to something close to them

¹¹This may be due to the fact that Ethiopic has never had any native grammarians (Dillmann 1974).

¹²Note that cross-linguistically demonstratives are generally considered to be definite (Lyons 1999).

(Kogan 1997). Distal demonstratives are illustrated in (12) and proximal demonstratives in (13).

- (12) **ʔit-i** wɛdi, **ʔit-a** g^wal, **ʔit-om** ʔawɛdat, **ʔit-ɛn** ʔawɛlid
 DIST-MS boy, DIST-FS girl, DIST-MP boys, DIST-FP girls
 ‘the/that boy, the/that girl, the/those boys, the/those girls’
- (13) **ʔiz-i** wɛdi, **ʔiz-a** g^wal, **ʔiz-om** ʔawɛdat, **ʔiz-ɛn** ʔawɛlid
 PROX-MS boy, PROX-FS girl, PROX-MP boys, PROX-FP girls
 ‘this boy, this girl, these boys, these girls’

Distal and proximal demonstratives can substitute the entire determiner phrase, as shown in (14) and (15).

- (14) **ʔiz-i/ʔit-i** dɛlji-ɛ
 PROX-MS/DIST-MS want.GER-SM.1S
 ‘I want this/that.’
- (15) **ʔit-i!**
 DIST-MS
 ‘That one!’ (from Nazareth 2011:29,(20c))

In Arabic (as well as Amharic) demonstratives co-occur with determiners, as shown in (16). It is therefore not surprising to see a similar phenomenon in Tigrinya. However, this occurs only when the speaker wants to emphasize the distance or proximity of the noun being qualified. This is somewhat similar to the difference between *ceci/celui-ci* ‘this one’ and *cela/celui-là* ‘that one’ in French, or the construction *questo X qui* ‘this X here’ and *quell’ X lì* ‘that X there’ in Italian (Conti Rossini 1940). In such cases, the demonstrative appears twice in the noun phrase with the second element realized in what has been called its “full” (Mason 1996) or “long” (Nazareth 2011) form. This is shown in (17).¹³

- (16) **ʔuhibb-u** **hað-a** l-kitāb-a
 IPFV.love-SM.1S DIST-MS DET.S-book.MS-ACC
 ‘I love this book.’ *Standard Arabic*

¹³Further details on double determination in Tigrinya are presented in Esayas 2003 and Nazareth 2011.

- (17) **ʔiz-a** məkina **ʔizi-a** kab-ta məkina **ʔiti-a**
 PROX-FS car.FS ‘FULL’_PROX-FS from-DIST-FS car.FS ‘FULL’_DIST-FS
 ti-ʕabi
 SM.3FS-be_big.IPFV
 ‘This car (here) is bigger than that car (there).’

Adjectives are inflected for the number and gender of the noun they modify. In Tigrinya, they are always placed between the determiner and the noun, as shown in (18).

- (18) a. ha-dε bəliḥ kara
 one-MS sharp.MS knife
 ‘a sharp knife’
 b. ha-nti belah səbəjiti
 one-FS smart.FS woman
 ‘a smart woman’

When several adjectives occur in the same clause, their order is not fixed (in contrast to languages such as English, French and Italian — cf. [Laenzlinger 2005](#), [Cinque 2010](#)). Regardless of the number of adjectives present in the same clause, their position is always prenominal, as shown in example (19) below.

- (19) a. ʔiz-a ts’ibiqti niʔift’o hats’inawit məkina
 PROX-FS beautiful.FS little metallic.FS car
 b. ʔiz-a niʔift’o ts’ibiqti hats’inawit məkina
 PROX-FS little beautiful.FS metallic.FS car
 c. ʔiz-a hats’inawit ts’ibiqti niʔift’o məkina
 PROX-FS metallic.FS beautiful.FS little car
 ‘this beautiful little metallic car’

Tigrinya does not have a special marker to express indefiniteness in nouns. Nevertheless, it is possible to use the numeral *hade* ‘one’ (*hanti* for feminine nouns) as a marker of indefiniteness and, more specifically, of specificity ([Nazareth 2011](#)), as in (18). With the exception of *hade/hanti*, numerals in Tigrinya are not inflected for gender. When they occur in a phrase along with adjectives, they always precede the adjectives.

To sum up, demonstratives, numerals and adjectives in Tigrinya appear before the

qualified noun in exactly this order. Nominal modifiers never appear postnominally. Examples (20) and (21) illustrate sentences containing all these elements.

- (20) ?it-om ?asertɛ killitɛ ?abejti sɛb-?ut dɛkim-om ?all-ɛwu
DIST-MP ten two old man-MP tired-MP BE.PRES-SM.3MP
‘The twelve old men are tired.’
- (21) ?it-om ?arbaʃetɛ niwaḥti zanta-tat ḥa qina-tat ?ijj-om
DIST-MP four long.P story-P true-P BE₂.PRES-SM.3MP
‘The four long stories are true.’

The Case system of Tigrinya comprises Nominative and Accusative.¹⁴ Nominative case is unmarked with both transitive and unaccusative verbs, as shown in (22).

- (22) a. ?it-a tʼirmuz tɛ-sɛbir-a
DIST-FS bottle.FS PASS-break.PFV-SM.3FS
‘The bottle broke.’ (from [Cacchioli & Overfelt 2023](#))
- b. Yonas nä-t-a tʼirmuz säbir-u-wa
Yonas.M DOM-that-FS bottle break.PFV-SM.3MS-OM.3FS
‘Yonas broke the bottle.’ (from [Nazareth 2011:56, \(55a–b\)](#))

In contrast to Nominative case, Accusative case is differentially marked ([Bossong 1991](#)) with the prefix *n(i)-*, as in the demonstrative *?ita* ‘the’ that modifies *tʼirmuz* ‘bottle’ in (22b).¹⁵ The prefix *n(i)-* can appear on demonstratives (with a consequent dropping of the first phoneme in the demonstrative [ʔ]) and on nouns. There are, however, restrictions on the marking of Accusative case in Tigrinya: an element is in fact marked with *n(i)-* only in a sentence in which that element is definite or specific ([Weldeyesus 2004, Kievit & Kievit 2009, Nazareth 2011, Overfelt 2022](#)).¹⁶

The *n(i)-* prefix is also used to mark Dative case, encompassing “a wide range of adverbial meanings, including direction, purpose, addressee” ([Kievit & Kievit 2009:56](#)).

¹⁴“If in a language the verb follows both the nominal subject and nominal object [SOV] as the dominant order, the language almost always has a case system” ([Greenberg 1963:Universal 41](#)). Also noted by [Gebregziabher 2013:37](#).

¹⁵According to [Overfelt 2022](#), this element is a preposition, and it in fact serves also the function of a directional preposition, for example in *ni-Kanada* ‘to Canada’ ([Gebregziabher 2013:37](#)).

¹⁶I have observed some optionality among my consultants in the application of this rule: some consultants use this prefix sometimes to mark an object which is indefinite or nonspecific. I do not entirely understand the reason; however, I think that the optionality is related to discourse-marking properties.

The uses of this prefix sometimes overlap with those of a pronominal suffix composed of the element *-l(i)* and object markers expressing ‘for the benefit of’ or ‘on behalf of’ (Mason 1996) (henceforth I gloss this element as OBL for ‘oblique’, but it could also be considered an applicative marker). Some uses of the *n(i)*- prefix and the *-l(i)* suffix to mark oblique cases are shown in (23) and (24), but they are not further analyzed in this dissertation.

- (23) a. ʔit-i mægbi **ni**-xelbi ʔijj-u
DIST-MS food DAT-dog BE₂.PRES-SM.3MS
‘This food is for the dog.’ (from Mason 1996:31)
- b. ʔit-i gɛnzɛb **ni**-ʔabo-ʔa hib-ɛ-jo
DIST-MS money DAT-father-POSS.3FS give.GER-SM.1S-OM.3MS
‘I gave the money to her father.’
- (24) a. maʔɛts'o kɛfit-u-**lu**
door open.GER-SM.3MS-OBL.3MS
‘He opened the door for him.’ (from Mason 1996:81)
- b. Tɛsfai timali ts'ihif-u-**llay**
Tɛsfay yesterday write.GER-SM.3MS-OBL.1S
‘Tɛsfay wrote to me yesterday.’

For further details on these and other nominal modifiers — such as quantifiers or possessives (whose properties are not relevant for this dissertation) — see Nazareth 2011.

The Verbal Domain - Verbs

As already mentioned, because Tigrinya is a Semitic language, words are formed from consonantal roots into which vowels templates are transfixed. To form verbs, these vowels templates carry TAM (tense, aspect and modality) information.

In Semitic, verbs are normally classified into two main aspects: Perfective and Imperfective.¹⁷ With regard to the form of these aspects, the Perfective is a “suffixal conjugation” (Hetzron 1997), meaning that subject-agreement morphology appears as a suffix on the verbal form; whereas the Imperfective is a “prefixal conjugation” (Hetzron 1997) in which subject morphology appears mainly as a prefix. Subject-agreement morphology

¹⁷Note that there is no consensus in Semitic literature regarding whether these two paradigms represent aspect, tense, both, or neither (Al-Aqarbeh & Al-Sarayreh 2017). I do not delve into this debate here.

consists of person, number and gender agreement with the grammatical subject in the sentence.¹⁸

Beside the Perfective and Imperfective - and unlike Arabic and Hebrew - Tigrinya possesses a third conjugation known as the Gerund.¹⁹ Like the Perfective, the Gerund is also a suffixal conjugation, but the forms follow different paradigms. This conjugation was probably introduced into the language through contact with Cushitic languages (Appleyard 2009, 2015, Edzard 2014).

Table 1.1 shows the paradigms of the Perfective, Imperfective and Gerund conjugations for the root /gbr/.²⁰

	Imperfective	Perfective	Gerund
SM.1S	jǐ-gǝbbǐr	gǝbǝr-ku	gǝir-ǝ
SM.2MS	tǐ-gǝbbǐr	gǝbǝr-ka	gǝir-ka
SM.2FS	tǐ-gǝbr-i	gǝbǝr-ki	gǝir-ki
SM.3MS	jǐ-gǝbbǐr	gǝbǝr-ǝ	gǝir-u
SM.3FS	tǐ-gǝbbǐr	gǝbǝr-ǝt	gǝir-a
SM.1P	nǐ-gǝbbǐr	gǝbǝr-na	gǝir-na
SM.2MP	tǐ-gǝbr-u	gǝbǝr-kum	gǝir-kum
SM.2FP	tǐ-gǝbr-a	gǝbǝr-kin	gǝir-kin
SM.3MP	jǐ-gǝbr-u	gǝbǝr-u	gǝir-om
SM.3FP	jǐ-gǝbr-a	gǝbǝr-a	gǝir-in

Table 1.1: Verbal Conjugations in Tigrinya.

With regard to the meaning of these verbal forms, the Imperfective expresses present habitual in the absence of auxiliaries: it denotes present actions that are incomplete or customary (Conti Rossini 1940, Leslau 1941, Mason 1996, Kogan 1997). The Perfective is used in narratives to express distant past (Conti Rossini 1940, Leslau 1941, Mason 1996) or present perfect (Nazareth 2011). However, unlike other Semitic languages, the

¹⁸Note that I wrote that subject morphology in Imperfective verb forms occur *mainly* as prefixes. This is because in plural second and third persons ϕ -features are split between prefixes and suffixes. It seems that number and gender are prefixes and person is a suffix; however, things are more complicated than this. In fact, which affix is dedicated to which feature is still an ongoing investigation in Semitic literature.

¹⁹The Gerund is also found in Amharic, Argobba and Ge'ez (Girma 2003:118).

²⁰Note that in Eritrean Tigrinya the first-person singular and the third-masculine singular forms of the Imperfective are syncretic (*jǐgǝbbǐr* 'I do/he does'). However, in Ethiopian Tigrinya and Classic Tigrinya the subject-agreement prefix of the first-person singular is *ǝ*, resulting in the form *ǝǐgǝbbǐr*. This phenomenon is also taking place in other Semitic languages, for example in spoken Hebrew (Ur Shlonsky, p.c.). Also, note that in the Gerund paradigm, C2 has disappeared: instead of *gǝbǝr-ǝ* there is *gǝir-ǝ*. This seems to happen when C2 is the consonant [b] (also with the verb *seberǝ* 'break'). Because phonology is outside the scope of this dissertation, I will not discuss this further.

function of expressing perfectivity is accomplished by the Gerund. In fact, the Gerund is used today in Tigrinya to express past actions in affirmative declarative clauses (Bulakh 2019, Cacchioli 2023); despite its original use as a converb in subordinate clauses to express anteriority or posteriority with respect to the matrix verb (Guss-Kosicka 2019). In the literature on Amharic, the Gerund has been considered a converb and a non-finite dependent form denoting subordinating events (Butts 2019, Edzard 2019, Kramer 2014, 2023). Quite interestingly for the focus of this dissertation, this is because the Gerund cannot be used in presence of clausal prefixes. In this case the Perfective is introduced as a sort of suppletive form (more on this, in the chapters to come). The meaning of these forms (Imperfective, Perfective and Gerund, in this order) is shown in the examples below. Note that due to its rich verbal morphology, Tigrinya is a subject *pro*-drop language.²¹

(25)	ji-gebbir SM.3MS-do.IPFV 'he does'	(26)	geber-ε do.PFV-SM.3MS 'he has done'	(27)	geir-u do.GER-SM.3MS 'he did'
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The terms *Perfective*, *Imperfective* (*accompli* and *inaccompli* in French Linguistics (Desclés & Guentchéva 2012)) and *Gerund* have been borrowed from Indo-European linguistics (Comrie 1976, Pat-El & Huehnergard 2019). One or more of these terms can, however, be considered misnomers (Nazareth 2011). Conti Rossini 1940, Leslau 1941 and Mason 1996 use “Imperfect” and “Perfect” rather than “Imperfective” and “Perfective”. Bulakh 2019 uses “New Suffixal Conjugation” rather than “Gerund”. For the sake of convention — following Leslau 1941 and Girma 2003 for Tigrinya and Kramer 2014 for Amharic — in this thesis I adopt the terms Imperfective, Perfective and Gerund.

The three verbal forms mentioned above are not the only ones found in Tigrinya. Semitic languages are known not to possess an infinitival form; however, in Tigrinya a (nominal) infinitive is obtained through the affixation of the prefix *m(i)-* to a CCaC pattern, such as *mibilaʕe* ‘to eat, eating’ for the root */blʕ/* (Conti Rossini 1940, Leslau 1941, Mason 1996, Bulakh 2019). There are also two participial forms: the present participle follows a CeCaCi pattern, such as in *negari* ‘that tells, teller’, and the past participle fol-

²¹Unlike Tigrinya, Amharic is said to be also object *pro*-drop (Girma 2001).

lows a CeCuC pattern, as in *sebur* ‘broken’, (Conti Rossini 1940). These participial forms no longer seem to be employed and their meanings have been taken over by relativized forms (Bulakh 2019).

The Verbal Domain - Auxiliaries

Tigrinya possesses two verbs that can be translated with the English ‘to be’: *?allo* — from the root /hlw/ — and *?ijju*.²² With regard to the form of these copulas, Table 1.2 illustrates their paradigms in present tense. Note that the former follows a Perfective conjugation, whereas the latter follows a Gerund conjugation.²³

	?allo	?ijju
SM.1S	?all-εxu	?ijj-ε
SM.2MS	?all-εxa	?ix-a
SM.2FS	?all-εxi	?ix-i
SM.3MS	?all-o	?ijj-u
SM.3FS	?all-a	?ijj-a
SM.1P	?all-εna	?i-na
SM.2MP	?all-εxum	?ix-um
SM.2FP	?all-εxin	?ix-in
SM.3MP	?all-εwu	?ijj-om
SM.3FP	?all-εwa	?ijj-in

Table 1.2: The paradigms of *?allo* and *?ijju*.

With regard to the meaning of these verbs, *?allo* has three functions. First, it serves as the copula in the context of stage-level predicates (terminology from Carlson 1977) and locatives, as shown in (28). However, note that with nominal events, such as *the meeting*, *?ijju* is used instead, as illustrated in (29).²⁴

²²I am not able to identify the root of *?ijju*, nor the phonological processes involved in its conjugation in Table 1.2 (note the difference in the occurrence of the glide [j] in all persons except the second, in which the velar fricative [x] appears).

²³As is traditional in the literature on Semitic languages, I use the 3MS form to label these auxiliaries. I also gloss *?allo* as BE₁ and *?ijju* as BE₂ to distinguish them.

²⁴I thank Richard Hughye for pointing out to me that this is true also for Spanish. Note that if an individual were used, *?allo* would be used, as in (i).

- (i) Tεsfai ?ab kifili ?arbaʕεε ?all-o /*?ijj-u
 Tεsfay LOC room four BE₁.PRES-SM.3MS BE₂.PRES-SM.3MS
 ‘Tεsfay is in room 4.’

A more in-depth analysis of copulas is left to future research.

- (28) a. ʔit-a dmu dɛxɛmɛ-t **ʔall-a**
DIST-FS cat tired-FS BE₁.PRES-SM.3FS
‘The cat is tired.’
- b. ʔit-om mɛts’hafti $\text{ʔab bet-mɛts’hafti}$ **ʔall-ɛwu**
DIST-MP book.MP LOC library BE₁.PRES-SM.3MP
‘The books are in the library.’
- (29) a. ʔit-i axeba sɛʔat ʔarbaʔɛtɛ **ʔijj-u**
DIST-MS meeting.MS hour.FS four BE₂.PRES-SM.3FS
/ $\{\text{*ʔall-o}\}$
BE₁.PRES-SM.3MS
‘The meeting is at 4 o’clock.’
- b. ʔit-i axeba ʔab kifili ʔarbaʔɛtɛ **ʔijj-u**
DIST-MS meeting.MS LOC room four BE₂.PRES-SM.3MS
/ $\{\text{*ʔall-o}\}$
BE₁.PRES-SM.3MS
‘The meeting is in room 4.’

Second, *ʔallo* functions as the auxiliary that marks progressive aspect (Conti Rossini 1940, Leslau 1941, Mason 1996, Kogan 1997), as in (30).²⁵

- (30) ʔindʒera ni-bɛliʃ **ʔall-ɛna**
injera SM.1P-eat.IPFV BE₁.PRES-SM.1P
‘We are eating *injera*.’

Third, when an object marker is attached to *ʔallo*, it also functions as a possessive copula (the lexical verb ‘to have’ in English), as shown in (31).

- (31) han-ti habti **ʔall-atɪ-ni**
one-FS sister.F BE₁.PRES-SM.3FS-OM.1S
‘I have one sister.’ (Lit.: ‘One sister is to me.’)²⁶

To sum up, the copula *ʔallo* performs the following three functions.

Stage-level Copula	Progressive Aspect	Possession
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Table 1.3: The functions of *ʔallo*.

²⁵This is in accordance with what has been noted by Ferguson 1969, namely that in an SOV language where the Existential verb and the Copula are lexically separate, the Existential also tends to function as an auxiliary to form the present continuous.

²⁶I suggest that the *ti* in *ʔallatini* may be an epenthetic syllable.

The verb *ʔijju* also has three functions. First, it is used as a copula in the context of individual-level predicates, as in (32).

- (32) mari ʔakim-a ʔijj-a
 Mary doctor-FS BE₂.PRES-SM.3FS
 ‘Mary is a doctor.’

Second, it is used in a sentence with a verb in the Imperfective form prefixed by *ki-* (the prefix investigated in Chapter 4) to express future tense (Kogan 1997, Bulakh 2019, Cacchioli 2024).

- (33) Tesfay gaʔat k-i-beliʃ ʔijj-u
 Tesfay gaʔat SBJ-SM.3MS-eat.IPFV BE₂.PRES-SM.3MS
 ‘Tsfay will eat *gaʔat*.’

Third, I have argued that *ʔijju* is also a marker of Verum Focus: it adds assertiveness and emphasis to the truth of a sentence — similarly to emphatic-*do* in English (Cacchioli 2024) — as in (34). In addition, it can also mark Narrow Focus (Contrastive and Exhaustive) by being right-adjacent to a focalized constituent (Zellou 2010), as in (35).

- (34) a. diṃu riεj-ε
 cat.MS see.GER-SM.1S
 ‘I saw a cat.’
 b. diṃu riεj-ε ʔijj-ε
 cat.MS see.GER-SM.1S VERUM-SM.1S
 ‘I **did** see a cat.’
- (35) diṃu ʔijj-ε riεj-ε
 cat.MS FOCUS-SM.1S see.GER-SM.1S
 ‘I saw A CAT (not a dog).’

To sum up, the copulas *ʔallo* and *ʔijju* perform the following functions.²⁷

²⁷Amharic also possesses two copulas with separate functions: the predicational copula *nāw* and the existential copula *ʔallä* (Wodem 2014). Spanish also distinguishes two verbs ‘to be’ that differ in the types of predicates they can occur with: the verb *ser* ‘to be’ is used as a copula with individual-level predicates (as *ʔijju*), and the verb *estar* ‘to be’ is used with stage-level predicates and marks progressive aspect (as *ʔallo*) (González-Vilbazo & Remberger 2005, Roy 2013).

<i>?allo</i>	<i>?ijju</i>
Stage-level Copula	Individual-level Copula
Progressive Aspect	Future Tense
Possession	(Verum) Focus

Table 1.4: The functions of *?allo* and *?ijju*.

So far, I have shown the forms of *?allo* and *?ijju* in the present tense. These two verbs share a common Perfective form, *nɛbɛɛ*, a common Gerund form, *nɛjiru* (sometimes *neru*), and common form used in both future constructions and in the presence of negation, *kone* (the lexical verb *become*).²⁸

Now, after this short overview of the functions of *?allo* and *?ijju*, let us return to the second function of *?allo*, namely marking progressive aspect. In (36), I repeat an example presented earlier.

- (36) *?indɕera ni-bɛliʃ* ***?all-ɛna***
 injera SM.1P-eat.IPFV BE₁.PRES-SM.1P
 ‘We are eating *injera*.’

Note that the auxiliary *?allɛna* agrees with the silent grammatical subject of the sentence *niɪhna* ‘we’, just as the verb *niɪbɛliʃ* does. A mismatch in subject-agreement between the verb and the auxiliary, as well as default 3MS subject-agreement, is ungrammatical, as shown in (37).

- (37) a. *hanti sɛbɛjiti may tɪ-seti* ***?all-a***
 one.FS woman water SM.3FS-drink.IPFV BE₁.PRES-SM.3FS
 ‘A woman is drinking water.’
 b. **hanti sɛbɛjiti may tɪ-seti* ***?all-o***
 one.FS woman water SM.3FS-drink.IPFV BE₁.PRES-SM.3MS
 c. **hanti sɛbɛjiti may jɪ-seti* ***?all-o***
 one.FS woman water SM.3MS-drink.IPFV BE₁.PRES-SM.3MS

²⁸I find the fact that they share a common past and Gerund form extremely interesting; however, I am not investigating this further in this work. Note nevertheless that [Ferguson 1969](#) claims that, typologically, if in a language the existential verb (in our case *?allo*) and the copula (in our case both *?allo* and *?ijju*) are lexically separate in the present tense, they share a single past tense. Ferguson also claims that if a language possesses a defective verb *to be* a verb *to become* often serves as its suppletive form. This is true, for example, in Hungarian, in which the verb *lesz* ‘become’ supplies the infinitive, the gerund, the past conditional and the imperative of *van* ‘to be’ ([Ferguson 1969:112](#)).

Example (37) shows that a verb in the Imperfective together with *?allo* expresses present progressive. If, however, *?allo* occurs in its past form, namely *nejiru*, with a verb in the Imperfective, it expresses past progressive (Conti Rossini 1940, Leslau 1941, Mason 1996, Kogan 1997), as in (38).

- (38) ?indɕera ni-beliʃ **nejir-na**
 injera SM.1P-eat.IPFV BE₁.GER-SM.1P
 ‘We were eating *injera*.’

If *?allo* occurs in its past form *nejiru* with a verb in Gerund, it expresses past perfect, as in (39).

- (39) ɤʔidmi ?indɕena b-ʔit-a dmu mi-sbar-u n-it-i
 before again by-DIST-FS cat NOM-break-SM.3MS DOM-DIST-MS
 mɛtsʔ ʔawɛri ʔaʃɛr-ɛ-jo **nejir-ɛ**
 toy repair.GER-SM.1S-OM.3MS BE₁.GER-SM.1S
 ‘I had repaired the toy before the cat broke it again.’
 (Lit. ‘Before again by the cat broken the toy repaired I have’)
 (from Cacchioli 2024:30,(22b))

Finally, if *?allo* occurs in its present form with a verb in Gerund, it expresses a “resultative state” (Shlomo 1980:244), in which the result of a past action in the present is said to be achieved (Leslau 1941, Kogan 1997). This periphrastic tense form is probably somewhat similar to the Present Perfect in English, hence the translation provided for it in (40b).

- (40) a. kab ɣɛza wɛtsʔi-a
 from house leave.GER-SM.3FS
 ‘She left the house.’
 b. kab ɣɛza wɛtsʔi-a **?all-a**
 from house leave.GER-SM.3FS BE₁.PRES-SM.3FS
 ‘She has left the house.’

My consultants explained that in (40b) there is emphasis on the fact that the person was able to leave the house or that whatever happened before led to the *result* of the person leaving the house.

The Verbal Domain - Object Marking

Let me now digress for a moment on the object-marking morphology that can appear on verbal forms in Tigrinya. Note that subject-marking morphology on verbs (either prefixes or suffixes) occur across Semitic languages; however, object-marking morphology is more specific to the Ethio-semitic branch. Table 1.5 shows object-marking morphology for all persons.

Person	OM
1S	-ni
2MS	-ka
2FS	-ki
3MS	-o
3FS	-a
1P	-na
2MP	-kum
2FP	-kin
3MP	-om
3FP	-in

Table 1.5: Object-marking morphology in Tigrinya according to persons.

In a previous section, I mentioned that objects are differentially marked by the prefix $n(i)-$. When an object is marked by $n(i)-$, it is obligatorily cross-referenced by a ϕ -agreeing suffixal object marker on the verb (Kievit & Kievit 2009, Overfelt 2022). These markers always follow subject agreement if it surfaces, as shown in (41).

- (41)
- a. Tesfaj seɓʔaj qɛtil-u
 Tesfay man kill.GER-SM.3MS
 ‘Tefay killed a man.’
 - b. Tesfaj **n-it-i** seɓʔaj qɛtil-w-**o**
 Tesfay DOM-DIST-MS man kill.GER-SM.3MS-OM.3MS
 ‘Tefay killed the man.’
 - c. *Tesfaj seɓʔaj qɛtil-w-**o**
 Tesfay man kill.GER-SM.3MS-OM.3MS
 - d. *Tesfaj **n-it-i** seɓʔaj qɛtil-u
 Tesfay DOM-DIST-MS man kill.GER-SM.3MS

It is important to point out that object markers in Tigrinya are not only “direct objects” (or Accusative markers) in English. They can also be used as “indirect objects” (or Dative

markers). This is shown in (42) and (43).

- (42) tufaḥ hib-u-ni
 apple give.GER-SM.3MS-OM.1S
 ‘He gave me an apple.’
- (43) ?it-a g^wal ‘sɛlam’ ?il-at-na
 DIST-FS girl hello say.GER-SM.3FS-OM.1P
 ‘The girl told us ‘hello’.’

The term ‘object marker’ is used as a neutral term to avoid taking a stand between claiming that it is the morphological result of agreement or clitic doubling. To my knowledge, there are only two works that have tried to establish which option is correct in Ethio-semitic languages: [Kramer 2014](#) argues that in Amharic object marking is a (clitic) pronoun, whereas [Gebregziabher 2021](#) arrives at the opposite conclusion, namely that in Tigrinya object marking is agreement. To reach their verdict, Kramer and Gebregziabher ran several diagnostics following the list of properties that separate pronouns from agreement proposed in [Zwicky & Pullum 1983](#) and in [Preminger 2009](#). I summarize the diagnostics in Table 1.6, and recommend that the reader consult the papers for further details. Where Amharic appears in the table, the diagnostics have been run for it in [Kramer 2014](#) if the language is followed by ^K; where Tigrinya appears in the table, the diagnostics have been run for it in [Gebregziabher 2021](#) if the language is followed by ^G, and by myself if the language is followed by ^C.

DIAGNOSTICS	OBJECT MARKING	LANGUAGE
Free Selection	Agreement (have to attach) Clitic Pronoun (can stand alone)	Tigrinya^G
Semantic Sensitivity	Agreement (does not affect meaning) Clitic Pronoun (affects meaning)	Tigrinya^G , Amharic ^K Tigrinya^G , Amharic ^K
Optionality	Agreement (not optional) Clitic Pronoun (optional)	Tigrinya^G Tigrinya^C , Amharic ^K
Tense Variability	Agreement (conditioned) Clitic Pronoun (not conditioned)	Tigrinya^G , Amharic ^K
Multiple OM per host	Agreement (no) Clitic Pronoun (yes)	Tigrinya^G , Amharic ^K
Dislocation (on Aux)	Agreement (no) Clitic Pronoun (yes)	Tigrinya^G , Amharic ^K
Referential Binding	Agreement (allowed) Clitic Pronoun (not allowed)	Tigrinya^G
Similarity with D	Agreement (no) Clitic Pronoun (yes)	Tigrinya^G Amharic ^K
Case marking alternations	Agreement (no) Clitic Pronoun (yes)	Tigrinya^G
Passive/Reflexive/Unaccusative	Agreement (not attested) Clitic Pronoun (attested)	Amharic ^K
Cooccurrence of D & OM in RCs	Agreement (not attested) Clitic Pronoun (attested)	Tigrinya^G Amharic ^K
Default OM	Agreement (grammatical) Clitic Pronoun (ungrammatical)	Tigrinya^C Amharic ^K
Tripling effects	Agreement (grammatical) Clitic Pronoun (ungrammatical)	Tigrinya^C

Table 1.6: Diagnostics for object marking morphology as clitic pronouns or agreement.

From the table above, one can see that object-marking morphology appears to be the result of Agreement in Tigrinya and of the cliticization of a pronoun in Amharic. A couple of comments are nevertheless in order here. First, according to [Anagnostopoulou 2017](#), the Tense Variability and the Referential Binding diagnostics are not reliable. Second, I disagree with [Gebregziabher 2021](#) regarding the Optionality and the Default OM diagnostics: object-marking morphology can be optional in some cases and sometimes appear as 3MS default agreement. Without going into too much detail, consider (44) which contains the perception verb /*rʔj*/ ‘see’ (that will be discussed in Chapters 3 and 4) and (45) which contains the factive verb /*rsʔ*/ ‘forget’ (that will be discussed in Chapter 3).

- (44) a. *?it-a sebei ti kemzi-xede-t re?i-na*
DIST-FS woman COMP-leave-PFV-SM.3FS see.GER-SM.1P
‘We saw that the woman left.’
- b. *?it-a sebei ti kemzi-xede-t re?i-na-ja*
DIST-FS woman COMP-leave-PFV-SM.3FS see.GER-SM.1P-OM.3FS
‘We saw that the woman left.’
- c. **?it-a sebei ti kemzi-xede-t re?i-na-jo*
DIST-FS woman COMP-leave-PFV-SM.3FS see.GER-SM.1P-OM.3MS
‘We saw that the woman left.’
- (45) a. **?it-a sebei ti kemzi-xede-t resifi-na*
DIST-FS woman COMP-leave-PFV-SM.3FS forget.GER-SM.1P
‘We forgot that the woman left.’
- b. *?it-a sebei ti kemzi-xede-t resifi-na-ja*
DIST-FS woman COMP-leave-PFV-SM.3FS forget.GER-SM.1P-OM.3FS
‘We forgot that the woman left.’
- c. *%?it-a sebei ti kemzi-xede-t resifi-na-jo*
DIST-FS woman COMP-leave-PFV-SM.3FS forget.GER-SM.1P-OM.3MS
‘We forgot that the woman left.’
- (from Overfelt & Cacchioli submitted)

In examples (44a) and (45a), there is no object morphology on the matrix verb. In (44b) and (45b), the grammatical subject of the embedded clause *?ita sebei ti* ‘the woman’ is cross-referenced on the verb of the matrix clause with 3FS object morphology. In (44c) and (45c), the matrix verb carries default 3MS object morphology. The grammaticality, partial acceptability, and ungrammaticality of the examples presented above will be addressed in future research.

Third, in Overfelt & Cacchioli 2024 it was suggested that object-marking morphology can give rise to tripling effects, such as those in (46).

- (46) *?it-om səb-at ?it-a farat ki-Ø-diqqis-u-la*
DIST-MP person-PL DIST-FS bed SBJ-SM.3MP-sleep.IPFV-SM.3MP-OBL.3FS
ji-gibba?-a
SM.3MS-need.IPFV-OM.3FS
‘Those people need to sleep in the bed.’
- (adapted from Overfelt & Cacchioli 2024)

In the example above, the nominal *?ita farat* ‘the bed’ is cross-referenced with object morphology on the embedded verb and the matrix verb (assuming that the construction is biclausal, as I show in Chapter 4). The example above shows that the possibility of object markers to be clitics is precluded by the fact that the nominal would require two separate clitics, from different morphological paradigms to be able to “triple” a nominal.

I now turn to the syntax of declarative clauses in Tigrinya.

1.6 Root Clauses

Before turning to more complex sentences that involve the investigated clausal prefixes and thus embedded clauses, it is important to provide a syntactic analysis of simple root clauses in Tigrinya. I first consider declarative root clauses with simple verb forms (in which the verb appears alone) and then I turn to declarative root clauses with periphrastic tense forms (in which the verb occurs along with an auxiliary).

1.6.1 Root Clauses with Simple Verb Forms

Let us start with simple verb forms. Remember from Section 1.5.2 that Tigrinya is an SOV language, that habitual present tense is expressed with a simple verb form conjugated in the Imperfective, and that past tense is expressed with a simple verb form conjugated in the Gerund. Consider the following two examples.

- (47) *Təsfai kullu giziə ?indǵera ji-bəliʃ*
 Təsfay all time injera SM.3MS-eat.IPFV
 ‘Təsfay always eats *injera*.’

- (48) *Təsfaj timali ?indǵera bəliʃ-u*
 Təsfay yesterday injera eat.GER-SM.3MS
 ‘Təsfay ate *injera* yesterday.’

How can we derive the correct word order in SOV sentences such as the ones above? There is a very limited number of previous accounts that have proposed a detailed syntax of Tigrinya simple declarative root clauses. To my knowledge, there is in fact only one

author who has proposed a detailed analysis: [Girma 2003](#). Girma takes an LCA and anti-symmetric approach to head-finality ([Kayne 1994](#)) and presents a more semantic analysis of the Perfective, Imperfective and Gerund in Ethio-semitic (not specifically Tigrinya) following Stowell’s Theory of Tense ([Stowell 1996](#)). As already explained in Section 1.3, unlike Girma, in this dissertation I assume a cartographic approach to syntax that follows the head-parameter. A similar approach is taken in [Tesfay 2016](#); however, that work does not provide a fully explicit syntax of simple clause structure in Tigrinya but offers sufficient theoretical discussion to allow it to be inferred.²⁹

In this dissertation, I aim to adopt a more straightforward approach. More importantly, I seek to determine whether verbs raise in Tigrinya. This question is particularly challenging in head-final languages because the surface word order (SOV) is uninformative. Consequently, diagnostics such as the insertion of adverbs — commonly used to detect verb movement in languages like French — cannot be applied, since the presence or absence of verb raising in an SOV language does not affect the surface order. Moreover, even if the verb were to raise to T° , this would not produce any visible effect in the linear order, as both the base position (V°) and the landing site (T°) occur at the end of the sentence.

I propose that Imperfective and Perfective verbs raise to T° in Tigrinya. To support this claim, I draw on indirect evidence from relative clauses, the details of which will be provided in Chapter 2. This phenomenon will also be revisited in the same chapter. To obtain a relative clause that expresses past tense, Tigrinya uses two strategies: either a Perfective (it will be shown later on that this form is used only as suppletive form of the Gerund when clausal prefixes are attached to it) is inserted to carry the clausal prefix $z(i)-$, as in (49), or a “dummy” auxiliary is inserted to accompany the Gerund and carry the $z(i)-$ prefix (the Gerund remains bare), as in (50). This is reminiscent of auxiliary insertion observed in languages such as English (*do*-support): in a language in which verbal forms do not raise and thus cannot provide a host for verbal morphology — as in English — a pleonastic auxiliary is inserted as a repair strategy ([Chomsky 1957](#)).

²⁹I could also mention [Overfelt 2009](#), which is more focused on relative clauses, and [Spadine 2020](#), which is more focused on embedded clause structure.

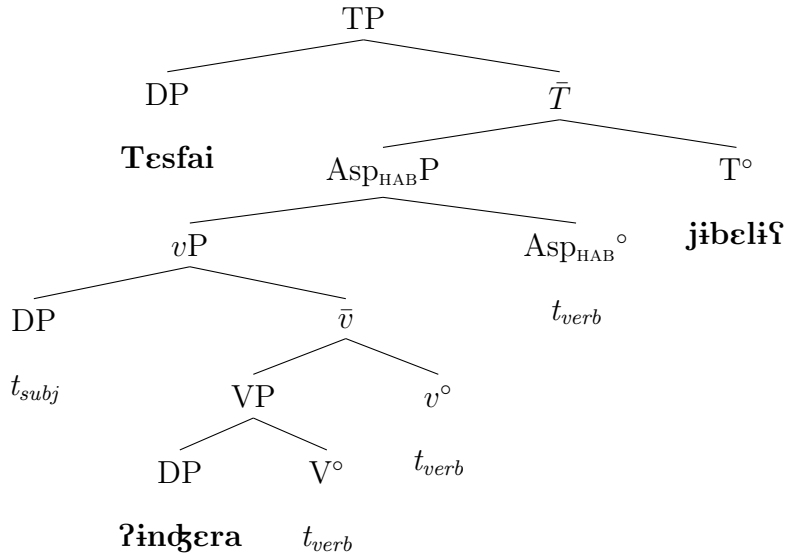
- (49) ?it-a ?it-i sɛbʔaj **zi**-ts'ihaf-i-wa dɛbdabɛ niwaḥ
DIST-FS DIST-MS man REL-write.PFV-SM.3MS-OM.3FS letter.FS long.FS
ʔijj-a
BE₂.PRES-SM.3FS
'The letter that the man wrote is long.'
- (50) ?it-a ?it-i sɛbʔaj ts'ihif-u-wa **z**-ɛ-llo
DIST-FS DIST-MS man write.GER-SM.3MS-OM.3FS REL-BE₁.PRES-SM.3MS
dɛbdabɛ niwaḥ ʔijj-a
letter.FS long.FS BE₂.PRES-SM.3FS
'The letter that the man wrote is long.'

The phenomenon exemplified in (49) and (50) has been noted and used as evidence for verb raising in Tigrinya by [Overfelt 2009](#).³⁰ Overfelt argues that Perfective (and Imperfective) verbs raise in Tigrinya and land in T°. From this position, they undergo a further I-to-C movement, which allows them to be prefixed with *z(i)-*, assumed to be base-generated in C°. In contrast, because the Gerund does not move, an auxiliary must be inserted in the derivation. Although, as will be explained in Chapter 2, I do not claim that *z(i)-* is base-generated in C°, I propose that a similar syntactic mechanism is at work: more specifically, the examples in (49) and (50) provide indirect evidence that Imperfective and Perfective verbs raise to T°, whereas the Gerund does not.

With all this in mind, I propose that the sentence in (47) (for the sake of simplicity, without the adverbial phrase) — which presents a verbal form in the Imperfective — can be syntactically represented as in (51).

³⁰The distribution of relative morphology is also used in [Baker & Kramer 2014](#) as evidence for verb raising in Amharic.

(51)



I follow [Chomsky 1995](#) in merging the subject in Spec,*v*P and the object as the complement of the verb. The verb moves to *v*° before reaching the aspectual layer, because the phase *v*P would prohibit its extraction ([Chomsky 2004, 2008](#)) if it did not move under its head (more on phases later, when analyzing *z(i)-* in Chapter 2). I suggest that before landing in T°, the verb moves through Asp_{HAB}°, where it gets its templatic form of an Imperfective, also proposed in [van Urk 2025](#) that suggest that vocalic melodies are the realization of AspP)³¹. The tree in (51) presents only one aspectual projection following a more Minimalist approach to syntax ([Chomsky 1995](#)), but I label it HAB (Habitual), thus hinting at a more Cartographic view of aspectual projections ([Cinque 1999](#)). Remember that the Imperfective in Tigrinya expresses habitual present tense when occurring without an auxiliary. I claim that the Imperfective verb form moves under the T° head, despite some alternative views, especially in the literature on Arabic, that claim that the Semitic Imperfective behaves more like a (Romance) participle and thus do not carry tense ([Shlonsky 1997, Benmamoun 2000, Hallman 2015, Alqassas 2019](#) a.o., contra [Ouhalla 1994](#)).

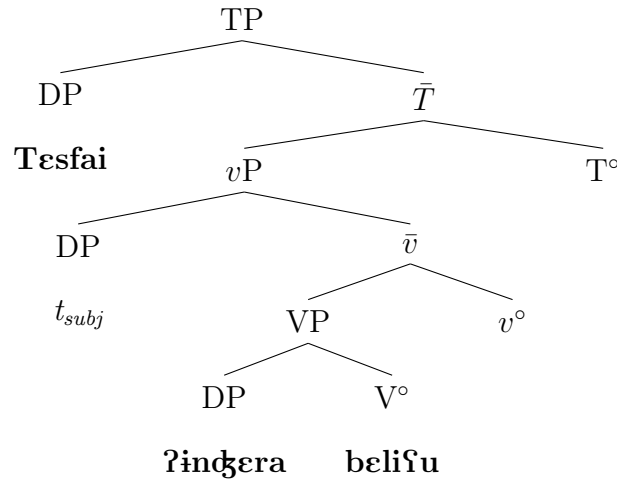
Before providing a derivation for a sentence containing a Gerund verb, it is necessary to briefly outline the formation of Semitic verbs. The tree shown above is in fact simpli-

³¹Note that both [Overfelt 2009](#) and [Spadine 2020](#) omit an aspectual layer in the structure of Tigrinya clauses.

fied. In most analyses - often framed within Distributed Morphology (Halle & Marantz 1993) - verbs are not base-generated as fully formed items. Instead, a consonantal root (typically represented as $\sqrt{CCC}$, e.g. /ktb/ ‘writing’) is inserted (Benmamoun 2000). The template (or *binyan* in Hebrew literature) is realized in v° , and the vocalic melody (the vowels transfixed into the consonantal root) is hosted by Voice $^\circ$ (Arad 2005, Tucker 2011). Head movement of the root through the structure derives the appropriate verb form, imperfective or perfective (cf. Godfrey 2011). For expository clarity, these derivational steps are not shown in the trees presented in this dissertation.

Now, let us turn to the sentence in (48) — again, without the adverbial phrase — that presents a Gerund verb form. Consider (52).

(52)



As mentioned, in contrast to a verb in the Imperfective, I propose that a verb in Gerund does not raise in the structure but remains in its first-merge position.³² I argue this on the basis of the indirect evidence from relative clauses discussed, but also on the basis of the original use of the Gerund as a converb, as explained in Section 1.5.2.³³ Moreover, I suggest that the Gerund occurs in a “bare” VP with no associated AspP (hence the lack of an aspectual layer in the structure above). If the Gerund does not move under the T°

³²One could argue that the Gerund might move to some extent, because otherwise it would be difficult to explain how it obtains its agreement morphology. However, the important point I want to emphasize here is that the Gerund does not move high enough to reach the aspectual layer.

³³The Gerund is considered a non-finite predicate in Amharic because it cannot occur independently as a predicate in a main clause (Kramer 2014) (in contrast to Tigrinya).

head, how can the sentence convey tense? I propose that the Gerund behaves similarly to English tensed verbs: they are able to express past without moving to TP (Emonds 1978, Pollock 1989). However, I do not assume that tense is phonologically null; rather, I argue that it is conveyed by the form of the Gerund itself — namely its vowel templates.

With regard to a verb in Perfective, I claim that it has the same structure of the Imperfective, with the only difference being the properties (and thus label) of the aspectual projection, which I suggest could be $\text{Asp}_{\text{PFV}P}$.

Let us now turn to periphrastic tense forms.

1.6.2 Root Clauses with Periphrastic Tense Forms

Things now become more complex. What happens when the verb is accompanied by an auxiliary in terms of the syntax of these constructions? Are these periphrastic tense forms underlyingly monoclausal (involving one CP and one TP) or biclausal (involving two CPs or two TPs (Wurmbrand 2004))? These questions have already been asked for other Semitic languages (a.o. Fassi Fehri 2004, Ouali & Fortin 2007, Al-Aqarbeh & Al-Sarayreh 2017).

Remember from section 1.5.2 that in Tigrinya there are four different periphrastic tense forms that express different aspects/tenses. Progressive aspect is expressed with a periphrastic tense form in which the verb is conjugated in the Imperfective and the auxiliary is *?allo* for present progressive and *nejiṛu* for past progressive. Perfect aspect is expressed with a periphrastic tense form in which the verb is conjugated in the Gerund and the auxiliary is *?allo* for present perfect (or the “resultative state” mentioned above) and *nejiṛu* for past perfect. These four different aspects/tenses are illustrated in (53) and (54) below.

- (53) a. Tesfai ?inǧera ji-beliṯ ?all-o
 Tesfay injera SM.3MS-eat.IPFV BE₁.PRES-SM.3MS
 ‘Tesfay is eating *injera*.’
- b. Tesfai ?inǧera ji-beliṯ nejiṛ-u
 Tesfay injera SM.3MS-eat.IPFV BE₁.GER-SM.3MS
 ‘Tesfay was eating *injera*.’

- (54) a. Segen wets'i-a ʔall-a
 Segen leave.GER-SM.3FS BE₁.PRES-SM.3FS
 'Segen has left.'
- b. Segen wets'i-a neʒir-a
 Segen leave.GER-SM.3FS BE₁.GER-SM.3FS
 'Segen had left.'

Now, as was asked for root clauses with simple verb forms, how can we derive the correct word order in SOVAux sentences such as those in (53a) and (54a)?³⁴ With respect to Arabic varieties, previous accounts propose different analyses. For example, Fassi Fehri 2004 suggests that Standard Arabic periphrastic tense forms involve two TPs, as Ouali & Fortin 2007 claim for Moroccan Arabic (however, *vP* is only present in the embedded clause containing the verb). In contrast, Al-Aqarbeh & Al-Sarayreh 2017 claim that periphrastic tense forms in Standard Arabic are monoclausal. With regard to Tigrinya, again, the previous accounts on this are extremely limited³⁵ and I mention only those of which I am aware: Girma 2003 and Tesfay 2016.

As already mentioned in the previous section, Girma 2003 takes an LCA approach to head-finality; in contrast, Tesfay 2016 takes a head-parameter approach. Girma 2003 distinguishes between two types of periphrastic tense forms in Ethio-Semitic: the first type consists of constructions in which the verb occurs along with an auxiliary which is suffixed to it, such as the future in Amharic or the simple present and future in Zay (Girma 2003:45-46), as shown in (55).

- (55) a. yi-mät't'-all
 3MS-come_{impf}-Aux_{pres}
 'he will come' *Amharic* (adapted from Girma 2003:45, (20a))
- b. yi-säbr-äl
 3MS-come_{impf}-Aux_{pres}
 'he breaks/will break' *Zay* (from Leslau 1999:83)

The second type consists of constructions in which the verb occurs along with an auxiliary

³⁴The syntax of the past counterparts of these sentences in (53b) and (54b) and more importantly the issue of the suppletive form of the past auxiliary *neʒiru* will be left to future research.

³⁵For example, Overfelt 2009 and Spadine 2020 in which the syntax of clauses in Tigrinya is analyzed, do not consider periphrastic verb forms.

that is not suffixed to it but appears as a free morpheme, such as the past progressive in Harari.

- (56) ti-käbt-i nār-ti
 3FS_S-write_{impf}-3FS_S Aux_{past}-3FS_S
 ‘she was writing’ *Harari* (from [Leslau 1958:21](#))

According to Girma, the first type of construction is monoclausal: the verb is in V° and the auxiliary is in T° ([Girma 2003:46](#)). To obtain the correct word order from a head-initial configuration, AspP containing the verb moves to Spec,TP to check an EPP feature ([Chomsky 1993](#)). In contrast, for the second type of construction Girma argues that it is biclausal: like verbs, auxiliaries head verbal projections ([Girma 2003:147](#)).

[Tesfay 2016](#) suggests that Tigrinya periphrastic tense forms are monoclausal: the verb remains in its first-merge position (in *v*°), and the auxiliary, as in Girma’s approach, is base-generated under the T° head ([Tesfay 2016:200](#), tree in (14) and 224, (31)).³⁶

It seems to me that these works do not provide enough arguments to support their choice between a monoclausal and biclausal syntax for periphrastic tense forms. I try to remedy this lacuna here. One could argue that periphrastic tense forms in Tigrinya are monoclausal because adjuncts like prepositional and adverbial phrases cannot appear between the verb and the auxiliary, as shown in (57) and (58).

- (57) a. **mis ʕark-ei** ji-beliʕ ʔall-ɛxu
 with friend-POSS.1S SM.1S-eat.IPFV BE₁.PRES-SM.1S
 ‘I am eating with a friend.’
 b. *ji-beliʕ **mis ʕark-ei** ʔall-ɛxu
 SM.1S-eat.IPFV with friend-POSS.1S BE₁.PRES-SM.1S
 ‘I am eating with a friend.’
- (58) a. **qelt’ifu** ji-beliʕ ʔall-ɛxu
 quickly SM.1S-eat.IPFV BE₁.PRES-SM.1S
 ‘I am eating quickly.’
 b. *ji-beliʕ **qelt’ifu** ʔall-ɛxu
 SM.1S-eat.IPFV quickly BE₁.PRES-SM.1S
 ‘I am eating quickly.’

³⁶[Kramer 2023](#) also claims that auxiliaries are in T° in Amharic.

However, it is not the case that nothing can intervene between the verb and the auxiliary. In fact, as will be shown in the following chapters, the clausal prefixes investigated in this dissertation, $z(i)-$, $k\epsilon mz(i)-$, $k(i)-$ and the prefixal part of the negative circumfix $?aj-...-(i)n$ can appear between the verb and the auxiliary. To my knowledge, beside clausal prefixes, the only element that can appear between the verb and the auxiliary is the interrogative and focus-sensitive particle *do*, as shown in (59).

- (59) a. $ti-s\epsilon mi\text{ʕ}a-ni$ $?all\epsilon-ka$
 SM.2MS-hear.IPFV-OM.1S BE₁.PRES-SM.2MS
 ‘You are hearing me.’
- b. $ti-s\epsilon mi\text{ʕ}a-ni$ **do** $?all\epsilon-ka?$
 SM.2MS-hear.IPFV-OM.1S INT BE₁.PRES-SM.2MS
 ‘Are you hearing me?’

Nevertheless, I think that the adverb argument, and especially the fact that in other Ethio-semitic languages such as Amharic the auxiliary is suffixed to the verb, are strong enough arguments to claim that periphrastic tense forms in Tigrinya are monoclausal.

Before turning to the syntactic derivation of these constructions, I would like to point out that I leave the discussion of the occurrence of ϕ -features agreement on both the verb and the auxiliary to further research. In the literature this phenomenon has already been investigated for Bantu languages. In fact, [Carstens 2001](#) explains iterated subject agreement in periphrastic tense forms in terms of multiple agreement: subject morphology results from a successive-cyclic *A*-movement of the subject through the specifier of every verbal heads before landing in Spec,TP. This approach has also been endorsed in [Tesfay 2016](#) for Tigrinya.

Now, for the syntactic structure of periphrastic tense forms in Tigrinya, I follow [Al-Aqarbeh & Al-Sarayreh 2017](#). I claim that these are underlyingly monoclausal constructions that — because the verb and the auxiliary are both inflected for ϕ -features — involve two VPs that project an “extended inflectional projection that is aspectual in nature” ([Al-Aqarbeh & Al-Sarayreh 2017:77](#)). Therefore, the embedding in these structures is maximally aspectual and TP and CP appear only once. This is also similar to what has

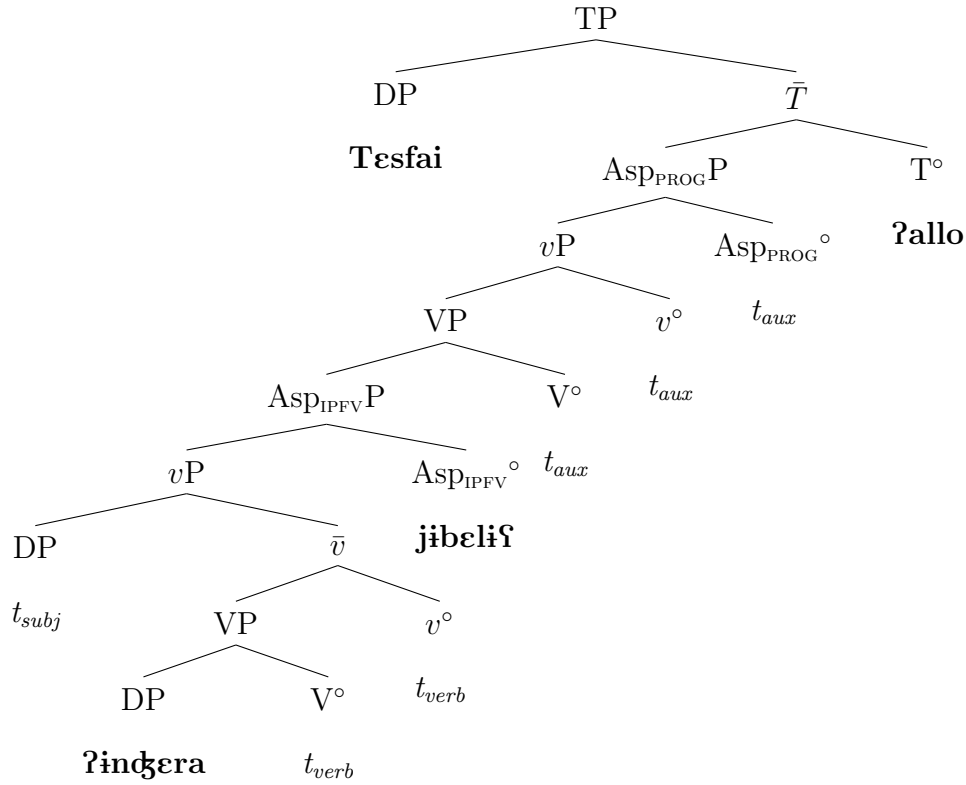
been proposed for English in [Harwood 2014](#); however, in that work there is a vP shell on top of every aspectual projection and not vice versa (cf. also [Deal 2009](#), [Bošković 2014](#)).

More concretely, for clauses in present progressive I propose again that the subject is merged in Spec,vP and the object is the complement of the verb. For the sake of simplicity I represent the subject A -moving from Spec,vP to Spec,TP , but one could imagine it landing in the specifier of each projection along the path as mentioned above according to [Carstens 2001](#). With regard to the verb, it moves to v° and to Asp° . Unlike simple verb forms, I label the aspectual projection in which the verb lands $\text{Asp}_{\text{IPFV}}P$.³⁷ With regard to the auxiliary, it is merged in a higher VP and moves to v° and Asp° before landing in T° . I label the higher aspectual projection in which the auxiliary lands $\text{Asp}_{\text{PROG}}P$, given that the auxiliary marks progressive when occurring with the verb in the Imperfective. Note that [Cinque 1999](#) suggests that $\text{Asp}_{\text{PROG}}P$ is actually lower than $\text{Asp}_{\text{HAB}}P$ (cf. also [Shlonsky & Bocci 2019](#)); however, ([Al-Aqarbeh & Al-Sarayreh 2017](#)) claim that the aspectual layer containing the auxiliary is higher than the one containing the verb.

The sentence in (53a) can thus be represented as the tree in (60).

³⁷This is to avoid having a sentence containing a verb in a habitual form and an auxiliary in a progressive form, which would not make much sense.

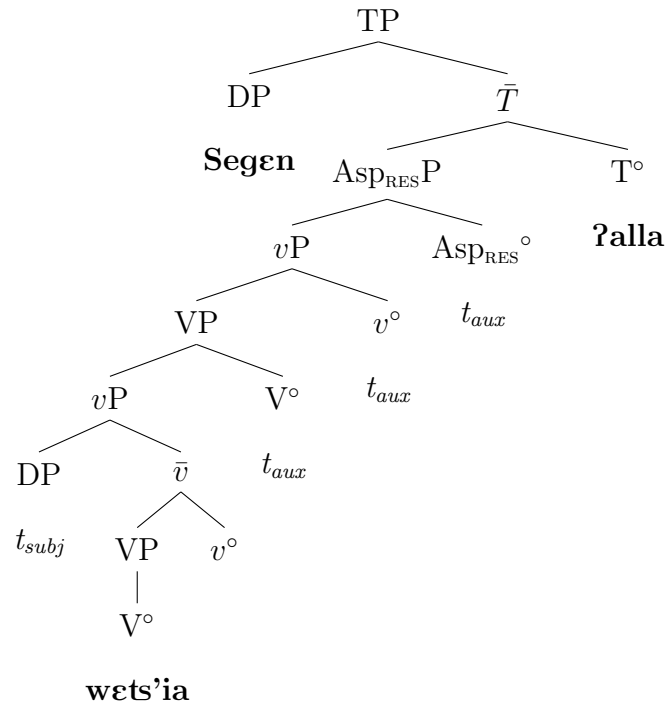
(60)



Now, let us turn to periphrastic tense forms that express the present perfect, in which a verb in the Gerund occurs along with an auxiliary. As in simple verb forms, in these constructions I propose that a verb in the Gerund does not raise in the structure, but remains in its first-merge position, in contrast to a verb in the Imperfective. As in the previous tree, the auxiliary is merged in a higher VP and moves to v° and Asp° before landing in T° . I label the higher aspectual projection in which the auxiliary lands $\text{Asp}_{\text{RES}}\text{P}$ from ‘resultative’ following Cinque 1999 given that the auxiliary marks a resultative state when occurring with the verb in the Gerund.

The sentence in (54a) can thus be represented as the tree in (61).

(61)



Now that we have the basic syntax for declarative root clauses in mind, we can turn to the investigation of subordinate clauses, starting with the clausal prefix $z(i)$ -.

Chapter 2

The Clausal Prefix *z(i)-*

2.1 Setting the Scene

The first clausal prefix investigated in this dissertation is *z(i)-* ([i] is an epenthetic vowel; the morpheme is *z*). In the Introduction in Chapter 1 I gave the example in (1) to show one of the contexts in which this prefix occurs in Tigrinya — in the following sections I guide the reader step by step through the morphosyntax of sentences such as the subject relative clause in (1), and I also present and describe other types of relative clauses.

- (1) n-it-a n-iz-a mets'ħaf **zi**-ts'əħaf-et səbejiti
 DOM-DIST-FS DOM-PROX-FS book.FS REL-write.PFV-SM.3FS woman
 rekib-ε-ya
 meet.GER-SM.1S-OM-.3FS
 'I met the woman who wrote this book.'

Zi- has been described in the literature as a relative marker (a.o. [Leslau 1941](#), [Palmer 1962](#), [Mason 1996](#), [Kogan 1997](#), [Nazareth 2011](#)). In Ethio-Semitic, relative markers behave differently across languages. For example, in Amharic there are two allomorphs of the same relative marker: the prefix *yä-* is attached to verbs in the Perfective form, the prefix *yämmə-* (or *?əmmə-*) to verbs in the Imperfective form ([Leslau 1995](#)). In Argobba, the allomorphs are *i-* and *imm-* ([Wetter 2023](#)). In the Chaha group, *jə-* is attached to verbs in the Perfective form, whereas there is no overt marking on Imperfective forms ([Menuta](#)

& Völlmin 2023). The Amharic case is shown in (2).¹

- (2) Kassa yayyaččäw ləjočc yəhedallu
 Kassa rel-he-saw-them children they-go-pres
 ‘The children whom Kassa saw are going.’ *Amharic* (from Hailu 1972:497,(1a))

Unlike Amharic, Argobba and the Chaha group, in Tigre the particle *la* functions as both the definite article and the relative marker. This is shown in (3).

- (3) la dəbäſ la ʔətbakom ʔäla
 the forest that around-them was
 ‘the forest that was around them’ *Tigre* (from Palmer 1961:24, (6))

Returning to Tigrinya, although $z(i)-$ has been labeled a relative marker, as will be shown in the following sections, this prefix does not only appear in relative clauses but also in other types of clauses, namely noun clausal complements, comparative clauses, *seem*-clauses, and superlatives. For this reason, the term “relative marker” is too reductive to describe the distribution of $z(i)-$. If $z(i)-$ is not a relative marker, then what is it?

I answer this question in this chapter. I claim that $z(i)-$ is a reflex of successive-cyclic movement that marks $\bar{A}$ -dependencies. I show that in every clause that $z(i)-$ appears, an element undergoes $\bar{A}$ -movement from its argument position to the left periphery of the clause. This is seen more clearly in clauses with periphrastic tense forms in which the verb occurs along with an auxiliary: in these clauses $z(i)-$ is obligatorily attached to both the verb and the auxiliary. This means that the element undergoing $\bar{A}$ -movement does not reach the C° domain in one step but stops along the way on the edges of two cyclic domains (the two aspectual projections).

This chapter is structured as follows. Before turning to the syntactic behavior of $z(i)-$ and its distribution, in Section 2.2.1 I discuss the word order of nominal modifiers and the position of relative clauses among them. I provide a structure for the Tigrinya DP and some tentative hypotheses to explain the word order of the elements in it. Then,

¹As already mentioned, glosses and translations of data from the literature are reproduced as faithfully as possible and are not adapted, unless specified.

I turn to $z(i)$ - and examine the constructions in which it appears: relative clauses are discussed in Section 2.2.2; noun clausal complements, comparative clauses, *seem*-clauses, and conditional clauses are discussed in Section 2.2.3. The analysis for each clause type is provided in Section 2.3.

2.2 $Z(i)$ -clauses

The first part of this section investigates the external syntax of relative clauses, specifically how these clauses interact distributionally with other nominal modifiers. The second part of this section focuses first on relative clauses with simple verb forms and then on relative clauses with periphrastic tense forms.

2.2.1 External Syntax of Relative Clauses

I begin by looking at clauses in which the *z(i)-* prefix is most famously found: relative clauses (cf. a.o. [Leslau 1941](#), [Palmer 1962](#), [Mason 1996](#), [Kogan 1997](#)). Before discussing the internal structure of relative clauses, however, I first examine their interaction and distribution within the Tigrinya DP. I then provide some hypotheses to explain the syntax of nominal clauses containing relative clauses along with demonstratives, adjectives and cardinal numerals (henceforth, ‘numerals’).

Consider (4) which shows *z(i)*- attached to a verb in the Imperfective², in a sentence containing an object relative clause.³

- (4) ʔit-om ʔanɛ z-i-nbib-om mets'ħafti ʔazenagaʔi
DIST-MP I REL-SM.1S-read.IPFV-OM.3MP book.MP amusing
ʔijj-om
BE₂.PRES-SM.3MP
'The books that I read are amusing.'

²As in Ge'ez, there is no specific verbal form for subordinate clauses; thus, the relativized verb does not formally differ from a main clause (Hasselbach 2013).

³The relativized nominal head is the subject of the relative clause in subject relative clauses and it is the object of the relative clause in object relative clauses.

The very first thing that needs to be clarified is where the boundaries of the relative clause are. This ensures not only that what we have in front of us is indeed a relative (and thus an *embedded*) clause but also helps us understand how it interacts with other nominal modifiers. Restrictive relative clauses are said to delimit the reference of the relativized head (Ungar 1986) and, at a more semantic level, are interpreted as “intersective modifiers” of the nominal head (Bianchi 1992:197).

In section 1.5.2 — example (20) — I already showed that demonstratives, numerals and adjectives in Tigrinya appear before the noun they modify in exactly this order. This is shown in (5).

- (5) ?it-ɛn kilite ts’ibwat dəbdabe-tat niwəh-ti ?ijj-ɛn
 DIST-FP two beautiful letter.F-P long-F BE₂.PRES-SM.3FP
 ‘The two beautiful letters are long.’

Let us now add a relative clause in the mix and see what happens.⁴

Dem-Num-RC-N

I first add a relative clause to a sentence such as the one in (6), containing a demonstrative and a numeral.

- (6) ?it-ɛn kilite dəbdabe-tat niwəh-ti ?ijj-ɛn
 DIST-FP two letter.F-P long-F BE₂.PRES-SM.3FP
 ‘The two letters are long.’

The relative clause (with a null subject pronoun⁵) can be placed in three different positions, as shown in (7).⁶

⁴This short study was conducted in 2018 with Ur Shlonsky and Bahareh Samimi on the basis of some data I collected for a home assignment during my Master’s. The data presented here was gathered later, since 2020.

⁵Remember that, as stated in the Introduction, Tigrinya is a subject *pro*-drop language.

⁶Note that the 3MS *ɛ* subject-marking morphology of the relativized Perfective form has phonologically merged with the 3FP *-in* object-marking morphology that cross-references the object *dəbdabetat* ‘letters’.

- (7) a. ʔit-ɛn **zi-ts'ɛħaf-in** $\text{kilitɛ dɛbdabe-tat niwəħ-ti}$
DIST-FP REL-write.PFV-SM.3MS-OM.3FP two letter.F-P long-F
 ʔijj-ɛn
BE₂.PRES-SM.3FP
‘The two letters that he wrote are long.’
- b. ʔit-ɛn kilitɛ **zi-ts'ɛħaf-in** $\text{dɛbdabe-tat niwəħ-ti}$
DIST-FP two REL-write.PFV-SM.3MS-OM.3FP letter.F-P long-F
 ʔijj-ɛn
BE₂.PRES-SM.3FP
‘The two letters that he wrote are long.’
- c. $\%ʔit-ɛn$ $\text{kilitɛ dɛbdabe-tat}$ **zi-ts'ɛħaf-in** niwəħ-ti
DIST-FP two letter.F-P REL-write.PFV-SM.3MS-OM.3FP long-F
 ʔijj-ɛn
BE₂.PRES-SM.3FP
‘The two letters that he wrote are long.’

The one order which is unacceptable is when the relative clause precedes the demonstrative, as in (8). The order RC-Dem is ungrammatical also in all the different sentences that I present from now on.

- (8) ***zi-ts'ɛħaf-in** ʔit-ɛn $\text{kilitɛ dɛbdabe-tat niwəħ-ti}$
REL-write.PFV-SM.3MS-OM.3FP DIST-FP two letter.F-P long-F
 ʔijj-ɛn
BE₂.PRES-SM.3FP

Note also that the relative clause with a phonologically realized subject, such as *ʔit-i sɛbʔai* ‘the man’ (placed right in front of the *z(i)*-marked verb) can also be placed in three different positions, as shown in (9).

- (9) a. ʔit-ɛn ʔit-i sɛbʔai **zi-ts'ɛħaf-in** $\text{kilitɛ dɛbdabe-tat}$
DIST-FP DIST-MS man REL-write.PFV-SM.3MS-OM.3FP two letter.F-P
 niwəħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP
- b. $\%ʔit-ɛn$ kilitɛ ʔit-i sɛbʔai **zi-ts'ɛħaf-in** dɛbdabe-tat
DIST-FP two DIST-MS man REL-write.PFV-SM.3MS-OM.3FP letter.F-P
 niwəħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP

- c. %ʔit-ɛn kilite dəbdabe-tat ʔit-i səbʔai **zi-ts'ɛħaf-in**
DIST-FP two letter.F-P DIST-MS man REL-write.PFV-SM.3MS-OM.3FP
niwaħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP
‘The two letters that the man wrote are long.’

In the examples above, we can see that relative clauses with an overt subject are slightly less acceptable in two positions than those with a non-pronounced subject.

Table 2.1 summarizes the possible orders seen above (excluding the order in which the relative clause precedes the demonstrative).

	null <i>pro</i> SUBJ	DP SUBJ
(a) Dem- RC -Num-N	✓	✓
(b) Dem-Num- RC -N	✓	%
(c) Dem-Num-N- RC	%	%

Table 2.1: Acceptability of the Distribution of a Relative Clause among Dem-Num-N.

The order that seems to be preferred by my consultants is the one in which the relative clause appears between the demonstrative and the numeral. The further it moves from this position, the less acceptable it becomes.⁷ In other words, the relative clause appears to be more acceptable in a prenominal position than a postnominal one. In fact, [Palmer 1962](#) also states that only *rarely* is the relative clause grammatical when following the noun. Moreover, note that the preferred order is considered to be the universal base structure of nominal phrases ([Cinque 2008:9](#)).

Dem-A-RC-N

I now add a relative clause to a sentence such as the one in (10), containing a demonstrative and an adjective.

- (10) ʔit-ɛn ts'ibuʁat dəbdabe-tat niwaħ-ti ʔijj-ɛn
DIST-FP beautiful letter.F-P long-F BE₂.PRES-SM.3FP
‘The beautiful letters are long.’

The relative clause (with a null subject pronoun) can be placed in three different po-

⁷Two consultants out of three accept the order (c) with a null *pro* subject and the orders (c) and (d) with an overt DP subject

sitions, as shown in (11). As mentioned above, the relative clause cannot precede the demonstrative.

- (11) a. ʔit-ɛn **zi-ts'ɛħaf-in** ts'ibuʁat dɛbdabe-tat niwəħ-ti
DIST-FP REL-write.PFV-SM.3MS-OM.3FP beautiful letter.F-P long-F
ʔijj-ɛn
BE₂.PRES-SM.3FP
'The beautiful letters that he wrote are long.'
- b. ʔit-ɛn ts'ibuʁat **zi-ts'ɛħaf-in** dɛbdabe-tat niwəħ-ti
DIST-FP beautiful REL-write.PFV-SM.3MS-OM.3FP letter.F-P long-F
ʔijj-ɛn
BE₂.PRES-SM.3FP
'The beautiful letters that he wrote are long.'
- c. ʔit-ɛn ts'ibuʁat dɛbdabe-tat **zi-ts'ɛħaf-in** niwəħ-ti
DIST-FP beautiful letter.F-P REL-write.PFV-SM.3MS-OM.3FP long-F
ʔijj-ɛn
BE₂.PRES-SM.3FP
'The beautiful letters that he wrote are long.'

If there are two different adjectives in the sentence, the relative clause can appear in four different positions, as in (12).

- (12) a. ʔit-ɛn **zi-ts'ɛħaf-in** ts'ibuʁat ts'aʕɛda dɛbdabe-tat
DIST-FP REL-write.PFV-SM.3MS-OM.3FP beautiful white letter.F-P
niwəħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP
- b. ʔit-ɛn ts'ibuʁat **zi-ts'ɛħaf-in** ts'aʕɛda dɛbdabe-tat
DIST-FP beautiful REL-write.PFV-SM.3MS-OM.3FP white letter.F-P
niwəħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP
- c. ʔit-ɛn ts'ibuʁat ts'aʕɛda **zi-ts'ɛħaf-in** dɛbdabe-tat
DIST-FP beautiful white REL-write.PFV-SM.3MS-OM.3FP letter.F-P
niwəħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP
- d. ʔit-ɛn ts'ibuʁat ts'aʕɛda dɛbdabe-tat **zi-ts'ɛħaf-in**
DIST-FP beautiful white letter.F-P REL-write.PFV-SM.3MS-OM.3FP
niwəħ-ti ʔijj-ɛn
long-F BE₂.PRES-SM.3FP
'The beautiful white letters that he wrote are long.'

If the adjective is heavier (a complex expression instead of a single word), the relative clause is preferred before the adjective (also when there is an overt subject), as in (13).

- (13) a. ?it-ɛn **zi-ts'ɛhaf-in** bizuh siʔelɛ-tat
DIST-FP REL-write.PFV-SM.3MS-OM.3FP a.lot image-P
z-ɛll-o-win dɛbdabe-tat niwah-ti ?ijj-ɛn
REL-BE₁.PRES-SM.3MS-OM.3FP letter.F-P long-F BE₂.PRES-SM.3FP
- b. ??it-ɛn bizuh siʔelɛ-tat z-ɛll-o-win
DIST-FP a.lot image-P REL-BE₁.PRES-SM.3MS-OM.3FP
zi-ts'ɛhaf-in dɛbdabe-tat niwah-ti ?ijj-ɛn
REL-write.PFV-SM.3MS-OM.3FP letter.F-P long-F BE₂.PRES-SM.3FP
- c. ??it-ɛn bizuh siʔelɛ-tat z-ɛll-o-win dɛbdabe-tat
DIST-FP a.lot image-P BE₂.PRES-SM.3FP letter.F-P
zi-ts'ɛhaf-in niwah-ti ?ijj-ɛn
REL-write.PFV-SM.3MS-OM.3FP long-F BE₂.PRES-SM.3FP
‘The letters full of images that he wrote are long.’

The sentences in (13) show that there are principles of processing independent of the preference for placing heavy modifiers after lighter ones. This phenomenon can also be seen in Italian. Consider (14) and (15) with the relative clause in bold.⁸

- (14) a. La lettera interessante **che mi ha scritto** è su-l tavolo.
the letter interesting that to-me has written is on-the table
- b. *La lettera **che mi ha scritto** interessante è su-l tavolo.
the letter that to-me has written interesting is on-the table
‘The interesting letter that he wrote to me is on the table.’
- (15) a. La lettera ricca di metafore **che mi ha scritto** è su-l tavolo.
the letter full_of_metaphors that to-me has written is on-the table
- b. La lettera **che mi ha scritto** ricca di metafore è su-l tavolo.
the letter that to-me has written full_of_metaphors is on-the table
‘The letter full of metaphors that he wrote to me is on the table.’

The sentences in (14) contain the simple adjective *interessante* ‘interesting’, which can precede but not follow the relative clause. In contrast, the sentences in (15) contain the complex adjective *ricca di metafore* ‘full of metaphors’ which can both precede and follow

⁸I thank Ivano Caponigro for pointing out these examples to me.

the relative clause. The weight of nominal complementizers therefore plays a role in the position of the relative clause in different languages.

We have seen a difference in the acceptability of the sentences above depending on the weight of nominal modifiers. What happens when the relative clause itself is heavier or lighter? Does this affect the acceptability of its distribution? To examine this, let us compare example (11) above with (16), which contains two different phonologically overt subjects: a monosyllabic subject *Dan* and a multisyllabic subject *?iti ?iziju niwiḥ seb?ai* ‘the very tall man’.

- (16) a. *?it-ɛn Dan/{?it-i ?iziju niwiḥ seb?ai}*
 DIST-FP Dan/DIST-MS very tall man
zi-ts’ɛḥaf-in ts’ibuʁat dɛbdabe-tat niwaḥ-ti
 REL-write.PFV-SM.3MS-OM.3FP beautiful letter.F-P long-F
?ijj-ɛn
 BE₂.PRES-SM.3FP
- b. *?it-ɛn ts’ibuʁat Dan/{%?it-i ?iziju niwiḥ seb?ai}*
 DIST-FP beautiful Dan/DIST-MS very tall man
zi-ts’ɛḥaf-in dɛbdabe-tat niwaḥ-ti *?ijj-ɛn*
 REL-write.PFV-SM.3MS-OM.3FP letter.F-P long-F BE₂.PRES-SM.3FP
- c. *?it-ɛn ts’ibuʁat dɛbdabe-tat Dan/{%?it-i ?iziju niwiḥ seb?ai}*
 DIST-FP beautiful letter.F-P Dan/DIST-MS very tall man
zi-ts’ɛḥaf-in niwaḥ-ti *?ijj-ɛn*
 REL-write.PFV-SM.3MS-OM.3FP long-F BE₂.PRES-SM.3FP
 ‘The beautiful letters that Dan/the very tall man wrote are long.’

The sentences in (16) show that the longer the subject, the less acceptable the relative clause becomes (only 1 consultant out of 3 accepts it) when it is farther from the demonstrative.

Table 2.2 summarizes the possible word orders seen with the relative clause appearing with one adjective (excluding the order in which the relative clause precedes the demonstrative).

	SUBJ			heavy A
	<i>pro</i>	short	long	
(a) Dem- RC -A-N	✓	✓	✓	✓
(b) Dem-A- RC -N	✓	✓	%	?
(c) Dem-A-N- RC	✓	✓	%	?

Table 2.2: Acceptability of the Distribution of a Relative Clause among Dem-A-N.

Once again, the order that seems to be preferred is the one in which the relative clause appears between the demonstrative and the numeral (the base universal word order in DPs). Note also that when the relative clause co-occurs with a normal-sized adjective, it is accepted in all positions with a *pro* and a short subject, in contrast to when it co-occurs with a numeral.

Dem-Num-A-RC-N

Finally, I now add a relative clause to a sentence such as the one in (17), containing a demonstrative, a numeral and an adjective.

- (17) ?it-ɛn kilite ts'ibuxat dɛbdabe-tat niwah-ti ?ijj-ɛn
DIST-FP two beautiful letter.F-P long-F BE₂.PRES-SM.3FP
'The two beautiful letters are long.'

The relative clause, with a null subject pronoun or an overt DP subject can be placed in four different positions (with an overt DP, 1 consultant out of 3 was more reticent about the third sentence), as shown in (18). As mentioned above, the relative clause cannot precede the demonstrative.

- (18) a. ?it-ɛn (?it-i seb?ai) **zi-ts'ɛhaf-in** kilite ts'ibuxat
DIST-FP DIST-MS man REL-write.PFV-SM.3MS-OM.3FP two beautiful
dɛbdabe-tat niwah-ti ?ijj-ɛn
letter.F-P long-F BE₂.PRES-SM.3FP
- b. ?it-ɛn kilite (?it-i seb?ai) **zi-ts'ɛhaf-in** ts'ibuxat
DIST-FP two DIST-MS man REL-write.PFV-SM.3MS-OM.3FP beautiful
dɛbdabe-tat niwah-ti ?ijj-ɛn
letter.F-P long-F BE₂.PRES-SM.3FP

- c. ?it-ɛn kilite ts'ibuʁat (%?it-i sɛbʔai) **zi-ts'ɛħɛf-in**
 DIST-FP two beautiful DIST-MS man REL-write.PFV-SM.3MS-OM.3FP
 dəbdabe-tat niwaħ-ti ?ijj-ɛn
 letter.F-P long-F BE₂.PRES-SM.3FP
- d. ?it-ɛn kilite ts'ibuʁat dəbdabe-tat (?it-i sɛbʔai)
 DIST-FP two beautiful letter.F-P DIST-MS man
zi-ts'ɛħɛf-in niwaħ-ti ?ijj-ɛn
 REL-write.PFV-SM.3MS-OM.3FP long-F BE₂.PRES-SM.3FP
 'The two beautiful letters that he/the man wrote are long.'

If the sentences in (18) are passivized, the relative clause can also be placed in four different positions, all perfectly grammatical. This is illustrated in (19).

- (19) a. ?it-ɛn b-it-i sɛbʔai **zi-ts'ɛħɛf-a** kilite ts'ibuʁat
 DIST-FP by-DEM.MS man REL-write.PASS-SM.3FS two beautiful
 dəbdabe-tat niwaħ-ti ?ijj-ɛn
 letter.F-P long-F BE₂.PRES-SM.3FP
 'The two beautiful letters that are written by the man are long.'
- b. ?it-ɛn kilite b-it-i sɛbʔai **zi-ts'ɛħɛf-a** ts'ibuʁat
 DIST-FP two by-DEM.MS man REL-write.PASS-SM.3FS beautiful
 dəbdabe-tat niwaħ-ti ?ijj-ɛn
 letter.F-P long-F BE₂.PRES-SM.3FP
 'The two beautiful letters that are written by the man are long.'
- c. ?it-ɛn kilite ts'ibuʁat b-it-i sɛbʔai **zi-ts'ɛħɛf-a**
 DIST-FP two beautiful by-DEM.MS man REL-write.PASS-SM.3FS
 dəbdabe-tat niwaħ-ti ?ijj-ɛn
 letter.F-P long-F BE₂.PRES-SM.3FP
 'The two beautiful letters that are written by the man are long.'
- d. ?it-ɛn kilite ts'ibuʁat dəbdabe-tat b-it-i sɛbʔai
 DIST-FP two beautiful letter.F-P by-DEM.MS man
zi-ts'ɛħɛf-a niwaħ-ti ?ijj-ɛn
 REL-write.PASS-SM.3FS long-F BE₂.PRES-SM.3FP
 'The two beautiful letters that are written by the man are long.'

The different positions illustrated in the preceding paragraphs are schematically summarized in Table 2.3.

	active		passive
	<i>pro</i> SUBJ	DP SUBJ	
(a) Dem- RC -Num-A-N	✓	✓	✓
(b) Dem-Num- RC -A-N	✓	✓	✓
(c) Dem-Num-A- RC -N	✓	%	✓
(d) Dem-Num-A-N- RC	✓	✓	✓

Table 2.3: Acceptability of the Distribution of a Relative Clause among nominal modifiers.

Interestingly, from Table 2.3, we can see that when a relative clause co-occurs with adjectives and numerals, it is accepted in four different positions.⁹ When the subject is a DP and it is overtly realized, the position of the relative clause between the adjective and the noun is less preferred.

As seen, relative clauses in Tigrinya can iterate among nominal modifiers and be placed in different positions without an apparent difference in meaning. If there is indeed a slight difference in meaning in the sentences provided, my consultants did not seem to grasp it during the fieldwork. In my view, a possible difference in meaning could be related to the focus properties of the different elements or to the independent processing principles already mentioned.

However, before proposing a hypothesis (but not a full analysis, due to lack of space and time in this dissertation) for the itinerant behavior of relative clauses, I propose an analysis for the internal structure of the Tigrinya DP to explain the rigid Dem-Num-A-N word order in the language. The structure and derivation of the DP are particularly relevant for the approach taken in this dissertation — namely, a head-final approach to head-finality — given that the order of the elements mentioned above would be straightforward under a head-initial approach to head-finality, as in [Esayas 2003](#).

The Tigrinya DP

It goes without saying that I follow [Abney 1987](#) and suggest that cross-linguistically noun phrases are DPs in which D° is a functional element (as C° is in CP). In the literature it has been claimed that demonstratives occupy different positions than definite articles because

⁹For a description of the possible word orders when a relative clause co-occurs with possessors, cf. [Gebregziabher 2013](#).

they can co-occur in certain languages — such as Arabic, Romanian and even Tigrinya, as seen in Section 1.5.2. It has therefore been claimed that if definite articles are in D° (or Q° , a.o. Shlonsky 1991), demonstratives are base-generated (or moved to) Spec,DP (Giusti 1997, Bernstein 1997, Brugè 2002). More specifically for the Ethio-semitic family, Baker & Kramer 2014 also suggest that in Amharic demonstratives are in Spec,DP.

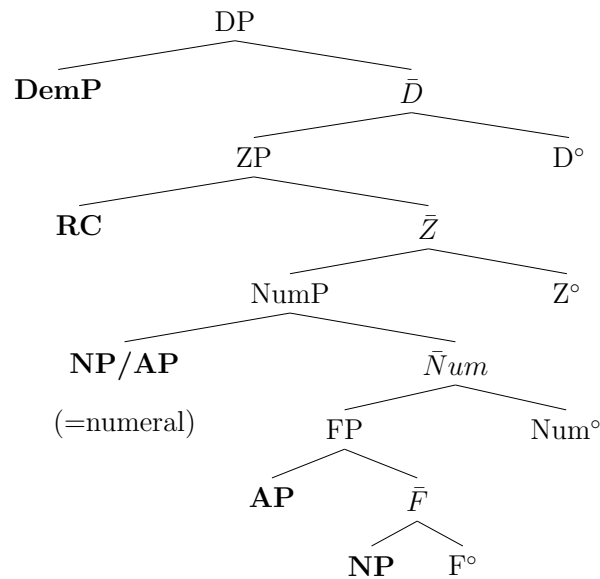
With respect to numerals, there is disagreement in the literature on where exactly they are merged: under a PlP (plurality phrase) head (Svenonius 2007a), in Spec,dP (Cinque 2015), in a D° head outside NP (Arregi 2013), under a NumP (number phrase) or QP (quantifier phrase) head (Ritter 1992, Zamparelli 2000), or in the specifier of an agreement or other functional projections (Jackendoff 1977, Giusti 1997, Shlonsky 2004), to name a few. More specifically for the Ethio-semitic family, Esayas 2003 proposes to treat numerals in Tigrinya as quantifiers base-generated in Spec,NP, and Baker & Kramer 2014 suggest that they are in the specifier of a functional projection FP.

Adjectives are the only nominal modifiers that can appear multiple times in the same nominal phrase. For this reason, adjectives are said to be stacked and generated in the specifier of different functional projections according to the category they belong to (Scott 2002). The hierarchy of these projections dictates the surface order of adjectives (Laenzlinger 2005, Svenonius 2007b, Cinque 2010). In the literature there is no clear agreement on the exact number of adjectival categories or on the universality of their order in the syntactic structure. Moreover, the nature of the functional projections also remains disputed. More specifically for the Ethio-semitic family, Esayas 2003 proposes to treat adjectives in Tigrinya as weak quantifiers base-generated in Spec,NP.

With all this in mind, I suggest that the Tigrinya DP containing a demonstrative, a numeral, an adjective, a relative clause and a noun has the following structure. The demonstrative is in Spec,DP. Because the first cardinal number agrees in gender and number with the noun, I claim that numerals are NP- or AP-like elements that occupy the specifier position of a NumP. I follow Baker & Kramer 2014 and suggest that adjectives are base-generated in a functional projection FP. Finally, I also follow Baker & Kramer 2014 and claim that relative clauses are generated in the specifier of a functional projection

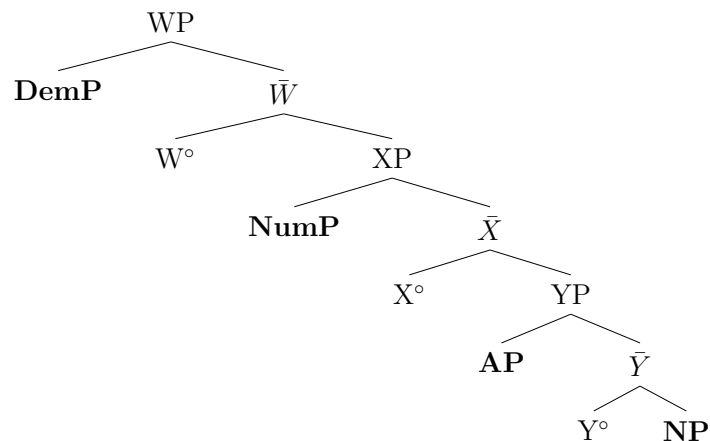
ZP that is the complement of D° . Note that with this configuration, relative clauses are positioned in a similar way of adjectives, emphasizing the fact that adjective modification shares properties with that of relative clauses (Leu 2015). This structure is illustrated in the tree in (20).

(20)



Note that the tree in (20) is not very different (all modifiers appears as phrases in specifiers) from what has been proposed by Cinque 2005 with a head-initial approach, as in the tree in (21) (adapted and without the RC).

(21)



Now, let us return to the itinerant behavior of relative clauses seen previously.

Itinerant Relative Clauses: Short Discussion

How can we explain the possibility for relative clauses in Tigrinya to move around nominal modifiers (with some restrictions in some cases)?

One could hypothesize that in Tigrinya, unlike in languages such as English, it is possible to distinguish cases in which a relative clause modifies only the noun or the noun plus one or two nominal modifiers, depending on the linear position of the relative clause itself. For example, in (18) the relative clause modifies the *two beautiful letters* in (18a), it modifies the *beautiful letters* in (18b), and it modifies the *letters* in (18c). What it modifies exactly in (18d) and, by consequence, what the semantic difference between a prenominal and a postnominal relative clause is in Tigrinya requires further investigation.

However, all this could perhaps be explained in terms of extraposition triggered by a need for emphasis. In other words, the four different word orders involve different information structures. More concretely, this would be in line with the literature that has argued that Focus movement can account for reordering of elements within the DP (Syed 2015). Assuming a Focus projection in the nominal domain (Aboh 2004), one could imagine that the relative clause moves according to which element(s) is/are focalized, perhaps right-adjoining them (a hint at the use of the copula *?ijju* as a focus marker mentioned in section 1.5.2). A more in-depth analysis that takes into consideration focus and prosody (Zubizarreta 1998) is in order here.

More specifically concerning the underlying syntax of these different word orders, another possibility is that we are not actually dealing with *restrictive* relative clauses. From the English translations it seems that the relative clauses under investigation in the previous sections are restrictive. However, remember from section 1.5.2 in the Introduction that demonstratives in Tigrinya are used both as determiners (hence the interpretation of the relative clauses as restrictive) and as distal demonstratives. In this second case the relative clauses would be non-restrictive (or appositive) relative clauses. These relative clauses are considered adjuncts and can therefore freely adjoin wherever they can semantically compose, namely in four different positions in Tigrinya.

At the same time, if these were truly *restrictive* relative clauses, they would be base-

- (24) [ʔit-i ʃekolata z-i-bɛliʃ wɛdi] Tɛkle ji-bhal
 DIST-MS chocolate.FS REL-SM.3MS-eat.IPFV boy Tɛkle.MS SM.3MS-call.IPFV
 ‘The boy who eats chocolate is called Tɛkle.’
- (25) [Sɛgɛn n-it-i wɛrɛqɛt z-i-bɛliʃ sɛbʔai]
 Sɛgɛn DOM-DIST-3MS paper REL-SM.3MS-eat.IPFV man
 rakib-a-wo
 find.GER-SM.3FS-OM.3MS
 ‘Sɛgɛn found the man who eats paper.’

Note that in (22)–(25) *z(i)-* does not inflect for ϕ -features or for aspect/tense. It is said to have probably developed etymologically from demonstrative roots (Dillmann 1974), such as the demonstrative masculine singular accusative *ze-* in Ge’ez (Butts 2019). In fact, the *z* morpheme, also found in proximal demonstratives in Tigrinya (e.g. *ʔiz* as in *ʔizi* for masculine singular), seems to convey the meaning of referring to the nearer object (Murtonen 1967:29) or to what is at hand and well-known (Dillmann 1974:330).¹¹

Z(i)- is the outermost prefix in the verb stem: it precedes both inflectional and derivational prefixes.¹² With regard to the latter, example (26) shows that *z(i)-* precedes the prefix *t(i)-* that marks passive.

- (26) ʔiti bi-Tɛwɛldɛ z-i-ti-sɛtɛj-ɛ may t’ɛʃum
 DIST-MS BY-Tewelde.MS REL-PASS-drink.PFV-SM.3MS water.MS delicious
 ʔijj-u
 BE₂.PRES-SM.3MS
 ‘The water that was drunk by Tewelde is delicious.’

With respect to inflectional prefixes, these are the prefixes found in verb forms in the Imperfective that mark the ϕ -features of the verb. When *z(i)-* attaches to these, the phonological realization of the first phoneme of the verb changes. If *z(i)-* occurs on an Imperfective verb starting with [j] (first person singular, third person masculine singular and third person plural), this consonant is changed into a vowel [i] or [ɪ], as in (24)¹³.

¹¹It is common to find languages in which demonstratives are reanalyzed as relative pronouns: this is because they are elements internal to the DP and relative pronouns are parts of the internal relativized head (Poletto & Sanfelici 2015).

¹²In Rayya Tigrinya *z(i)-* is realized as *d(i)-* (Niguss 2021:111).

¹³Note that this does not happen only with *z(i)-* but also with other prefixes such as *k(i)-* and negation. There is variation among speakers for the choice of the vowel. Phonology being outside the scope of this dissertation, I did not investigate a possible phonological rule.

If $z(i)$ - occurs on a verb starting with $[ʔ]$ (such as the auxiliary *ʔallo*, as will be shown later), the glottal stop is dropped and a vowel $[ɛ]$ appears instead.

It is important to point out that in grammars of Tigrinya, $z(i)$ - is said to appear under a different form in some cases: when it precedes a $[t]$ phoneme of the Imperfective second person and third feminine singular person, and the $[n]$ phoneme of first person plural, $z(i)$ - is realized as $ʔ(i)$ - (Leslau 1941, Mason 1996, Nazareth 2011, Bulakh 2019). However, my consultants do not produce $ʔ(i)$ - (either when speaking or when writing) and instead leave the verb bare, as shown in (27).¹⁴

- (27) ʔiti nissxa **ti**-seti-jo may t'eʃum
 DIST-MS you.MS SM.2MS-drink.IPFV-OM.3MS water.MS delicious
 ʔijj-u
 BE₂.PRES-SM.3MS
 'The water that you drink is delicious.'

- (28) ʔiti niħna **ni**-seti-jo may t'eʃum ʔijj-u
 DIST-MS we SM.1P-drink.IPFV-OM.3MS water.MS delicious BE₂.PRES-SM.3MS
 'The water that we drink is delicious.'

This phenomenon could be considered an instance of allomorphy — $ʔ(i)$ - in front of second person, 3FS person and 1P person, and $z(i)$ - elsewhere — or phonological dropping. In my view, the phenomenon illustrated above cannot be accounted solely by phonology but must be explained in morphological terms. This is because if the absence of $z(i)$ - were only a consequence of phonology, we would also expect $z(i)$ - to drop in the presence of the passive prefix $t(i)$ -, but this is not the case — cf. example (26) above. Moreover, $z(i)$ - is not dropped in front of quadriliteral roots that always start with $[ti]$ in Tigrinya. This is shown in (29).

- (29) ʔit-i **zi**-tizareb-ε sebʔaj ʔabo-j ʔijj-u
 DIST-MS REL-speak.PFV-SM.3MS man father-POSS.1S BE₂.PRES-SM.3MS
 'The man that spoke is my father.'

Now, moving away from phonology and morphology and back to syntax, I showed in

¹⁴This is also the case for Jason Overfelt's and Coppe van Urk's consultants (p.c.).

- zi**-dele-xa-jo ki-ti-bil ti-xiʔil
REL-want.PFV-SM.2MS-OM.3MS SBJ-SM.2MS-say.IPFV SM.2MS-can.IPFV
'You can say whatever you want(ed).'

- ʔit-i z-i-gebr-o ts'ibuʁ ʔijj-u
DIST-MS REL-SM.3MS-do.IPFV-OM.3MS good BE₂.PRES-SM.3MS
'What he does is great.'

- ʔit-i niɪna **zi**-dali-na-jo ni-beliɬ-jo
DIST-MS we REL-prepare.PFV-SM.1P-OM.3MS SM.1P-eat.IPFV-OM.3MS
'We eat what we prepared.'

- n-it-a zɪ-wedɛ-qa-ya wɛini k-i-sɛti
DOM-DIST-FS REL-drop.PFV-SM.2MS-OM.3FS wine.FS SBJ-SM.1S-drink.IPFV
mi-dɛlɛ-xu
NMZ-want.PFV-SM.1S
'I would like to drink the wine that you dropped on the floor.'¹⁹

- kibur z-ɛj-xon-ɛ mɛts'ħaf ji-dɛli
expensive REL-NEG-BE₂.FUT-SM.3MS book.MS SM.1S-want.IPFV
'I want a book that is not expensive.'

¹⁸Amount/maximalizing relative clauses do not modify the Head; instead, the Head provides a degree variable via the operation of maximalization (Bianchi 2002).

¹⁹Unfortunately, the use and properties of the *m(i)-* prefix - probably a nominalizer - will not be investigated in this dissertation due to lack of space and time. However, this will be an interesting topic for further research. Note that in Amharic the element *mä-* seems to behave exactly like *m(i)-* in Tigrinya. [Leung & Girma 2017](#) suggest that this element is a nominalizer positioned within the VP domain.

²⁰ Postnominal relative clauses are also possible but rare in Amharic (Hailu 1972:497) and Tigre (Palmer 1961:38).

- (39) a. ?it-om **zi**-nigeri-ku-wom zanta-tat haqinat-at
DIST-MP REL-tell.PFV-SM.1S-OM.3MP story.MP true.P
?i-jjom
BE₂.PRES-SM.3MP
‘The stories that I told are true.’
- b. ?it-om zanta-tat **zi**-nigeri-ku-wom haqinat-at
DIST-MP story.MP REL-tell.PFV-SM.1S-OM.3MP true.P
?i-jjom
BE₂.PRES-SM.3MP
‘The stories that I told are true.’

In the case where two relative clauses modify the same noun, [Palmer 1962:38](#) claims that one precedes and the other follows the noun. This is true; however, it is not the only option. In fact, the sentences in (40) show that the two relative clauses can also be prenominal (the order between the two may be restricted by their interpretation).

- (40) a. ?it-a **zi**-derbe-ka-ja mets’ haf ?ab li?eli t’awla
DIST-FS REL-throw.PFV-SM.2MS-OM.3FS book.FS LOC table
zi-neber-et bt’a?emi ?agedasi ?ijj-a
REL-BE₁.PFV-SM.3FS very important BE₂.PRES-SM.3FS
‘The book that you throw away that was on the table was very important.’
- b. ?it-a **zi**-derbe-ka-ja ?ab li?eli t’awla
DIST-FS REL-throw.PFV-SM.2MS-OM.3FS LOC table
zi-neber-et mets’ haf bt’a?emi ?agedasi ?ijj-a
REL-BE₁.PFV-SM.3FS book.FS very important BE₂.PRES-SM.3FS
‘The book that you throw away that was on the table was very important.’
- c. ??it-a ?ab li?eli t’awla **zi**-neber-et
DIST-FS LOC table REL-BE₁.PFV-SM.3FS
zi-derbe-ka-ja mets’ haf bt’a?emi ?agedasi
REL-throw.PFV-SM.2MS-OM.3FS book.FS very important
?ijj-a
BE₂.PRES-SM.3FS
‘The book that you throw away that was on the table was very important.’

Let us now turn to more complicated cases.

Relative Clauses with periphrastic tense forms

So far, I have presented data on the occurrence of *z(i)*- in relative clauses when it is attached to a simple verb form (in which the verb is either the Perfective or the Imper-

fective occurs alone). Now, what happens in periphrastic tense forms, in which the verb is accompanied by an auxiliary? In these cases, *z(i)-* is obligatorily prefixed to both the verb and the auxiliary, appearing twice in the clause, as shown in (41) and (42). This phenomenon was first mentioned by [Palmer 1962](#) and later noted also by [Bulakh 2019](#).

- (41) ?it-i ?anɛ z-i-nbiḃ-o z-ell-exu mɛts'ħaf
DEM.MS I REL-SM.1S-read.IPFV-OM.3MS REL-BE₁.PRES-SM.1S book.MS
ʔazenagaʔi ʔijj-u
amusing BE₂.PRES-SM.3MS
'The book that I am reading is amusing.'
- (42) ?it-i fɛkolata z-i-beliʕ z-ell-o wɛdi
DIST-MS chocolate.FS REL-SM.3MS-eat.IPFV REL-BE₁.PRES-SM.3MS boy
Tɛkle ji-bhal
Tekle.MS SM.3MS-call.IPFV
'The boy who is eating chocolate is called Tekle.'

The phenomenon presented in the two examples above is quite surprising: in more well-studied languages such as English, French and Arabic, the element that marks relative clauses appears only once. Despite having been mentioned, this phenomenon has never received proper attention. In the following section I provide an explanation of why $z(i)$ -can occur on both the verb and the auxiliary in periphrastic tense forms. For now, let us continue looking at the different types of clauses in which this clausal prefix occurs.

Before doing this, there is one last point I would like to mention in this section. This has already been mentioned in Section 1.6 when discussing the possibility of verb raising in Tigrinya. The prefix *z(i)-* cannot be attached to a Gerund verb (cf. also Palmer 1962), hence the ungrammaticality of (43).

- (43) *ʔit-a ʔit-i seɓʔaj **zi**-ts'ɛhif-u-wa dɛbdabɛ niwah
DIST-FS DIST-MS man REL-write.GER-SM.3MS-OM.3FS letter.FS long.FS
ʔijj-a
BE₂.PRES-SM.3FS
'The letter that the man wrote is long.'

To obtain a relative clause that expresses past tense (such as those in examples (32), (34), (36), (37) and (39)) Tigrinya uses two strategies: either a suppletive Perfective

form is inserted instead of the Gerund and $z(i)$ - is prefixed to it, or a “dummy” auxiliary (cf. also Overfelt 2009) is inserted to accompany the Gerund and carry the $z(i)$ - prefix (the Gerund remains bare). Although the auxiliary marks progressive aspect when co-occurring with a verb in the Imperfective and expresses a resultative state (or perfect) when co-occurring with a verb in the Gerund (as seen above), in this case the auxiliary marks neither progressive aspect nor perfect tense, hence the translation. These two strategies are illustrated in (44) and respectively (45) below.

- (44) ?it-a ?it-i sɛbʔaj **zi**-ts'ihaf-i-wa dɛbdabɛ niwəh
DIST-FS DIST-MS man REL-write.PFV-SM.3MS-OM.3FS letter.FS long.FS
?ijj-a
BE₂.PRES-SM.3FS
‘The letter that the man wrote is long.’
- (45) ?it-a ?it-i sɛbʔaj ts'ihif-u-wa **z-ɛ-llo**
DIST-FS DIST-MS man write.GER-SM.3MS-OM.3FS REL-BE₁.PRES-SM.3MS
dɛbdabɛ niwəh ?ijj-a
letter.FS long.FS BE₂.PRES-SM.3FS
‘The letter that the man wrote is long.’

I have presented and described several properties of the $z(i)$ - prefix and the first type of clauses in which it occurs, namely relative clauses. Let us now turn to the other clause types.

2.2.3 Other Clauses with $z(i)$ -

Beside relative clauses, $z(i)$ - is also found in noun clausal complements, comparative clauses, *seem*-clauses and superlatives. I present data for all these constructions in the following pages.

(Subject) Noun Clausal Complements

Noun clausal complements²¹ in Tigrinya take the clausal prefix $z(i)$ - when they function as subjects. Consider (46).

²¹Term taken from Krapova & Cinque 2016. These constructions are also sometimes referred to as *noun complement clauses* or *complex noun phrases*.

- (46) ?it-i Solomon feres gezi?-u z-i-bil zereba
DIST-MS Solomon horse buy.GER-SM.3MS REL-SM.3MS-say.IPFV story.MS
hasot ?ijj-u
false BE₂.PRES-SM.3MS
‘The story that Solomon bought a horse is false.’
- (47) ?it-i merfat sefiḥ ?ijj-u z-i-bil hasab
DIST-MS Earth.MS flat.MS BE₂.PRES-SM.3MS REL-SM.3MS-say.IPFV idea.MS
?agedasi ?ijj-u
interesting BE₂.PRES-SM.3MS
‘The idea that the Earth is flat is interesting.’
- (48) ?it-i ?iz-i p’resident mojit-u z-i-bil
DIST-MS PROX-MS president die.GER-SM.3MS REL-SM.3MS-say.IPFV
were haqi ?ijj-u
rumor.MS true BE₂.PRES-SM.3MS
‘The rumor that this president has died is true.’

In (46)–(48), the structure of noun clausal complements is very similar to that of relative clauses: the head nouns *zereba* ‘story’, *hasab* ‘idea’ and *were* ‘rumor’ are preceded and modified by a demonstrative followed by a relative clause. This relative clause consists of the conjugated verb /bhl/ ‘to say’, to which *z(i)-* is attached, forming *zibil*.²² This relativized verb is absent in the English translations. In English (as shown in the translations) the noun is followed by a clause introduced by *that*; semantically, the type of noun affects the status (the truth value) of the proposition. Returning to Tigrinya, the sentences above could therefore be translated more literally as ‘The story/idea/rumor *that says* that...’.²³ The difference between English and Tigrinya could lie in the fact that in Tigrinya it is not possible for propositional content — e.g. ‘that this president has died’ in (48)— to follow nominal content — e.g. ‘the rumor’ in (48)— directly without an intervening predicate — e.g. the verb ‘say’ in (48).

The examples above illustrate noun clausal complements functioning as subjects of the main clause. When these constructions serve as objects of the main clause, the situation becomes more complex. If such clauses are the object of, for example, a perception verb

²²One informant produced the form *zibhal* in (46). This is still a form of the verb ‘to say’; however, I cannot explain the difference between the two forms.

²³The translation cannot be ‘The story/idea/rumor *says* that...’ (without the first *that*) because, as shown in Chapter 3, in that case the clausal prefix *kemz(i)-* would appear instead of *z(i)-*.

such as /smʕ/ ‘hear’, then *z(i)-* is replaced by the clausal prefix *kemz(i)-*, because these kind of verbs trigger its appearance (as explained in Chapter 3). Consider (49) and compare it with (48).

- (49) ?it-i ?iz-i p’resident **kem-zi**-mot-ε zejna səmiʕ-ε
DIST-MS PROX-MS president kemzi-die.PFV-SM.3MS rumor hear.GER-SM.1S
‘I heard the rumor that this president has died.’

- (50) ?it-i Tesfai **kem-zi**-tebarer-ε ts’ebts’ab ?anbib-ε
DIST-MS Tesfay kemzi-be_fired.PFV-SM.3MS report read.GER-SM.1S
‘I read the report that Tesfay was fired.’

The presence of the prefix *kemz(i)-* in this type of sentences will be analyzed later. For now, let us continue to focus on the types of clauses in which *z(i)-* appears.

Comparative Clauses & Superlatives

The third type of clauses in which *z(i)-* appears is the comparative clause. Let us first examine how comparatives (i.e. comparative adjectives in English) are formed in Tigrinya. Consider the comparatives in (51).

- (51) a. Tesfai kab Kidane ji-niwεh
Tsfay.MS from Kidane SM.3MS-be_tall.IPFV
‘Tsfay is taller than Kidane.’

b. Ts’ega kab hεbt-a ti-niwεh
Tsega.FS from sister-POSS.3FS SM.3FS-be_tall.IPFV
‘Tsega is taller than her sister.’ (from [Mason 1996:92](#))

In comparatives, the preposition *kab* ‘from’ functions as the counterpart of ‘than’ in English. Instead of using a comparative adjective as in the English translation, Tigrinya employs an Imperfective form of the adjective *niwεh* ‘tall/long’ (from the root /nwh/). This verb agrees with the grammatical subject, ‘Tsfay’ and ‘Tsega’ respectively, in the two examples above.

Let us now turn to comparative clauses; consider (52) and (53).

- (52) Tɛsfay kab-t-i ʔanɛ zɪ-ħasibɛ-xu-wo
 Tɛsfay.MS from-DIST-MS I REL-think.PFV-SM.1S-OM.3MS
 jɪ-ʕabi
 SM.3MS-be_old.IPFV
 ‘Tɛsfay is older than I thought.’
- (53) Segen kab-t-i ʔanɛ zɪ-ħasibɛ-xu-wo
 Segen.FS from-DIST-MS I REL-think.PFV-SM.1S-OM.3MS
 ti-ʕabi
 SM.3FS-be_old.IPFV
 ‘Segen is older than I thought.’

Comparative clauses also present the preposition *kab* ‘from’. This preposition attaches to a masculine singular demonstrative (note that it remains masculine even when the subject of the sentence, such as ‘Segen’, is feminine). In this case too the matrix verb carries subject morphology agreeing with the grammatical subject. The *z(i)*-marked verb carries subject morphology that agrees with the embedded subject and default 3MS object morphology.²⁴

The structure of the sentences in (52) and (53) closely resembles the light-headed relative clause (Citko 2004) presented in example (36) in a previous section: the relativized verb is accompanied by a demonstrative, but the relativized element itself is absent. A more detailed and developed syntactic analysis of comparative clauses will be provided later.

The clausal prefix *z(i)*- also appears in superlatives. This is unexpected, since superlatives in English are not clauses but rather adjectives expressing the highest degree of a quality (marked by the suffix *-(e)st* attached to an adjective or the particle *most* preceding an adjective). Let us examine the phenomenon in Tigrinya. Consider (54) and (55).

- (54) ʔaman (ʔit-i) zi-niwɛħ sɛbʔaj ʔijj-u
 Aman DIST-MS REL-long.MS man BE₂.PRES-SM.3MS
 ‘Aman is the tallest man.’

²⁴My consultants were confused when presented with 3FS object morphology *wa* in the example with *Segen* ‘Segen’ (a woman’s name): they found it not acceptable, but also suggested it could work if the speaker was strongly focusing on the subject.

Seem-Clauses

Z(i)- occurs also in clauses that present the verbal root /*msl*/ ‘seem’. Example (58) illustrates a root clause and example (59) shows a *z(i)*-clause, both with the verb /*msl*/ ‘seem’.

- (58) Daniʔel lɛgas ji-mɛsɪl
 Daniel generous SM.3MS-seem.IPFV
 ‘Daniel seems generous.’
- (59) ʔit-om waga-tat kɛbir-om **zi**-nɛbɛr-u ji-mɛsl-u
 DIST-MP price.MP big.MP REL-BE₁.PFV-SM.3MP SM.3MP-seem.IPFV-SM.3MP
 ‘Prices seem to have risen.’

As in relative clauses, when the sentence contains a periphrastic tense form, *z(i)*- appears on both the verb and the auxiliary, as in (60).

- (60) a. Kidanɛ ʔit-i mɛts’ ʔhaf **z**-ɛ-nbɪb-o
 Kidane.MS DIST-MS book.MS REL-SM.3MS-read.IPFV-OM.3MS
 ji-mɛsɪl
 SM.3MS-seem.IPFV
 ‘Kidane seems to read the book.’
- b. Kidanɛ ʔit-i mɛts’ ʔhaf **z**-ɛ-nbɪb-o
 Kidane.MS DIST-MS book.MS REL-SM.3MS-read.IPFV-OM.3MS
z-ɛll-o ji-mɛsɪl
 REL-BE₁.PRES-SM.3MS SM.3MS-seem.IPFV
 ‘Kidane seems to be reading the book.’

It is important to note that the 3MS subject morphology carried by the matrix verb in the example above cross-references the grammatical subject of the clause and is not default agreement. This can be seen in (61) with a subject that is not 3MS.

- (61) ʔalkol ti-seti **z**-ɛll-ɛxa ti-mɛsɪl
 alcohol SM.2MS-drink.IPFV REL-BE₁.PRES-SM.2MS SM.2MS-seem.IPFV
 ‘You seem to be drinking alcohol.’

Unlike English, Tigrinya cannot express these sentences with an overt expletive, such as *It seems that...* This is because Tigrinya lacks overt expletives. For example, the subject of

meteorological verbs such as *rain* and *snow* is not an expletive; their meaning is expressed through a periphrastic construction, as in (62).

- (62) zinab ji-zenab
rain SM.3MS-rain.IPFV
'It rains.' (Lit.: Rain rains.)

However, as shown in Chapter 3, Tigrinya can express sentences such as *It seems to me/him/us that...* without an overt expletive, using the clausal prefix *kɛmz(i)-* instead of *z(i)-*.

Finally, note that in Amharic, *seem*-clauses also involve the clausal prefix used in relative clauses *yə-*, as shown in (63).

- (63) yosef makina **yə**-gəzz-a yi-məslal
 Josef car CM-buy-3MS seems
 ‘It seems that Yosef has bought a car.’ (from [Desalegn 2021:5](#), (8))

In the next section I turn to the syntactic analysis of the other types of constructions with $z(i)$ - presented in the previous pages.

2.3 The Analysis of $z(i)$ -clauses

In the previous section, I presented data on the types of structures in which $z(i)$ - occurs. I began by showing that this clausal prefix appears in relative clauses. As already noted in the introduction to this Chapter, in the literature $z(i)$ - has indeed been described as a relative marker. However, because it also appears in other types of clauses — namely noun clausal complements, comparative clauses, *seem*-clauses and superlatives — this label is not sufficiently precise.

In the following sections, I provide a syntactic analysis of all the clauses seen so far and propose that $z(i)$ - is in fact a marker of $\bar{A}$ -dependencies. The analyses developed below will not only clarify the properties of $z(i)$ - but also address the question of how a head-final language can possess such a *prefix*.

2.3.1 Toward an Analysis of $z(i)$ -

Before providing an analysis of $z(i)$ -clauses, I first investigate what $z(i)$ - is.

What is $z(i)$ -?

We have seen that this prefix does not carry ϕ -features and can attach to verbs in the Imperfective and Perfective, but not on verbs in the Gerund. Most importantly, it appears on both the verb and the auxiliary in periphrastic tense forms. The question is therefore: what kind of element is $z(i)$ - and how can it appear twice in a periphrastic tense form?

I previously mentioned that $z(i)$ - developed from demonstrative roots, which would make it a determiner-like element rather than a complementizer. Its occurrence on both the verb and the auxiliary in periphrastic tense forms could thus resemble determiner spreading (Kayne 1994, Alexiadou et al. 2006, Panagiotidis & Marinis 2011, Alexiadou 2014, Danon 2008), as found in Arabic/Hebrew or Greek (with some differences). It could also be compared to preposition doubling in Amharic (Baker & Kramer 2014), where in a clause with multiple adjectives prepositions are required on the first adjective and may also be repeated on the second. However, unlike determiner spreading and preposition doubling, $z(i)$ - is not doubled on nouns or adjectives, but on verbal elements.

Another possibility is that the double occurrence of $z(i)$ - is an instance of syntactic doubling similar to that of a *wh*-pronoun or a (strong) subject pronoun in Dutch dialects (Barbiers et al. 2010). In these dialects, there are cases of (identical and partial) syntactic doubling in *wh*-dependencies where more than one member of the same chain is spelled out, as in (64) and (65).

- (64) **Wie** denk je **wie** ik gezien heb?
who think you who I seen have
'Who do you think I have seen?' *Drenthe* (from Barbiers et al. 2010:2,(1))

- (65) **Zij** heeft **zij** daar niets mee te maken
she has she there nothing with to do
'She has got nothing to do with it.' *Flemish Brabant*
(from Barbiers et al. 2010:2,(2))

However, $z(i)$ - is not a pronoun. Yet the idea of “dependencies” helps address the problem. To find a solution, it is important to remember that this prefix occurs in, among other clauses, relative and comparative clauses. What do relative clauses and comparative clauses have in common? $\bar{A}$ -movement (Chomsky 1965, 1977).

With regard to relative clauses, the literature offers a variety of proposals for the syntactic processes at work in their derivation. The debate began with Chomsky 1965, who proposed a *matching* analysis of relative clauses, later discussed and extended in Sauerland 1998, 2003 and more recently reworked, modified and relabelled as the *double-headed* analysis in Cinque 2011, 2013, 2015, 2020. In the matching analysis, relative clauses have an internal (i.e. generated inside CP) and an external (i.e. generated outside CP) head (Carlson 1977). These two heads match, meaning that they are identical (or highly similar) but are not part of the same chain. Under identity, the internal head can be deleted following an $\bar{A}$ -movement of the external head in a position that c-commands it. The matching analysis was challenged by Vergnaud 1974, who proposed a *raising* analysis later revised by Kayne 1994. In this analysis, relative clauses have only one head, which is internal and $\bar{A}$ -moves from its argument position to the left periphery of the clause. The raising analysis was also disputed, leading to yet another proposal claiming that the head of relative clauses is actually generated outside CP, leaving an Operator in its argument position to $\bar{A}$ -move to the C° domain instead. This *operator* analysis was proposed by Montague 1974, Partee 1975, Chomsky 1977, Jackendoff 1977 among others. Today, these analyses are generally accepted as alternative derivations available in some languages but not in others. In other words, what emerges from the literature is that both matching and raising relative clauses exist and, crucially, they both involve $\bar{A}$ -movement.

With regard to comparative clauses, since Bresnan 1973 and Chomsky 1977, it has been established that $\bar{A}$ -movement is involved in comparative constructions. In his paper, Chomsky claims that comparative clauses contain a null *wh*-operator that undergoes $\bar{A}$ -movement to the front of the clause. This is supported by the fact that clauses such as *John is taller than Bill (is)* (Chomsky 1977:88,(58)) involve a phenomenon called Comparative Deletion (where part of the comparative clause is omitted — in this case

tall); this deleted element is thus a gap that must be bound by a null element (Chomsky 1977). Other works have also proposed that in English comparative clauses there is $\bar{A}$ -movement of the compared element (Donati 1997) or of the comparative element (Kennedy 2002) to Spec,CP, with consequent deletion under identity of the raised phrase with the first term of comparison. Moreover, it has been proposed that comparative clauses are actually relative-type clauses: Kayne 1994 claims that they are restrictive relative clauses in which the adjective raises, Donati 1997 suggests that they are free relatives selected by *more* in which a *wh*-element raises, and McCoy 2017 proposes that they are degree-phrase relative clauses in which the adjective is headed by a degree marker such as *-er*, *as* and *how* and this DegP (à la Abney 1987) raises.

With this in mind, $z(i)$ - could thus be a marker of $\bar{A}$ -movement. To substantiate this claim, one must show that there is indeed movement in all $z(i)$ -clauses. I do so a little later. For now, I would like to digress briefly on *wh*-interrogatives.

***Wh*-interrogatives**

If $z(i)$ - is indeed a marker of $\bar{A}$ -movement, one would expect it to appear also in *wh*-interrogative clauses if these involve movement in Tigrinya. Let us therefore look at how *wh*-interrogative clauses are formed in Tigrinya. Consider (66) and (67).

- (66) a. *nisxa ?intay belifi-ka*
 you.MS what eat.GER-SM.2MS
 ‘What did you eat?’
 b. **?intay nisxa belifi-ka*
 what you.MS eat.GER-SM.2MS
- (67) a. *nisxa ni-men ri?i-ka-yo*
 you.MS DOM-who see.GER-SM.2MS-OM.3MS
 ‘Who did you see?’
 b. **ni-men nisxa ri?i-ka-yo*
 DOM-who you.MS see.GER-SM.2MS-OM.3MS

Examples (66) and (67) show that the *wh*-element *?intay* ‘what’ cannot precede the subject *nisxa* ‘you’. Thus the only possible word order in *wh*-interrogative clauses is SOV

(as in declarative clauses), not OVS. However, the literature presents a different picture from the one obtained in my fieldwork. Consider (68).

- (68) a. Selam *ni-mən riʔ-a?*
 Selam ACC-who see.GER-3FSG.S
 ‘Who did Selam see?’ (from Gebregziabher & Duguine 2024:2, (2b))
- b. *ni-mən riʔ-a Selam?*
 ACC-who see.GER-3FSG.S Selam
 ‘Who did Selam see?’ (from Gebregziabher & Duguine 2024:2, (3b))
- c. *ni-mən Selam riʔ-a?*
 ACC-who Selam see.GER-3FSG.S
 ‘Who did Selam see?’ (from Gebregziabher & Duguine 2024:2, (4))

According to Gebregziabher & Duguine 2024, in Tigrinya *wh*-elements can occur in situ, as in (68a), but can also precede the subject and the verb, as in (68c), and can precede the verb when it precedes the subject, as in (68b).

The difference between the findings in Gebregziabher & Duguine 2024 and those of my fieldwork could reflect regional variation (to my knowledge, Gebregziabher is Ethiopian). But can one say that there is movement in interrogative clauses in Tigrinya or not? This question is interesting and a satisfactory answer would require further investigation. Gebregziabher & Duguine 2024 claim that the word order in example (68b) involves *wh*-movement of *ni-mən* followed by T-to-C movement of the verb *riʔa* ‘she sees’, and that (68c) again involves movement of *ni-mən* but this time without a consequent T-to-C movement of the verb. The authors provide several arguments in favor of movement, considering different phenomena like long-distance movement, reconstruction effects and islands.

However, the possibility for *wh*-elements in Tigrinya to appear in the positions illustrated above does not necessarily involve *wh*-movement. It may be that in Tigrinya *wh*-elements undergo clause-internal scrambling as in Japanese, Korean and Xining Mandarin Chinese. Japanese and Korean are SOV (like Tigrinya) and *wh*-in-situ languages in which *wh*-words can undergo scrambling (Grewendorf & Sabel 1999 and Law 2010 for Japanese; Beck & Kim 1997 for Korean). In Xining Mandarin Chinese (a dialect of Mandarin spoken in the Qinghai province), which is an SVO and *wh*-in-situ language, it has

also been argued that *wh*-elements can undergo scrambling (Bell 2019). It may also be that the OVS order involves right dislocation of the subject rather than *wh*-movement. Moreover, in my view, the examples provided contain sentences so short that they could be accounted for by alternative analyses not considered in the paper (such as topicalization or focalization). Therefore, for the current proposal that $z(i)$ - is a marker of $\bar{A}$ -movement, the absence of this prefix in *wh*-interrogatives is not problematic.

Moreover, it is possible that $z(i)$ - occurs only in embedded contexts and cannot appear in a root clause such as a *wh*-interrogative. In fact, as will be shown in more details in Chapter 3, $z(i)$ - *does* appear in embedded interrogatives, such as those in (69) and (70) (this anticipates the proposal of Chapter 3, namely that the prefix $k\epsilon m z(i)$ - is a bimorphemic element composed of the particle $k\epsilon m$ and the clausal prefix $z(i)$ -).

- (69) $\text{ʔaman ʔintay } k\epsilon m z\text{-}i\text{-}f\acute{e}ttu \quad \quad \quad \text{hatit-u-ni}$
 Aman what COMP-REL-SM.1S-like.IPFV ask.GER-SM.3MS-OM.1S
 ‘Aman asked me what I like.’

- (70) a. $\text{T\acute{e}sfai ni-Kidane ʔaman } k\epsilon m z\text{-}i\text{-}b\acute{e}liʃ$
 Tɛsfay to-Kidane Aman COMP-REL-SM.3MS-eat.IPFV
 nɛgir-u-wo
 tell.GER-SM.3MS-OM.3MS
 ‘Tɛsfay told Kidane that Aman ate.’
 b. $\text{T\acute{e}sfai ni-Kidane ʔaman ʔintay } k\epsilon m z\text{-}i\text{-}b\acute{e}liʃ$
 Tɛsfay to-Kidane Aman what COMP-REL-SM.3MS-eat.IPFV
 nɛgir-u-wo
 tell.GER-SM.3MS-OM.3MS
 ‘Tɛsfay told Kidane what Aman ate.’

Now, let us return to the two main questions addressed in this chapter: what is $z(i)$ - and how can it appear twice in a periphrastic tense form? So far, my first approximation is:

(First approximation): $Z(i)$ - is a marker of $\bar{A}$ -movement.

I still need to show that all $z(i)$ -clauses involve $\bar{A}$ -movement. To do so, it is important to establish first where exactly $z(i)$ - is located in the syntax. If this prefix were only a relative marker, as the literature has traditionally claimed, several options would be available: for example, Kayne 1994 and de Vries 2002 claim that relative markers occupy

Spec,CP; [Cinque 2015](#) suggests that they head C° ; and [Saito 1985](#) argues for a position in T° . However, as shown, $z(i)$ - does not only mark relative clauses and, crucially, it appears on both the verb and the auxiliary in periphrastic tense forms. Therefore, what syntactic position could account for this phenomenon?

Suppose an element — whose nature I have not yet identified — raises in the structure of a $z(i)$ -clause from a low argument position to a higher $\bar{A}$ -position, perhaps in the left periphery. This element would cross both the verb and the auxiliary, the elements to which $z(i)$ - attaches. Why, then, is $z(i)$ - attached to both? Possibly because this element cannot move directly from a low position to a higher one without intermediate stopping points. In fact, this scenario resembles what happens in long-distance movement.

Reflexes of Successive-Cyclic Movement

Since the early 1970s it has been claimed that long-distance movement of an element base-generated in an embedded clause to an $\bar{A}$ -position in the left periphery of a matrix clause is not carried out in one step but proceeds through smaller steps and intermediate positions. In other words, *wh*-movement (or $\bar{A}$ -movement) must be cyclic ([Chomsky 1973, 1977, 2000c, Müller 1995, 2011](#)). Consider (71).

- (71) **Who_i** did Mary hope [that Tom would tell Bill [that he should visit t_i]] ?
 (from [Chomsky 1977:74,\(10\)](#))

In the sentence above, the *wh*-word *who* has not moved directly from its base-generated position (where it left a trace) to its surface position at the beginning of the sentence but has stopped along the way. In more technical terms, in the context of long-distance $\bar{A}$ -movement, XP (*who* in (71)) moves (and merges) from Spec,CP to Spec,CP, establishing a relation between itself and the head C° ; this relation is known as a Spec-head AGREE relation ([Chomsky 2000a](#)). These intermediate steps leave no phonologically overt traces in English. However, this is not the case in all languages. In some languages, successive-cyclic $\bar{A}$ -movement leaves reflexes along the structure ([Georgi 2017](#)).

In Irish, for example, finite complement clauses are introduced by the particle *go*, as in

(72). Finite clauses out of which elements move to an $\bar{A}$ -position (such as relative clauses) are instead introduced by the particle *aL*, as in (73).

(72) Creidim gu-r inis sé bréag
 I-believe go-[PAST] tell he lie
 ‘I believe that he told a lie.’ (from McCloskey 2002:189, (8))

(73) an ghirseach a ghoid na síogaí
 the girl aL stole the fairies
 ‘the girl that the fairies stole away’ (from McCloskey 2002:189, (9a))

In Passamaquoddy (Algonquian), when a relative clause is formed by long-distance movement, optional participial agreement can appear on the verbs along the path of the dependency (Bruening 2001). In Wolof (Niger-Congo), *wh*-movement triggers class markers (called *u-forms* because they appear as *-u* suffixes) on the verbs between the extraction site and the landing position of the *wh*-element (Torrence 2012). This is similar to Seereer (Niger-Congo), where verbal morphology is also extraction sensitive Baier 2014. Finally, van Urk 2015 and van Urk & Richards 2015 show that in Dinka (Nilo-Saharan), extraction of a plural DP from its argument position triggers the appearance of the element *ke* next to every verb crossed along the path of movement.

In the literature these reflexes are generally considered to be markers of an agreement relationship between the extracted phrase XP (a nominal, a *wh*-element, an operator) and the head in whose specifier the XP has landed. For example, in Irish, “the moved phrase in Spec,CP deposits a mark on the head C° ... forc[ing] C° to be realized as *aL*” (McCloskey 2002:186). The order in which the operations MERGE and AGREE are applied is crucial: according to Georgi 2017, if in a language MERGE takes place before AGREE there are no reflexes of movement, whereas if AGREE takes place *before* MERGE, reflexes of movement arise. In the latter case, the probe on a projection F is not valued because at the point where it probes there is no goal in its search domain (Spec, FP), XP moving there only afterwards. Without a value on F there can be no morphological realization of it and hence no reflex of movement (Georgi 2017:596).

But what does all this have to do with *z(i)*- and the fact that it appears twice in a

monoclausal periphrastic tense construction? The answer lies in what the intermediate positions mentioned above actually delineate: phases.

Phases

According to Phase Theory, launched in [Chomsky 1999, 2000a, 2001](#) and expanded in [Chomsky 2004, 2008](#), $\bar{A}$ -movement is successive-cyclic because the output of syntactic operations is sent to the Sensorimotor (Phonological Form, PF) and the Conceptual-Intentional (Logical Form, LF) interfaces not all at once but in stages ([Bruening 2001](#)). These stages are called *phases*. Phases are thus domains that “exhibit a degree of syntactic, semantic, and phonological independence from the rest of the computation” ([van Craenenbroeck et al. 2020](#)). In other words, phases are specific points during a syntactic derivation when the language faculty interacts with the two cognitive systems PF and LF ([Citko 2014](#)). Once a phase is sent to the interfaces, its content is no longer accessible to the derivation. This is the Phase Impenetrability Condition (PIC) ([Chomsky 2000a](#)). However, the PIC states that elements merged at the *edge* of a phase - the head and the specifier - remain visible and accessible to computation until the transfer of the subsequent phase. In other words, the idea of phases is based on the fact that the syntactic operation TRANSFER (whose function is to send the syntactic output of each phase to the interfaces) has a “memory”, which requires phases to be as small as possible ([Chomsky 2004:116](#)). Therefore, the only possible intermediate landing sites for elements that raise from their argument positions and undergo $\bar{A}$ -movement are phase edges ([Chomsky 2004, Abels 2012](#)). Is Spec,CP, mentioned in the previous section as the landing site of the movement of *who* in example (71), a phase edge? Yes.

CP is accepted as a phase by the majority of scholars ([Chomsky 2008](#)).²⁵ That said, the literature abounds with works on phase theory (a.o. [Gallego 2010, Bošković 2014, Citko 2014, van Urk 2020](#)) and there is not a perfect consensus on which phrases constitute a phase. For example, regarding v*P (v* being the functional head associated with

²⁵However, there is no consensus in the literature on which projection marks the edge of this phase between Rizzi’s left-peripheral ForceP and FinP ([Rizzi 1997](#)), leading Chomsky to describe the phase CP as a *region* rather than a specific projection ([Chomsky 2008:143](#)).

argument structure (Kratzer 1996)), there is no clear agreement: v^*P is treated as a phase by Chomsky 2000b, 2001, Sato 2012, van Urk & Richards 2015, Georgi 2017, but not by Keine & Zeijlstra 2021. There is also ongoing debate on the phasal status of TP (Deal 2016, Wurmbrand 2017) and even DP, NP, QP and NumP (van Craenenbroeck et al. 2020). Some authors claim that different aspectual layers belong to different syntactic cycles: for example, Perfect aspect would belong to the CP phase, while Progressive aspect to the v^*P phase (Wurmbrand 2017:346). Moreover, some scholars argue that each aspectual projection constitutes its own phase: for instance, Harwood 2014, Ramchand 2017, Sheehan & Cyrino 2024 claim that VoiceP/ProgP is a phase.

In section 1.6 at the beginning of this chapter, I proposed a syntactic analysis of declarative root clauses in Tigrinya. Regarding periphrastic tense forms, where the verb occurs with an auxiliary, I argued that these are monoclausal constructions that nonetheless contain two aspectual layers, two vP layers, and two VP layers, following (and adapting) Al-Aqarbeh & Al-Sarayreh 2017. In the previous paragraph I mentioned that vP is considered (by most scholars) to be a phase. Moreover, some authors suggest that aspectual layers are themselves phases.

These claims and findings in the literature can be used to explain the occurrence of $z(i)$ - on both the verb and the auxiliary in periphrastic tense forms: in Tigrinya the verb and the auxiliary in periphrastic tense forms each define a phase boundary through which elements undergoing $\bar{A}$ -movement must pass. The element that moves in $z(i)$ -clauses thus undergoes $\bar{A}$ -movement by stopping at the edge of the phasal domains of the verb and the auxiliary, resulting in $z(i)$ - attaching to both. The phonological realization of this clausal prefix is therefore the result of a Spec-Head agreement relation between an $\bar{A}$ -moved element in the specifier position of the two aspectual phrases and the verb and the auxiliary moving in the head of these two projections.²⁶

²⁶Note that when I claim that the requirement for $z(i)$ - to appear is a Spec-Head agreement, I am not proposing that the aspectual projection is a criterial position in the sense of (Rizzi 1996, Bocci 2007, Shlonsky & Rizzi 2018), because such positions are dedicated to scope-discourse properties and because the phrase that lands in the specifier is frozen and is unavailable to further movement operations (Shlonsky & Rizzi 2018:29).

In other words, I can update the definition of this prefix as follows:

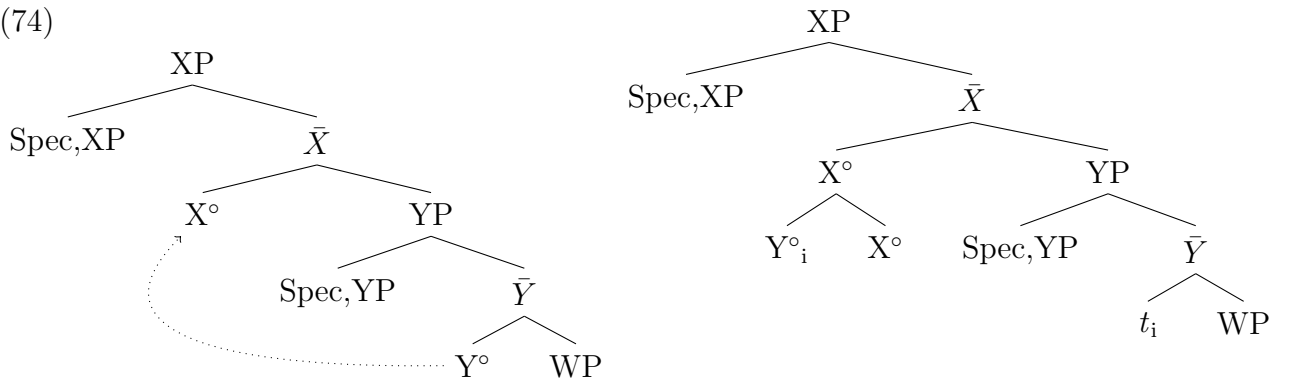
$Z(i)$ - is a reflex of successive-cyclic $\bar{A}$ -movement.

The definition of $z(i)$ - above seems complete; however, it remains crucial to explain why this element is a clausal *prefix* and what “reflex of successive-cyclic movement” means syntactically in Tigrinya.

$Z(i)$ - as a prefix

I mentioned above that $z(i)$ -clauses involve the movement of an element from its argument position to the left periphery of the clause. However, the appearance of $z(i)$ - is also triggered by the movement of the verb (and the auxiliary in periphrastic tense forms), and it is precisely on this that I wish to focus for a moment. As shown in section 1.6, where I proposed a syntactic derivation for declarative clauses in Tigrinya, I suggest that the verb raises in the structure via head-movement. The literature abounds with works on head-movement (both for and against), from the seminal studies [Koopman 1984](#), [Travis 1984](#), [Baker 1985](#), [Pollock 1989](#) to more recent approaches ([Brody 2003](#), [Matushansky 2006](#), [Dékány 2018](#)). I do not provide a summary of this literature but point out only that this type of movement has been argued to result obligatorily in adjunction to a head ([Matushansky 2006](#), [Harley 2011](#), [Baker 1988](#)), with the consequent morphological growth of the head ([Dékány 2018](#)), as shown in (74) (adapted from [Matushansky 2006:72,\(2\)](#)).

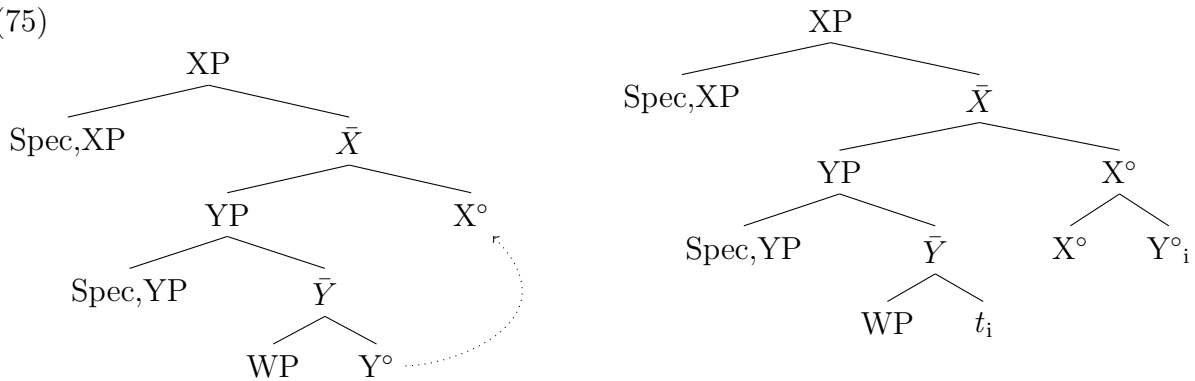
(74)



Y° forms a complex head with X° by landing on its left. Note also that the two trees above are head-initial. What happens when the structure is head-final? I suggest that

we find the exact mirror image: when Y° raises in a head-final structure, it lands on the right of X° , as in the trees below.

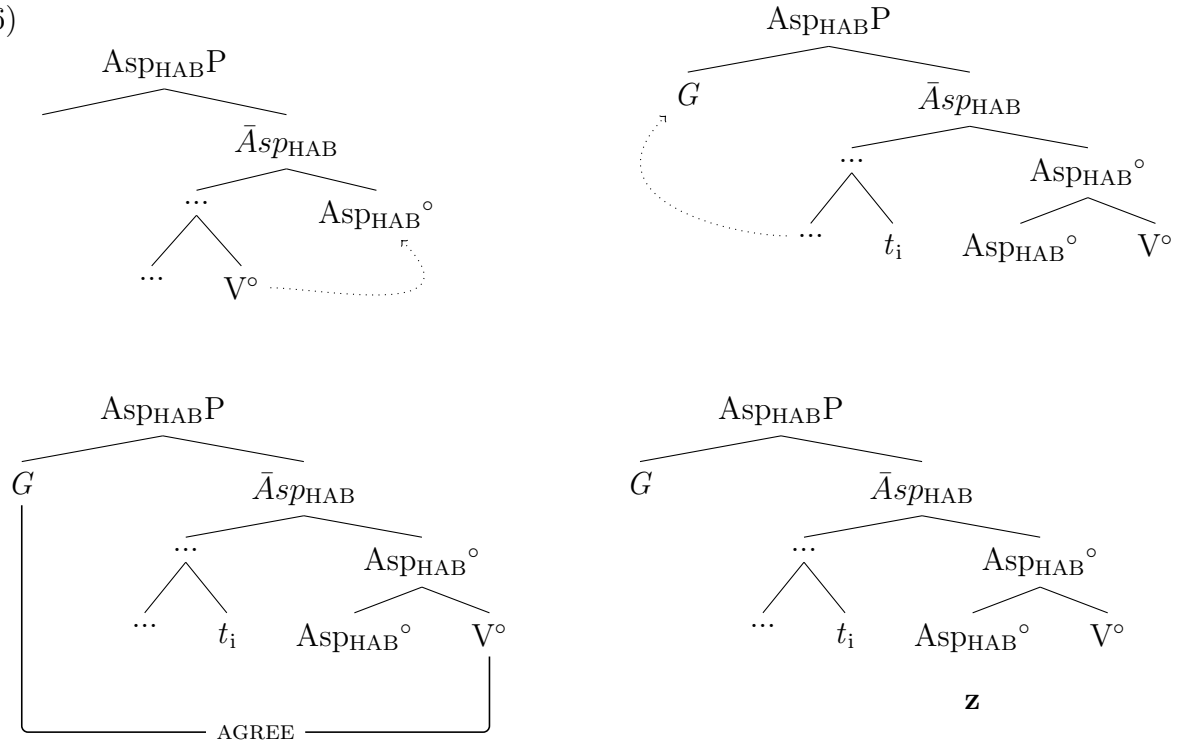
(75)



Returning to $z(i)$ -, the trees above help explain the process underlying its realization. An element (not yet defined, let us call it G) in $z(i)$ -clauses $\bar{A}$ -moves from its argument position to the left periphery. The verb (and the auxiliary if present) moves from V° to $\text{Asp}_{\text{HAB}}^\circ$ as in declarative clauses. When the $\bar{A}$ -moved element lands in the specifier of $\text{Asp}_{\text{HAB}}P$, which I consider an $\bar{A}$ -position, a Spec-head agreement is established and $z(i)$ - is phonologically realized. Importantly, the triggering of $z(i)$ - occurs only after a Spec-Head agreement relationship established by the head of an $\text{Asp}_{\text{HAB}}P$. This head must be filled by the verb; empty heads or other phrases cannot trigger $z(i)$ -. This distinction is crucial, because otherwise, as will become clearer later, CP could also potentially trigger it — which, in fact, does not occur.

This mechanism is illustrated in (76) (the complete syntactic derivation of $z(i)$ -clauses will be provided later).

(76)



Beside explaining how $z(i)$ - is prefixed onto the verb, the trees above also show that $z(i)$ - is a head (a X° element or terminal node) and not a phrase (an XP element). This aligns with the literature on relative markers in languages closely related to Tigrinya, which also claim that these markers are heads: for example, [Baye 1987](#) suggests that the relative marker *kan* in Oromo (Cushitic) is a X° in C° ; [Alemayehu 1990](#) proposes that the relative marker *yä-* in Chaha (Semitic) is also a X° in C° and [Den Dikken 2007](#) and [Baker & Kramer 2014](#) claim that the relative marker *yä-* in Amharic is also a X° in C° .

At this point, the last possible question before turning to the syntactic derivation of $z(i)$ -clauses, is whether $z(i)$ - carries any particular feature. Given its origin as a demonstrative root, perhaps $z(i)$ - carries a $[D]$ feature. However, these features typically encode gender, number, case, or location (DIST or PROX) specifications ([Leu 2015](#)), which are not overtly realized on $z(i)$ -. I therefore leave this question open.

Let us now turn to the syntactic analyses of $z(i)$ -clauses, starting with relative clauses.

2.3.2 Analysis of Relative Clauses

Before taking a stand with regard to the structure of relative clauses in Tigrinya as a whole, it is important to review what has been proposed in the literature so far. Note that, as shown in section 2.2.2, $z(i)-$ appears in all kinds of relative clauses, ranging from object restrictive to amount/maximalizing, and from subject non-restrictive to free and light-headed. For reasons of space, in this dissertation I provide a syntactic analysis only for subject and object *restrictive* relative clauses leaving the other types for future research.

Previous Accounts

To my knowledge, there are only two fairly detailed analysis of relative clauses in Tigrinya: Overfelt 2009 and my own previous article, Cacchioli 2023.²⁷ In Overfelt 2009, it is argued that $z(i)-$ is a complementizer, or at least heads a CP projection. Apart from relative clauses, Overfelt does not discuss other constructions in which $z(i)-$ occurs, except for a brief mention of clauses introduced by $k\epsilon m z(i)-$ — which, as already stated, I claim in Chapter 3 to contain $z(i)-$. In his study, Overfelt suggests that prefixation of $z(i)-$ in relative clauses is obtained through I-to-C head-movement of the verb (Overfelt 2009:50): because Imperfective and Perfective verbs move under C° , they are able to carry the relative prefix. What is missing in Overfelt 2009 is an account of the presence of $z(i)-$ on both the verb and the auxiliary in periphrastic tense forms.²⁸

In my previous work, Cacchioli 2023, written within a Cartographic framework (Rizzi 1997, Cinque 1999), I analyzed relative clauses (and declarative clauses) in Tigrinya within an LCA and antisymmetric framework (Kayne 1994). I claimed that $z(i)-$ heads what I called — for lack of a better term — $z(i)P$ projections merged above the aspectual projections Asp_{PROG} (where the auxiliary is base-generated) and $Asp_{IMPERFECTIVE}$ (where the verb moves). Looking back, I think that this paper presented several problems, which I mention here to motivate the analysis provided later in this dissertation. First, the two

²⁷See also Kayne 1994, Ouhalla 2004 and Den Dikken 2007 for detailed analysis of relative clauses in Amharic.

²⁸Unfortunately Jason Overfelt did not encounter this phenomenon, presumably because clauses in progressive aspect were not investigated in his fieldwork (p.c.).

$z(i)$ P projections were rather *ad hoc*, despite being labelled as such only provisionally. If $z(i)$ - could project a phrase, then by analogy all prefixes and elements in Tigrinya could project their own phrases, which is implausible. Second, there is no principled reason to posit two identical projections — the two $z(i)$ Ps — when the working hypothesis of the Cartographic approach holds that there should be only “one feature, one head” (Rizzi & Cinque 2016:155). Third, the analysis involved unnecessarily complex syntax with multiple movements, including extraction of remnant material from CP (also allowed in Cinque 2020:68), which does not seem computationally economical or syntactically plausible. In this dissertation I therefore distance myself from the analysis proposed in Cacchioli 2023 and argue for a simpler syntax.

The two works mentioned above also differ in where they place the relativized head in the structure. According to Overfelt 2009 the head of the relative clause in Tigrinya is outside CP and what raises in the structure is an operator, whereas in Cacchioli 2023 the head is internal and moves from its argument position to the C° domain, following a raising analysis of relative clauses. Moreover, previous accounts of relative clauses in Ethio-semitic place the relativized head in an external position: Baker & Kramer 2014 for Amharic and Alemayehu 1990 for Chaha. In Cacchioli 2023 I did not provide supporting arguments for an internal position of the relativized head. Here I do.

Where is the Relativized Head?

Previous analyses of Ethio-Semitic relative clauses propose that the relativized head is merged in an external position (Alemayehu 1990, Baker & Kramer 2014), likely due to the prenominal nature of relative clauses. However, these studies do not provide syntactic evidence to support this claim.

As mentioned above, Overfelt 2009 suggests that the relativized head in Tigrinya relative clauses is placed externally to CP. This work offers two arguments for this: licensing of Negative Polarity Items (NPI) and Case Assignment. The first argument relies on the fact that, in English, NPIs such as *any* must be locally licensed by negation (Jespersen 1917, Klima 1964), as in (77).

- (77) a. I didn't see **anything**.
 b. *I saw **anything**.

Overfelt 2009 claims that this is also true for the Tigrinya element *wala* 'any', as shown in (78). Note that if this element is followed by the feminine numeral *hanti* not followed by a noun, it means 'anything'. Sentential negation, that will be investigated in Chapter 5, is expressed by the circumfix *?aj-...-(i)n* attached the verb in Tigrinya, as in (78).

- (78) a. Tesfai **wala** hanti ?aj-Ø-bel?i-n
 Tesfay any one.FS NEG-SM.3MS-eat.IPFV-NEG
 'Tesfay does not eat anything.'
 b. *Tesfai **wala** hanti ji-beli?
 Tesfay any one.FS SM.3MS-eat.IPFV
 'Tesfay does not eat anything.'

Building on Cole & Hermon 1994, Overfelt 2009 argues that NPIs in Tigrinya must be c-commanded by negation; therefore, if the embedded verb in a relative clause is negative and the relativized head contains an NPI, that NPI can be licensed only if it is internal to CP and not external (Overfelt 2009:39-40). Example (79) is given as evidence.

- (79) *Nahor **wala** hanti ?it-i sebbaj z-i-ts' əħafə-n mets' haf
 Nahor any one.F DIST-MS man REL-NEG-write.PFV-SM.3MS-NEG book.FS
 ʕadəgə
 write.PERF.SM.3MS
 'Nahor bought any books that the man did not write.'
 (from Overfelt 2009:44, (51))

Because the sentence in (79) is ungrammatical, the conclusion is that the relativized head must be external. However, it is important to point out that the example in (79) might be ungrammatical for other reasons: as explained in Chapter 5, the suffix *-(i)n* of the negative circumfix does not appear in relative clauses; the prefix *?aj-* does, but it is absent in Overfelt's sentence. In addition, according to my consultants, the NPI *wala hanti* 'any (FS)' cannot be detached from the noun it modifies, *mets' haf* 'book' (this may reflect regional variation). Consider example (80), which is the "corrected" version of (79), and

(81) which is its grammatical counterpart.

- (80) *Nahor ?it-i seɓ?aj z-aj-ts'ɛħafɛ **wala** ħanti mets'ħaf
 Nahor DIST-MS man REL-NEG-write.PFV-SM.3MS-NEG any one.F book.FS
 ʔadɛgu
 write.GER.SM.3MS
 'Nahor bought any books that the man did not write.'
- (81) Nahor ?it-i seɓ?aj zi-ts'ɛħaf-a **wala** ħanti mets'ħaf
 Nahor DIST-MS man REL-write.PFV-SM.3MS-OM.3FS any one.F book.FS
 ʔaj-ʔadɛg-i-n
 NEG-write.PFV.SM.3MS-NEG
 'Nahor didn't buy any books that the man wrote.'

Following Overfelt's logic, example (80) and (81) support an external-head analysis.

The argument in [Overfelt 2009](#) assumes that NPIs must be c-commanded by negation to be licensed. This relies on English-like behavior in object position, as in (78). However, unlike English, Tigrinya *wala* 'any' can also occur in subject position, as shown below.

- (82) a. ***Anybody** didn't see me.
 b. ***Anybody** saw me.
 c. Nobody saw me.
- (83) a. **wala** ħadɛ seɓ ʔinɕɛra ʔaj-Ø-bɛlʔi-n
 any one.M human injera NEG-SM.3MS-eat.IPFV-NEG
 'Nobody eats injera.'
- b. ***wala** ħadɛ seɓ ʔinɕɛra ji-bɛliʔ
 any one.M human injera SM.3MS-eat.IPFV
 'Nobody eats injera.'

Since English NPIs are ungrammatical in subject position, Tigrinya *wala* 'any' element is likely not an NPI (also contra [Cao & Liotta 2023](#) and [Abraham 2024](#)). It patterns more like a negative quantifier ([Zeijlstra 2020](#)), comparable to Italian *nessuno* 'nobody', which is grammatical in subject position and requires negation only postverbally ([Zeijlstra 2016](#)), as in (84).

- (84) a. **Nessuno** vuole andare a casa.
 nobody wants go to home
 ‘Nobody wants to go home.’
- b. Non vuole andare a casa **nessuno**.
 not wants go to home nobody
 ‘Nobody wants to go home.’
- c. Non ho visto **nessuno**
 not have seen nobody
 ‘I didn’t see anybody.’

This means that the ungrammaticality of (80) cannot be straightforwardly attributed to lack of c-command and hence lack of licensing of *wala*. If negation can license *wala* in subject position without c-command, then it should also license a negative relative head without c-command.

Another potential problem with the argument in Overfelt 2009 is that even if *wala* were an NPI, it would be unsuitable for testing headedness of relative clauses in Tigrinya because NPIs do not reconstruct. For example, English NPIs cannot be topicalized (Heycock 1995) because in a topic position they would be outside the scope of c-commanding negation and cannot be licensed in their base-generated position below negation, as illustrated in (85).

- (85) a. I don’t like any country song.
 b. *Any country song, I don’t like.

To determine whether Tigrinya relativized heads are internal or external to CP, we must therefore test elements that *do* reconstruct. To locate the head of relative clauses in Tigrinya, I use binding principles, namely Condition A and Condition C (a.o. Chomsky 1993). According to Condition A, a reflexive pronoun (an anaphor) must be c-commanded by a locally merged antecedent (i.e., within the same CP), as in (86).

- (86) Səɣen ʔab məsɪtəwat nɪnɛbs-a rɪʔij-a
 Səɣen LOC mirror self-FS see.GER-SM.3FS
 ‘Səɣen saw herself in the mirror.’

- (87) a. $\text{ʔit-i} \quad \text{q}^{\text{w}}\text{olʔa} \text{ʔit-a} \quad \text{ʔade} \quad \text{kemzi-ħats'eb-et-o}$
DIST-MS child DIST-FS mother kemzi-wash.PFV-SM.3FS-OM.3MS
ħasib-u
think.GER-SM.3MS
‘The child_i thought that the mother washed him_{i/j}.’
- b. $*\text{ʔit-i} \quad \text{q}^{\text{w}}\text{olʔa} \text{ʔit-a} \quad \text{ʔade} \quad \text{ninεbs-u}$
DIST-MS child DIST-FS mother self-MS
kemzi-ħats'eb-et-o ħasib-u
kemzi-wash.PFV-SM.3FS-OM.3MS think.GER-SM.3MS
Intended: ‘The child thought that the mother washed himself.’

In (86), *ninεbsa* ‘herself’ is licensed by the local antecedent *Segen*; in (87) *ninεbsu* ‘himself’ lacks a local antecedent and is ungrammatical. Consider now relative clauses.

- (88) *Context: In Ruth’s dream there are many reflecting surfaces. She sees herself in a window and looks younger; she sees herself in a mirror and looks older,...*

$\text{Rut} \quad \text{ʔab} \quad \text{mesitewat} \quad \text{zi-riʔij-et-a} \quad \text{ninεbs-a} \quad \text{ʔaregit}$
Ruth LOC mirror REL-see.PFV-SM.3FS-OM.3FS self-FS old
nεjir-a
BE₂.GER-SM.3FS

‘Herself that Ruth saw in the mirror was older.’

This example is grammatical. Therefore the antecedent of *ninεbsa* ‘herself’, *Ruth*, must be in the same clause as the reflexive. If both are in the same CP, the relativized head is internal to CP.

Now consider Condition C, which states that an R-expression cannot be bound; its antecedent (a pronoun) cannot c-command it.

- (89) $\text{nissu} \quad \text{n-iz-i} \quad \text{sεbʔai} \quad \text{jī-fiqir-o}$
he DOM-PROX-MS man SM.3MS-love.IPFV
‘He_i loves this man_{*i/j}.’
- (90) $\text{nissa} \quad \text{mis} \quad \text{kεid-a}, \quad \text{Sεgen} \quad \text{ħadim-a}$
she when leave.GER-SM.3FS Segen flee.GER-SM.3FS
‘When she_{i/j} left, Segen_i ran away.’

In (89), *nizi sεbʔai* ‘this man’ is c-commanded by *nissu* ‘he’, so coreference is impossible;

in (90), *Segen* is not c-commanded by *nissa* ‘she’, so coreference is possible. Now consider a relative clauses.

- (91) ʔit-i nissu z-i fiqir-o sɛbʔai tʰibur ʔijj-u
DIST-MS he REL-SM.3MS-love.IPFV man nice BE₂.PRES-SM.3MS
‘The man_i that he_{*i/j} loves is nice’

Coreference between *nissu* ‘he’ and *sebʔai* ‘man’ is ungrammatical, which means the pronoun c-commands the noun. Hence the relativized head must be internal to CP (or possibly merged internally and raised out but able to reconstruct). The binding facts in Tigrinya thus support the claim that the relativized head is internal to CP.

Now, let us consider another phenomenon proposed by [Overfelt 2009](#) in support of an externally-headed analysis of relative clauses in Tigrinya: Case assignment. By assuming that Case is assigned via AGREE ([Chomsky 1995](#)), [Overfelt 2009](#) claims that in sentences such as (92) the head *ʔita gobije* ‘the turtle’ must be external to CP because, if base-generated internally, it would receive accusative case as the complement of the verb *gezapikuwa* ‘I bought her’, making the *n(i)*- marker grammatical.

- (92) (*n)-ʔit-a zi-gezapi-ku-wa gobije ʕimbaba ti-fettu
DOM-DIST-FS REL-buy.PFV-SM.1S-OM.3FS turtle flowers SM.3FS-like.IPFV
'The turtle that I bought likes flowers.'

The demonstrative *ʔita* ‘the’ must therefore be merged outside CP, dominating it (Bianchi 1999) and thus, receives Nominative and not Accusative case. Case assignment facts in Tigrinya thus support the claim that the relativized head is external to CP.

Because binding principles have shown that the head of relative clauses in Tigrinya must be internal and case assignment facts have shown the opposite, namely that the head must be external, I adopt a matching analysis of relative clauses following a.o. [Chomsky 1965](#) and [Sauerland 1998](#), which claim that the relativized head is both internal and external to CP.

I now turn to the syntactic analyses.

We begin with restrictive subject and object relative clauses containing simple verb forms, such as those in (93) and (94).

- I propose that in Tigrinya restrictive relative clauses are c-commanded by the demonstrative associated with the relative head — for instance, *ʔitom* ‘the’ in (93) and *ʔiti* ‘the’ in (94)— following Bianchi 1999. This has also been proposed for Amharic, where relative clauses are argued to be CPs (Den Dikken 2007:302 and Baker & Kramer 2014; contra Kayne 1994). As noted earlier, relative clauses are primarily prenominal, like attributive adjectives. I therefore follow Baker & Kramer 2014 and suggest that they are generated in the specifier of a functional projection FP à la Cinque 2010.²⁹

²⁹I set aside the question of whether relative clauses are complements (a.o. [Kayne 1994](#), [Arsenijević 2009](#)) or adjuncts (a.o. [Bianchi 1999](#)).

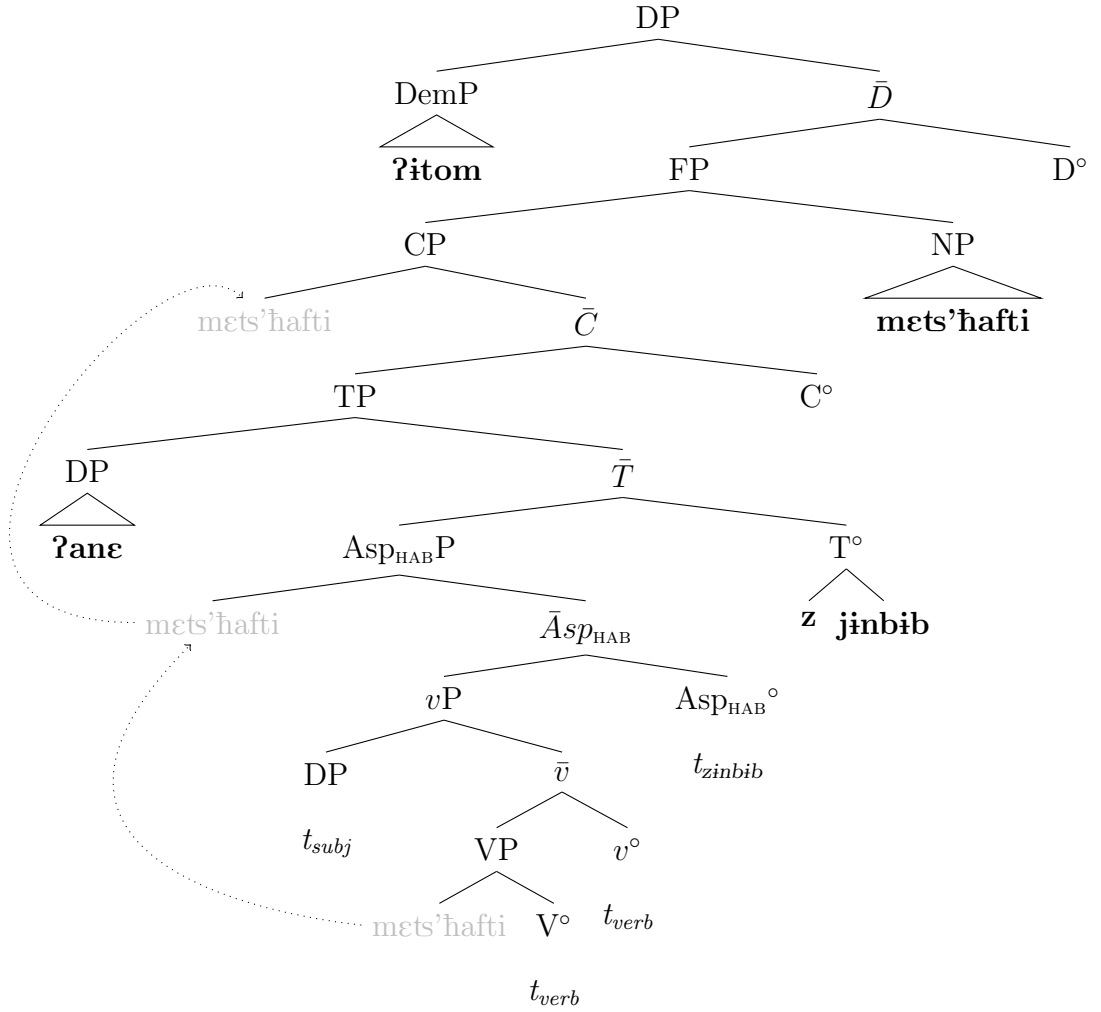
functional projection hosting the relative CP in its specifier, without affecting the matching analysis. To acknowledge these possibilities while keeping the representations in the trees manageable, I will depict the external head as an NP attached to an FP projection, without committing to a single structural alternative.

Finally, note that in the object relative clause in (93), object morphology appears on the relativized verb, whereas it does not appear in the subject relative clause in (94). In the literature on Ethio-Semitic, object markers have been claimed to be the result of agreement between the ϕ -features on v° and an internal argument (Kramer 2014, Baker & Kramer 2018, Gebregziabher 2021, Overfelt 2022), realized as agreement morphology at PF. I leave the investigation of object morphology in relative clauses to future research.³⁰

A restrictive object relative clause, such as that in (93) can be represented as in the tree in (95).

³⁰This is part of an ongoing project on the syntax and semantics of headless and light-headed relative clauses in Tigrinya with Ivano Caponigro.

(95)

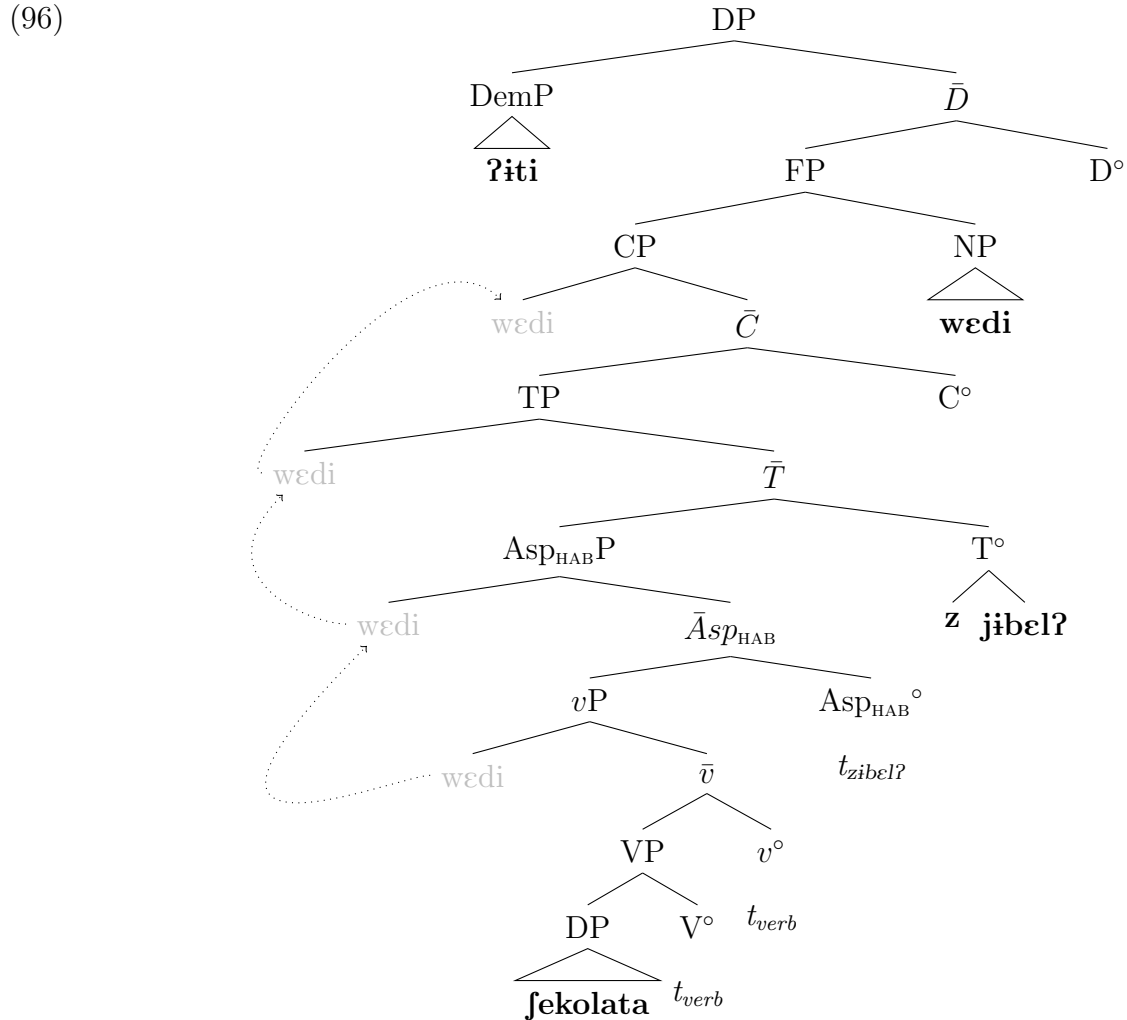


In the derivation above, the relativized head *mɛts'hafti* ‘books’ is base-generated both internally to the relative CP and externally. The internal copy undergoes $\bar{A}$ -movement from its argument position to the left periphery of the clause. This movement can be explained in terms of feature checking (Chomsky 1995): the C° head bears an uninterpretable [uRel] (Rizzi & Bocci 2017) or [u $\bar{A}$]³¹ feature that can only be checked by an operator carrying an interpretable counterpart. The internal copy moves successively-cyclically, stopping in the specifier of the cyclic domain $Asp_{HAB}P$, which I take to be an $\bar{A}$ -position. An AGREE relationship between the relativized head and the verb, which has already moved to Asp_{HAB}° , is established, and *z(i)-* surfaces as a prefix on the verb. The complex verb then moves under T° . When the relativized head reaches Spec,CP it is

³¹Probably not [uwh] given that we have seen that *z(i)-* does not appear in matrixinterrogative clauses where movement also (might) occurs.

deleted under identity with the external copy that c-commands it.

A restrictive subject relative clause, such as that in (94) can be represented as in the tree in (96).



In the derivation above, the relativized head *wɛdi* ‘boy’ is base-generated both internally to the relative CP and externally. The internal copy undergoes $\bar{A}$ -movement from its argument position to the left periphery of the clause. This movement proceeds successively-cyclically, stopping in the specifier of the cyclic domain $\text{Asp}_{\text{HAB}}\text{P}$. An AGREE relationship between the relativized head and the verb, which has already moved to $\text{Asp}_{\text{HAB}}^\circ$, is established, and *z(i)*- surfaces as a prefix on the verb. The complex verb then moves under T° . When the relativized head reaches Spec,CP (having also passed through Spec,TP as subjects do) it is deleted under identity with the external copy that c-commands it.

The relative clauses represented above express present habitual aspect. I now turn to

those expressing progressive aspect and, finally, past tense.

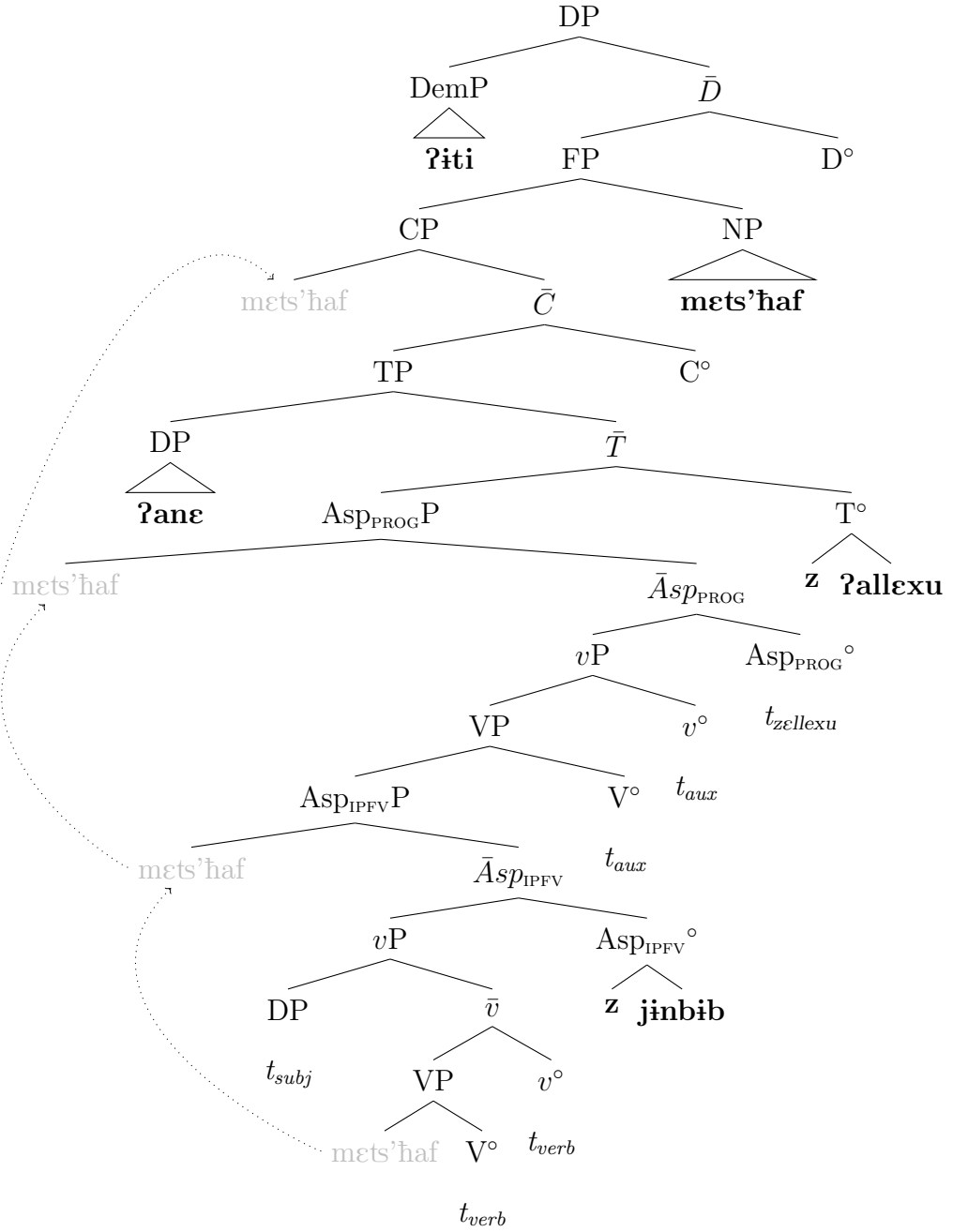
Analysis of Relative Clauses with Periphrastic Tense Forms

We now turn to relative clauses with periphrastic tense forms expressing progressive aspect, such as those below.

- (97) ?it-i ?ane z-i-nbib-o z-ell-exu mets'haf
DEM.MS I REL-SM.1S-read.IPFV-OM.3MS REL-BE₁.PRES-SM.1S book.MS
'The book that I am reading...'
- (98) ?it-i fokolata z-i-belif z-ell-o wedi
DIST-MS chocolate.FS REL-SM.3MS-eat.IPFV REL-BE₁.PRES-SM.3MS boy
'The boy who is eating chocolate...'

Recall from Section 1.6 that I proposed that these forms, in which a verb appears with an auxiliary, are monoclausal but display two aspectual layers. An object relative clause, as that in (97) is illustrated in the tree in (99).

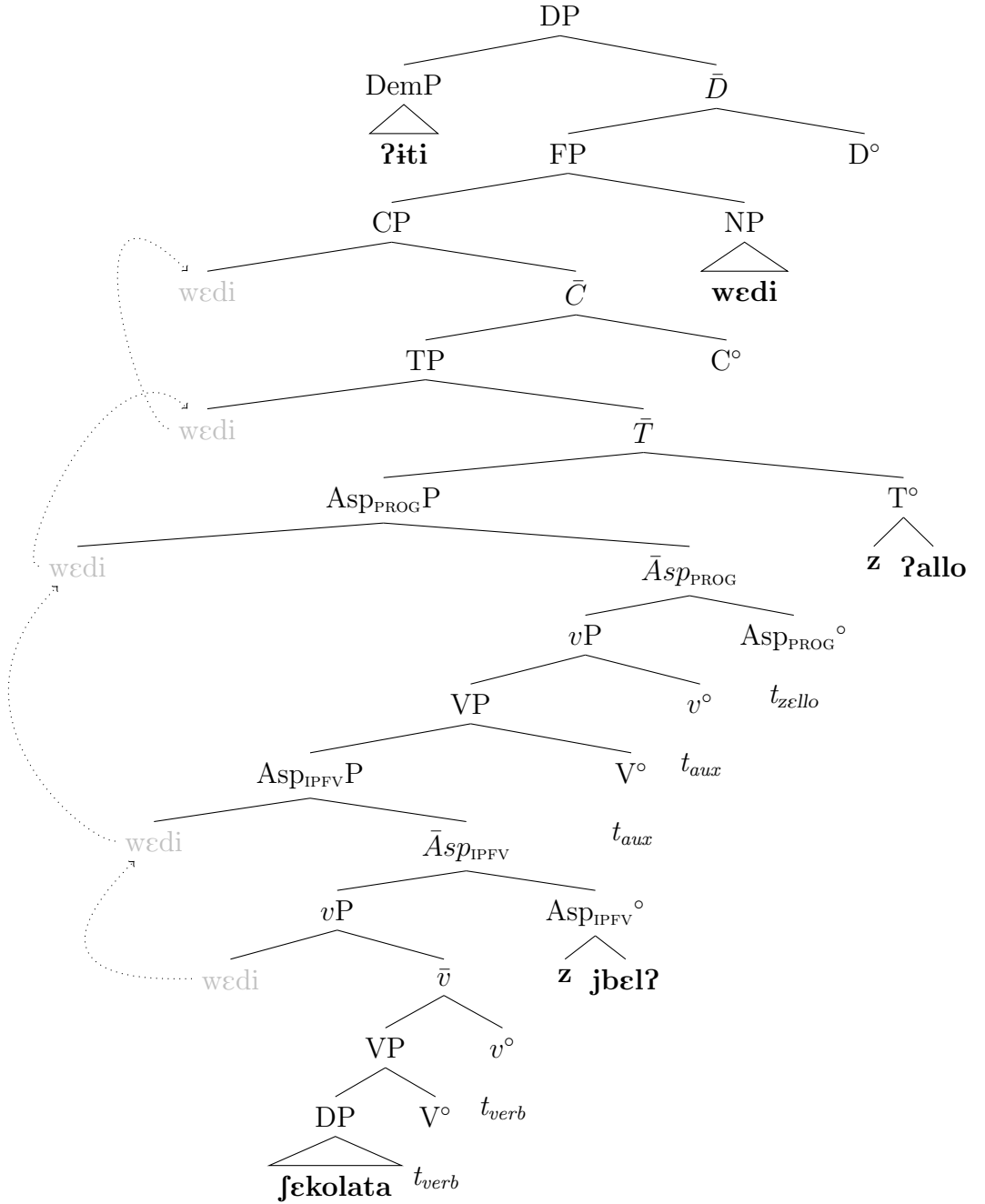
(99)



In the derivation in (99), the relativized head *mɛts'haf* ‘book’ is base-generated both internally to the relative CP and externally. The internal copy undergoes $\bar{A}$ -movement from its argument position to the left periphery of the clause. This movement proceeds successive-cyclically, stopping in the specifier of the cyclic domains $\text{Asp}_{\text{IPFV}}\text{P}$ and $\text{Asp}_{\text{PROG}}\text{P}$. AGREE relationships between the relativized head and both the verb and the auxiliary are established, and $z(i)$ - surfaces as a prefix on each. The complex consisting of $z(i)$ - and the auxiliary then moves under T° . When the relativized head reaches Spec,CP it is deleted under identity with the external copy that c-commands it.

A subject relative clause, as that in (98), is represented in the tree in (100).

(100)



In the derivation in (100), the relativized head *wɛdi* ‘boy’ is base-generated both internally to the relative CP and externally. The internal copy undergoes $\bar{A}$ -movement from its argument position to the left periphery of the clause. This movement proceeds successively-cyclically, stopping in the specifier of the cyclic domains $\text{Asp}_{\text{IPFV}}\text{P}$ and $\text{Asp}_{\text{PROG}}\text{P}$. AGREE relationships between the relativized head and both the verb and the auxiliary are established, and *z(i)*- surfaces as a prefix on each. The complex consisting of *z(i)*- and the

auxiliary then moves under T° . When the relativized head reaches Spec,CP (after having landed also in Spec,TP as subjects do) it is deleted under identity with the external copy that c-commands it.

We now turn, finally, to relative clauses expressing past tense.

Analysis of Relative Clauses involving the Gerund

I now address the impossibility for $z(i)$ - to prefix a verb in the Gerund. As already argued in the syntactic derivation of declarative root clauses in Section 1.6, the Gerund occurs too low in the structure to host $z(i)$ -.

I propose that a Gerund verb does not raise but remains in its first-merge position, as claimed for declarative clauses. Consequently, it cannot bear $z(i)$ -, which appears only in aspectual layers. A similar analysis has been proposed in Overfelt 2009: in this work it is argued — as already explained in Section 1.6 when discussing verb raising — that the Gerund does not move to T° ; by consequence, it cannot be selected by C° in which $z(i)$ - is base-generated.

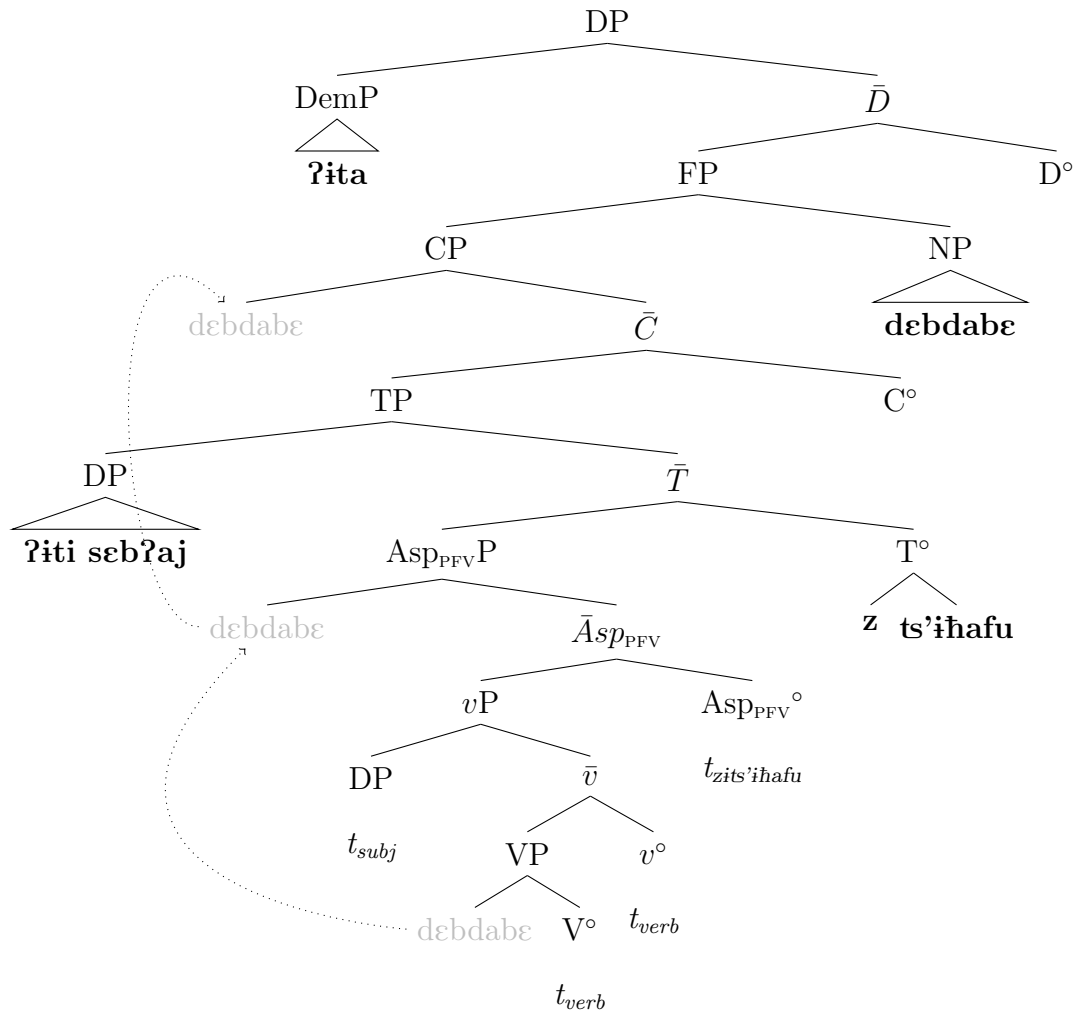
As seen, there are two strategies for a relative clause to express past tense: either a suppletive perfective form appears instead of the Gerund, as in (101), or a “dummy” auxiliary is inserted to rescue the derivation, as in (102).

- (101) ʔit-a ʔit-i səbʔaj **$z\dot{i}$** -ts’ihaf-i-wa dəbdabə
DIST-FS DIST-MS man REL-write.PFV-SM.3MS-OM.3FS letter.FS
‘The letter that the man wrote...’

- (102) ʔit-a ʔit-i səbʔaj ts’ihif-u-wa **z** -ε-llo
DIST-FS DIST-MS man write.GER-SM.3MS-OM.3FS REL-BE₁.PRES-SM.3MS
 z
 z
letter.FS
‘The letter that the man wrote...’

The first strategy, namely a relative clause with the suppletive perfective, as in (101), is represented in the tree in (103).

(103)

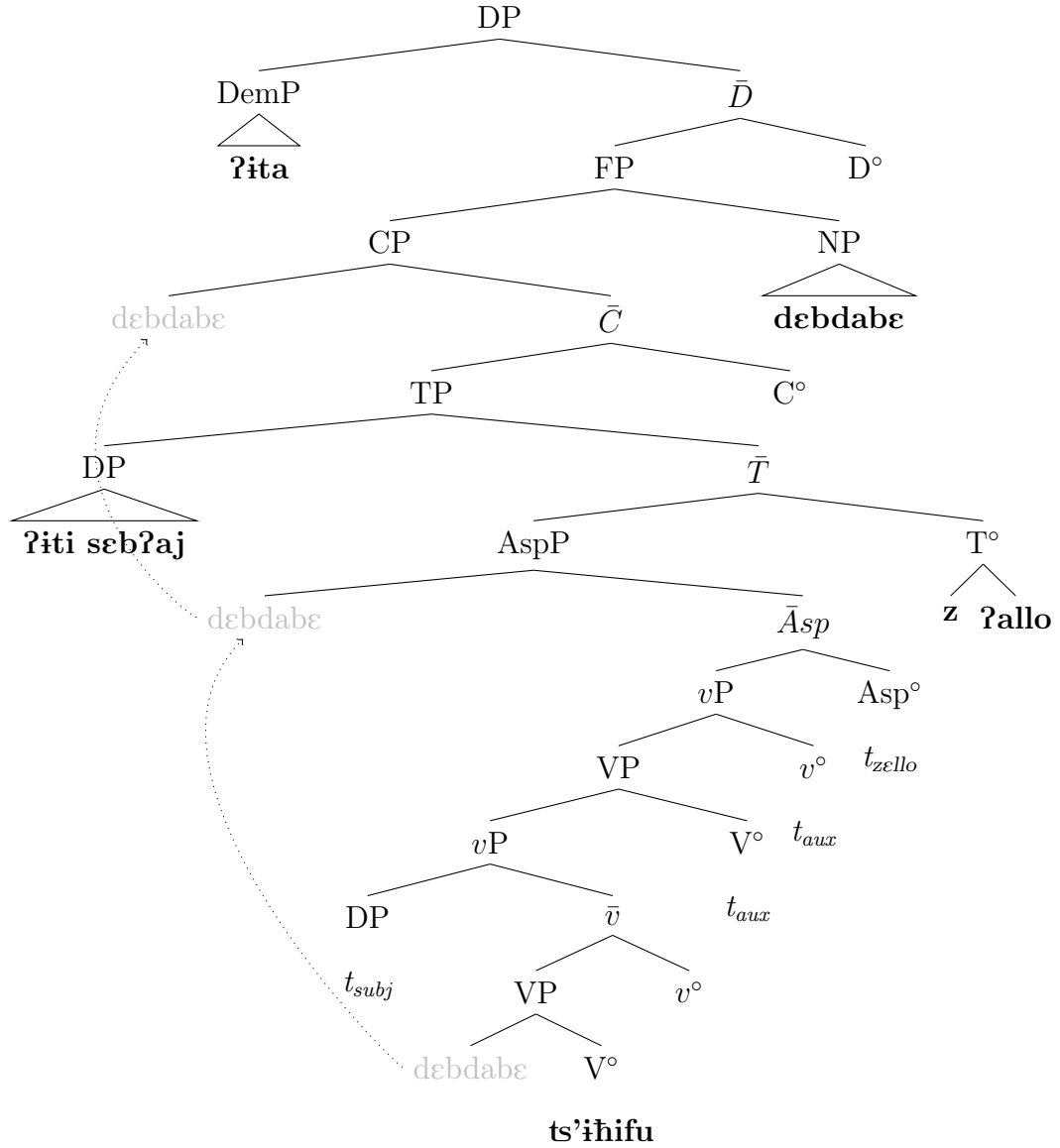


The derivation in (103) is identical to (95); the only difference is the aspectual projection involved.

The second strategy, namely a relative clause involving a dummy auxiliary, as in (102), is represented in the tree in (104).³²

³²One might argue that the auxiliary is not truly “dummy”, but instead encodes a resultative aspect, as discussed in Section 1.5.2, rather than past tense. I was unable to determine conclusively which interpretation is correct. I thank Coppe van Urk for bringing this issue to my attention.

(104)



In the derivation above, the relativized head *dεbdabe* ‘letter’ is base-generated both internally to the relative CP and externally. The Gerund verb does not raise from its first-merge position, so no Spec-Head agreement takes place in the verbal domain, and therefore *z(i)-* does not appear on the Gerund verb. However, the internal copy undergoes $\bar{A}$ -movement, stopping in the specifier of the only cyclic domain present, AspP. An AGREE relationship between the relativized head and the auxiliary is established, and *z(i)-* surfaces as a prefix on the auxiliary. The complex consisting of *z(i)-* and the auxiliary then moves under T° . When the relativized head reaches Spec,CP it is deleted under identity with the external copy that c-commands it.

It is now time to turn to the analysis of Noun Clausal Complements.

2.3.3 Analysis of (Subject) Noun Clausal Complements

We now move away from relative clauses and focus on the other types of clauses in which $z(i)$ - occurs. In Section 2.2.3, I showed that this prefix appears in noun clausal complements such as that in (105), repeated here.

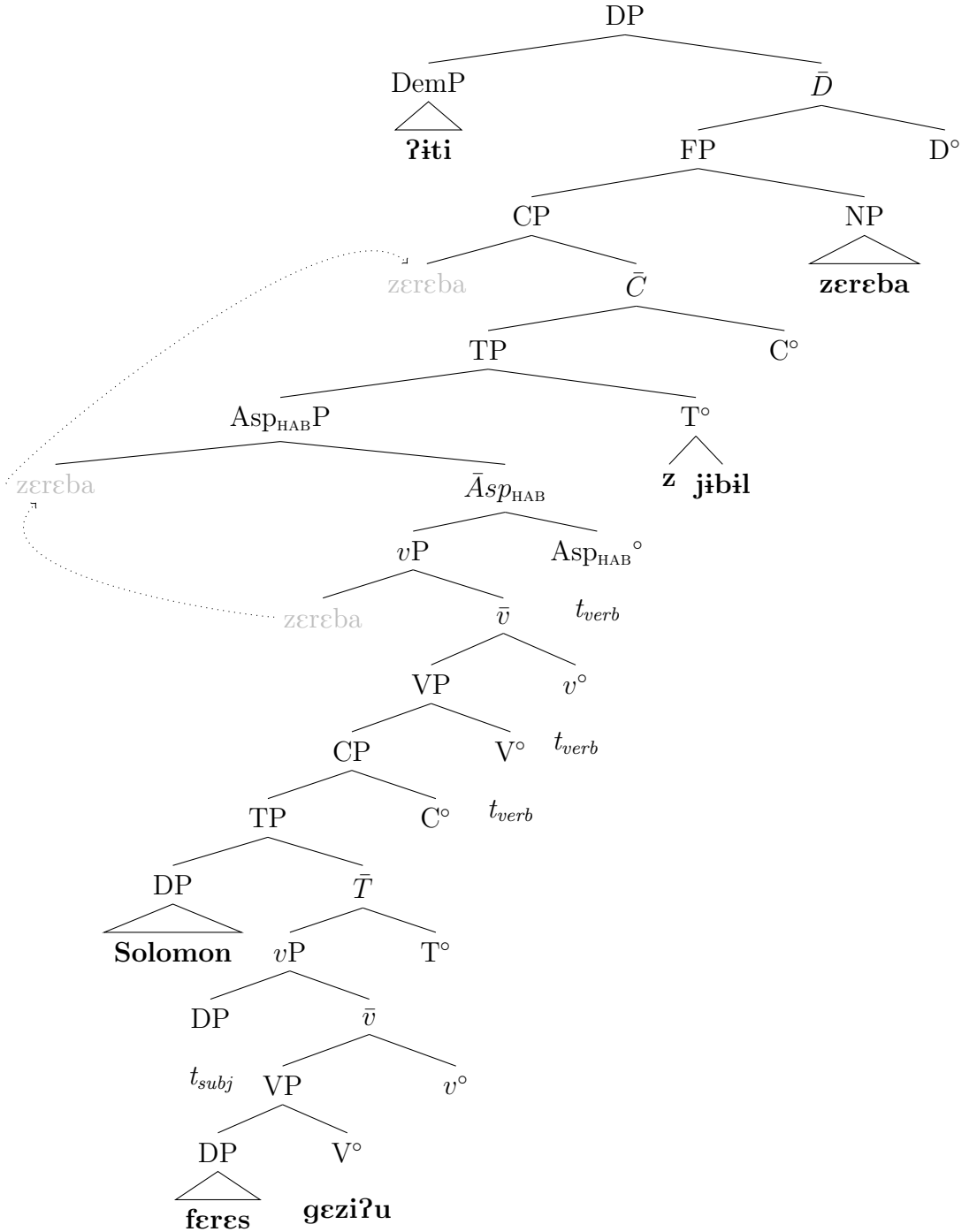
- (105) ʔit-i Solomon feres geziʔ-u **z-i**-bił zereba
 DIST-MS Solomon horse buy.GER-SM.3MS REL-SM.3MS-say.IPFV story.MS
 ʔasot ʔijj-u
 false BE₂.PRES-SM.3MS
 ‘The story that Solomon bought a horse is false.’

As shown in the example above, and already noted, Tigrinya noun clausal complements differ from English ones in containing an additional predicate, *jibil* ‘says’, which links the propositional content and the nominal content. $Z(i)$ - is obligatorily prefixed to this verb, which precedes the noun *zereba* ‘story’. The example above can therefore be more literally translated as *The story that says that Solomon bought a horse is false*.

Because of the presence of $z(i)$ -, I suggest that noun clausal complements in Tigrinya involve an underlying relative clause structure. Although noun clausal complements are not nominal modifiers (in (105) *that Solomon bought a horse* is not the kind of *story* but rather its content)³³, the idea that they are (underlyingly) similar to relative clauses has also been proposed for several other languages (Manzini & Savoia 2003, Aboh 2005, Kayne 2008b,a, Arsenijević 2009, Haegeman 2010b), especially because some of them (such as Italian, French, English, Bosnian, Croatian and Serbian) use the same complementizer to introduce both types of clauses (Krapova & Cinque 2016:81).

I propose that the noun clausal complement in (105) (without the matrix clause) can be represented as in (106).

³³(Stowell 1981) argues, however, that the relation between the clause and the noun (*story*, *claim*, *belief*, etc.) is one of adjunction/modification.



In the tree above, the relativized head *zεɛba* ‘story’ is base-generated both internally to the relative CP and externally. The syntactic derivation is the same as that proposed for subject relative clauses. The only difference is that the complement of the verb is not an object but a clause. Moreover, the analysis above support the view that clausal complements to nouns are not arguments (Stowell 1981, Kratzer 2006, Moulton 2009).

I now turn to comparative clauses.

2.3.4 Analysis of Comparative Clauses

The current proposal predicts that $z(i)$ - appears only in embedded clauses with $\bar{A}$ -movement. This means that in the comparative clause in (107) some element must undergo this type of movement.

- (107) Tesfay kab-t-i ?anɛ **zi**-hasibɛ-xu-wo
 Tesfay.MS from-DIST-MS I REL-think.PFV-SM.1S-OM.3MS
 ji-fabi
 SM.3MS-be_old.IPFV
 ‘Tefay is older than I thought.’

The question, then, is which element undergoes $\bar{A}$ -movement. As noted in section 2.2.3, comparative clauses have been analyzed as involving the $\bar{A}$ -movement of various elements (Chomsky 1977, Kayne 1994, Donati 1997, Kennedy 2002, McCoy 2017). In (107) we first see that the counterpart of the English adjective *taller* is actually a verb in Tigrinya. Second, (107) contains the masculine demonstrative *?iti* ‘that/the’ exactly where it appears in a relative clause - before the subject of the relative clause and the embedded verb prefixed by *z(i)-*. Third, a masculine object marker is suffixed to the embedded verb. As noted when presenting comparative clauses in the data section, these clauses strongly resemble light-headed relative clauses. These facts, in my view, point to the existence of a silent head in the structure. So what is “missing” in (107)?

If we consider a literal translation of (107), roughly *Tesfay than-the I that-thought-it is-tall*, we see that a degree marker such as English suffix *-er* does not appear. This is not unexpected, since some languages also lack a degree marker in their comparative clauses. For example, the comparative marker *te* ‘more’ in Korean is optional in both phrasal and clausal comparatives (Kim 2010). In Japanese, as shown in (108), comparative clauses lack degree morphology and the comparison arises by inference (Beck et al. 2004).

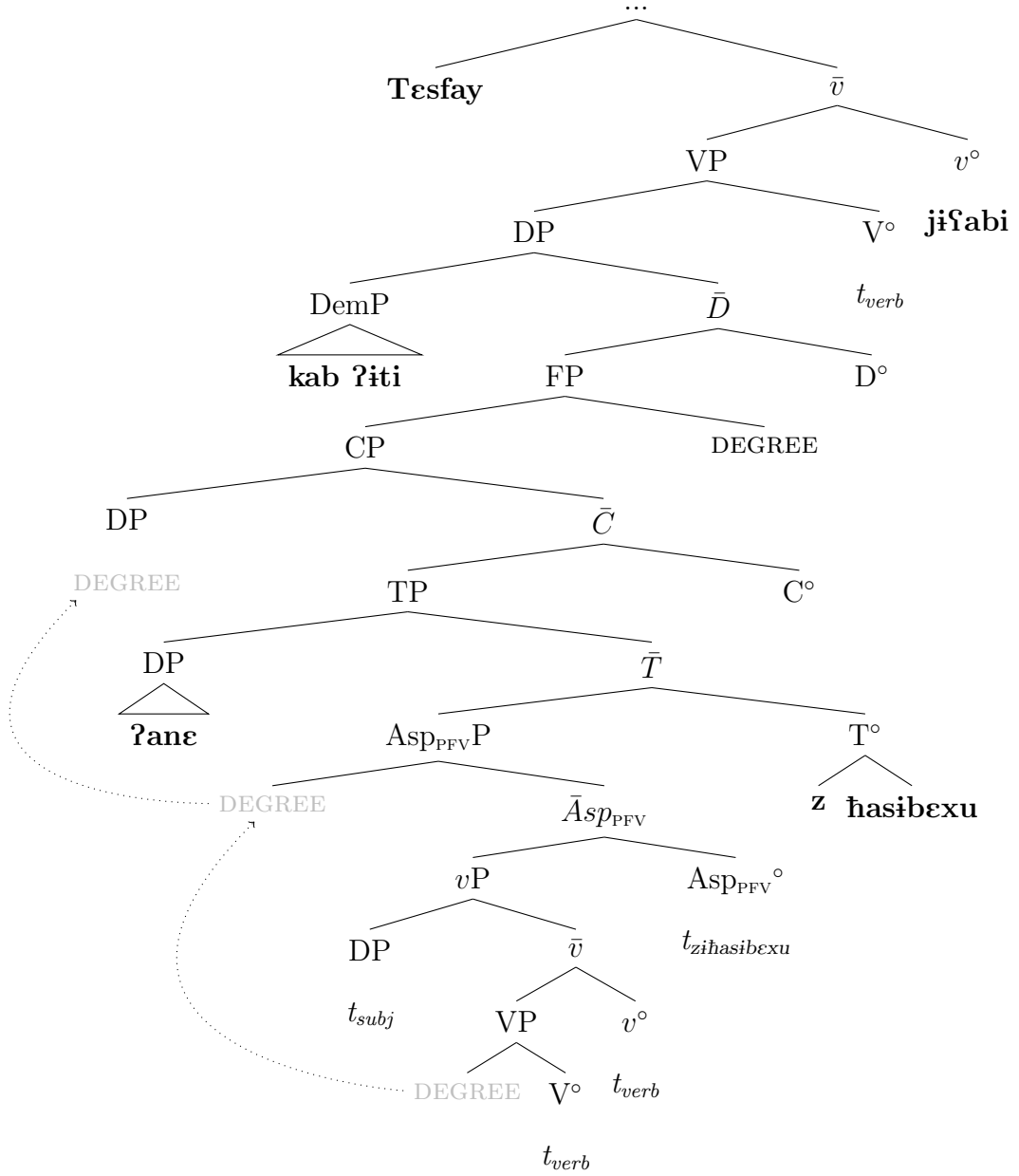
- (108) John-wa Mary-ga kitaisita-yori kasikoi
John-TOP Mary-NOM expected-than smart
'John is smarter than Mary expected.'
(from Sudo 2015:4,(6b))

With this in mind, I tentatively propose that comparative clauses in Tigrinya involve an

underlying relative structure and a silent element that $\bar{A}$ -moves in the clause, triggering the appearance of $z(i)$ - This silent element, which I label DEGREE for lack of a better term, is a null degree operator comparable to English *-er*, *as* and *too* (Heim 2000). In the literature, degree operators have been proposed to be the elements that raise in comparative clauses, such as in the example below.

- (109) a. John was climbing higher trees than Bill was
 b. [-*er* than_{wh} Bill was ~~climbing *t*-high trees~~] John was climbing *t* high trees
 (from Heim 2000:53-54)

The degree/comparative operator is usually associated with the adjective in comparative clauses, often merged in Spec,AP (Matushansky 2002). Because there is no adjective in sentences such as (107), and because there is an object marker on the $z(i)$ -marked verb as in object relative clauses, I propose that the silent nominal DEGREE is the complement of the embedded verb (conceptually something like *Tesfay is taller than the degree that I thought*). The syntactic derivation of the comparative clause in (107) is represented in the tree in (110).



In the tree above, the relativized silent element **DEGREE** is base-generated both internally to the relative CP and externally. The syntactic derivation is the same as that proposed for object relative clauses. Note that because the syntax of prepositions is outside the scope of this dissertation, I represent the prepositional phrase *kab ?iti* ‘from-the/that’ as a simple Demonstrative phrase. As mentioned in the Introduction, the status of prepositions in Tigrinya remains unsettled, despite analyses of Amharic treating them as case markers (Tremblay & Kabbaj 1990, Baker & Kramer 2014).

I now turn to the last construction analyzed in this chapter: *seem*-clauses.

2.3.5 Analysis of *seem*-Clauses

It is now time to examine clauses that contain the verbal root /msl/ ‘seem’. I presented the sentence in (111) in the data section.

- (111) Kidane ?it-i mɛts’haf z-ɛ-nbɪb-o
Kidane.MS DIST-MS book.MS REL-SM.1S-read.IPFV-OM.3MS
ji-mɛsil
SM.3MS.seem.IPFV
‘Kidane seems to read the book.’

According to the proposal defended here, $z(i)$ - appears in clauses that involve $\bar{A}$ -movement. In the literature, however, *seem*-clauses have been analyzed as resulting from *A*-movement (Rosenbaum 1967, Postal 1974, Chomsky 1993, Hornstein 1999), not $\bar{A}$ -movement. Nothing precludes the possibility that Tigrinya *seem*-clauses involve both $\bar{A}$ - and *A*-movement. If so, how do we account for the presence of $z(i)$ - in this clause type? In other words, what undergoes $\bar{A}$ -movement?

First, we need to determine whether a sentence like (111) in Tigrinya involves *A*-movement. If (111) were analyzed on a par with English *seem*-construction, hence as involving Subject Raising, the subject *Kidane* ‘Kidane’ would move from its argument position in the embedded clause to a *A*-position (Spec,TP) in the matrix clause, crossing the matrix verb *ji-mɛsil* ‘seems’. Although raising out of finite clauses is impossible in languages like English (Rizzi 1982), crosslinguistic work shows that many languages allow it, a phenomenon known as Hyperraising (Nunes 2008, Carstens 2011, Den Dikken 2017, Zyman 2023).

In the literature, Tigrinya sentences like (111) have been labelled - though not analyzed - as raising constructions (Girma 2001, Gebregziabher 2013, Spadine 2020). A complication of this is that examples in the literature often involve a 3MS grammatical subject that triggers 3MS subject morphology on the verb, raising the question of whether this morphology is default agreement (which I have argued it is not in Tigrinya). Consider

the Amharic example in (112).

- (112) $[_{IP} e [_{CP} \text{ saba worq-u-n } \text{ yä-šät'-äčč-iw }] \text{ yi-mäsl-all }]$
 Saba gold-DEF-ACC COMP-sell.PERF-3FS-3MS 3MS-seemAUX.PRES
 ‘It seems that Saba sold the gold.’ (from [Girma 2001:196](#))

Girma translates the sentence with an expletive, posits an empty category in the matrix clause, and takes the 3MS subject morphology on the matrix verb to be default agreement. This approach follows analyses of Amharic *seem*-clauses, whose structure is slightly different from Tigrinya. Consider another Amharic pair in (113).

- (113) a. säw-očč-u bäg yä-gäzz-u yi-mäsl-al
 man-PL-DEF sheep COMP-REL-buyPF-3PL 3MS-seem-AUX
 ‘It seems that the men have bought a sheep.’
 b. säw-očč-u bäg yä-gäzz-u yi-mäsl-all-u
 man-PL-DEF sheep COMP-REL-buyPF-3PL 3P-seem-AUX-pl
 ‘The men seem that (they) have bought a sheep.’
 (from [Baye 1991:111-112](#), (14a-b))

In Amharic the auxiliary *all* (counterpart of Tigrinya *?allo*) is suffixed to the verb. Baye argues that in (113a), with both the verb and auxiliary in 3MS, there must be a matrix expletive; whereas in (113b), with a 3MS verb but 3MP auxiliary, there is no expletive and the verb agrees with the grammatical subject ([Baye 1991:112](#)). However, [Lumsden & Girma 2011](#) argue that the two sentences have different interpretations: (113a) states that “something about the general situation” indicates that the men bought a sheep, whereas (113b) states that “something about the men” suggests this ([Lumsden & Girma 2011:133](#)). Given the semantic contrast, they conclude that Amharic *seem* is not a raising predicate.

By contrast, as noted in Section 2.2.3, in Tigrinya the verb *seem* must agree with the grammatical subject, as in (114).

- (114) a. ?alkol ti-seṭi z-ell-εxa ti-mεsil
 alcohol SM.2MS-drinkIPFV REL-BE₁.PRES-SM.2MS SM.2MS-seem.IPFV
 ‘You seem to be drinking alcohol.’
- b. *?alkol ti-seṭi z-ell-εxa ji-mεsil
 alcohol SM.2MS-drinkIPFV REL-BE₁.PRES-SM.2MS SM.3MS-seem.IPFV
 ‘You seem to be drinking alcohol.’

Therefore the Amharic-based claims do not carry over to Tigrinya.

Consider now Hyperraising. In several languages that show Hyperraising, including Brazilian Portuguese, we find the following:

- (115) [_{DP} Os meninos]₁ T_{φ-complete} parecem [_{CP} que₋₁ T_{φ-defective} fizeram a
 the boys seem that did the
 tarefa]
 homework
 ‘The boys seem to have done their homework.’ (from Nunes 2008:99,(31c))

Here the embedded T° is φ-defective (Chomsky 2000a) and cannot assign nominative case to the subject of the clause; however, the matrix T° is φ-complete. It probes the embedded subject *Kidane* ‘Kidane’, and attracts it to matrix Spec,TP (Nunes 2008, Zyman 2023). Applying this to Tigrinya requires that *Kidane* ‘Kidane’ in (111) be in the matrix clause. To check whether it is actually in the matrix clause, let us consider low modifiers, as in (116).

- (116) a. Kidane **bi-bikēri** maj z-i-seṭi ji-mεsil
 Kidane with-glass water REL-SM.3MS-drink.IPFV SM.3MS-seem.IPFV
 ‘Kidane seems to drink water with a glass.’
- b. ***Bi-bikēri** Kidane maj z-i-seṭi ji-mεsil
 with-glass Kidane water REL-SM.3MS-drink.IPFV SM.3MS-seem.IPFV
 ‘Kidane seems to drink water with a glass.’
- c. **Bi-bikēri**, Kidane maj z-i-seṭi ji-mεsil
 with-glass Kidane water REL-SM.3MS-drink.IPFV SM.3MS-seem.IPFV
 ‘Kidane seems to drink water with a glass.’

An embedded low modifier such as *bi-bikēri* ‘with a glass’ cannot precede the subject *Kidane* ‘Kidane’ unless topicalized (indicated by the comma). Since low modifiers cannot precede the subject, the subject must be in the matrix clause, not the embedded clause.

The question, then, is whether *Kidane* ‘Kidane’ is base-generated in the matrix clause or raises into it. I argue that it raises for two related reasons. First, the matrix verb /msl/ ‘seem’ does not assign θ -roles. Second, *Kidane* ‘Kidane’ bears the AGENT θ -role of the embedded verb *jiseti* ‘drink’.

Now, if *seem*-clauses involve Hyperraising, does the *A*-movement of the subject trigger $z(i)$ -? No. If it did, $z(i)$ - should also appear on the matrix verb, which it does not. There must therefore be another element in (111) that undergoes $\bar{A}$ -movement. One possibility is a phonologically null operator, by analogy with *tough*-movement (Chomsky 1977, Wilder 1991, Rezac 2006, Hicks 2009):

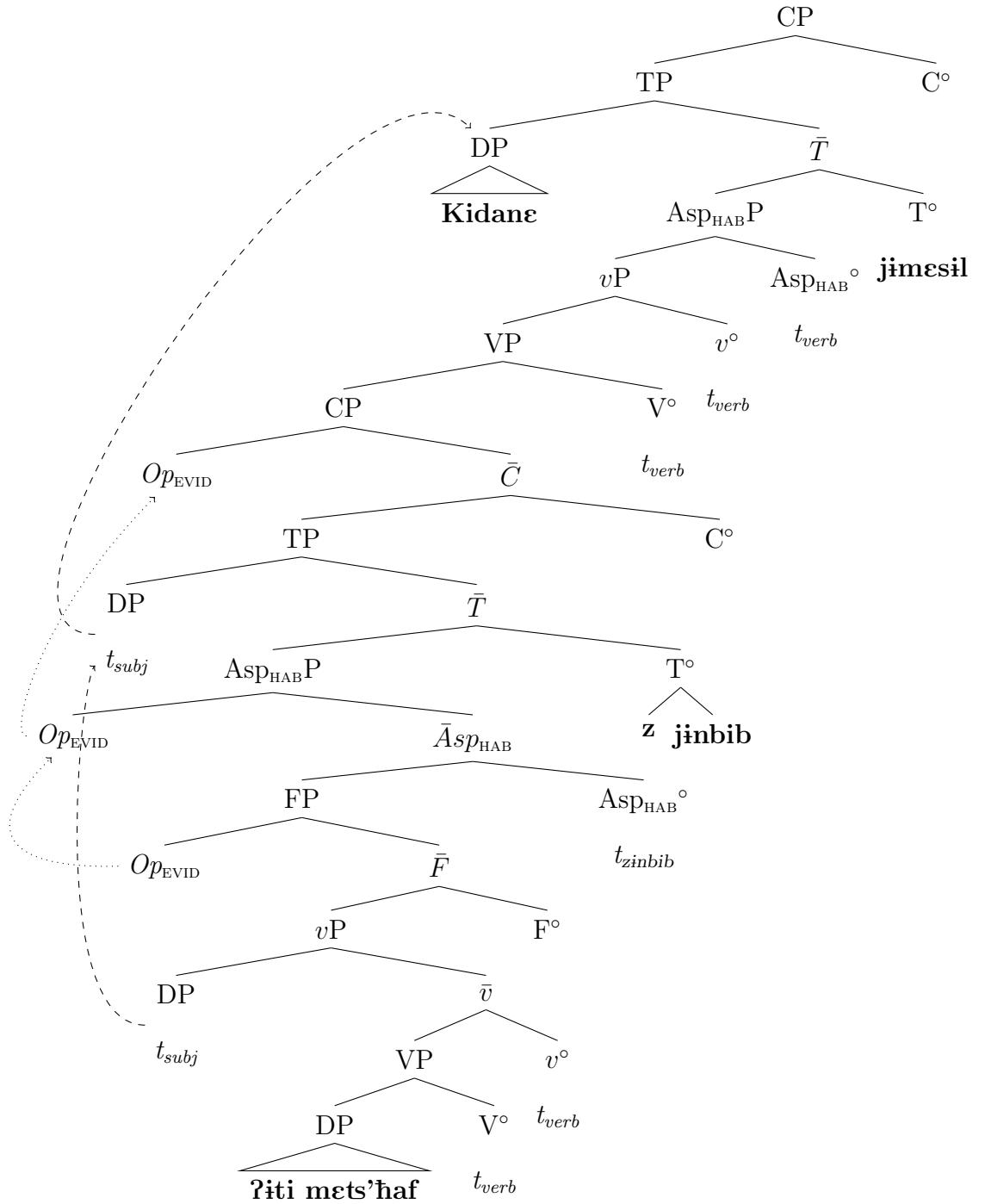
(117) John_i is tough [_{CP} *Op*_i [_{TP} PRO to please *t*_i]] (from Hicks 2009:536, (4))

This approach would incorrectly imply that *seem* assigns a thematic role to *Kidane* ‘Kidane’, which, as just noted, is not the case.

I therefore propose an alternative analysis in which *seem*-clauses involve a phonologically null evidential operator. The verb *seem* encodes perceived or indirect evidence, and I assume that this evidential meaning is syntactically represented by an operator that is base-generated within the argument structure of the embedded predicate, plausibly in the specifier position of a functional projection immediately above *vP*. This position allows the operator to be interpreted in close connection with the event described by the embedded verb.

From this embedded position, the operator undergoes $\bar{A}$ -movement to the left periphery of the clause, ultimately targeting the C° domain. This movement is consistent with the widely attested association between evidentiality and discourse-related functional structure, as well as with analyses that locate operators encoding speaker-oriented or epistemic content in the clausal periphery. Crucially, this movement takes place within the embedded clause and does not involve the matrix predicate. On this view, /msl/ ‘seem’ selects a CP that contains an evidential operator and licenses its interpretation. This approach thus provides a syntactic implementation of the evidential semantics of *seem*. A *seem*-clause such as that in (111) can thus be represented as in (118).

(118)



In this structure, the subject *Kidane* hyperraises from the embedded clause to the matrix clause via *A*-movement, while the evidential operator raises via $\bar{A}$ -movement. The derivation is similar to that proposed for Brazilian Portuguese. The embedded T° is ϕ -defective and cannot assign nominative case. The matrix T° is ϕ -complete and probes for *Kidane* ‘Kidane’ to value its unvalued ϕ -features, yielding /msl/ to surface as a finite verb.

2.4 Interim Summary of Chapter 2

In this Chapter I investigated the clausal prefix $z(i)$ - in Tigrinya. I showed that this prefix appears in several clause types - relative clauses of various kinds, comparative clauses, noun clausal complements and *seem*-clauses - and is also used to form superlatives. I proposed that $z(i)$ - is a reflex of successive-cyclic $\bar{A}$ -movement of an operator that raises from an argument position to the C° domain. I argued that aspectual layers in Tigrinya are cyclic domains (phase-like), and that $\bar{A}$ -moving elements stop in their specifiers (phase edges) before moving higher. When an element lands in Spec,AspP after the verb has moved to Asp° , a Spec-head agreement relation is established and $z(i)$ - is triggered and appears on the left of the verb looking like a prefix. I also provided a syntactic analysis of restrictive relative clauses with periphrastic tense forms, in which $z(i)$ - appears on both the verb and the auxiliary, explained by the element halting in the specifiers of two AspP projections, yielding appearance of $z(i)$ - on both the verb and the auxiliary.

Chapter 3

The Clausal Prefix *kɛmz(i)-*

3.1 Setting the Scene

The second clausal prefix investigated in this dissertation is *kɛmz(i)-* (the [i] vowel is an epenthetic vowel; the morpheme itself is *kɛmz*). In the introduction, I provided the example in (1) to illustrate one of the contexts in which this prefix occurs in Tigrinya:

- (1) tufaḥ **kɛmzi**-bɛliʔ-u ji-fellit'
 apple COMP-eat.PFV-SM.3MP SM.1S-know.IPFV
 ‘I know that they ate an apple.’

Kɛmz(i)- has been described in the literature as a conjunction (Leslau 1941, Mason 1996, Kogan 1997, 2007) and as a complementizer (a.o. Tesfay 2016, Spadine 2020) meaning ‘that’. In Ethio-Semitic, complementizers that introduce complement clauses derive from simulative morphemes (such as ‘like’ in English) and occur either as prefixes, as in Amharic *ində* and Argobba *mäss*, or as suffixes, as in Zay *hum*. Examples of complement clauses in these languages are given in (2)–(4).

- (2) ləmma **ində**-mətt’a səmma-hu
 Lemma COMP-REL-arrive.PFV.3SM hear.PFV-1SG
 ‘I heard that Lemma has arrived.’ *Amharic* (from Amberber 2023:438, (40))

- (3) *abbila dräs ib-bät'-u* **mäss** *sämmäf-ew*
 Abbila up_to 2-come_out.IPFV-PL COMP hear.PFV-1S
 'I heard that you come out up to Abbila'

Argobba (adapted from [Wetter 2023:469](#), (45))

- (4) *ma:mmo: dernä jä-bäl-e-j-hum* *tʃa:l*
 Mamo lunch REL-eat.PFV-3SM-OBJ.3SM-COMP know.PFV.3SM
 'He knows that Mamo ate lunch.' *Zay*(adapted from [Meyer 2023:662](#), (43))

In this chapter I argue that *kɛmz(i)-* is a bimorphemic element composed of the particle *kɛm* and the prefix *z(i)-*, the same clausal prefix examined in Chapter 2. *Kɛm* is one of several particles in the language that require *z(i)-* to attach to the embedded verb, but it is the only one that introduces complement clauses. I show that *kɛmz(i)-* occurs in complement clauses selected by factive, cognitive non-factive, fiction, utterance/saying and perception verbs. I claim that all *kɛmz(i)-* clauses are underlyingly relative clauses involving $\bar{A}$ -movement of a null element from its argument to the left periphery, hence the presence of *z(i)-*. *Kɛm* is a similative morpheme associated with a low projection in the argument structure of the embedded verb and is paired with the null element. I also show that in periphrastic tense forms *kɛmz(i)-* exhibits a distinctive syntactic behavior, being able to prefix either the verb or the auxiliary under certain constraints.

This chapter is structured as follows. In Section 3.2.1 I first situate *kɛmz(i)-* within the broader class of subordinators that are actually bimorphemic elements composed of a preposition and the clausal prefix *z(i)-* introducing subordinate clauses in Tigrinya. I then present *kɛmz(i)-* clauses with periphrastic tense forms (Section (13)). Next, I turn to the data in Section 3.2.2, providing examples of *kɛmz(i)-* clauses with a range of verbs to show which verb classes can select them. I then briefly compare *kɛmz(i)-* with the head-final complementizer *?ilu* 'that' in Section 3.2.3. The second part of the chapter (Section 3.3) is devoted to the syntactic analysis of *kɛmz(i)-* clauses.

3.2 *Kεmz(i)*-clauses

This section presents the data on clauses in which *kεmz(i)*- appears.

3.2.1 What is *kεmz(i)*-?

The first question that must be addressed in order to understand and analyze the syntax of *kεmz(i)*-clauses is: what is *kεmz(i)*-? The most striking feature of this clausal prefix is its morphological similarity to the prefix *z(i)*-, examined in Chapter 2. Could it be that the *kεmz(i)*- prefix actually contains the prefix *z(i)*-? As shown, *z(i)*- occurs in relative clauses, comparative clauses (and superlatives), noun clausal complements and *seem*-clauses. I therefore argued that this clausal prefix marks $\bar{A}$ -movement. This would imply that if *z(i)*- appears in *kεmz(i)*-clauses, then these clauses involve $\bar{A}$ -movement.

The first argument supporting the claim that *kεmz(i)*- is in fact a bimorphemic element is the existence of particles that require *zi*- to be prefixed to the embedded verb they introduce. These elements are summarized Table 3.1.¹

<i>kεm</i> ('like') + <i>z(i)</i> -	'that'	Complement Clauses
<i>?intε</i> + <i>z(i)</i> -	'if'	Conditional Clauses
<i>silε</i> ('for') + <i>z(i)</i> -	'because'	Adverbial Clause
<i>kisab/kisaʕ</i> + <i>z(i)</i> -	'until'	Adverbial Clause
<i>kindi</i> + <i>z(i)</i> -	'instead'	Adverbial Clause
<i>?inkab</i> + <i>z(i)</i> -	'since'	Adverbial Clause

Table 3.1: Particles that require *z(i)*-clauses.

Table 3.1 shows that *kεmz(i)*- is the only element that introduces complement clauses. This chapter therefore focuses mainly on it rather than the other elements. However, in the next three subsections I briefly discuss these adverbial clauses before returning to *kεmz(i)*-.

¹There might be other particles that require *zi*-marked verbs that I am not aware of. For example, [van Urk 2025](#) states that the particle *mis* — an alternative to *mεʕas* 'when' — also precedes a *zi*-marked verb.

Conditional Clauses

The $z(i)$ - prefix appears in some types of conditional clauses. Conditional constructions with $z(i)$ - are always optional, and other strategies without it can apparently express the same meaning.²

To express a possible condition (Mason 1996), $z(i)$ - attaches to a verb in the Imperfective and co-occurs with the particle $?int\epsilon$ ‘if’.³ This is shown in (5) with a present-tense matrix clause and in (6) with a future-tense matrix clause. My consultants sometimes write $?int\epsilon$ attached to $z(i)$ - and sometimes separated.

- (5) bun **?intε** z-i-seti, ħaili ?all-o-ni
 coffee if REL-SM.1S-drink.IPFV energy.MS BE₁.PRES-SM.3MS-OM.1S
 ‘If I drink coffee, I have energy.’
- (6) nissu **?intε** z-i-xεjjid, k-i-beki ?ijj-ε
 he if REL-SM.3MS-leave.IPFV SBJ-SM.1S-cry.IPFV BE₂.PRES-SM.1S
 ‘If he leaves, I’ll cry.’

Nothing can intervene between the particle $?int\epsilon$ ‘if’ and $z(i)$ -, as shown in (7).

- (7) a. nigihō bun **?intε** z-i-seti, ħaili
 morning coffe if REL-SM.1S-drink.IPFV energy.MS
 ?all-o-ni
 BE₁.PRES-SM.3MS-OM.1S
 ‘If I drink coffee in the morning, I have energy.’
- b. *bun **?intε** nigihō z-i-seti, ħaili
 coffe if morning REL-SM.1S-drink.IPFV energy.MS
 ?all-o-ni
 BE₁.PRES-SM.3MS-OM.1S
 ‘If I drink coffee in the morning, I have energy.’

To express a past counterfactual, two different strategies are used (Mason 1996). Either, as with a possible condition, $z(i)$ - attaches to a verb in the Imperfective and co-occurs with the particle $?int\epsilon$ ‘if’; or a Gerund verb form is followed by $?int\epsilon$, which is followed by $z(i)$ - prefixed to the verb *kone* — the lexical verb ‘become’ and the suppletive form of

²For an overview of all the possible constructions used to express conditionals, see Mason 1996.

³According to van Urk 2025, $?int\epsilon$ can also appear without $z(i)$ - in embedded interrogatives.

the auxiliaries *?allo* and *?ijju* used in negation and future. The first strategy is shown in (8) and the second in (9).

- (8) gizie ʔinte z-i-hiliw-ini, ʔerefti mi-ked-ku
time.MS if REL-SM.3MS-BE₁.IPFV-OM.1S vacation NMZ-go.PFV-SM.1S
'If I had time, I would go on vacation.'
- (9) gizie neir-u-ni ʔinte zi-xewun, ʔerefti
time.MS BE.GER-SM.3MS-OM.1S if REL-become.SM.3MS vacation
mi-ked-ku
NMZ-go.PFV-SM.1S
'If I had time, I would go on vacation.'

Although I do not present a full syntactic analysis of conditional clauses in Tigrinya, I briefly sketch a tentative account to explain their structure and especially the occurrence of *z(i)-*. In the literature, conditional clauses have been claimed to have a syntax similar to that of correlatives (Geis 1985, von Stechow 1994) and (free) relative clauses (a.o. Bhatt & Pancheva 2002, 2017, Arsenijević 2009, Haegeman 2010a). It has been proposed to analyze conditional *if*-clauses as involving a world operator that raises to Spec,CP, as in the following representation.

- (10) [CP Op_w C° [John arrives late in w]]
 ‘If John arrives late...’ (from Haegeman 2010a:599,(11))

It is natural to think of conditional clauses as relative clauses also because conditional conjunctions such as *if* can be paraphrased by expressions like ‘in case that’ or ‘in the event that’ (Haegeman 2010b:637). It is commonly assumed that *if* is a complementizer, base-generated in C° (Kayne 1991) or Int° (Rizzi 2001).

However, conditional clauses are semantically very different than relative clauses. Relative clauses are intersective modifiers: in *The lady that I saw...* the relative clause restricts the set of all possible ladies to the one that I saw. By contrast, conditional clauses are not intersective. If they were, in *If I drink coffee, I have energy* we would need to take all the worlds where *I have energy* is true and intersect them with all the worlds where *I drink coffee* is true. It is therefore difficult, from a semantic perspective, to treat

conditional clauses as underlyingly be relative clauses.⁴

Adverbial Clauses

The *z(i)*- prefix is also involved in some types of adverbial (adjunct) clauses. These are illustrated in (11)-(13).

- (11) a. bizuh **sile** **z-i**-nbib ji-fetwi-ni
a.lot for REL-SM.3MS-read.IPFV SM.3MS-like.IPFV-OM.1S
‘He likes me because I read a lot.’
- b. Segen **sile** **zi**-hagez-ku-wa genzeb hib-et-ni
Segen for REL-help.PFV-SM.1S-OM.3FS money give.GER-SM.3FS-OM.1S
‘Segen gave me money because I helped her.’⁵
- (12) a. ?it-i wedi kullu gizie **kisab** **z-i**-sihi? ji-beki
DIST-MS boy all time until REL-SM.3MS-laugh.IPFV SM.3MS-cry.IPFV
‘The boy always cries until he laughs.’
- b. n-it-a geza **kisab** **z-i**-ri?i-a tezawir-e
DOM-DIST-FS house until REL-SM.1S-see.IPFV-OM.3FS walk.GER-SM.1S
‘I walked until I saw the house.’
- c. *n-it-a geza **kisab** **z-i**-ri?i-ku-wa tezawir-e
DOM-DIST-FS house until REL-see.PFV-SM.1S-OM.3FS walk.GER-SM.1S
- (13) a. hibisti **?ab** **kindi** **zi**-beli?i-ku mai setej-e
cake instead REL-eat.PFV-SM.1S water drink.GER-SM.1S
‘I drank water instead of eating a cake.’
- b. hibisti **?ab** **kindi** **z-i**-beli? mai setej-e
cake instead REL-SM.1S-eat.IPFV water drink.GER-SM.1S
‘I drank water instead of eating a cake.’

Two points follow from the examples above. First, my consultants sometimes write the particles *sile*, *kisab* and *?ab kindi* as prefixed to *z(i)*- and sometimes as free morphemes. Second, the *z(i)*-marked verb may be Imperfective or Perfective with *sile*, only Imperfective with *kisab* and either Imperfective or Perfective without a change in meaning with *?ab kindi*. I do not yet have an account for these differences.

The parallel with *kemz(i)*- becomes clearer in periphrastic tense forms.

⁴I thank Ivano Caponigro for pointing this out to me.

⁵Grammars describe *sile* as preposition meaning ‘for’ (cf. Leslau 1941:135). Note that ‘why’ in Tigrinya is *silemintay*, literally ‘for what’.

***kɛmz(i)-* in Periphrastic Tense Forms**

This section provides data on *kɛmz(i)-* in complement clauses with periphrastic tense forms, where an auxiliary occurs with the verb. I compare these with the adverbial clauses in the preceding section. This is crucial for understanding the syntactic behavior and position of *kɛmz(i)-*. Only after this section will I detail which verb classes select *kɛmz(i)-*-clauses.

As explained in Section 1.5.2, periphrastic tense forms express either progressive aspect or present perfect/resultative. In the progressive, an Imperfective verb occurs with an auxiliary; in the present perfect, a Gerund verb occurs with the same auxiliary.

Given the behavior of *z(i)-* in periphrastic tense forms discussed in Chapter 2 and my claim that *kɛmz(i)-* is bimorphemic, it is important to know what happens when a verb occurs along with an auxiliary: does *kɛmz(i)-* appear on both verbal elements? Or maybe *z(i)-* appears twice and *kɛm* only once? And if *kɛm* appears only once, does it appear before the verb or before the auxiliary? Let us take a look at all of this.

Consider (14), that shows *kɛmz(i)-*-clauses with a progressive, and (15) which shows the present perfect. According to my consultants, (14a) and (14b) are truth-conditionally equivalent, as are (15a) and (15b).⁶

- (14) a. Solomon tɛlɛviziɔn ji-riʔi **kɛmz-ɛll-o**
 Solomon TV SM.3MS-watch.IPFV COMP-BE₁.PRES-SM.3MS
 ji-fillit'
 SM.1S-know.IPFV
 'I know that Solomon is watching TV.'
- b. Solomon tɛlɛviziɔn **kɛmz-i-riʔi** **z-ello**
 Solomon TV COMP-SM.3MS-watch.IPFV REL-BE₁.PRES-SM.3MS
 ji-fillit'
 SM.1S-know.IPFV
 'I know that Solomon is watching TV.'
- c. *Solomon tɛlɛviziɔn **z-i-riʔi** **kɛmz-ello**
 Solomon TV REL-SM.3MS-watch.IPFV COMP-BE₁.PRES-SM.3MS
 ji-fillit'
 SM.1S-know.IPFV

⁶Also noted by van Urk 2024. I thank Coppe Van Urk for discussion.

- d. *Solomon television **kɛmz-i-riʔi** **kɛmz-ɛllo**
 Solomon TV COMP-SM.3MS-watch.IPFV COMP-BE₁.PRES-SM.3MS
 ji-fellit'
 SM.1S-know.IPFV
- (15) a. Tesfay nsa kab geza wɛts'iʔ-a **kɛmz-ɛll-a**
 Tesfay she from house leave.GER-SM.3FS COMP-BE₁.PRES-SM.3FS
 hasib-u
 think.GER-SM.3MS
 'Tesfay thought that she left the house.'
- b. Tesfay nsa kab geza **kɛmzi**-wɛts'iʔ-ɛt **z-ɛll-a**
 Tesfay she from house COMP-leave.PFV-SM.3FS REL-BE₁.PRES-SM.3FS
 hasib-u
 think.GER-SM.3MS
 'Tesfay thought that she left the house.'
- c. *Tesfay nsa kab geza **zi**-wɛts'iʔ-ɛt **kɛmz-ɛll-a**
 Tesfay she from house REL-leave.PFV-SM.3FS COMP-BE₁.PRES-SM.3FS
 hasib-u
 think.GER-SM.3MS
- d. *Tesfay nsa kab geza **kɛmzi**-wɛts'iʔ-ɛt **kɛmz-ɛll-a**
 Tesfay she from house COMP-leave.PFV-SM.3FS COMP-BE₁.PRES-SM.3FS
 hasib-u
 think.GER-SM.3MS

Examples (14a) and (15a) show that *kɛmz(i)-* can appear on the auxiliary, leaving the verb bare. Examples (14b) and (15b) show that it can also appear on the verb (a suppletive Perfective form appears in the present perfect), in which case the auxiliary is prefixed by *z(i)-*. It is not possible for *z(i)-* to appear on the verb when *kɛmz(i)-* is on the auxiliary, as in (14c) and (15c), nor for *kɛmz(i)-* to prefix both the verb and the auxiliary, as in (14d) and (15d).⁷

⁷However, *z(i)-* can appear on the verb while *kɛmz(i)-* appears on the auxiliary when the clause is negative, as in (i) (the other two configurations are also possible).

- (i) Tesfai z-ɛj-Ø-dərrif kɛmz-ɛll-o ji-fellit'
 Tesfay REL-NEG-SM.3MS-sing.IPFV COMP-BE₁.PRES-SM.3MS SM.1S-know.IPFV
 'I know that Tesfay is not singing.'

This was also noted by Coppe van Urk (p.c.). There is speaker variation: some accept the sentence as fully grammatical, others are reluctant. I leave this configuration for future research.

For clarity, the possible and impossible configurations are summarized in Table 3.2.

V $k\epsilon m z(i)$-Aux	✓
$k\epsilon m z(i)$-V $z(i)$-Aux	✓
$z(i)$-V $k\epsilon m z(i)$-Aux	✗
$k\epsilon m z(i)$-V $k\epsilon m z(i)$-Aux	✗

Table 3.2: Possible configurations of $k\epsilon m z(i)$ -clauses with periphrastic tense forms.

Now consider adverbial clauses. Recall that *?intε z(i)-* introduces *if*-clauses, *sile z(i)-* introduces *because*-clauses, *kisab z(i)-* introduces *until*-clauses and *?ab kindi z(i)-* introduces *instead*-clauses. Examples (16) and (17) illustrate *sile z(i)*-clauses and *kisab z(i)*-clauses with periphrastic tense forms.

- (16) a. Segen ji-higz-a **silɛ** z-ɛlle-xu genzɛb
 Segen SM.1S-help.IPFV-OM.3FS for REL-BE₁.PRES-SM.1S money
 hib-ɛt-ni
 give.GER-SM.3FS-OM.1S
 ‘Segen gave me money because I’m helping her.’
 b. Segen **silɛ** **zi**-higz-a z-ɛlle-xu genzɛb
 Segen for REL-help.IPFV-SM.1S-OM.3FS REL-BE₁.PRES-SM.1S money
 hib-ɛt-ni
 give.GER-SM.3FS-OM.1S
 ‘Segen gave me money because I’m helping her.’
- (17) a. n-ita mɛts’haf mi-nbab ?aj-Ø-halits’i-n ?ajin-ɛj
 DOM-DIST-FS book.FS NMZ-read NEG-SM.1S-stop.IPFV-NEG eye.S-POSS.1S
 ji-ri?i **kisab** z-ɛll-ɛxu
 SM.1S-see.IPFV until REL-BE₁.PRES-SM.1S
 ‘I don’t stop reading the book until my eyes can see.’⁸
 b. n-ita mɛts’haf mi-nbab ?aj-Ø-halits’i-n ?ajin-ɛj
 DOM-DIST-FS book.FS NMZ-read NEG-SM.1S-stop.IPFV-NEG eye.S-POSS.1S
kisab z-i-ri?i z-ɛll-ɛxu
 until REL-SM.1S-see.IPFV REL-BE₁.PRES-SM.1S
 ‘I don’t stop reading the book until my eyes can see.’

Like *kɛmz(i)-*, *silɛ z(i)-* and *kisab z(i)-* in periphrastic tense forms can attach to the auxiliary, leaving the verb bare, or attach to the verb with *z(i)-* on the auxiliary. Although I have not presented ungrammatical counterparts for space, *silɛz(i)-* and *kisab z(i)-* cannot

⁸Datapoint and translation offered spontaneously by one of my consultants. According to him, this is a (sort of) saying in Tigrinya.

attach to the auxiliary while $z(i)$ - prefixes the verb, nor can they attach to both the verb and the auxiliary, exactly like $k\epsilon m z(i)$ -.

Now consider $?int\epsilon z(i)$ -clauses and $?ab\ kindi z(i)$ -clauses with periphrastic tense forms in (18) and (19).

- (18) a. *bun ji-seti **?int\epsilon z-**ell-εxu,
 coffee SM.1S-drink.IPFV if REL-BE₁.PRES-SM.1S
 ?aj-ti-zaribu-ni
 NEG-SM.2MS-speak.IPFV-OM.1S
 ‘If I’m drinking coffee, do not talk to me!’
- b. *bun **?int\epsilon z-i**-seti **z-**ell-εxu,
 coffee if REL-SM.1S-drink.IPFV REL-BE₁.PRES-SM.1S
 ?aj-ti-zaribu-ni
 NEG-SM.2MS-speak.IPFV-OM.1S
 ‘If I’m drinking coffee, do not talk to me!’
- (19) a. *bani ji-beli? **?ab kindi z-**ell-εxu mai
 bread SM.1S-eat.IPFV instead REL-BE₁.PRES-SM.1S water
 ji-seti ?all-εxu
 SM.1S-drink.IPFV BE₁.PRES-SM.1S
 ‘I am drinking water instead of eating a cake.’
- b. *bani **?ab kindi z-i**-bel? **z-**ell-εxu mai
 bread instead REL-SM.1S-eat.IPFV REL-BE₁.PRES-SM.1S water
 ji-seti ?all-εxu
 SM.1S-drink.IPFV BE₁.PRES-SM.1S
 ‘I am drinking water instead of eating a cake.’

Unlike $k\epsilon m z(i)$ -, $sil\epsilon z(i)$ - and $kisab z(i)$ -, $?int\epsilon z(i)$ - and $?ab\ kindi z(i)$ - cannot express progressive aspect, hence the ungrammaticality of (18) and (19).

For clarity, the possibilities are summarized in Table 3.3 (X stands for the element preceding $z(i)$ -).

	$k\epsilon m$	$sil\epsilon$	$kisab$	$?int\epsilon$	$?ab\ kindi$
V $Xz(i)$-Aux	✓	✓	✓	✗	✗
$Xz(i)$ - V $z(i)$-Aux	✓	✓	✓	✗	✗
$z(i)$ - V $Xz(i)$-Aux	✗	✗	✗	✗	✗
$Xz(i)$ - V $Xz(i)$-Aux	✗	✗	✗	✗	✗

Table 3.3: Possible configurations of adverbial and complement clauses with periphrastic tense forms.

Let us refocus on *kɛmz(i)-* now.

Why *kɛmz(i)-* is bimorphemic

The aim of the discussion on conditional clauses, adverbial clauses and periphrastic tense forms in the previous sections is to provide a first argument for analyzing *kɛmz(i)-* as a bimorphemic element. The pattern seen with particles that appear with *z(i)-* suggests that *z(i)-* is an independent morpheme in *kɛmz(i)-* as well. If *z(i)-* is independent, then *kɛm* must be independent too.

As a matter of fact, a second argument in support of *kɛmz(i)-* as a bimorphemic element is that *kɛm* is a lexical word in the language and is used outside complement clauses. This particle has the meaning of the adverbial/prepositional *like* (cf. [Leslau 1941:133](#)). *Kɛm* is likely cognate with the Akkadian preposition and adverbial conjunction *kīma*, composed of the preposition *kī-* and the emphatic particle *-ma* ([Deutscher 2000](#)). Cognates of *kīma* (and/or *kī-*) are used across Semitic languages to introduce finite complement clauses ([Deutscher 2000:63](#)). It is therefore unsurprising that *kɛm* introduces complement clauses in Tigrinya. When it does not introduce complement clauses, *kɛm* is used as in the examples below.

- (20) **kɛm** ʔinisisa-tat ji-mɛlɛs-u
like animal-P SM.3MP-behave.IPFV-SM.3MP
'They behave like animals.'

- (21) nissa **kɛm**-ay tɛmaharit ʔijj-a
she like-1S student.FS BE₂.PRES-SM.3FS
'She is a student like me.'

At this point, one could ask whether the *z(i)-* morpheme in *kɛmz(i)-* is actually the *z(i)-* clausal prefix investigated in Chapter 2. If this were the case, it would be a third argument in support of *kɛmz(i)-* as a bimorphemic element. I claim that this is the case, and offer two morphological arguments.

First, in Chapter 2, I showed that *z(i)-* is not realized when prefixed to Imperfective stems beginning with [t] in 2S and 3FS and with [n] in 1P. For convenience, consider (22).

(22) ?it-i nsxa ti-se?i-jo z-e?l-e-xa may
DIST-MS you.MS SM.2MS-drink-IPFV-OM.3MS REL-BE₁.PRES-SM.2MS water
t'i?um ?ijj-u
delicious BE₂.PRES-SM.3MS
'The water that you are drinking is delicious.'

In clauses with *kɛmz(i)-*, when the embedded subject is 2nd person, 3FS or 1P, the *z(i)-* portion of *kɛmz(i)-* is likewise dropped, as in (23) and (24) (cf. also [Tesfay 2006](#)).

(23) bzuḥ gizie nissa sinema **kem**-ti-xejjid ji-fillit'
often time she cinema COMP-SM.3FS-go.IPFV SM.1S-know.IPFV
'I know that she often goes to the cinema.'

(24) kab met'en ni-laʕeli **kem**-ni-seti ji-ʔimen
from proportion DOM-up COMP-SM.1P-drinkIPFV SM.1S-admit.IPFV
'I admit that we drink too much.'

Second, as detailed in Chapter 5, sentential negation in Tigrinya is expressed by the circumfix *?aj-...-(i)n*, as in (25), but in negative relative clauses the negative suffix is dropped, as in (26).

(25) mɛts'hafti ʔaj-Ø-nbibi-n
book.MP NEG-SM.1S-read.-IPFV-NEG
'I do not read books.'

(26) ʔit-om ʔanɛ z-ɛy-Ø-nbib-om mɛts'hafti ʔab-t-i
DEM.MP I REL-NEG-SM.1S-read.-IPFV-OM.3MP book.MP in-DIST-MS
armadio ʔall-ɛwu
cabinet BE₁.PRES-SM.3MP
'The books that I do not read are in the cabinet.'

When $k\epsilon m z(i)$ attaches to a negative verb, the same pattern holds: the negative suffix is dropped and only the negative prefix appears, as in (27).

(27) n-it-i hijab **kemz-ey-Ø-bibɛl-o** wesin-ε
DOM-DIST-MS gift COMP-NEG-SM.1S-accept.IPFV-OM.3MS decide.GER-SM.1S
‘I decided that I would not accept the gift.’
(Lit. ‘I decided to not accept the gift’)

These two morphological facts support identifying the *z(i)-* in *kɛmz(i)-* with the same prefix discussed in Chapter 2, rather than a homophonous morpheme.

Finally, fieldwork notes show orthographic variation: consultants sometimes write *kɛmz(i)-* as one word and sometimes as two, *kɛm-* and *zi-* — as it is also the case for the elements that introduce adverbial clauses previously discussed. In the latter case, the former is a free morpheme while the latter is bound, as in (28). To my knowledge, nothing can intervene between *kɛm* and *zi-*.

- (28) tufaḥ **kɛm** **zi**-bəliʔ-u ji-fəllit'
 apple COMP REL-eat.PFV-SM.3MP SM.1S-know.IPFV
 ‘I know that they ate an apple.’

This variation is also found in the literature. Some sources write *kɛmz(i)-* as one word (Leslau 1941, Kogan 1997, Overfelt 2009, Spadine 2020); others write it as two (Mason 1996, Bulakh 2019). Taken together, these facts support the conclusion that *kɛmz(i)-* is bimorphemic, composed of *kɛm* and *z(i)-*. Henceforth, in the examples, I represent *kɛm-z(i)-* with the two morphemes separated and gloss it as COMP-REL-.

We now turn to the verb classes that select *kɛmz(i)-*-clauses.

3.2.2 *Kɛmz(i)-* in Simple Verb Forms

This section presents how *kɛmz(i)-* is used in complement clauses with simple verb forms selected by factive and cognitive non-factive verbs, the verb /msl/ ‘seem’, fiction, utterance and perception verbs.

Factive Verbs

Kɛmz(i)- introduces complement clauses with factive verbs such as ‘know’, ‘forget’, ‘admit’, ‘accept’. This is illustrated in (29)–(32).

- (29) bzuḥ gizie nisu sinema **kɛm-z-i**-xəjjid ji-fəllit'
 often he cinema COMP-REL- SM.3MS-go.IPFV SM.1S-know.IPFV
 ‘I know that he often goes to the cinema.’

- (30) fadus gɪziɛ **kɛm-z-i-mɛts'iʔ** rɛsiŋ-ɛ
 noon time COMP-REL-SM.3MS-come.IPFV forget.GER-SM.1S
 'I forgot that he comes at noon.'
- (31) **kɛm-z-i-fɛt-wa** ji-ʔɛmin
 COMP-REL-SM.3MS-like.IPFV-OM.3FS SM.3MS-admit.IPFV
 'He admits that he likes her.'
- (32) haʃɛɾɛ-tat **kɛm-z-i-fettu** ji-qɛbɪl
 spider.P COMP-REL-SM.1S-like.IPFV SM.1S-accept.IPFV
 'I accept that he likes spiders.'

These examples show that *kɛmz(i)-* is prefixed to embedded verbs and that *kɛmz(i)-*clauses precede the matrix clauses (which here contain only the matrix verb, given Tigrinya *pro*-drop property). Also note that *kɛmz(i)-* can attach to Imperfective and Perfective verbs.

Cognitive Non-Factive Verbs

Kɛmz(i)- also introduces complement clauses with cognitive non-factive verbs (term from Baunaz & Puskás 2014) such as 'think', 'believe' and 'suspect', as in (33)–(35).

- (33) ʔaman Tesfay mɛts'hafti **kɛm-z-i-ʃɛjit** ji-ħasib
 Aman Tesfay book.FP COMP-REL-SM.3MS-sell.IPFV SM.3MS-think.IPFV
 'Aman thinks that Tesfay sells books.'
- (34) ʔaman Tesfay mɛts'hafti **kɛm-z-i-ʃɛjit** ji-ʔamin
 Aman Tesfay book.FP COMP-REL-SM.3MS-sell.IPFV SM.3MS-believe.IPFV
 'Aman believes that Tesfay sells books.'
- (35) nsu genzeb **kɛm-z-i-sɛrriɪx** ji-t'irit'r-o
 he money COMP-REL-SM.3MS-steal.IPFV SM.3MS-suspect.IPFV-OM.3MS
 'I suspect that he steals money.'

In (33), the root used for 'think' is /ħsb/. Another verbal root that conveys a similar meaning is /msl/ 'seem', already discussed in Chapter 2; it translates more closely as *it seems to X that...*

Consider (37) and (38), and compare them with (36) from Chapter 2.

- (36) Kidane ?it-i mets'haf z-ε-nbib-o ji-mesil
Kidane.MS DIST-MS book.MS REL-SM.1S-readIPFV-OM.3MS SM.3MS-seem.IPFV
'Kidane seems to read the book.'
- (37) Segen **kem-zɨ**-Ø-mesl-a ?aman sga ji-beli?
Segen COMP-REL-SM.3MS-seem.IPFV-OM.3FS Aman meat SM.3MS-eat.IPFV
'It seems to Segen that Aman eats meat.'
- (38) **Kem-zɨ**-Ø-mesl-inni, nissa ti-fetwi-nni
COMP-REL-SM.3MS-seem.IPFV-OM.1S she SM.3FS-love.IPFV-OM.1S
'It seems to me that she loves me.'

In (37), the verb bearing *kemz(i)-* shows 3MS subject agreement and does not agree with the feminine subject ‘Segen’. The same verb also shows object morphology (*-a*) agreeing with ‘Segen’. Similarly, in (38) the verb */msl/* does not agree with the null pronominal *?ane* ‘I’ but carries object morphology (*-nni*) cross-referencing it. As noted in Section 1.5.2, despite being called *object* morphology, these suffixes can index either direct or indirect objects in Tigrinya. It is therefore natural to interpret *-a* in (37) as ‘to her’ and *-nni* in (38) as ‘to me’ (a dative experiencer rather than an accusative patient). Consequently, (37) is more literally *Segen, it seems to her that Aman eats meat*. The subject in these clauses is a null 3MS expletive, while ‘Segen’ and the null pronominal ‘I’ are experiencers (see the analysis section).

Given the syncretism between 1S and 3MS Imperfective subject morphology, one might wonder whether */msl/* ‘seem’ really bears 3MS default agreement rather than 1S in (38). This is resolved by (39a) and (39b).

- (39) a. ***kem-zi**-Ø-mesl-inni z-ell-oxu, nissxa nifuɣe
COMP-REL-SM.3MS-seem.IPFV-OM.1S REL-BE₁.PRES-SM.1S you.MS smart
ʔix-a
BE₂.PRES-SM.2MS
'It seems to me that you are smart.'
- b. **kem-zi**-Ø-mesl-inni z-ell-o, nissxa
COMP-REL-SM.3MS-seem.IPFV-OM.1S REL-BE₁.PRES-SM.3MS you.MS
nifuɣe ʔix-a
smart BE₂.PRES-SM.2MS
'It seems to me that you are smart.'

On another note, the occurrence of the auxiliary *ʔallo* with /msl/ in the Imperfective in (39b) should yield progressive aspect. The sentence should therefore be translated more precisely as *It is seeming to me that you are smart* (ungrammatical in English), even though stative verbs such as ‘know’ and ‘love’ do not occur in the progressive aspect in Tigrinya.

(40) *ʔaman sga Segen **kɛm-zɪ-Ø**-mesl-a ji-beliʔ
Aman meat Segen COMP-REL-SM.3MS-seem.IPFV-OM.3FS SM.3MS-eat.IPFV
'It seems to Segen that Aman eats meat.'

¹⁰I thank Jason Overfelt and Ivano Caponigro for discussion on this.

Finally, in Amharic - where the verb *seem* triggers the clausal prefix *yä-* found in relative clauses to appear on the embedded verb - no additional element is required to form the adverbial clause, as in (41).

- (41) Kassa **yä**-hedä yə-mäslä-ññ-al
 Kassa RM-he-went it-seem-(to-)me-PR
 ‘It seems to me that Kassa went.’ (from Hailu 1972:510, (44))

Fiction Verbs

Fiction verbs — a term from Giannakidou 2009 — such as ‘dream’ and ‘imagine’, also require embedded verbs to be marked with *kemz(i)-*, as shown in (42) and (43).

- (42) Tesfay ?aman kelbi **kem-zī**-geze?-ε halim-u
 Tesfay Aman dog COMP-REL-buy.PFV-SM.3MS dream.GER-SM.3MS
 ‘Tesfay dreamt that Aman bought a dog.’
- (43) ?aman KidanE **kem-zī**-dekes-ε gemit-u
 Aman Kidane COMP-REL-sleep.PFV-SM.3MS imagine.GER-SM.3MS
 ‘Aman imagined that Kidane was sleeping.’

Note that the interpretation of ‘imagine’ above is epistemic: the context is one in which Aman and Kidane are classmates who have an early-morning course. Since Aman does not see Kidane in class, he *infers* that Kidane must still be in bed. The same applies for the interpretation of ‘dream’: Tesfay sees Aman buying a dog in a dream, but this does not mean that Tesfay dreams (in the sense of wanting or wishing) that Aman buys a dog.

Utterance verbs

Kemz(i)- is also used to introduce speech (Leslau 1941), with verbs such as ‘say’, ‘explain’, ‘read’, and ‘write’, as shown in (44)–(47).

- (44) dum-ay ?ab-z-i **kem-z**-ell-a negir-a
 cat-POSS.1S LOC-PROX-MS COMP-REL-BE₁.PRES-SM.3FS tell.GER-SM.3FS
 ‘She said that my cat is here.’

- (45) *derho kɛm-z-ɛj-ti-nifir gelits'-ɛ*
 chicken COMP-REL-NEG-SM.3FS-fly.IPFV explain.GER-SM.1S
 'I explained that a chicken doesn't fly.'
- (46) *dɛmamɯ ʔanats'u kɛm-z-i-bɛlif-u ʔanbib-ɛ*
 cat.MP mouse.MP COMP-REL-SM.3MP-eat.IPFV-SM.3MP read.GER-SM.1S
 'I read that cats eat mice.'
- (47) *Tɛsfai sirah kɛm-z-i-dɛli ts'ihif-u-llai*
 Tɛsfay job COMP-REL-SM.3MS-look_for.IPFV write.GER-SM.3MS-OBL.1S
 'Tɛsfay wrote to me that he is looking for a job.'

Kɛmz(i)- also occurs in indirect questions with verbs such as 'ask' and 'wonder', and in indirect polar questions under 'ask' illustrated in (48)–(50).

- (48) *mɛʔas (nissu) nab bet timihirti kɛm-z-i-xɛjjid ninebs-ɛi*
 when he to school COMP-REL-SM.3MS-go.IPFV self-POSS.1S
ħatit-ɛ-ja
 ask.GER-SM.1S-OM.3FS
 'I wondered (lit. asked myself) when he goes to school.'¹¹
- (49) *ʔaman ʔintay kɛm-z-i-fettu ħatit-u-ni*
 Aman what COMP-REL-SM.1S-like.IPFV ask.GER-SM.3MS-OM.1S
 'Aman asked me what I like.'
- (50) *ʔaman qobiʔ-u kɛm-z-i-fett-wa*
 Aman hat-POSS.3MS COMP-REL-SM.1S-like.IPFV-OM.3FS
ħatit-u-ni
 ask.GER-SM.3MS-OM.1S
 'Aman asked me if I like his hat.'

Perception verbs

Kɛmz(i)- is also used to introduce complement clauses with perception verbs as matrix verbs, such as 'see' and 'hear', as illustrated in (51) and (52).

¹¹I am not certain about the correct translation of *ninebs*. It appears to be a nominal (note the suffixal possessive attached to it) and its gender is consistently feminine (note the 3FS object marker on the matrix verb). It may be a cognate with Hebrew *nefesh* 'soul', which is also feminine (Ur Shlonksy, p.c.).

- (51) qεbaʔi **kεm-zi-kon-it** sεmiʔ-ε
 painter COMP-REL-become.PFV-SM.3FS hear.GER-SM.1S
 ‘I heard that she became a painter.’
- (52) a. nisatom **kεm-zi-Ø-tεmεʔa-wu** { *ʔil-ε }
 they COMP-REL-SM.3MP-marry.IPFV-SM.3MP COMP-SM.1S
 riʔεy-ε
 see.GER-SM.1S
 ‘I saw that they got married.’
- b. nsu n-iti ʔinɕεra **kεm-zi-Ø-beliʔ-o**
 he DOM-DIST-MS injera COMP-REL-SM.3MS-eat.IPFV-OM.3MS
 { *ʔil-ε } riʔεy-ε-yo
 COMP-SM.1S see.GER-SM.1S-OM.3MS
 ‘I saw that he ate *injera*.’

The examples above involve what has been called *indirect evidence/perception* (Frajzyngier 1991): it has been argued that the English translations contain finite clause complements that express an “epistemically non-neutral (factive) attitude” (Moulton 2009:9). In other words, the speaker infers the content of the complement clause rather than witnessing it directly. In (52), the speaker was not present at the wedding and did not witness the eating of the *injera*; for example, she may have seen an announcement of the couple’s marriage in a newspaper or a picture of their wedding in their house, and noticed either the absence of the *injera* from the fridge in her and her brother’s apartment or some crumbs on the kitchen counter.

Table 3.4 below summarizes the types of verbs discussed so far that require *kεmz(i)-* to attach to embedded verbs.

<i>kεmz(i)-</i>	
FACTIVES	e.g. ‘know’, ‘forget’, ‘admit’, ‘accept’
COGNITIVE NON-FACTIVES	e.g. ‘think’, ‘believe’, ‘suspect’
FICTION VERBS	e.g. ‘dream’, ‘imagine’
UTTERANCE VERBS	e.g. ‘say’, ‘explain’, ‘read’, ‘write’, ‘ask’, ‘wonder’
PERCEPTION VERBS	e.g. ‘see’, ‘hear’

Table 3.4: Classes of verbs that select for *kεmz(i)-*clauses.

I now compare the uses of *kεmz(i)-* and the clause-final complementizer *ʔilu*.

3.2.3 *Kemz(i)-* vs. *?ilu*

The clausal prefix *kemz(i)-* is not the only element that introduces subordinate clauses with cognitive non-factive, fiction and utterance verbs. In fact, the complementizer *?ilu* can appear in exactly (and only) these contexts, as illustrated in (53), (54) and (55).¹²

- (53) nsu genzeb ji-serrix **?il-ε** ji-t'irit'r-o
 he money SM.3MS-steal.IPFV COMP-SM.1S SM.1S-suspect.IPFV-OM.3MS
 'I suspect that he steals money.' *Cognitive Non-Factive*
- (54) Tesfay ?aman kelbi gezi?-u **?il-u** halim-u
 Tesfay Aman dog buy.GER-SM.3MS COMP-SM.3MS dream.GER-SM.3MS
 'Tesfay dreamt that Aman bought a dog.' *Fiction*
- (55) demamu ?anats'u ji-beli?-u **?il-ε** ?anbib-ε
 cat.MP mouse.MP SM.3MP-eat.IPFV-SM.3MP COMP-SM.1S read.GER-SM.1S
 'I read that cats eat mice.' *Utterance*

From the examples above, *?ilu* always agrees with the grammatical subject of the matrix sentence and occurs between the matrix and the embedded verb. Its clause-final position is also visible in CP coordination. Coordination in Tigrinya is expressed by the suffix *-(i)n* attached to each coordinated element (Mason 1996).¹³ Coordinated elements can be nominals, as in (56), but also clausal, as in (57). In the latter case, *?ilu* cannot be repeated and cannot bear the coordination marker. This suggests that *?ilu* is embedded under the matrix verb and selects the coordinated elements.

- (56) dmu-**n** gobijε-**n**
 cat-COORD turtle-COORD
 'a cat and a turtle'
- (57) a. ?aman Tesfai mets'hafti ji-ʃejit'i-n Kidane tigrina
 Aman Tesfay book.MP SM.3MS-sell.IPFV-COORD Kidane Tigrinya
 ji-mehiri-n **?il-u** ji-hasib
 SM.3MS-teach.IPFV-COORD COMP-SM.3MS SM.3MS-think.IPFV
 'Aman thinks that Tesfay sells books and Kidane teaches Tigrinya.'

¹²For a detailed semantic but also syntactic investigation of this complementizer, see Spadine 2020.

¹³Realized as [y] in Rayya Tigrinya (Niguss 2021:222).

without any coordination or subordination elements” (Nazareth 2011:169). According to Spadine, *?ilu* is a clause-final complementizer that can appear embedded with a matrix verb, or in monoclausal “truncated *?ilu* constructions” when no matrix verb is present (as in (58c)).

?ilu also occurs in a circumlocution expressing the meaning of *sitting*, as shown in (59) (the meaning of *kof* alone is unknown to me).

- (59) ?it-i q^welʕa kof ?il-u ?all-o
 DIST-MS child.MS sitting COMP-SM.3MS BE₁.PRES-SM.3MS
 ‘The child is sitting.’

It can also express “goodwill or good intention” in sentences with benefactives (Nazareth 2011), as in (60). Benefactives are marked with the BEN marker *n(i)-* (note that in the same example the same marker is used as a DOM marker).

- (60) Saba n-it-i mɛts’haf ni-yonas (?il-a)
 Saba DOM-DIST-MS book.MS BEN-Yonas COMP-SM.3FS
 gɛziʔ-a-jo
 buy.GER-SM.3FS-OM.3MS
 ‘Saba bought the book (as she intended) for Yonas.’
 (adapted from Nazareth 2011:168,(167))

If both *kɛmz(i)-* and *?ilu* can be used with cognitive non-factive, fiction and utterance verbs, what distinguishes sentences that use one or the other? According to my consultants (and most speakers (cf. Spadine 2020)), there is no perceptible difference in meaning between *kɛmz(i)-* and *?ilu* in (58). However, closer analysis reveals an important difference. Spadine 2020 argues that, unlike *kɛmz(i)-* clauses, *?ilu* clauses can receive a shifted interpretation (Spadine 2020:29), or in other words, trigger indexical shift.

Indexical shift is the phenomenon in which the reference of indexicals is determined by a context different from the utterance context (Deal 2020). For example, in Amharic indexical pronouns (*I* and *you*) can refer to the speaker of reported speech rather than of the actual discourse (Schlenker 2003). This is also true for Tigrinya, as shown below (the

index _k refers to an individual outside the discourse).¹⁴

- (61) a. Kidane ni almaz ?issu niʔaʔa **kimzi**-räʔaj-ä-ya
 Kidane.M DOM Almaz.F he ACC.3FS COMP-see-SM.3MS-OM.3FS
 nägar-u-wa
 tell-SM.3MS-OM.3FS
 ‘Kidane_i told Almaz_j that he_{i/k} saw her_{j/*k}.’
- b. Kidane ni almaz **?anä niʔaki** **kimzi**-räʔaj-ku-ki
 Kidane.M DOM Almaz.F I ACC.2FS COMP-see-SM.1S-OM.2FS
 nägar-u-wa
 tell-SM.3MS-OM.3FS
 ‘Kidane_i told Almaz_j that I_{speaker/*k} saw you_{addressee/*k}.’
- (adapted from Spadine 2020:101, (139))

- (62) Kidane ni almaz **?anä niʔaki** räʔaj-ä-ki **?il-u**
 Kidane.M DOM Almaz.F I ACC.2FS see-SM.1S-OM.2FS COMP-SM.3MS
 nägar-u-wa
 tell-SM.3MS-OM.3FS
 ‘Kidane_i told Almaz_j that he_{i/*k} saw her_{j/*k}.’ (adapted from Spadine 2020:102,
 (140))

In (61b), with *kemz(i)-*, the indexical pronoun *?anä* ‘I’ can only refer to the speaker of the utterance context. In (62), with *?ilu*, the indexical pronoun *?anä* ‘I’ refers to *Kidane*. Thus, in the *?ilu*-clause the indexicals shift. These examples show that *kemz(i)-* clauses are always “interpreted with respect to the utterance context” and never shift (Spadine 2020:30). In contrast with *?ilu*, *kemz(i)-* does not contribute an attitude-report interpretation and does not allow indexical shift.

Although the syntax of *?ilu* will not be investigated here (because the focus is on clausal prefixes rather than clause-final complementizers, and because it has already been studied (cf. Spadine 2020)), mentioning this complementizer is important to understand better the syntax of *kemz(i)-* and its function within the language.

Table 3.5 summarizes the types of verbs requiring *kemz(i)-* and *?ilu* with embedded verbs.

¹⁴Note a typo in the glossing of Spadine’s example (139-b) regarding subject and object marking on the embedded verb, corrected here.

	<i>kɛmz(i)-</i>	<i>?ilu</i>
FACTIVES	✓	✗
COGNITIVE NON-FACTIVES	✓	✓
FICTION VERBS	✓	✓
UTTERANCE VERBS	✓	✓
PERCEPTION VERBS	✓	✗

Table 3.5: Classes of verbs that select for *kɛmz(i)-*-clauses and *?ilu*-clauses.

It is now time to turn to the syntactic analysis of *kɛmz(i)-*-clauses.

3.3 The Analysis of *kɛmz(i)-*-clauses

In the previous section I presented extensive data on the structures in which *kɛmz(i)-* occurs. I began by comparing *kɛmz(i)-*-clauses to conditional and adverbial clauses, supporting the view that *kɛmz(i)-* is bimorphemic. I also discussed how, in periphrastic tense forms, *kɛmz(i)-* can be placed in different positions. I then showed that this clausal prefix appears in clauses introduced by factive, cognitive non-factive, fiction, utterance, and perception verbs.

In the next sections, I provide a syntactic analysis for *kɛmz(i)-*-clauses, drawing on the concept of “subordination as relativization”. The analyses below not only shed light on the properties of *kɛmz(i)-* but also address how a head-final language can possess such a *prefix*.

3.3.1 Complement Clauses as Relative Clauses

Let us focus again on *kɛm*. We have seen that this particle is etymologically a preposition and, beyond *kɛmz(i)-*-clauses, is also used in Tigrinya as a simulative morpheme (term from [Haspelmath & Buchholz 1998](#)) meaning ‘like’. In some languages, complementizers derive historically from prepositions — sometimes also analyzed as adverbs — ([Noonan 2007:57](#)). For example, some languages use simulative morphemes to introduce complement clauses; in English, ‘like’ has grammaticalized into a complementizer used to introduce reported

speech, as shown in (63).¹⁵

(63) And she’s **like**, “Um... Well, that’s cool.” (from Romaine & Lange 1991:227)

Similarly, Ethio-Semitic languages make use of similitive morphemes to introduce certain complement clauses and purpose clauses (Treis 2017:91). At the beginning of this chapter I have already shown in (2), (3) and (4) that this holds for Amharic, Argobba and Zay. The types of complement clauses marked by ‘like’ morphemes are summarized in Table 3.6.

Knowledge (‘know’, ‘forget’)	Tigrinya , Tigre, Amharic, Harari, Wolane, Zay, Gumer
Utterance (‘tell’)	Tigrinya , Tigre, Amharic, Wolane, Gumer, Argobba, Inor
Manipulative (‘do/tell, cause s.o. to V’)	Amharic, Wolane, Zay, Argobba, Inor
Perception (‘hear’, ‘see’)	Tigrinya , Tigre, Amharic, Argobba, Muher
Desiderative (‘want s.o. to V’)	Tigre, Amharic, Argobba, Wolane

Table 3.6: Types of complement clauses marked with ‘like’ across Ethio-Semitic (from Treis 2020:23).

In the literature, similitive complements are said to be uncommon with propositional attitude verbs (such as ‘think’ and ‘believe’) and unattested with modals (Treis 2020). I have shown that the latter is true for Tigrinya (among the classes of verbs that select *kɛmz(i)*-clauses there are no modals; in fact, in Chapter 4 I show that modals select *k(i)*-clauses), but the former is false (the verbs ‘think’ and ‘believe’ do trigger *kɛmz(i)*- to attach to the embedded verb). Introducing complement clauses with a similitive morpheme is therefore not specific to Tigrinya.

Now consider *z(i)*-. As seen, *kɛm* is a particle, likely a preposition-like element, that must occur with *z(i)*-. In Romance languages some complement and adverbial clauses are introduced by particles that resemble *kɛmz(i)*- in being composed of a preposition-like element plus a subordinator or complementizer. These include Italian *affinché* ‘so that’ from *a* ‘to’ + *fin(e)* ‘end’ + *che* ‘that’; French *puisque* ‘since/because’ from *puis* ‘then’ + *que* ‘that’; and Spanish *porque* ‘because’ from *por* ‘for’ + *que* ‘that’. The complementizers

¹⁵In some varieties of English a sentence introduced with ‘like as how’, such as *John was telling like as how he went to the movies.*, is also grammatical (Richard Kayne, p.c.).

che, *que* and *que* in these languages are also used in relative clauses, exactly like *z(i)*-.¹⁶

Within the Semitic family there are comparable cases in which complement and/or adverbial clauses are introduced by a preposition-like (or adverb-like) element plus a subordinator. In Levantine Arabic the adverbs *ʔəbil* ‘before’ (Classical Arabic *qabla*) and *baʔd* ‘after’ are followed by the particle *mā* before a verb (McCarus et al. 1978). In Standard Arabic several complementizers traditionally known as *ʔinna* and *its sisters* appear to be composed, like *kəməz(i)*-, of a subordinator *ʔanna* plus the prepositions *li* (in *liʔanna* ‘because’) and *ka* (in *kaʔanna* ‘as if’) (Ahmed 2015, Mudhafar Ali 2015). In Akkadian, when *kīma* (a cognate of *kəmə*; see the next section) occurs in a clause, a subordinate suffix *-u* is attached to the verb (Deutscher 2006:161). Returning to the Ethio-semitic family, in several languages complement clauses are introduced by a preposition-like element plus a subordinator, as already illustrated in (3) and (4). Thus, introducing complement clauses with a preposition is not specific to Tigrinya, and the use of a single element to introduce both relative clauses and complement clauses is cross-linguistically attested.

The literature contains many demonstrations that the same element introduces both complement clauses and relative clauses in a wide range of languages (a.o. Kayne 1976, Ouhalla 2004, Franco 2012, Manzini & Roussou 2020, Poletto & Sanfelici 2021). This observation underlies the theoretical idea that complement clauses are in fact relative clauses. On this view, complementation and relativization derive from a single syntactic derivation (Franco 2012) involving the same introducing element. Kratzer 2006 also argues that complement clauses are predicates, like relative clauses. A common line of research holds that complementizers are relative pronouns (Manzini & Roussou 2020) belonging to a determiner-like category (Poletto & Sanfelici 2021). For instance, Kayne 1976 proposes that in French the complementizer *que* and the relative *que* instantiate the same item.¹⁷ In English, sentential *that* has been analyzed as a sub-case of relative *that* (Kayne 2010:192),

¹⁶Consider also *tale quale* in Italian - as in *Tua sorella è tale quale a te* ‘Your sister is like you.’ - containing the comparative element *tale* ‘such’ accompanied by the *wh*-element *quale* ‘which’ (Emanuela Sanfelici, p.c.).

¹⁷By contrast, for *qui*, Sportiche 2011 argues it is a pronoun, contra Kayne 1976, who suggests it is a special form of *que* triggered by subject extraction (Manzini 2014).

and factive complement clauses are introduced by “the fact that” i.e., partly invisible relative clauses (Kratzer 2006, Kayne 2008a, 2010, Haegeman 2012). Arsenijević 2009 further suggests that complement clauses contain a noun (e.g., *claim* below) that is a nominal realization of a force feature which moved from Spec,ForceP of the embedded clause and incorporated into the verb, making complement clauses akin to noun clausal complements. This is illustrated in (64).

- (64) a. John claimed that Mary came late.
 b. John [_{VP} made [_{DP} the [_{NP} claim [_{CP} that Mary came late]]]]
 (adapted from Arsenijević 2009:43, (8))

The subordination-as-relativization idea has been applied to many languages: Italian (Manzini & Savoia 2003, Manzini 2014, Manzini & Roussou 2020), Gbe (for factive complement clauses) (Aboh 2005), Adyghe (Caucasian) (Caponigro & Polinsky 2008, 2011), Serbo-Croatian (Arsenijević 2009), Bulgarian (Krapova 2010), Washo¹⁸ (Hanink & Bochnak 2017), and Nez Perce (Native American) (Deal 2024), among others. Some authors claim that complement clauses are relative clauses cross-linguistically - for example Kayne 2008a and Arsenijević 2009 - while others allow for cross-linguistic variation (Caponigro & Polinsky 2008, 2011).¹⁹ In this dissertation I claim that, at least in Tigrinya, complement clauses introduced by *kɛmz(i)*- are underlyingly relative clauses, or more precisely, that both constructions involve $\bar{A}$ -movement. The two constructions nevertheless differ in properties, notably relative clauses contain a gap and thus an $\bar{A}$ -dependency, unlike complement clauses.

Regarding the $\bar{A}$ -dependency, I propose that *z(i)*- appears only in clauses with $\bar{A}$ -movement. Since *z(i)*- occurs in *kɛmz(i)*-clauses, something must move. What moves? In the semantics literature on subordination as relativization there is a tendency to treat complement clauses as gapless (Kratzer 2006, Moulton 2014), lacking $\bar{A}$ -dependencies. Syntactic accounts, however, posit a gap in complement clauses (a.o. Arsenijević 2009,

¹⁸An endangered Native American language isolate spoken around Lake Tahoe (California, USA).

¹⁹Kayne 1994 and Arsenijević 2009 further suggest that if relative clauses are generated as complements rather than adjuncts, the difference between finite complement clauses and restrictive relative clauses disappears across languages.

Caponigro & Polinsky 2008, 2011, Haegeman 2012) corresponding to a null nominal such as FACT (Kayne 2010) or EVENT (Poletto & Sanfelici 2021), or to an event variable bound by a null operator (Haegeman 2012). The argument for a null nominal EVENT draws on the connection between relative clauses and sentential complements with perception verbs like *see*: in the relative clause in (65), the speaker saw someone (the child), and in the complement clause in (65) the speaker saw something (an event).

- (65) a. I saw the child [that was hanging in a cradle].
 b. I saw [that the child was hanging in a cradle] .
 (adapted from Dixon 2006:36)

In the semantics literature it is claimed that there is a “hidden argument place for events” (Higginbotham 1985:555); that is, the thematic grid of verbs contains an “open position for events or situations” (Higginbotham 1989:467), a view originating with Davidson 1967. Following Higginbotham, Poletto & Sanfelici 2021 propose that an event argument is base-generated in the highest projection of the *vP* layer, specifically in the specifier of an unlabeled FP. The question is what licenses the null nominal EVENT. Poletto & Sanfelici 2021 do not elaborate. Higginbotham 1985 argues that the licenser is tense: “the position *E* of the thematic grid of the verb is discharged at the point where VP meets Infl.” (Higginbotham 1985:561). In a different approach, Borer 2005 posits an EventP projection; she suggests that this phrase can be licensed by various elements (Poletto & Sanfelici 2021:387, fn. 15), but mainly the verb (more precisely *vP*) is responsible (Borer 2005:36).

However, not all complement clauses are introduced by *see* and therefore not all complement clauses necessarily involve an event. In fact, claiming that *kemz(i)*-clauses involve an event nominal is problematic for two reasons. First, as seen, not all *kemz(i)*-clauses describe events. These clauses also occur with states, as in (66).

- (66) Tesfai Kidanɛ hawɛb-u **kɛm-zi-xon-ɛ**
 Tesfay Kidane cousin-POSS.3MS COMP-REL-become.PFV-SM.3MS
 rɛsiʃi-w-o
 forget.GER-SM.3MS-OM.3MS
 ‘Tesfay forgot that Kidane is his cousin.’

Second, *kɛmz(i)*-clauses are also selected by matrix verbs that cannot take events as complements. For example, ‘believe’ can take an individual (*I believe you.*) or a proposition (*I believe that they left.*), but not an event.

I therefore adopt the idea that complement clauses involve the relativization of null nominals (but not necessarily events) and I refine the nature of these elements below.

3.3.2 Null Nominals

I suggest that *kɛmz(i)*-clauses involve the $\bar{A}$ -movement of a null nominal (FACT, STATE, EVENT,...) inferred by the speaker and hearer. That the interpretation of a nominal is inferred and not present in the syntax is not unusual. Consider the sentences in (67) that involve the phenomenon of copredication ([Pustejovsky 1995](#)).

- (67) a. The book is heavy/red/made of wood.
 b. The book is interesting/too long/well written.

The predicates in (67a) target the concrete object *book*; those in (67b) target the story/content. Nothing in the string distinguishes the two readings, but the difference is inferable.²⁰ I propose that a similar mechanism guides the choice of the null nominal in *kɛmz(i)*-clauses.

The type of null nominal is inferred, but this inference depends on the embedded verb. For this reason, the null nominal is merged low, within the argument structure of the verb. Why should it depend on the embedded verb rather than the matrix one? One might hypothesize that verbs encode semantic and syntactic information about their complements. Consequently, with matrix *see* one expects EVENT, with *know* FACT, with *think* THOUGHT, and so on.

However, some semanticists reanalyze the selection of finite complements by shifting

²⁰I thank Gillian Ramchand for pointing this out to me.

relevant meaning from the matrix verb to the embedded clause (Kratzer 2006, Moulton 2009). Consider (68) with ‘discover’.

- (68) a. I discovered that he went to Paris.
b. I discovered that he is a spy.

In (68a) the complement denotes an event; in (68b) it denotes a state. The verb ‘discover’ can take either, and the complement type is determined by the embedded predicate, which denotes an event or a state. An alternative would treat ‘discover’ as polysemous, but “if we resorted to several different lexical items all pronounced [discover], we would still need to account for the correlation between the meaning of [discover] with its complement type” (Moulton 2009:130).

By positing a null nominal, I suggest that $k\epsilon m z(i)$ -clauses are akin to noun clausal complements. Their syntax is not identical, however, since in the latter only $z(i)$ - appears, without $k\epsilon m$. As proposed in Chapter 2, I assume the following for Tigrinya:

- (69) a. The story that Solomon bought a horse is false. *Noun Clausal Complement*
→ The STORY that says that Solomon bought a horse is false.
b. I know that Solomon bought a horse. *Complement Clause*
→ I know (the) FACT that Solomon bought a horse.

What remains to be determined is the position of $k\epsilon m$.

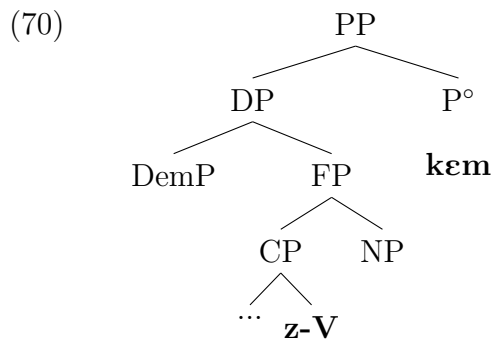
3.3.3 Where is $k\epsilon m$?

We have seen that $k\epsilon m$ is a similative, preposition-like element that has likely undergone grammaticalization and is now used to introduce complement clauses (Hernández 2024). If $k\epsilon m$ were indeed a preposition, and thus a lexical element, it would be difficult to explain how it could attach only to the auxiliary in one of the strategies for complement clauses with periphrastic tense forms.²¹ It must therefore be a functional element. But where exactly? Recall that the embedded subject and object precede $k\epsilon m$ in $k\epsilon m z(i)$ -clauses.

²¹I thank Amy-Rose Deal for pointing this out to me.

Under a head-initial approach to head-finality placing *kem* in C° or Spec,CP, the subject and the object would have to be higher than CP, which is implausible. Under a head-final approach (as adopted so far) placing *kem* in Spec,CP would still require the subject and object to move above CP. One could instead claim that *kem* is in C° , or at least, in the head of a functional projection in the left periphery.

One could adopt Haegeman 2007, 2012’s analysis of prepositional temporal clauses, which are introduced by prepositions such as *before*, *after*, *since* and *until* and involve a CP with an operator raising in the structure. Because CPs cannot be the complement of a preposition (Stowell 1981, Alrenga 2005) (contra Borsley 1986, 2001), the preposition must select a nominal complement. If we adapt Haegeman’s approach to the idea of *kemz(i)*-clauses as noun clausal complements, we obtain the following:



The problem with this derivation is that *kem* cannot be prefixed (or precede) the *z(i)*-marked verb because there is a ban on extraction of elements from the clausal complements of a noun (Larson 1990, Haegeman 2012).

Another approach could place *kem* in the C° head of the clausal complement itself. This is problematic, however, for the various configurations of *kemz(i)*-clauses with periphrastic tense forms. In particular, the configuration in which *kem* is prefixed onto the *z(i)*-marked verb while the auxiliary is only marked by *z(i)*- (S-O-*kemzi*-V-*zi*-Aux) would be difficult to derive: the *z(i)*-marked verb would have to move over the *z(i)*-marked auxiliary to be prefixed by *kem*, and the *z(i)*-marked auxiliary would then need to move higher than CP without motivation.

With this in mind, I suggest, following my previous work (Cacchioli & Shlonsky 2023), that *kem* is actually low in the structure. Several authors have already argued that

prepositional complementizers merge lower than CP: [Pesetsky & Torrego 2004](#) place them under T^0 ; [Kayne 1999](#) base-generates prepositional complementizers in Romance above VP; [Rizzi 1982](#) reanalyzes them as VP adjuncts. This makes sense because prepositions help structure the event or state of the predicate and contribute to its interpretation in terms of its temporal and spatial interpretation ([Verkuyl 1972](#)) (compare *I sat* vs. *I sat on the chair*).

Moreover, I claim that *kem* is paired with the null nominal. This idea is based on [Poletto & Sanfelici 2021](#), who propose that in complement clauses (sharing the syntax of relative clauses) the null nominal EVENT is paired with the Italian complementizer *che* ‘that’. They do not define ‘paired with’ but they suggest that the relativized head has the form *che* EVENT ‘that EVENT’, since the complementizer modifies the null nominal ([Poletto & Sanfelici 2021:386](#)).

We now have all the pieces of the puzzle. Before turning to the analysis of the syntax of *kemz(i)*-clauses, I discuss two previous accounts in the following section.

3.3.4 Previous Accounts

To my knowledge, only one work briefly proposes a possible syntactic analysis of *kīma*-clauses in Akkadian, [Franco 2012](#), and there is one study on Tigrinya *kemz(i)*-clauses: namely, [van Urk 2025](#) (a manuscript developed from [van Urk \(2024\)](#), a workshop talk).

For Akkadian *kīma*-clauses, since they involve suffixation of the subordinator *-u*, as already mentioned, it is apparently possible to analyze their syntax within a Cartographic approach: *kīma* and *-u* would head the projection dedicated to finiteness (FinP) and illocutionary force (ForceP), following [Rizzi 1997](#)’s proposal of a stretched complementizer field ([Franco 2012:580](#)). Franco suggests that the structure *kīma*-clause-*u* (i.e. ForceP-clause-FinP) is straightforward to derive if the clause moves to a Spec,XP position in the Left Periphery ([Franco 2012:581](#)).

For Tigrinya, [van Urk 2025](#), adopting a head-final approach to head-finality, focuses on the behavior of *kemz(i)*- in periphrastic tense forms expressing present perfect (a Gerund verb co-occurring with an auxiliary), as illustrated in (15), where *kemz(i)*- can attach to

either the verb or the auxiliary. Van Urk asks how this is possible and answers as follows. $K\epsilon m z(i)-$ is composed of $k\epsilon m$ and $z(i)-$. $K\epsilon m$ is base-generated at the right edge of the clause and is thus a head-final complementizer. Given the scenario in periphrastic tense forms, the cliticization of $k\epsilon m$ can be non-local (van Urk 2024:2). To explain this non-local cliticization, $k\epsilon m$ is analyzed as a proclitic element undergoing Local Dislocation (lowering) in the postsyntactic component, à la Embick & Noyer 2001. This lowering mechanism in Tigrinya is triggered by a morphological constraint that forces complementizers to immediately precede a V head (van Urk 2025:16). This lowering involves inversion of adjacent elements, in this case of $k\epsilon m$ and the $z(i)-$ -marked auxiliary (van Urk 2024:3), resulting in (15a).

Moreover, van Urk proposes that the occurrence of $z(i)-$ in $k\epsilon m z(i)-$ clauses is the result of a rule of node insertion - the operation in Distributed Morphology that allows a morpheme to impose vacuous morphological marking on another (Noyer 1998) - triggered by $k\epsilon m$. This is articulated as in (71) in van Urk 2024 and as in (72) in van Urk 2025.

(71) *Rule of $z(i)-$ insertion:* Copy the features of $C_{k\epsilon m}$ onto a preceding V/Aux.
(from van Urk 2024:4,(20))

(72) Rel Insertion: Insert a node Rel in the context $[_C \text{ DECL } [___ X_{[\text{CAT:V}]}]]$
(from van Urk 2025:20,(48))

If I understood van Urk correctly, the rule in (71) states that $k\epsilon m$ triggers the appearance of $z(i)-$ on V and Aux by copying its own features onto them. In addition, van Urk suggests that $z(i)-$ has a violable requirement to be preceded by a complementizer — violable because in relative clauses there is no complementizer — (van Urk 2024:6), which explains the ungrammaticality of (15c) and the grammaticality of (15b). In contrast, the rule in (72) forces the insertion of a relative node on all verbal heads that follows a declarative complementizer.

I find the Distributed Morphology approaches taken by van Urk intriguing but slightly weak on three grounds. First, with regard to the rule in (71), if the features of $k\epsilon m$ are copied, how does this yield $z(i)-$ rather than a second $k\epsilon m$ — which is ungrammatical, as

(15d) shows? Second, and more importantly, I showed that *kɛm* is not the only particle that requires the verb to be *z(i)*-marked (cf. table 3.1); this would mean that the exact same rule in (71) and (72) would have to apply for every particle, missing a generalization of the language. Third, I find the rule in (72) too vague, as it would, in principle, also apply to the declarative complementizer *?ilu*, discussed in Section 3.2.3. I therefore take a different approach here.

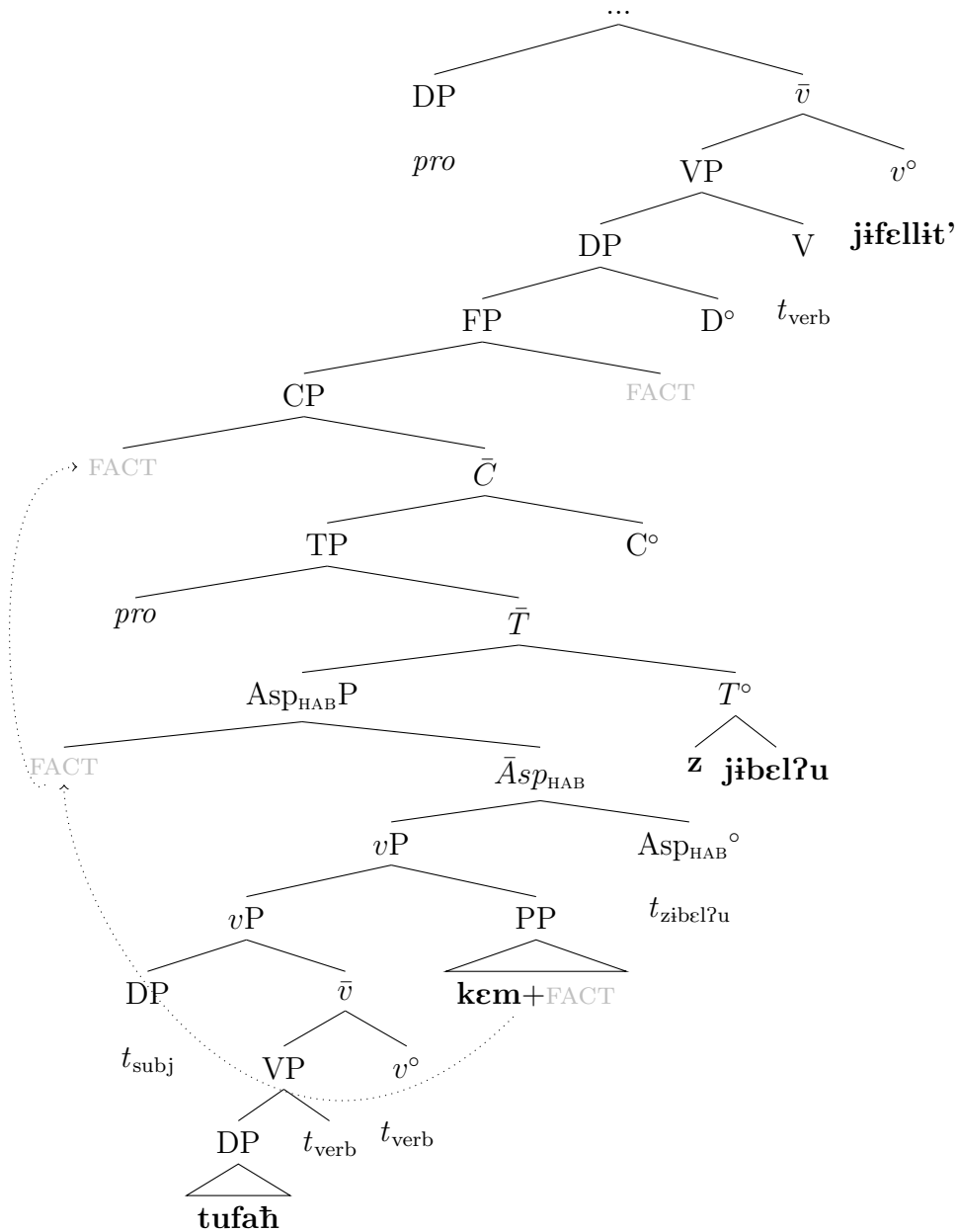
It is now time to turn to the analysis of the distribution of *kɛmz(i)*-. I first discuss *kɛmz(i)*-clauses with simple verb forms, then *kɛmz(i)*-clauses involving the verb /*msl*/ ‘seem’ and finally *kɛmz(i)*-clauses with periphrastic tense forms.

3.3.5 Analysis of *kɛmz(i)*-clauses with simple verb forms

I propose that *kɛmz(i)*-clauses in Tigrinya, such as that in (73) repeated from an example in the first section of this Chapter, can be syntactically derived as shown in the tree in (74).

- (73) tufah **kɛm zi**-bɛliʔ-u ji-fellit’
 apple KɛM zi-eat.PFV-SM.3MP SM.1S-know.IPFV
 ‘I know that they ate an apple.’

(74)



In the tree above, the null nominal FACT is paired with $k\epsilon m$ in a vP adjunct position. The null nominal raises to the specifier of $\text{Asp}_{\text{HAB}}P$, while $k\epsilon m$ remains in its base position. The verb also raises in the structure and lands in the $\text{Asp}_{\text{HAB}}^\circ$ head. When the silent nominal phrase and the verb are in a Spec-head agreement configuration, they trigger the appearance of $z(i)-$. When the internal copy of the null nominal FACT reaches Spec,CP the external copy deletes it under identity. However, this has no phonological effect, since the nominals are not phonologically overt.

3.3.6 Analysis of *kɛmz(i)*-clauses with /msl/ ‘seem’

In the previous sections, I showed that a special case of *kemz(i)*- involves the verb /msl/ ‘seem’, already discussed in Chapter 2. An example given above is repeated in (75).

- (75) Segen **kem-zi**-Ø-mesl-a ?aman sga ji-beli?
Segen COMP-REL-SM.3MS-seem.IPFV-OM.3FS Aman meat SM.3MS-eat.IPFV
'Segen, as it seems to her, Aman eats meat.'

I have already established several facts, which I repeat here. First, although *kɛmz(i)-* clauses have so far appeared to be complement clauses, the sentence above is actually an adverbial clause, given its interpretation. This is not surprising, since I show in Chapter 4 that the clausal prefix *k(i)-* can introduce both complement and adverbial clauses. Second, I have pointed out the fact that the *kɛmz(i)-* clause is left-dislocated (i.e., placed at the beginning of the sentence), again supporting the idea that it is an adverbial clause.²² Third, I have shown that the verb /*msl*/ ‘seem’ is marked with default 3MS subject morphology and carries an object marker *-a* ‘to her’ that cross-references the experiencer ‘Segen’.

Now, what is ‘Segen’? It cannot be a subject, because the verb /msl/ ‘seem’ carries default subject morphology, and the verb /blʕ ‘eat’ agrees with ‘Aman’. Even if I claimed that the translation of the example should be something like *As it seems to Segen...*, making ‘Segen’ an experiencer, it cannot be an indirect object, since it does not carry the prefix that marks Dative in Tigrinya, as explained in section 1.5.2 in the introduction, as shown in (76).

- (76) *n-Segen kem-zi-Ø-mesl-a ʔaman sga
 DAT-Segen COMP-REL-SM.3MS-seem.IPFV-OM.3FS Aman meat
 ji-beliʔ
 SM.3MS-eat.IPFV
 ‘Segen, as it seems to her, Aman eats meat.’

One could argue that ‘Segen’ is a hanging topic (Cinque 1977). A hanging topic is a clause-

²²However, there are instances of left-dislocated complement clauses, such as *Why she did it, I don't know* or *That she did it, I didn't know*.

initial, adjunct-like element that is not part of the syntactic structure of the predicate (i.e., it is neither a subject nor an object). Hanging topics have been investigated since the 1970s, mainly in European languages (to my knowledge: Italian, French, Catalan, Occitan, German), and it has been proposed that they are generated in the specifier position of a discourse-related projection labelled DiscP (Benincà 2001). An example in Italian is given in (77), with the hanging topic in bold.

- (77) **Giorgio**, ieri ho conosciuto la ragazza che gli ha scritto quelle
 Giorgio yesterday have met the girl who to_him has written those
 insolenze
 insults
 ‘About Giorgio, yesterday I met the girl who wrote those insults to him.’
 (from Cinque 1977:408, (33), gloss and translation are mine)

By definition, hanging topics are resumed by a coreferential element in the following clause, as with *gli* ‘to him’ in the example above, which is a strong island. If this were the case in Tigrinya, then we could be certain that ‘Segen’ is a hanging topic in the sentences above. Although we see in (75) the coreferential element *-a* is suffixed onto the *kemz(i)*-marked verb, it is important to test this by embedding the sentence and checking whether the results is grammatical. Example (78) illustrates that ‘Segen’ cannot be extracted from an embedded clause.

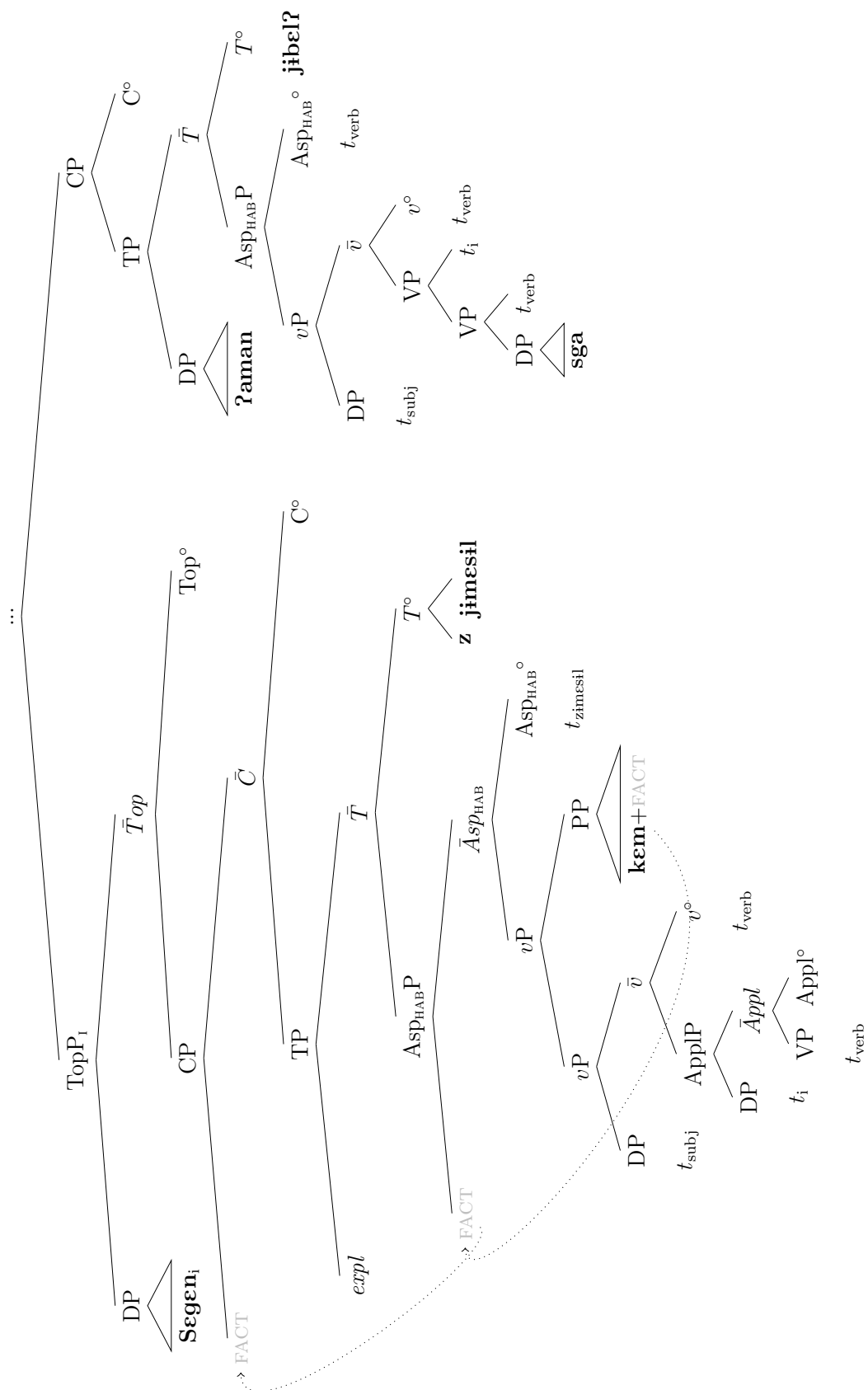
- (78) a. *Segen mis-t-a **kem-zi**-Ø-mesl-a ?aman sga
 Segen with-DIST-FS COMP-REL-SM.3MS-seem.IPFV-OM.3FS Aman meat
 ji-beli? **zi**-beliʔ-et səbejiti terakib-Ø-a
 SM.3MS-eat.IPFV REL-say.PFV-SM.3FS woman meet.GER-SM.1S-OM.3FS
 ‘I met the woman who said that it seems to Segen that Aman eats meat.’
 b. mis-t-a Segen **kem-zi**-Ø-mesl-a ?aman sga
 with-DIST-FS Segen COMP-REL-SM.3MS-seem.IPFV-OM.3FS Aman meat
 ji-beli? **zi**-beliʔ-et səbejiti terakib-Ø-a
 SM.3MS-eat.IPFV REL-say.PFV-SM.3FS woman meet.GER-SM.1S-OM.3FS
 ‘I met the woman who said that it seems to Segen that Aman eats meat.’

Thus, ‘Segen’ is not a hanging topic. What is it, then? I follow Tesfay 2016, who provides the Amharic example in (79).

- (79) ine innässu wädä roma yäm-yi-hed-u yi-mäsl-änn-al
 I they.3PL to-Rome COMP-REL-3-go-PL 3.appear-1s-exist
 ‘It appears to me that they are leaving for Rome.’ (from [Tesfay 2016:245,\(48h\)](#))

The example above presents the Amharic cognate of the verb /msl/ ‘seem’ in Tigrinya. Tesfay claims that this sentence shows how “applicative arguments may be focused or topicalized [moving] somewhere above the subject position” [Tesfay 2016:246](#). In other words, expletive subjects can appear with topicalized, nominative marked applicative objects. I therefore propose that ‘Segen’ in (75) above is a topicalized applicative object, namely an element with a peripheral semantic role that has topicalized during the derivation.

With all this in mind, I claim that (75) can be represented by the tree in (80). Note that, as with relative clauses in Chapter 2, I leave the investigation of object morphology on /msl/ to future research.



As in the other *kɛmz(i)*-clauses discussed above, the null nominal FACT is paired with *kɛm* in a *vP* adjunct position. ‘Segen’ is merged in the specifier of an Applicative phrase dominating VP (Pylkkänen 2000, McGinnis 2001). The null nominal raises to the specifier of *Asp_{HAB}P*, while *kɛm* remains in its base position. The verb also raises in the structure and lands in the *Asp_{HAB}°* head. When the silent nominal phrase and the verb are in a Spec-head agreement configuration, they trigger the appearance of *z(i)*-. Being topicalized, ‘Segen’ moves from its first-merge position to the specifier of TopP. The entire *kɛmz(i)*-clause is an adverbial clause that has been dislocated from an adjunct position (I placed it above VP, but it could also be above *AspP*) to a position higher than CP.

It is now time to turn to *kɛmz(i)*-clauses with periphrastic tense forms.

3.3.7 Analysis of *kɛmz(i)*-clauses with periphrastic tense forms

I have shown that in *kɛmz(i)*-clauses expressing progressive aspect with a verb occurring alongside an auxiliary, the word order is unusual. Either *kɛmz(i)*- appears on the auxiliary and the verb is bare, or it occurs on the verb and the auxiliary carries *z(i)*-. It is not possible for *kɛm* to appear on both the verb and the auxiliary, nor for the auxiliary to carry *kɛmz(i)*- while the verb is prefixed with *z(i)*-. This is shown in (81), repeated from a previous section.

- (81) a. Solomon televizion *ji-riʔi* **kɛm-z-ell-o**
Solomon TV SM.3MS-watch.IPFV COMP-REL-BE₁.PRES-SM.3MS
ji-ħasib
SM.1S-think.IPFV
‘I think that Solomon is watching TV.’
- b. Solomon televizion **kɛm-z-i-riʔi** **z-ello**
Solomon TV COMP-REL-SM.3MS-watch.IPFV REL-BE₁.PRES-SM.3MS
ji-ħasib
SM.1S-think.IPFV
‘I think that Solomon is watching TV.’
- c. *Solomon televizion **z-i-riʔi** **kɛm-z-ello**
Solomon TV REL-SM.3MS-watch.IPFV COMP-REL-BE₁.PRES-SM.3MS
ji-ħasib
SM.1S-think.IPFV

- d. *Solomon television **kɛm-z-i-rɪʔi**
 Solomon TV COMP-REL-SM.3MS-watch.IPFV
kɛm-z-ɛllo jɪ-ħasib
 COMP-REL-BE₁.PRES-SM.3MS SM.1S-think.IPFV

In the last ungrammatical clause in (81), *kɛm* appears twice, once on the verb and once on the auxiliary. The current proposal correctly predicts this clause to be ungrammatical. This is because I have suggested that *kɛm* is paired with a null nominal EVENT, FACT, etc.: the complement clause above, *that Solomon is watching TV* expresses only one fact/event - namely, Solomon’s watching of the TV - despite containing both a verb and an auxiliary. Thus, *kɛm*, which is paired with this single null nominal, can only appear once.

In the penultimate ungrammatical clause in (81), *kɛmz(i)-* appears once on the auxiliary, while the verb carries only the *z(i)-* prefix. The current proposal also predicts this clause to be ungrammatical. Given that *kɛm* is located low in the structure and paired with the null nominal that raises higher and triggers the appearance of *z(i)-*, the latter can never syntactically precede *kɛm*.

To my knowledge, the only work that investigates the phenomenon under discussion in this section is the already mentioned [van Urk 2025](#) (and [van Urk 2024](#)). The author claims that the placement of *kɛm* is actually “difficult to capture using purely syntactic mechanisms” given that head movement to C°, where he places *kɛm*, “should only allow [it] to surface on the highest verb/auxiliary” ([van Urk 2024:3](#)). In [van Urk 2024](#) it is claimed, following [Wiklund 2001](#), that the appearance of *z(i)-* on both the verb and the auxiliary in periphrastic *kɛmz(i)-*clauses is the result of “iterative feature copying optionally applying throughout the cluster” ([van Urk 2024:5](#)). [van Urk 2025](#) suggests that the appearance of *kɛmz(i)-* on the verb and that of *z(i)-* on the auxiliary in sentences such as (81b) is a consequence of a closeness rule and the possibility of excorporation in lowering operations ([Weisser 2024, 2025](#)) that allow *kɛm* to further lower in a position preceding the verb. Because the theoretical framework of this dissertation does not allow postsyntactic operations, I must distance myself from van Urk’s proposal and find another explanation for the phenomenon above.

(82)



In the tree above, the null nominal *THOUGHT* is paired with *kem* in a *vP* adjunct position. The null nominal raises to the specifier of $\text{Asp}_{\text{IPFV}}\text{P}$, while *kem* remains in its base position. The verb also raises in the structure and lands in the $\text{Asp}_{\text{IPFV}}^\circ$ head. When the silent nominal phrase and the verb are in a Spec-head agreement configuration, they trigger the appearance of *z(i)-*. This also occurs in the domain of the auxiliary, hence the appearance of *z(i)-* on both the verb and the auxiliary.

Let us now focus on the other possible configuration of *kemz(i)-*-clauses expressing progressive aspect, namely the one in which *kemz(i)-* appears on the auxiliary while the verb is bare. This seems to be the most complex scenario. Within the current proposal, if the null nominal raised higher than the auxiliary domain, one would expect it to cross over the verb and thus trigger the appearance of *z(i)-* on it, rather than leaving it bare.

To address the issue, I suggest that this specific case involves restructuring (a.o. Rizzi 1982, Manzini 1983). Restructuring is a syntactic process commonly assumed to explain clause-union phenomena such as clitic climbing in Romance languages. As illustrated in (83), in Italian “an unstressed pronoun originating in the infinitival complement can be cliticized either to the main or to the embedded verb” (Rizzi 1982:1).

- (83) a. Gianni deve presentar-**la** a Francesco.
 Gianni have_to introduce-her to Francesco
 ‘Gianni has to introduce her to Francesco.’
 b. Gianni **la** deve presentare a Francesco.
 Gianni her have_to introduce to Francesco
 ‘Gianni has to introduce her to Francesco.’

To explain how it is possible for the clitic *la* ‘her’ to cross over CP and appear before the modal, authors have proposed a rule that reanalyzes a terminal substring $V_x (P) V$ as a single unit, converting the biclausal structure into a monoclausal structure without altering the word order of the terminal nodes (Rizzi 1982:2, 5).²³ In other words, the rule of restructuring renders the CP boundary transparent (Cardinaletti & Shlonsky 2004).

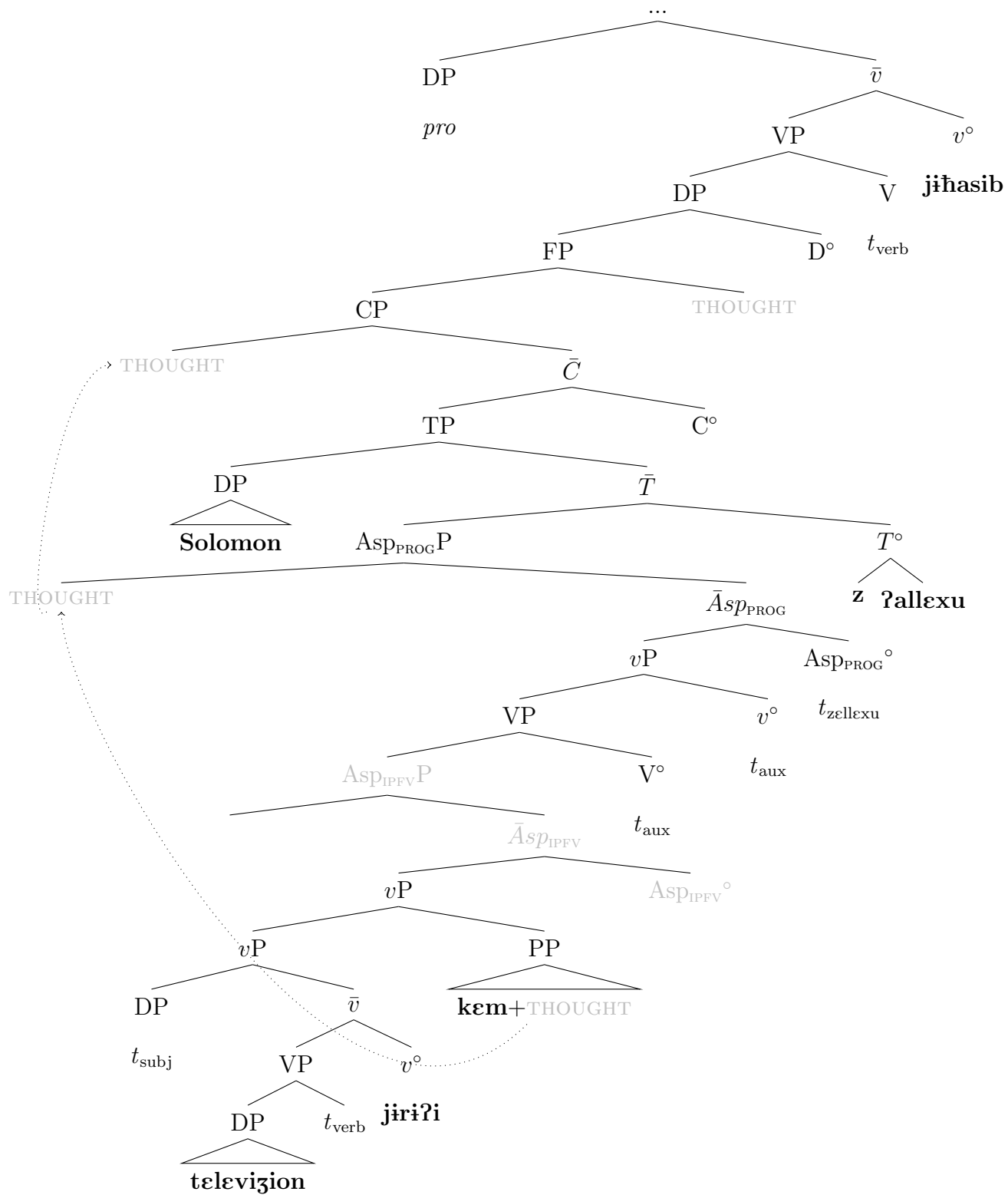
Although restructuring is presented as a rule that transforms a biclausal construction

²³Müller 2020 suggests that the restructuring rule involves a REMOVE operation, the counterpart of MERGE.

The syntax underlying the sentence in (84) is illustrated in the tree in (85). Note that the Asp_{IPFV} projection shown in gray represents the portion that has undergone restructuring.

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(85)



In the tree above, a rule of restructuring has applied, removing the AspP layer in the domain of the verb, and thereby blocking the prefixation of *z(i)-* on it. The null nominal

THOUGHT then raises and lands directly in the specifier of the AspP projection in the auxiliary domain. Because of the restructuring process, *kem* appears on the auxiliary rather than on the verb, since the verb cannot move from its argument position (no projections are available for it to move into, except the v° head).

Before concluding this Chapter with a short summary, I briefly address the case of *kemz(i)*-clauses expressing present perfect and thus presenting a Gerund form. As presented in Section 3.2.1 there are two strategies for this case: either *kemz(i)*- appears on the auxiliary, leaving the Gerund verb form bare, as in (a) in (86) below, or a suppletive perfective form carrying *kemz(i)*- is inserted in the derivation and the auxiliary is prefixed only by *z(i)*-, as in (b) in (86) below.

- (86) a. Tesfay nissa kab geza wets'i?-a **kem-z-ell-a**
 Tesfay she from house leave.GER-SM.3FS COMP-REL-BE₁.PRES-SM.3FS
 hasib-u
 think.GER-SM.3MS
 'Tesfay thought that she left the house.'
- b. Tesfay nissa kab geza **kem-zi-wets'i?-et**
 Tesfay she from house COMP-REL-leave.PFV-SM.3FS
z-ell-a hasib-u
 REL-BE₁.PRES-SM.3FS think.GER-SM.3MS
 'Tesfay thought that she left the house.'

The sentences above can be derived exactly as proposed for progressive aspect in the tree in (82). In clause (a), the Gerund verb does not move in the structure but remains in its first-merge position. As a result, *z(i)*- does not appear, because there is no Spec-head configuration with the raised null nominal. An auxiliary is inserted in the derivation, and it carries the prefix *z(i)*- after moving into the AspP domain of the auxiliary. In clause (b), the perfective verb raises and lands in the head of CP, where together with the null nominal it triggers the appearance of *z(i)*-.

A clarification regarding the adjunction site of *kem* plus the null nominal is in order here. I propose that these elements can occur only within the verbal domain of the lexical verb, and not within that of the auxiliary. This restriction follows from the fact that the interpretation of the null nominal depends, as discussed above, on the semantics of the

embedded verb. The auxiliary in the periphrastic tense forms under consideration here — and in those examined in the following paragraphs — marks aspect and does not play a role in determining the choice or interpretation of the null nominal.²⁴

Before concluding this chapter, recall that I mentioned other elements that appear to behave like *kɛmz(i)-*: *silez(i)-* ‘because’, *kisab zi-* ‘until’, *?inkab zi-* ‘since’ and *?ab kindi zi-* ‘because’, among others. I claim that the syntax of the clauses introduced by these elements is the same as that proposed for *kɛmz(i)-*. The only difference, I suggest, is that these are adjunct clauses rather than complement clauses.

3.4 Interim Summary of Chapter 3

In this Chapter, I have investigated the prefix *kɛmz(i)-* in Tigrinya. I showed that this prefix appears in several types of clauses, namely complement clauses introduced by factive verbs as well as by cognitive non-factive verbs, utterance verbs, perception verbs, and fiction verbs. I demonstrated that in some of these cases *kɛmz(i)-* can be replaced by the clause-final complementizer *?ilu* and highlighted the differences between the two. I also presented an interesting case that involves the verb /msl/ ‘seem’. After presenting data on all these clauses, I proposed that *kɛmz(i)-* is a bimorphemic element composed of the particle (a grammaticalized preposition) *kɛm* and the prefix *z(i)-*, and that *kɛmz(i)-* clauses are underlyingly relative clauses. I suggested that *kɛm* is merged low in the structure and paired with a null nominal whose interpretation varies according to the meaning of the sentence. This null nominal raises in the structure and lands in the left periphery, triggering the appearance of *z(i)-*. In this chapter, I used this syntactic derivation to explain not only the word order of *kɛmz(i)-* clauses with simple verb forms but also the possible and impossible configurations with periphrastic tense forms, and I concluded by noting that the syntax proposed for *kɛmz(i)-* clauses can be extended to adverbial clauses involving *z(i)-*.

²⁴Note that if *kɛm* plus the null nominal could be merged within the verbal domain of the auxiliary, restructuring would not be required to derive a configuration such as that in example (84) or it would be only an alternative. I thank Jason Overfelt for discussion of this point.

Chapter 4

The Clausal Prefix *k(i)-*

4.1 Setting the Scene

The third clausal prefix investigated in this dissertation is *k(i)-* ([i] is an epenthetic vowel; the morpheme itself is *k*). In the introduction, I presented the example in (1) to illustrate one of the contexts in which this prefix occurs in Tigrinya (specifically, control constructions).

- (1) *ʃay ki-Ø-seti* *dələj-ε*
 tea SBJ-SM.1S-drink.IPFV want.GER-SM.1S
 ‘I wanted to drink tea.’

K(i)- has been described in the literature as a conjunction ([Leslau 1941](#)), a marker of the adverbial infinitive ([Mason 1996](#)), a purposive particle ([Nazareth 2011](#)), a complementizer indicating mood and/or modality ([Tesfay 2016:199-200](#)), a goal conjunction ([Bulakh 2019](#)), and a future tense marker ([Gebregziabher 2021](#)). Notably, the functions expressed by *k(i)-* in Tigrinya are realized by different elements in other Ethio-semitic languages. This is, for example, the case in Amharic: in this language the elements *li-* and *?indi-* are used in contexts that require *k(i)-* in Tigrinya (in control constructions). Instances of these two elements are given in (2) and (3).

4.2 *K(i)*-clauses

In this section, I present data on *k(i)*-clauses. I show that this clausal prefix occurs in several types of clauses headed by directive, desire, emotive factive, fiction, control, and ECM verbs. Before presenting these data, I briefly review the literature on the subjunctive mood.

4.2.1 Subjunctive Mood

Although this dissertation focuses on Tigrinya, it is crucial in theoretical linguistics to compare new data and findings with other languages. Such comparisons not only facilitate the search for cross-linguistic generalizations that can help us understand better the faculty of language, but also clarify patterns in the specific language under investigation - in this case, Tigrinya.

In the introductory sections of Chapter 2 and Chapter 3, I pursued this approach by comparing Tigrinya with other Ethio-semitic languages, showing that counterparts (and/or cognates) of the clausal prefixes *z(i)*- and *kemz(i)*- found in Tigrinya also occur in other languages. In this chapter, it is similarly essential to consider other languages in order to understand the nature of the clausal prefix *k(i)*-. I noted that this prefix appears in many different contexts, some of which seem unrelated at first glance. What happens in other languages? Is there an element that consistently occurs in the same contexts? I argue that the answer is yes, and that this element is the subjunctive mood.

Subjunctive is a mood typically used in subordinate clauses.¹ Unlike indicative mood, subjunctive mood marks a clause as uncertain or non-factive (Matthews 2007). In semantic terms, subjunctive mood “designates that a proposition is intensionally anchored” (Gianakidou 1995). The literature shows that subjunctive mood is expressed in two main ways. Some languages mark it overtly on the verb, as in (4) for HUNGarian.

¹In some languages it also appears in root clauses. This is, for example, the case in French with sentences such as *Advienne que pourra!* ‘Come what may!’ (Baunaz & Puskás 2022:14,(12a)).

- (4) Gyuri azt akarja hogy Lajos egy regényt írjon
 Gyuri that want.3S that Lajos a novel write.3S.SBJ
 ‘Gyuri would like Lajos to write a novel.’

(from Baunaz & Puskás 2022:2, (1d))

Other languages express subjunctive mood through particles that resemble complementizers. This is the case for Balkan languages (Landau 2004). For example, Modern Greek uses the particle *na*, as in (5).

- (5) Thelo na kerdisi o Janis
 want.1S SBJ win.3S the John
 ‘I want John to win.’

(from Giannakidou 2009:1887,(10))

The Salentino variety of Italian shows both verbal morphology and a special particle, as in (6).

- (6) Speru cu began faccincumu
 hope.1S CSBJ comes-SBJ someone
 ‘I hope someone comes.’

(from Shlonsky 2021:18,(10))

Scholars have also observed that the choice of mood in subordinate clauses varies depending on the class of the matrix verb. For example, the verb *believe* triggers indicative mood in Spanish but subjunctive mood in Italian, as in (7) and (8).

- (7) Creo que llega/*llegue hoy
 believe.1S that arrive.3S.IND/SBJ today
 ‘I believe she/he arrives today.’

(adapted from Baunaz & Puskás 2022:13, (11))

- (8) Credo che %arriva/arrivi oggi
 believe.1S that arrive.3S.IND/SBJ today
 ‘I believe she(/he) arrives today.’

Subjunctive mood also occurs in Semitic languages. In Standard Arabic, it is expressed by the morpheme *-a* suffixed to jussive/imperfective verb forms (Birnstiel 2019:385), as in (9). Particles such as *ʔan* ‘to/that’ select for a subjunctive-marked verb, as in (10).

- (12) I Maria_i theli pro_{i/j} na fai.
 the Mary wants pro SBJ eats
 ‘Mary wants(him/her) to eat.’ (from (Landau 2004:856,(54b)))

Modern Greek lacks infinitival verb forms. The difference in subject obviation between subjunctive clauses may thus be a consequence of this (both infinitives and subjunctives occur in non-root contexts and their temporal interpretation depends on the root verb).²

In Tigrinya, subject obviation is not obligatory in embedded clauses, neither in those introduced by *kɛmz(i)-* (as in finite Romance clauses) nor in those introduced by *k(i)-* (as in subjunctive Modern Greek clauses). This is illustrated in the English translations of the Tigrinya examples below. Recall from Chapter 3 that the cognitive non-factive verb /ft/ ‘think’ selects for *kɛmz(i)-* clauses and, as shown at the beginning of this chapter, the verb /dlj/ ‘want’ selects for *k(i)-* clauses:

- (13) a. Tɛwɛldɛ **k-i-xɛjjid** ji-dɛli
 Tewelde SBJ-SM.3MS-goIPFV SM.3MS-want.IPFV
 ‘Tewelde_i wants that he_{i/j} leaves.’
 b. Tɛwɛldɛ **kɛm-z-i-xɛjjid** ji-fɛllit’
 Tewelde COMP-REL-SM.3MS-goIPFV SM.3MS-think.IPFV
 ‘Tewelde_i thinks that he_{i/j} leaves.’

As in Modern Greek, Tigrinya lacks infinitival verb forms. Imperfective, Perfective, and Gerund verbs are all finite.

Balkan and Romance subjunctives are both dependent moods, which is why they trigger obviation effects (Landau 2004). It has been argued that subjunctive mood depends on the matrix tense, whereas indicative mood is independent (Picallo 1984, Landau 2004, Giannakidou 2009, Baunaz & Puskás 2022). In other words, the tense of an embedded clause with subjunctive mood depends on (more precisely, is constrained by, though not necessarily identical to (Landau 2004)) the matrix tense. Whether the embedded tense is dependent or independent is determined by the semantic class of the matrix verb. In some languages the same matrix verb can select either indicative or subjunctive, yielding a difference in meaning (as in Italian or French). For Italian, (14) illustrates this contrast.

²For an in-depth explanation see Landau 2004.

Using indicative mood in (14a) presents “they don’t have time” more as a fact; using subjunctive mood in (14b) expresses a hypothesis regarding whether “they don’t have time”³

- (14) a. Immagin-o che non ha-nno tempo
 imagine-1S that NEG have.IND-1S time
 ‘I imagine that they don’t have time.’
 b. Immagin-o che non abbia-no tempo
 imagine-1S that NEG have.SBJ-1S time
 ‘I imagine that they don’t have time.’

This is also the case for Tigrinya: as I show in a later section, some verbs can select either *k(i)*- or *kəmz(i)*-clauses, with a corresponding difference in meaning.

Besides subject obviation, another important property of the subjunctive mood is its semantic association with *irrealis* contexts (Landau 2004, Giannakidou 2009, Baunaz & Puskás 2022). The term *irrealis* refers to states of affairs that are possible or conceivable - what could happen but has not happened or is not happening (Matthews 2007). *Irrealis* thus encodes that “some state of affairs is not real or only possible, or at least that the speaker does not express his commitment to the reality of the state of affairs” (van der Auwera & Devos 2012:171). This perspective will help explain why certain constructions contain *k(i)*- in the following sections.

Let us now examine which verbs select for *k(i)*-clauses.

4.2.2 *K(i)*-clauses with Subject Obviation

As noted at the beginning of the preceding section, the mood of a subordinate clause varies across languages. However, Baunaz & Puskás 2022 show that there is a core set of verb classes that select for the subjunctive mood cross-linguistically, namely directive and desire verbs. Let us begin with the former in Tigrinya.

When directive verbs such as ‘order’, ‘insist’, ‘suggest’ and ‘advise’ occur in the matrix clause, they select embedded verbs marked for the subjunctive or introduced by subjunc-

³In Standard Italian today, subjunctive mood is prescribed with verbs such as *immaginare* ‘imagine’, *credere* ‘believe’ and *supporre* ‘suppose’; however, indicative mood is widespread in speech and informal registers.

tive particles, depending on the language. In Tigrinya, embedded verbs selected by this class of verbs must carry the prefix *k(i)-*, as shown in (15) and (16).

- (15) **k***i*-t*i*-xɛjɪd ʔaziz*i*-na
 SBJ-SM.3FS-leave.IPFV order.GER-SM.1P
 ‘We ordered that she leave.’
- (16) **k***i*-t*i*-xɛjɪd ɸasab ʔaqirebi-na
 SBJ-SM.3FS-leave.IPFV suggestion give.GER-SM.1P
 ‘We suggested that she leave.’

Note that *k(i)-* is obligatory, which accounts for the ungrammaticality of (17), and that, as already mentioned in Section 3.2.2 of the previous chapter, it can attach only to verbs conjugated in the Imperfective paradigm, hence the ungrammaticality of (18).⁴

- (17) *t*i*-xɛjɪd ʔaziz*i*-na
 SM.3FS-leave.IPFV order.GER-SM.1P
- (18) ***k***i*-kɛd-ɛt ʔaziz*i*-na
 SBJ-leave.PFV-SM.3FS order.GER-SM.1P

Note also that in all the clauses presented in this chapter, *k(i)-* cannot be replaced by either the clausal prefix *kɛmz(i)-* or the final complementizer *ʔilu* discussed in Chapter 3.

With respect to desire verbs, ‘prefer’, ‘wish’ and ‘want’ select embedded verbs marked with *k(i)-* in Tigrinya. This is illustrated in (19)-(21).

- (19) **k**-i-zɛwɪr ji-mɛrrits’
 SBJ-SM.1S-walk.IPFV SM.1S-prefer.IPFV
 ‘I prefer walking.’⁵
- (20) **k***i*-t*i*-xɛjɪd tɛmɛni-na
 SBJ-SM.3FS-leave.IPFV wish.GER-SM.1P
 ‘We wished that she left.’

⁴I show later that there is an exception: *k(i)-* can also attach to the auxiliary *ʔallo*, which, despite its present-tense interpretation, follows the Perfective conjugation. This is likewise the case for the Amharic *li-*, the counterpart of *k(i)-* (Leung & Girma 2017). Moreover, in Ge’ez moods are said to be formed only from the Imperfective (Dillmann 1974).

⁵There is no subject obviation in this example.

- (21) ʔinɕera **ki**-ti-bɛliʃ dɛɛj-ɛ-ja
 injera SBJ-SM.3FS-eat.IPFV want.GER-SM.1S-OM.3FS
 ‘I wanted her to eat injera.’

The desire verbs ‘want’ and ‘wish’ are also classified by some authors as future-referring verbs, together with the verb ‘expect’ (Baunaz & Puskás 2014). This latter verb selects the subjunctive mood in Balkan languages and *k(i)*-clauses in Tigrinya, as shown in (22).

- (22) Tɛwɛldɛ ʔanɛ **k-i**-higiz-o ji-ts’ bɛji-ni
 Tewelde I SBJ-SM.1S-help.IPFV SM.3MS-expect.IPFV-OM.1S
 ‘Tewelde expects me to help him.’

Note that in (22), the embedded verb *jiḥigiz* ‘help’ is prefixed by *k(i)*- and agrees with the grammatical subject of the embedded clause *ʔanɛ* ‘I’. This subject is cross-referenced with object morphology on the matrix verb. Sentences containing the verb ‘expect’ in English have been called Exceptional Case Marking (ECM) constructions (Chomsky 1993), because the embedded subject appears in the matrix clause marked with accusative case. In Tigrinya, the ϕ -features of the embedded subject appear in the matrix clause as object morphology on the matrix verb. By analogy, I follow Cacchioli & Overfelt submitted using the term Exceptional Object Marking (EOM) to describe constructions containing the verb ‘expect’ in Tigrinya.

Beyond the core verb classes that select the subjunctive mood cross-linguistically, there are other classes that select it in some languages but not in others. These include emotive-factive and fiction verbs. Emotive-factive verbs such as ‘regret’, ‘be happy’ and ‘be surprised’ require the indicative mood in English and Hungarian but the subjunctive mood in Romance languages (Baunaz & Puskás 2022:22). In Tigrinya, emotive-factive verbs select *k(i)*-marked verbs, as shown in (23)–(25).

- (23) **ki**-ti-xɛjɪd tet’aʃisi-na
 SBJ-SM.3FS-leave.IPFV regret.GER-SM.1P
 ‘We regretted that she left.’

- (24) **k**i-ti-xɛjjid higos-at ʔi-na
 SBJ-SM.3FS-leave.IPFV happy.P BE₂.PRES-SM.1P
 ‘We are happy that she leaves.’
- (25) **k**i-ti-xɛjjid tiβɛsir-na
 SBJ-SM.3FS-leave.IPFV be_surprised.GER-SM.1P
 ‘We are surprised that she left.’

Note that when the embedded verb expresses past, as in (23) and (25) above, it is also possible for the nominalizer prefix *m*i to appear on the verb and itself be prefixed by the preposition *b*(i)- ‘by’, as in (26).

- (26) bi-mi-xid-a higos-at ʔi-na
 BY-NMZ-leave-SM.3FS happy.P BE₂.PRES-SM.1P
 ‘We are happy that she left.’

This specific case - the appearance of *b*(i)- on top of *m*(i)- - will not be investigated in this dissertation due to constraints of space and time.

Fiction verbs such as ‘dream’ and ‘imagine’ also show cross-linguistic variation in their association with the subjunctive mood (Baunaz & Puskás 2022). For example, in Italian the verb *sognare* ‘dream’ selects the indicative mood, whereas in French the verb *rêver* ‘dream’ can take either indicative or subjunctive mood. This is particularly interesting for Tigrinya. As shown in Chapter 3, fiction verbs can select *kɛmz*(i)-marked verbs, yielding an epistemic interpretation. When fiction verbs select *k*(i)-clauses, however, their interpretation differs. Consider (27) and (28) involving fiction verbs and *k*(i)-.

- (27) Daniʔel nab-pʼaris **k**-i-xɛjjid halim-u
 Daniel DIR-Paris SBJ-SM.3MS-go.IPFV dream.GER-SM.3MS
 ‘Daniel dreamt of going to Paris.’
- (28) Nataniʔel nab-pʼaris **k**-i-xɛjjid tɛmɛni-u
 Nataniel DIR-Paris SBJ-SM.3MS-go.IPFV imagine.GER-SM.3MS
 ‘Nataniel imagined going to Paris.’

The sentence in (27) is grammatical only in a context where Daniel wishes to go to Paris or has the dream of going to Paris, not in a context where he dreams while sleeping of going

to Paris; such a context would trigger $k(\varepsilon m z(i))$ -. The sentence in (28) is grammatical only in a context where Nataniel envisions himself going to Paris, not in a context where he infers that someone else is going to Paris; such a context would trigger $k(\varepsilon m z(i))$ - (and subject obviation).

Before turning to the analysis of sentences containing directive, desire, emotive-factive, and fiction verbs that select $k(i)$ -clauses, I briefly review what the literature has previously stated about $k(i)$ -. I have already suggested that $k(i)$ - is a subjunctive marker, but let us pause to consider where exactly it is merged in the syntax.

4.2.3 What is $k(i)$ -?

In Chapter 3 I showed that the clausal prefix $k\varepsilon m z(i)$ - derives etymologically from the Akkadian conjunction $k\bar{m}a$, which is composed of the preposition $k\bar{m}$ and the emphatic particle $-ma$. Today, among Semitic languages, $k(i)$ - complementizers/subordinators are said to originate from that same preposition (Hernández 2024). Some authors further note that $k(i)$ - originally meant ‘thus/truly/so’ and developed into a comparative preposition meaning ‘like’ (Tesfay 2006:876), similar to what is reported in the literature for $k\varepsilon m$.⁶

If we think of the English counterparts of the $k(i)$ -clauses discussed so far, they would involve the complementizer *to*, which - like $k(i)$ - - is also a preposition in English. Regardless of the etymology of $k(i)$ -, the question now is what kind of syntactic element it is, where it is base-generated, and what the syntax of the clauses in which it appears looks like.

Previous accounts - some offering detailed syntactic analyses, others adopting a more descriptive approach - treat this clausal prefix in different ways. Gebregziabher 2021 glosses $k(i)$ - in modal clauses as FUT (future tense marker), while Nazareth 2011 glosses it as PURP (purposive marker). Both approaches seem too narrow: the former reduces the prefix to its role in constructing future tense, and the latter limits it to its function in purpose or goal clauses (to be discussed later). Tesfay’s suggestion that $k(i)$ - encodes

⁶The difference between $k(i)$ - and $k\varepsilon m$ is that the former does not “require” $z(i)$ - to be marked on embedded verbs (Tesfay 2006:878).

“mood and/or modalities” (Tesfay 2016:205) is, I argue, more accurate.

Throughout the previous pages, I have compared the use of *k(i)-* in Tigrinya with the use of subjunctive mood in comparable clauses in other languages. I have noted that some languages express subjunctive mood through verbal morphology, while others do so through specific particles. Baunaz & Puskás 2022 argue that in the first case the subjunctive mood is licensed by a predicate carrying an [EMOTIVE] feature. For Spanish, Villalta 2008 suggests that the subjunctive mood is the morphological realization of an operator that evaluates alternatives and, since alternatives are often expressed through different types of Focus (e.g. contrastive), the subjunctive is the realization of a focus-sensitive operator. Giannakidou 2009 claims that the particle *na*, which triggers subjunctive mood in Modern Greek, has a dual function as a mood marker and a subordinator and heads a Mood projection dominating TP.

For Tigrinya, I follow Giannakidou 2009 and claim that, like *na* in Modern Greek, *k(i)-* is a subjunctive (or irrealis) marker that heads a Mood projection dominating the aspectual layer, but I propose that this projection is lower than TP - as also argued by Cacchioli & Overfelt 2023. This analysis of *k(i)-* is similar to what has been proposed for the particle *li-* in Amharic, namely that it is a prospective aspectual marker (functioning semantically as a posterior modal operator) (Leung & Girma 2017).

Moreover, I have shown that the subjunctive - and therefore *k(i)-* - enforces temporal dependence on the matrix clause, given that subjunctive clauses are dependently anchored. It has been suggested that the temporal specification of the subjunctive mood and that of its main clause “may be compared to the relation between an anaphor and its antecedent”, and, in more formal terms, the hypothesis advanced in the 1980s was that the Infl° node of a subjunctive clause lacks the feature [+Tense] (Picallo 1984:88). In the 1990s, Varlokosta and Hornstein developed this idea, proposing that embedded verbs in (a type of) subjunctive mood raise to C so that their tense can be licensed by the matrix verb (Varlokosta & Hornstein 1993).

I suggest that embedded verbs in *k(i)-* clauses raise for this reason to the Mood° head where *k(i)-* is merged. To see the temporal dependency of *k(i)-* clauses even more clearly,

consider (29) and (30) which contain an emotive-factive and a perception verb as matrix verbs (perception verbs will be discussed in more detail later).

- (29) a. **ki**-ti-xɛjjid ni-t'aʃɛs
 SBJ-SM.3FS-leave.IPFV SM.1P-regret.IPFV
 'We regret that she leaves.'
- b. **ki**-ti-xɛjjid tɛt'aʃisi-na
 SBJ-SM.3FS-leave.IPFV regret.GER-SM.1P
 'We regretted that she left.'
- (30) a. nissatom **k-i**-sɛhaq-u ji-sɛmiʃ
 they SBJ-SM.3MP-laugh.IPFV-SM.3MP SM.1S-hear.IPFV
 'I hear them laughing.'
- b. nissatom **k-i**-sɛhaq-u sɛmiʃ-ɛ
 they SBJ-SM.3MP-laugh.IPFV-SM.3MP hear.GER-SM.1S
 'I heard them laughing.'

Changing the tense of the matrix verbs in the examples above alters the temporal interpretation of the embedded verbs as well. For instance, in (29) the event of laughing takes place in the present in (a) but in the past in (b). This shows that the temporal interpretation of *k(i)*-clauses is dependent on the tense of the matrix clause.

Another point, indirectly made in the previous sections and linked to the fact that *k(i)*-clauses are dependently anchored to matrix clauses, is that they appear to have impoverished structures - as also claimed for *li*-clauses in Amharic (Leung & Girma 2017) - and thus constitute what the literature calls reduced clauses. This is supported by the fact that *k(i)*-clauses do not allow periphrastic tense forms except when expressing simultaneity (discussed in more detail later). This is illustrated in (31).

- (31) a. **ki**-ti-xɛjjid ʔazɛzi-na
 SBJ-SM.3FS-go.IPFV order.GER-SM.1P
 'We ordered that she leaves.'
- b. ***ki**-ti-xɛjjid ʔall-a ʔazɛzi-na
 SBJ-SM.3FS-go.IPFV BE₁.PRES-SM.3FS order.GER-SM.1P
- c. **ki**-ti-xɛjjid **k**-ell-a ʔazɛzi-na
 SBJ-SM.3FS-go.IPFV SBJ-BE₁.PRES-SM.3FS order.GER-SM.1P
 'While she left, we ordered/recommended (something).'

The sentences in (31) show that auxiliaries are ungrammatical in $k(i)$ -clauses unless they are themselves prefixed with $k(i)$ -, which yields a different meaning - specifically, an event simultaneous with that of the matrix clause (discussed in more detail later).

For this reason, I propose that $k(i)$ -clauses are reduced clauses lacking both a Tense projection and a CP projection.⁷ This absence explains why their temporal interpretation depends on that of the matrix clause.⁸

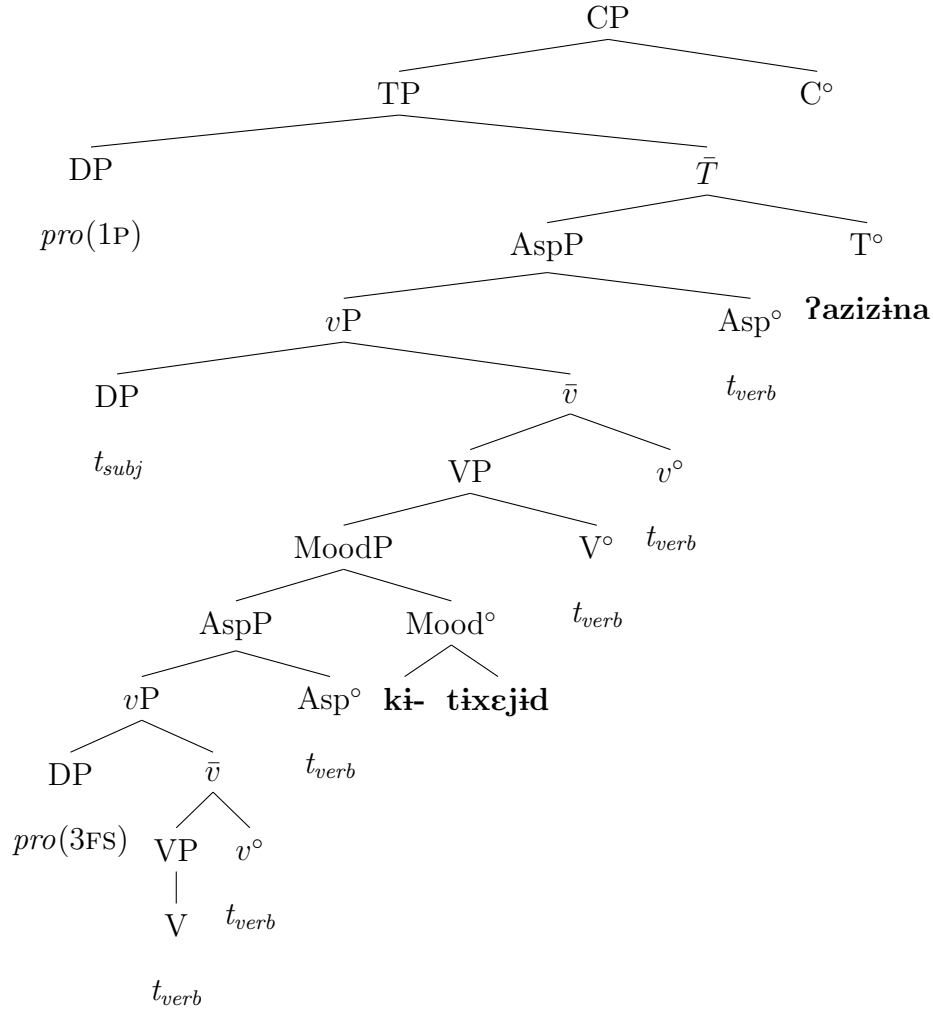
With this in mind, I propose the following syntax for sentences containing $k(i)$ -clauses selected by directive verbs (e.g. *order*), desire verbs (e.g. *want*), emotive-factive verbs (e.g. *regret*), and fiction verbs (e.g. *dream*). For now, I provide an analysis of the sentence in (32), which involves subject obviation.

- (32) **ki-ti-xɛjɪd** ʔazɛzi-na
 SBJ-SM.3FS-leave.IPFV order.GER-SM.1P
 ‘We ordered that she leave.’

⁷An alternative analysis is that $k(i)$ -clauses are nominalized clauses, given the possibility for the DOM prefix $n(i)$ - to appear on $k(i)$ -, as it will be shown for purpose clauses. This hypothesis emerged during a discussion with Jason Overfelt. I do not pursue this possibility further in this thesis.

⁸However, [Wurmbbrand 2017](#) argues that the temporal interpretation of a clause (containing an event) can be derived independently of TP, so a tense feature is not indispensable to express temporal events.

(33)



The tree above illustrates that in a structure containing a *k(i)*-clause, the embedded verb raises from its first-merge position to the Mood° head so that its tense/aspect can be licensed by the matrix verb. The structure is not biclausal in the sense of having two CPs, since the *k(i)*-clause is reduced to MoodP. For simplicity, the matrix clause is shown without a MoodP, though this projection may always be present.⁹

The clausal prefix *k(i)-* is base-generated in the Mood projection, and when the verb moves to its head, *k(i)-* is prefixed to it. Note that this tree represents the syntax of sentences containing *k(i)*-clauses with subject obviation; sentences containing *k(i)*-clauses without subject obviation are the focus of the next section.

⁹An open question is whether MoodP can also take the value of indicative or whether it appears only with the subjunctive. In the latter case, the projection might be better labeled SubjP. I thank Ur Shlonsky for pointing this out to me.

4.2.4 *K(i)*-clauses without Subject Obviation

Two other contexts in which *k(i)*- appears in Tigrinya are what have been labeled in English and other languages Control ([Chomsky 1993](#)) and (the already partly discussed) Exceptional Case Marking (ECM) constructions. To my knowledge, the latter do not involve the subjunctive mood in other languages, whereas the former involves the subjunctive particle *na* in Modern Greek, as shown in (34).

- (34) O Yanis elpizi **na** figi
 the John hopes SBJ win
 ‘John hopes to win.’ (from [Varlokosta & Hornstein 1993:510](#), (13))

Let us begin with Control constructions. These clauses typically involve verbs such as ‘want’ (already discussed in a previous section), ‘try’, ‘promise’, ‘hope’ and ‘choose’. Example (35) shows that *k(i)*- is prefixed to the lexical verbs selected by these verbs.

- (35) a. Tesfay ḡalba **k-i**-siʔel fettin-u
 Tesfay boat SBJ-SM.3MS-draw.IPFV try.GER-SM.3MS
 ‘Tsfay tried to draw a boat.’
 b. Natnaʔel **k-i**-wets’ iʔ tɛmɛbats’ iʕ-u-lley
 Nathaniel SBJ-SM.3MS-leave.IPFV promise.GER-SM.3MS-OBL.1S
 ‘Nathaniel promised me to leave.’
 c. ʔaman sirah **k-i**-rikeb ji-ʔemin
 Aman work SBJ-SM.3MS-find.IPFV SM.3MS-hope.IPFV
 ‘Aman hopes to find a job.’ (from [Cacchioli 2025:10](#), (42))
 d. **ki**-ti-xɛjid mɛrits’-a
 SBJ-SM.3FS-leave.IPFV choose.GER-SM.3FS
 ‘She chose to leave.’

Another control verb that requires *ki*- to be prefixed on the embedded verb is /ħtt/ ‘ask’, as shown in (36).

- (36) **ki**-ti-xɛjid hatit-a
 SBJ-SM.3FS-leave.IPFV ask.GER-SM.3FS
 ‘She asked to leave.’

Note that the verb ‘ask’ above has the meaning *ask for permission*. Recall example (37) from the previous chapter.

- (37) ʔaman ʔintay {*ki-} kɛm-z-i-fettu {ʔil-u}
 Aman what SBJ- COMP-REL-SM.1S-like.IPFV COMP-SM.3MS
 ʔatit-u-ni
 ask.GER-SM.3MS-OM.1S
 ‘Aman asked me what I like.’

In the example above, $k(i)$ - is not available, despite its occurrence being possible in the previous example. This is because the subordinate clause here is an embedded interrogative containing the element *?intay* ‘what’, which is in a $\bar{A}$ -dependency, hence the appearance of the prefix $k\epsilon m z(i)$ -. Indirect questions in Tigrinya were briefly presented in Section 3.2.2 of Chapter 3.

Let us now turn to what have been labeled ECM constructions in English. I have already shown in a previous section that the ECM verb ‘expect’ selects a $k(i)$ -marked embedded verb. The perception and ECM verb ‘see’ can likewise select a $k(i)$ -marked embedded verb, as shown in (38) and (39).

- (38) nɪsatom **ki**-Ø-mərɿa-wu rɪʔɛj-ɛ-jom
they SBJ-SM.3MP-marry.IPFV-SM.3MP see.GER-SM.1S-OM.3MP
'I saw them getting married.'
- (39) nɪsu n-it-i ʔɪndʒɛra **ki**-Ø-bɛlɪf-o
he DOM-DEM.MS injera SBJ-SM.3MS-eat.IPFV-OM.3MS
rɪʔɛy-ɛ-yo
see.GER-SM.1S-OM.3MS
'I saw him eating *injera*.'

As with the verb ‘expect’, the embedded verbs in the two examples above are prefixed with $k(i)$ - and agree with the grammatical subjects of the embedded clauses. These subjects are cross-referenced by object morphology on the matrix verbs.

Recall from Chapter 3 that the clausal prefix *kɛmz(i)-* introduces subordinate clauses with perception verbs when the interpretation is one of indirect perception or inference. In contrast, when these verbs select a *k(i)-*clause, the interpretation is that of direct

perception. My consultants report that in (38) and (39) the speaker witnessed the events (the wedding and the eating of the *injera*) first-hand, being *present* when they occurred. Note that the construction with the Gerund in English conveys exactly the same meaning: the speaker must have been present and have seen the two events with her own eyes.

That perception verbs trigger different meanings depending on the type of clause they embed is cross-linguistically common (Moulton 2009). This is also the case in Romance languages: finite complement clauses of perception verbs yield an indirect-perception interpretation, whereas infinitival clauses yield a direct-perception interpretation. This is shown for French in (40).

- (40) a. J' ai vu qu' il a mangé le camembert.
 I have seen that he has eaten the camembert
 'I saw that he ate the camembert.' Indirect Perception
- b. Je l'ai vu manger le camembert.
 I him-have seen eat the camembert
 'I saw him eating the camembert.' Direct Perception

Returning to Tigrinya, it is also possible to change the tense/aspect of the matrix verb to present progressive by adding an auxiliary (a pattern difficult to replicate with perception verbs in English). In this case, the speaker is a first-hand witness of the event with both *k(i)-* and *kemz(i)-*, as in (41) and (42). According to my consultants, there is no perceptible difference in meaning between the two sentences.

- (41) nisu n-it-i ?indʒera ji-beliʃ **kem-z-ell-o**
 he DOM-DIST-MS injera SM.3MS-eat.IPFV COMP-REL-BE₁.PRES-SM.3MS
 ji-riʔ-o ?allɛ-xu
 see.IPFV-SM.1S BE₁-SM.1S
 'I see that he is eating *injera*.' (Lit. I am seeing that he is eating *injera*)
- (42) nisu n-it-i ?indʒera **ki**-beliʃ-o **k-ell-o**
 he DOM-DIST-MS injera SBJ-SM.3MS-eat.IPFV-OM.3MS SBJ-BE₁-SM.3MS
 ji-riʔ-o ?allɛ-xu
 SM.1S-see.IPFV-OM.3MS BE₁.PRES-SM.1S
 'I see that he is eating *injera*.' (Lit. I am seeing that he is eating *injera*)

Note that in the example above, *k(i)-* is prefixed to both the verb and the auxiliary. In

Section 4.4, I examine this syntactic configuration and explain that it expresses simultaneity. I therefore suggest that the difference in meaning between (41) and (42) lies in the fact that the latter should be more precisely translated as *I am seeing him/it while he is eating injera*.

Now that I have presented data on the ECM verbs ‘expect’ and ‘see’, consider the two data points in (43) and (44) containing the strong modals /gbʔ/ ‘need/should’ and /hlw/ ‘must/have to’.¹⁰

- (43) ʔit-a səbeʒti **ki**-ti-xejid ji-gibbaʔ-(a)
DIST-FS woman SBJ-SM.3FS-leave.IPFV SM.3MS-need_to.IPFV-OM.3FS
‘The woman needs to leave.’ (from [Cacchioli & Overfelt submitted:14](#), (34a))
- (44) ʔit-a səbeʒti **ki**-ti-xejid ʔall-ε-*(wa)
DIST-FS woman SBJ-SM.3FS-leave.IPFV BE₁.PRES-SM.3MS-OM.3FS
‘The woman has to leave.’ (from [Cacchioli & Overfelt submitted:14](#), (34b))

Note that the verb /hlw/ is actually the auxiliary *ʔallo* (referred here according to its triconsonantal root). Recall from Section 1.5.2 in the Introduction that *ʔallo* is also used to express possession. It is therefore not surprising that the verb meaning ‘have’ in Tigrinya is also used to express ‘have to’, just as in English.

In Tigrinya, *k(i)-* is also used with certain modals (cf. [Tesfay 2016](#), [Gebregziabher 2021](#), [Cacchioli & Overfelt submitted](#)).¹¹ When clauses contain /gbʔ/ ‘need/should’ and /hlw/ ‘must/have to’, the embedded verb must carry the *k(i)-* prefix and agree with the grammatical subject of the sentence - *ʔita səbeʒti* ‘the woman’ in the examples above. However, the modals themselves do not agree with the grammatical subject of the sentence but carry 3MS default subject morphology.¹² They are also suffixed with object

¹⁰By *strong* modals I refer to the degree of their modal force, ranging from weak possibility (with verbs such as ‘may/might’ and ‘could’ in English) to strong necessity (with verbs such as ‘must’ and ‘have to’ in English). There is variation among speakers and authors regarding the precise translation of these root modals. For example, my Eritrean consultants claim that /gbʔ/ is weaker than /hlw/ (‘should’ vs. ‘must’), whereas Jason Overfelt’s Ethiopian consultant claims exactly the opposite, namely that /gbʔ/ is stronger.

¹¹The data and observations presented here are drawn from joint work on the syntax of verbal modality in Tigrinya with Jason Overfelt.

¹²This pattern recalls the contrast between *Elle doit venir* ‘she must come’ and *Il faut qu’elle vienne* ‘she must come’ in French, where the latter example contains an expletive *il* with the impersonal modal verb *falloir*.

morphology (/gbʔ/ optionally and /hlw/ obligatorily) that cross-references the grammatical subject of the sentence.¹³

The two examples above can be interpreted with different modal types - either epistemic or root - depending on the context. The contexts supporting this point are provided in [Cacchioli & Overfelt submitted](#).

The syntax of the strong modals discussed above thus appears very similar to that of the verb ‘expect’. However, the syntax differs with modals of weak force. These include the verb /kʔl/ ‘can’ and the verb /kwn/ ‘may/might’, as illustrated in (45) and (46) below. Note that the verb /kwn/ is actually the suppletive form of the copulas and the lexical verb *kone* ‘become’ (referred here according to its triconsonantal root).

- (45) ʔit-a səbejti **ki**-ti-xejid ti-xiʔil
 DIST-FS woman SBJ-SM.3FS-leave.IPFV SM.3FS-can.IPFV
 ‘The woman can leave.’ (from [Cacchioli & Overfelt submitted:30](#), (76))

- (46) ʔit-a səbejti ti-xejid ti-xiwwin
 DIST-FS woman SM.3FS-leave.IPFV SM.3FS-might.IPFV
 ‘The woman might leave.’ (from [Cacchioli & Overfelt submitted:14](#), (33))

Note from the two examples above that the embedded verb *ti-xejid* ‘leave’ is prefixed with *k(i)*- only with /kʔl/ ‘can’ and not with /kwn/ ‘may/might’.¹⁴ Both the embedded verbs and the modals are marked with subject morphology cross-referencing the grammatical subject of the sentence *ʔita səbejti* ‘the woman’. Note also that, unlike the two previous examples involving /gbʔ/ ‘need/should’ and /hlw/ ‘must/have to’, the two modals do not carry object-marking morphology.

¹³In sentences involving the modal /hlw/ ‘have to’, it is also possible to add the nominal element *giddin* ‘obligation’ to emphasize the strength of the modal, as in (i).

- (i) giddin **ki**-ti-zarib ʔall-exa
 obligation SBJ-SM.3FS-speak.IPFV BE₁.PRES-SM.2MS
 ‘You must speak.’

However, note that in this example there is no object-marking morphology suffixed to the modal. This example remains puzzling and is not included in the analysis below; I mention it here only for the sake of completeness.

¹⁴By contrast, in Amharic ‘may’ selects *li*-marked verbs ([Leung & Girma 2017](#)).

It is important to note that the root /xʔl/ ‘can’ can be interpreted as either an epistemic or a root modal depending on the context, whereas the root /kwn/ ‘might’ can be used only in an epistemic context (like the verb *might* in English).

In the past tense (whose syntax will be left for future research), verbal modality becomes slightly more complex. I present some examples here in (47)–(49) for completeness.

- (47) ʔit-a sɛbɛjti **ki**-ti-xɛjid nɛjɪr-u-*(wa)
 DIST-FS woman SBJ-SM.3FS-leave.IPFV BE₁.GER-SM.3MS-OM.3FS
 ‘The woman had to leave./The woman should have left.’
- (48) ʔit-a sɛbɛjti **ki**-ti-xɛjid ti-xiʔil nɛjɪr-a
 DIST-FS woman SBJ-SM.3FS-leave.IPFV SM.3FS-can.IPFV BE₁.GER-SM.3FS
 ‘The woman could have left.’
- (49) ʔit-a sɛbɛjti kɛid-a ti-xiwwin
 DIST-FS woman leave.GER-SM.3FS SM.3FS-might.IPFV
 ‘The woman might have left.’

The construction in (47) expresses both epistemic and deontic modality. It is conveyed by the verb *nɛjɪru*, the past form of the modal *ʔallo*. Example (48) illustrates past deontic modality with the weak modal /kʔl/ and *nɛjɪru*, and example (49) shows past epistemic modality with the epistemic modal /kwn/ and *nɛjɪru*.

Returning to the comparison between *k(i)*- and the subjunctive mood, note that modal verbs in impersonal constructions such as ‘be possible’ and ‘be necessary’ select the subjunctive mood in subordinate clauses across languages (Baunaz & Puskás 2022:21). Examples of these constructions are given in (50) for Italian and English.

- (50) a. È possibile che ven-ga con sua moglie
 BE.PRES.3S possible that come-SUBJ.3S with his/her wife
 ‘It is possible that he (will) come with his/her wife.’
- b. It’s necessary that you be/*are on time.
- (from (Baunaz & Puskás 2022:21, (28c))

With respect to modal verbs (outside impersonal constructions), in Romance, Germanic, and Hungarian these verbs do not select the subjunctive mood. By contrast, in Balkan

languages the modals ‘must’, ‘should’, and ‘can’ select the subjunctive (Socanac 2017), and in Modern Greek the verbs ‘must’, ‘should’, ‘can’, and ‘may’ do so as well (Giannakidou 2009, Roussou 2010, Baunaz & Puskás 2022).

Before moving to the analysis, let me summarize the discussion in this section for clarity. The data show that $k(i)$ -clauses are selected by verbs classified in English as Control verbs (e.g. ‘want’, ‘try’) and ECM verbs (e.g. ‘see’, ‘expect’). With respect to ‘see’, I showed that this verb can select either $k(i)$ -clauses (when direct perception is involved) or $k\epsilon m z(i)$ -clauses (when indirect perception is involved). I also showed that the weak modal ‘can’ in Tigrinya behaves like a Control verb, while the strong modals ‘must’ and ‘have to’ behave like ECM verbs.

In the literature, $k(i)$ -clauses have been labeled control constructions (Nazareth 2011, Spadine 2020), but no in-depth analysis has been provided to support this claim. Are they therefore truly control constructions, or could they be raising constructions? The answer is given in the following section.

4.2.5 Control vs. Raising

Control constructions have been investigated since the early stages of generative linguistics (Rosenbaum 1967, Chomsky 1973). These biclausal constructions involve a lexical verb whose complement is an infinitival clause containing a PRO element. The interpretation of this element is controlled by an argument in the main clause (the subject in subject-control and the object in object-control). An example of a subject-control sentence is given in (51).

(51) Mark_i is anxious [PRO_i to receive the prize]. (from Landau 2013:10, (22b))

Over the years, control has been reconceptualized as raising, a view that remains highly debated (Manzini 1983, Hornstein 1999, Manzini & Roussou 2000, Boeckx et al. 2010, Polinsky 2013). Raising constructions involve a lexical verb — or a raising predicate, as in (52)— whose complement is an infinitival clause containing a subject that moves from the embedded clause to the matrix clause, leaving a trace.

(52) Mark_i is likely [*t_i* to receive the prize]. (from Landau 2013:10, (22a))

Authors who argue for a reconceptualization of control as raising have suggested that subject-control constructions do not actually involve the null element PRO (and its associated null case) but rather an element that moves from the embedded clause to the matrix clause. For example, the moving element is not a DP (subject or object) as in standard raising, but a feature associated with it (Manzini & Roussou 2000). Other authors, who maintain that control and raising involve completely different syntactic mechanisms, have proposed that raising complements are smaller than control complements: the former are TPs — and therefore not introduced by a complementizer —, whereas the latter are CPs — and therefore possibly introduced by a complementizer — (Landau 2013:18). Clauses lacking a CP are thus more permeable, allowing extraction and raising of a DP from the embedded to the matrix clause without violating the Phase Impenetrability Condition (PIC) (Chomsky 2000a).

A few paragraphs above, I claimed that *k(i)*-clauses are reduced clauses that lack both a TP and a CP projection. This might suggest that *k(i)*-clauses are raising constructions. However, I argue here that *k(i)*-clauses actually involve subject control. This claim is based on two reasons. The first reason draws on conclusions from previous accounts discussing Ethio-Semitic languages. To my knowledge, only a few studies have investigated whether there is independent evidence for raising constructions in Amharic, and they have concluded that there is none. Although Baye 1991 claimed that raising exists in Amharic, the cases he cites reduce to raising out of finite clauses — which is impossible in languages such as English and French (Rizzi 1982) — and out of noun phrases. More importantly, Lumsden & Girma 2011 state that “the contrasting word orders discussed by Baye have semantic interpretations that are incompatible with a raising account of their relationships” (Lumsden & Girma 2011:132), and therefore raising constructions do not exist in the language.

Leung & Girma 2017 also argue that because *A*-movement is typically triggered by incompleteness of ϕ -features, the Amharic counterparts of *k(i)*-clauses in Tigrinya, which

are fully inflected for ϕ -features, provide no motivation for a raising operation.¹⁵

The second reason is empirical. In $k(i)$ -clauses with verbs such as ‘try’ and ‘hope’, the matrix verb cannot be inflected with default 3MS agreement; it must agree with the subject of the sentence, and null expletives are not possible, as shown for the verb /ftn/ ‘try’ in (53). This pattern is consistent with a control syntax for $k(i)$ -clauses.

- (53) a. *Səgen ɕalba **ki**-ti-siʔel fəttin-u
 Səgen boat SBJ-SM.3FS-draw.IPFV try.GER-SM.3MS
 ‘Səgen tried to draw a boat.’
 b. Səgen ɕalba **ki**-ti-siʔel fəttin-a
 Səgen boat SBJ-SM.3FS-draw.IPFV try.GER-SM.3FS
 ‘Səgen tried to draw a boat.’

Secondly, raising and control constructions behave differently under passivization. When a sentence is transformed from active to passive voice, the meaning is preserved in raising constructions but not in control constructions. Consider (54) and (55), which illustrate passivization with the standardly assumed raising verb /msl/ ‘seem’ (cf. Chapter 2) and the verb /ftn/ ‘try’.

- (54) a. həde həkim ni-Təsfai **z**-i-mirimir-o
 one.M doctor DOM-Təsfay REL-SM.3MS-examine.IPFV-OM.3MS
 ji-məsıl
 SM.3MS-seem.IPFV
 ‘A doctor seems to examine Təsfay.’
 b. Təsfai bi-həkim **zi**-tə-mərəmər-ε ji-məsıl
 Təsfay by-doctor REL-PASS-examine-SM.3MS SM.3MS-seem.IPFV
 ‘Təsfay seems to be examined by a doctor.’
 (55) a. həde həkim ni-Təsfai **k**-i-mirimir-o
 one.M doctor DOM-Təsfay SBJ-SM.3MS-examine.IPFV-OM.3MS
 fəttin-u
 try.GER-SM.3MS
 ‘A doctor tried to examine Təsfay.’

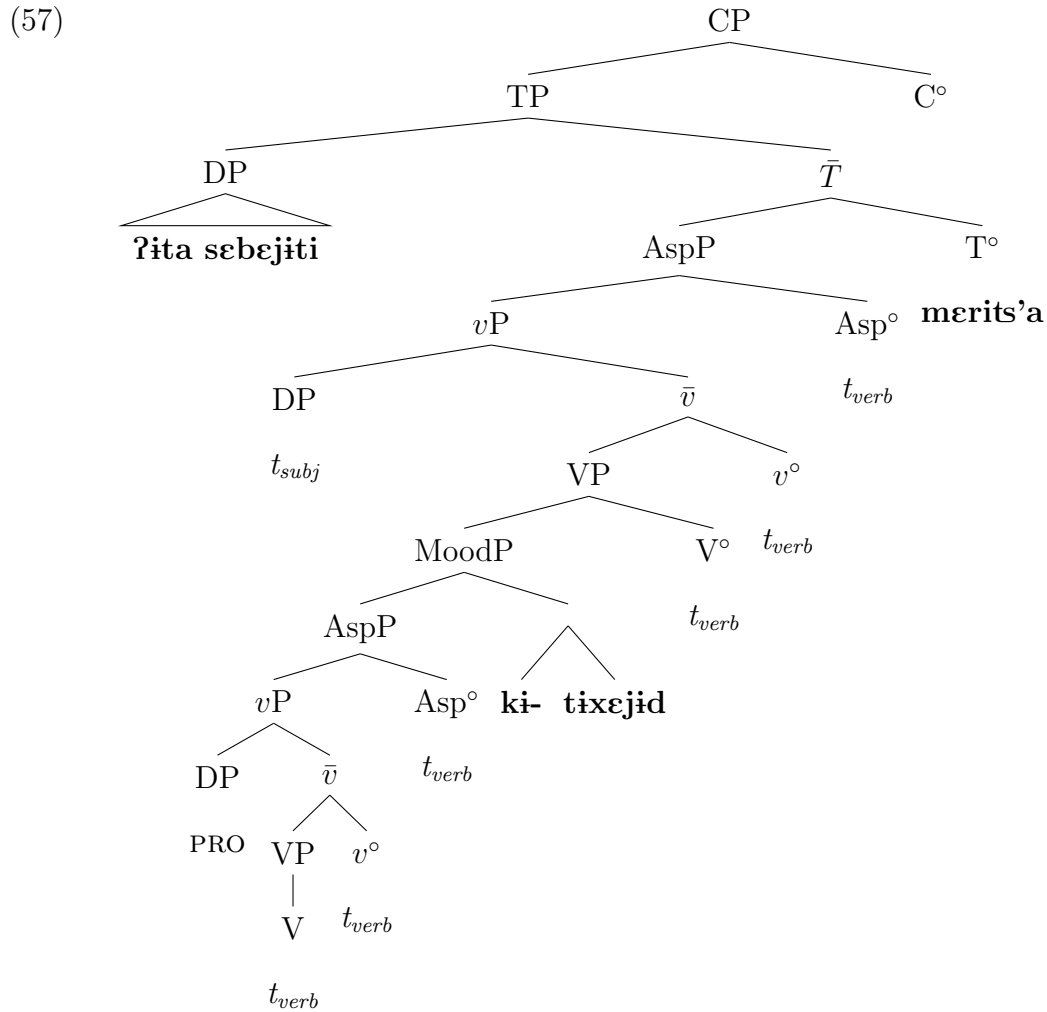
¹⁵Leung & Girma 2017 claim that desire verbs (‘intend’, ‘hope’, ‘prefer’), modal verbs (‘can’, ‘should’) and try-predicates (‘try’, ‘struggle’, ‘strive’, ‘pray’) are control predicates. Unlike Tigrinya, the verbs ‘think’ and ‘remember’ in Amharic also involve control syntax (in Tigrinya they select $kemz(i)$ -clauses).

- b. Təsfai bi-ħakim **k-i**-mirmir fəttin-u
 Tesfay by-doctor SBJ-SM.3MS-examine.PASS try.GER-SM.3MS
 ‘Təsfay tries to be examined by a doctor.’¹⁶

In (54), which contains a raising predicate, the meaning does not change between the active and the passive sentence; however, in (55), which contains the verb /ftn/ ‘try’, the meaning does change. I therefore suggest that this verb is a control predicate.

With this in mind, I claim that sentences containing *k(i)*-clauses without subject obviation can be represented as in the tree below.

- (56) ?it-a səbejiti **ki-ti-xejid** mərits’a-a
 DIST-FS woman SBJ-SM.3FS-leave.IPFV choose.GER-SM.3FS
 ‘The woman chose to leave.’



¹⁶Note that in Imperfective the *ti-* prefix that marks passive is assimilated by the first radical of the root (Mason 1996).

In the tree in (57), the matrix subject *?ita sebejiti* ‘the woman’ controls the PRO element in the embedded clause.¹⁷ As in the tree in the previous section, the embedded verb raises to Mood° so that its tense/aspect is licensed by the matrix verb.

I suggest that the tree also captures the syntax of the modal /kʔl/ ‘can’, which patterns identically. By contrast, the modals /gbʔ/ ‘should’ and /hlw/ ‘have to’ align more closely with the ECM verb ‘expect’. I propose that these modals instantiate an ECM syntax (or, as noted earlier, an EOM syntax), as also proposed in [Cacchioli & Overfelt submitted](#). I proceed step by step.

To my knowledge, only two accounts provide analyses of the syntax of verbal modality in Tigrinya: [Tesfay 2016](#) and [Gebregziabher 2021](#). [Tesfay 2016](#) claims that /kwn/ ‘may/might’ and /hlw/ ‘must/have to’ are auxiliaries merged in T° in a monoclausal construction, whereas /gbʔ/ ‘need/should’ and /kʔl/ ‘can’ are main verbs in a biclausal configuration with an unaccusative CP embedding that involves optional raising-to-subject. [Gebregziabher 2021](#) likewise proposes a raising-to-subject analysis for /hlw/ ‘must/have to’ preceded by a raising-to-object configuration in which the object moves through a high Applicative position. In this dissertation, I depart from these analyses, as previously done in [Cacchioli & Overfelt submitted](#).

As already noted in a preceding section, in clauses containing the modals /gbʔ/ ‘should’ and /hlw/ ‘have to’, the subject of the sentence is cross-referenced by subject morphology on the embedded verb and by object morphology on the matrix verb (obligatorily with /hlw/ ‘have to’ and optionally with /gbʔ/ ‘should’). Recall that the matrix verb (the modal) carries 3MS default subject morphology. The example involving /gbʔ/ ‘should’ is repeated in (58) for clarity.

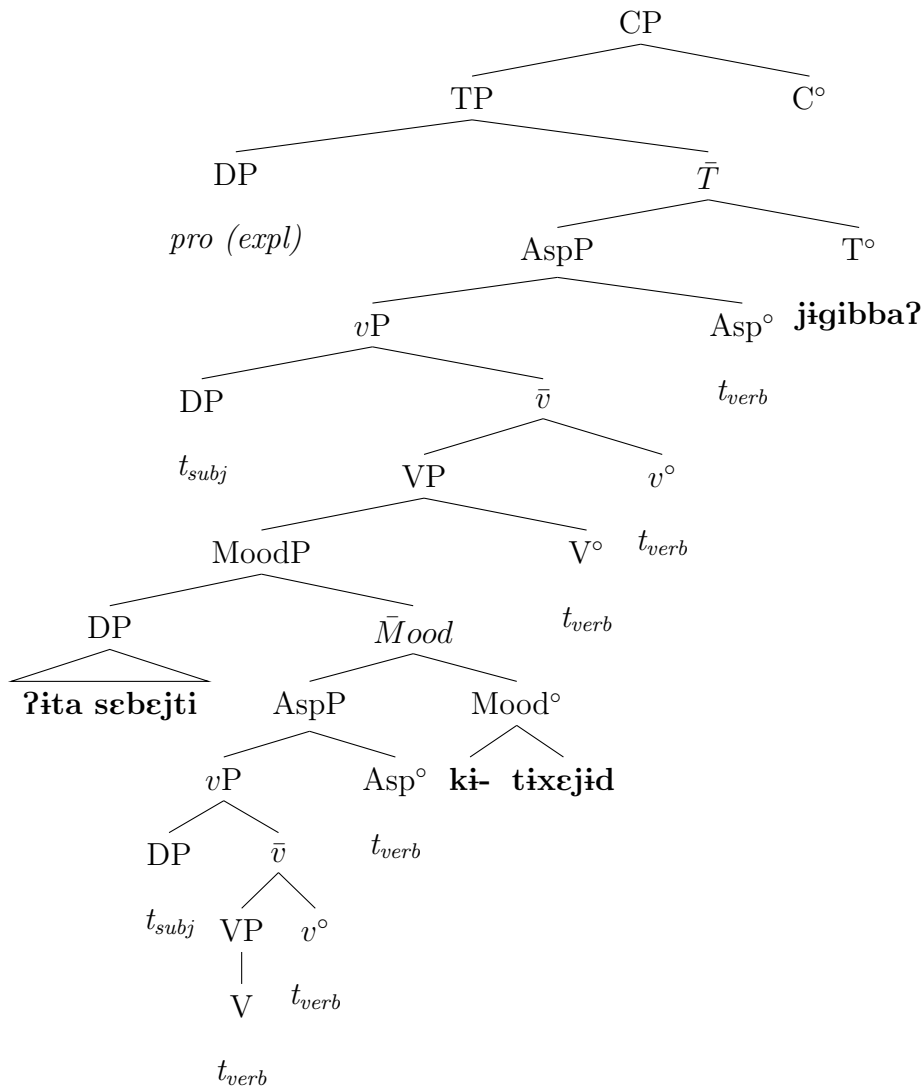
- (58) ?it-a sebejti **ki**-ti-xejid ji-gibbaʔ-(a)
 DIST-FS woman SBJ-SM.3FS-leave.IPFV SM.3MS-need.to.IPFV-OM.3FS
 ‘The woman should/needs to leave.’

This pattern cannot be explained by a control construction, since control constructions

¹⁷One could argue that the null element is actually *pro* rather than PRO, since [Landau 2004](#) suggests that in finite control constructions there is a controlled *pro*. However, according to Landau these constructions involve an embedded CP, which is not the case in my proposal for Tigrinya.

do not allow expletives. Could it be explained by a raising syntax? No, because in that case the subject would be in the matrix clause and the matrix verb should agree with it. I therefore suggest that the matrix subject in these clauses is actually a phonologically null expletive and that the logical subject (the overt subject, *?ita səbejti* ‘the woman’ in the example above) is embedded. Taking all this into account, the syntax of /gb?/ ‘should’ and /hlw/ ‘have to’ is represented in the tree in (59).

(59)



(62), *jɨnbɨb* ‘I read’, is prefixed with *k(i)-*. In contrast, when the same verb is selected by /kwn/ ‘may/might’ in (60), it is not prefixed with *k(i)-* but with *z(i)-*. This is exactly what one would expect in a periphrastic tense form in a relative clause where a verb occurs with an auxiliary. This pattern thus suggests that /kwn/ ‘may/might’ is an auxiliary.

Cacchioli & Overfelt submitted propose that /kwn/ ‘may/might’ is base-generated in the head of a ModP projection merged between the aspectual layer and TP. I do not address the exact position of /kwn/ ‘may/might’ here.

4.3 *K(i)* in other constructions

In this section, I present other constructions that involve the clausal prefix *k(i)-*, namely purpose clauses, sentential subjects, and the future tense. For each construction, I provide the relevant data and then propose syntactic analyses to account for their word order and their requirement for *k(i)-*.

4.3.1 Purpose Clauses

Continuing the comparison between the use of *k(i)-* in Tigrinya and the use of subjunctive morphology or subjunctive particles in other languages, purpose clauses constitute a construction that differs across languages with respect to mood selection. In Tigrinya, these clauses are introduced by *k(i)-* (recall that it has been proposed that *k(i)-* is actually a purposive marker (Nazareth 2011)), as illustrated in (63)–(66).

- (63) (miʔinti) gɛza (miʔinti) **k-i**-gɛziʔ mɛkin-ay
 in_order_to house for_the_sake_of SBJ-SM.1S-buy.IPFV car-POSS.1S
 fɛt'i-ɛ
 sell.GER-SM.1S
 ‘I sold my car in order to buy a house.’

- (64) (miʔinti) fɛtɛna (miʔinti) **k-i**-ɬalif ʔatalil-u
 in_order_to exam for_the_sake_of SBJ-SM.3MS-pass.IPFV cheat.GER-SM.3MS
 ‘He cheated in order to pass the exam.’

- (65) Mihiret kidawinti miʔinti **ki**-ti-hets'ib nab-t-i ruba
 Meheret clothes in_order_to SBJ-SM.3FS-wash.IPFV DIR-DIST-MS river
 werid-a
 descend.GER-SM.3FS
 'Meheret descended the river in order to wash her clothes.'

(adapted from [Mason 1996:99](#),(5))

- (66) miʔinti ʔad-εj frε-tat **ki**-ti-beliʃ nab ʃidaga
 in_order_to mother-POSS.1S fruit.P SBJ-SM.3FS-eat.IPFV DIR market
 k-i-ʃidagi-lla keid-ε
 SBJ-SM.1S-buyIPFV-OBL.3FS go.GER-SM.1S
 'So that my mother eats fruits, I went to the market to buy her some.'

(adapted from [Leslau 1941:142](#))

The sentences in (63)–(66) contain the element *miʔinti*, translated as ‘in order to, for the purpose of’ in [Mason 1996](#), as ‘pour que, afin que’ (‘in order to’ in English) in [Leslau 1941](#), and as ‘for the sake of’ in [Nazareth 2011](#). According to the first two authors, *miʔinti* is a subordinate conjunction, whereas according to the third author it is a preposition. This element can appear either in clause-initial position or immediately before the *k(i)*-marked verb, following the subject and the object. According to my consultants, there is no difference in meaning between the two positions. More importantly, if *miʔinti* does not appear in the sentence, purpose is still expressed by *k(i)*- alone.

If *miʔinti* truly means ‘in order to’, one might think that the sentences above are not purpose clauses but rather rationale clauses (term coined by [Faraci 1974](#)), which in English may optionally contain ‘in order to’. However, I argue that the examples above are purpose clauses. This is because, unlike purpose clauses, rationale clauses do not allow object control ([Landau 2013:222](#)). The difference between purpose and rationale clauses is illustrated with English examples in (67). In line with this point, consider (68), which shows that *k(i)*-clauses containing *miʔinti* are not rationale clauses but purpose clauses, since they can involve object control.

- (67) a. I bought a dog in order to bark at the neighbors. (= I bark at the neighbors.)
 b. I bought a dog to bark at the neighbors. (= The dog barks at the neighbors.)

(68) kelbi miʔinti n-it-om gorababti **k-i-nebbih**
 dog for_the_sake_of DOM-DIST-MP neighbor.P SBJ-SM.3MS-bark.IPFV
 geziʔ-e
 buy.GER-SM.1S
 ‘I bought a dog to bark at the neighbors.’

Although purpose clauses can be expressed with *k(i)-* alone, my consultants also use (and somewhat prefer) another strategy, whose syntactic analysis I leave for future research: the prefix *k(i)-* is preceded by the preposition *n(i)-* (already mentioned in Section 1.5.2 in the discussion of differential object marking), as shown below. This strategy is reported to be more productive than the one involving *miʔinti* (Nazareth 2011:170).

(69) ʔab dirar (ni)-k-i-xədin-o ɕʰaket geziʔ-ε
at dinner PURP-SBJ-SM.1S-wear.IPFV-OM.3MS jacket buy.GER-SM.1S
'I bought a jacket to wear at dinner.'

(70) babur (ni)-k-i-wəsɪd qeltʔifu tɛnsiʔ-u
train PURP-SBJ-SM.3MS-catch.IPFV early wake_up.GER-SM.3MS
'He woke up early to catch a train.'

With respect to the subjunctive mood, at least two languages (to my knowledge) require it in purpose clauses: Standard Arabic and Latin. In (71) and (72), the particle *li-* ‘to/for’ in Standard Arabic and the particle *ut* ‘(in order) to/that’ in Latin are obligatorily followed by a verb in the subjunctive.

(71) yaʃ-a li-ja-nḡaḥ-a fī ʔal-imiṭhān
cheat.PFV-SM.3MS to-SM.3MS-pass-SBJ in DET-exam
'He cheated in order to pass the exam.'
Standard Arabic

(72) Ven-it ut eam vid-ea-t
come.PRES.3MS in_order_to her see-SBJ-3S
'He comes in order to see her.'
Latin

Because purpose clauses are future-oriented and refer to events that have not yet occurred, they can be viewed as irrealis contexts and, consequently, as subjunctive contexts. From this perspective, it is not surprising to see *ki-* playing a role in the formation of purpose

clauses in Tigrinya.¹⁹

Note also that the purposive prefix *n(i)-* can optionally occur in other *k(i)-*clauses, namely those with directive verbs such as ‘order’, and control verbs such as ‘try’, ‘hope’, ‘persuade’, and ‘promise’. However, *n(i)-* cannot occur with ‘want’. This is shown in (73)–(75).

- (73) *ni-Dani?el (ni-)k-i-si?isi? ?a?imin-ε-jo*
 ACC-Daniel PURP-SBJ-SM.3MS-dance.IPFV persuade.GER-SM.1S-OM.3MS
 ‘I persuaded Daniel to dance.’
- (74) *Tesfai (ni-)k-i-bεgis ?azi?u-ni*
 Tesfay PURP-SBJ-SM.1S-leave.IPFV order.GER-SM.3MS-OM.1S
 ‘Tesfay ordered me to leave.’
- (75) *% (ni-)k-i-zarib ji-deli*
 PURP-SBJ-SM.1S-talk.IPFV SM.1S-want.IPFV
 ‘I want to talk.’²⁰

Let us now turn to the syntactic analysis of these clauses. In the literature, it has been claimed that purpose clauses are adjuncts and involve an operation of control (Faraci 1974, Bach 1982, Jones 1991, Landau 2013), with the null PRO subject controlled by the matrix theme (Grosz 2014). This aligns well with the current proposal, which suggests that, as seen earlier, *k(i)-*clauses can have a control syntax.

The remaining question is where *mi?inti* ‘in order to’ is placed in the structure of sentences such as those in examples (63) and (64). I have shown that this preposition - as labeled in Conti Rossini 1940 and Nazareth 2011 - appears either at the beginning of the sentence or after the embedded subject. Given that *mi?inti* ‘in order to’ can move within the clause, it may be more accurately treated as an adverb. This is for two reasons. First, in Tigrinya, adverbs can appear between the subject and the complement of the verb, after the complement of the verb, or before the subject (when topicalized), but not after the verb, as shown in (76).

¹⁹Leung and Girma claim that all *li-*clauses in Amharic (which I mentioned could be seen as the counterpart of *ki-* in Tigrinya) always express an “irrealis yet imminent event” (Leung & Girma 2017:6).

²⁰Probably a regional variation: my Eritrean consultants do not accept the sentence as grammatical, but Jason Overfelt’s Ethiopian consultant finds it perfectly fine.

- (76) a. *haft-εj* **sahti** nab bet migbi *ti-xεjjid*
 sister-POSS.1S rarely DIR restaurant SM.3FS-go.IPFV
 ‘My sister rarely goes to a restaurant.’
- b. *haft-εj* nab bet migbi **sahti** *ti-xεjjid*
 sister-POSS.1S DIR restaurant rarely SM.3FS-go.IPFV
 ‘My sister rarely goes to a restaurant.’
- c. **sahti**, *haft-εj* nab bet migbi *ti-xεjjid*
 rarely sister-POSS.1S DIR restaurant SM.3FS-go.IPFV
 ‘Rarely, my sister goes to a restaurant.’
- d. **haft-εj* nab bet migbi *ti-xεjjid* **sahti**
 sister-POSS.1S DIR restaurant SM.3FS-go.IPFV rarely

Second, there is no preposition stranding in Tigrinya; therefore, a preposition appearing in the positions in which *mi?inti* occurs is unexpected. The absence of preposition stranding in Tigrinya is illustrated in (77).

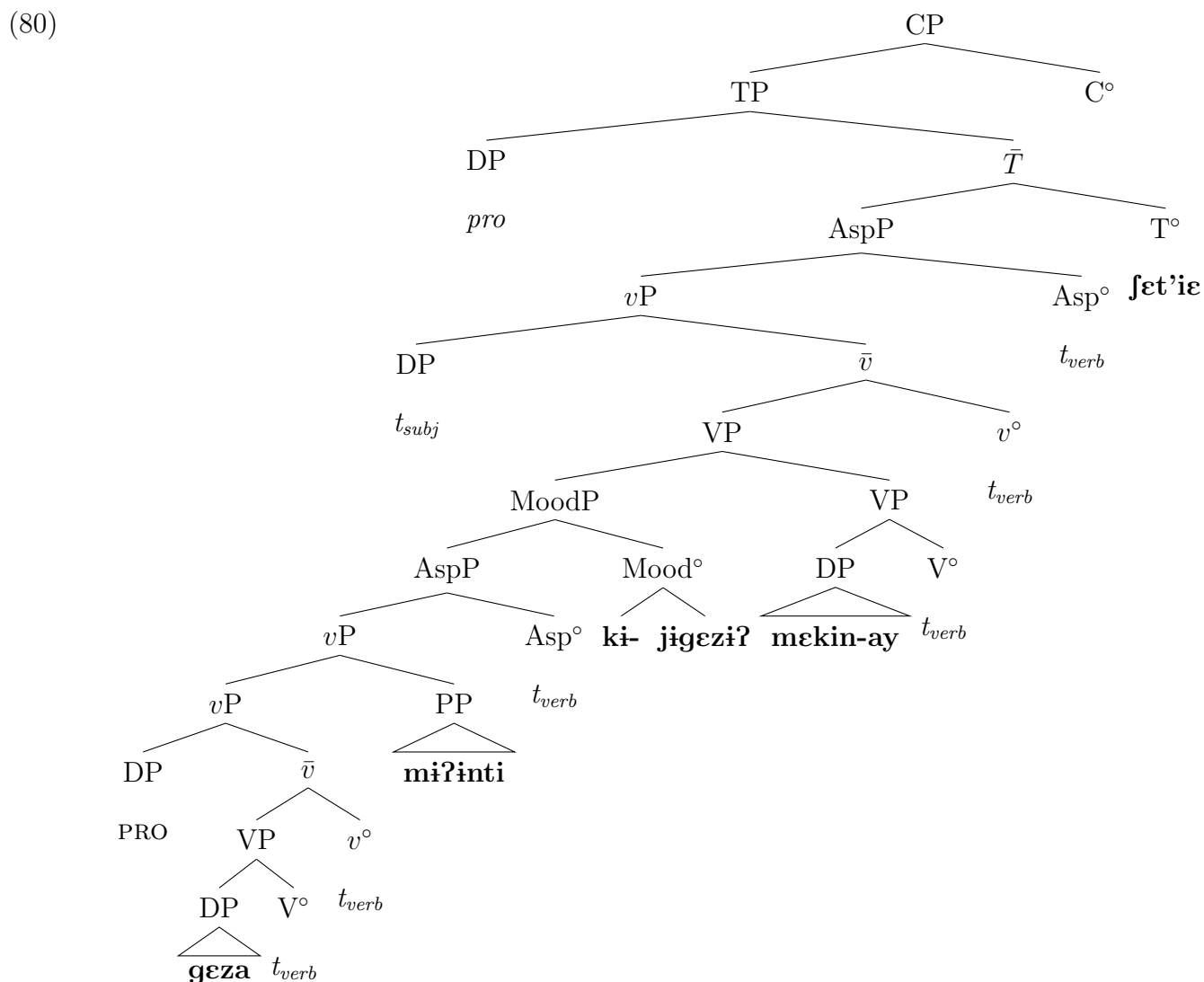
- (77) a. *mən mis tɛzarɛbi-xa*
 who with speak.GER-SM.2MS
 ‘With whom did you speak?’
- b. **mən tɛzarɛbi-xa* *mis*
 who speak.GER-SM.2MS with

However, *mi?inti* ‘in order to’ appears to behave like the prepositional elements discussed in Chapter 3, such as *kem* ‘like’, *sile* ‘for’, or *?inte* ‘if’. As shown, these prepositions require the verb to be marked with the *z(i)-* prefix. Similarly, *mi?inti* requires the verb to be marked with the *k(i)-* prefix and cannot occur alone, as shown in (78).

- (78) *(*mi?inti*) *fɛtɛna* (*mi?inti*) *ji-ħalif* *?atalil-u*
 in_order_to exam for_the_sake_of SM.3MS-pass.IPFV cheat.GER-SM.3MS
 ‘He cheated in order to pass the exam.’

Whether *mi?inti* ‘in order to’ should be categorized as a preposition or an adverb clearly requires further research. Another issue that warrants future investigation is its ability to appear in clause-initial position. Leaving these two points aside, I offer a tentative and preliminary syntactic analysis of purpose clauses such as the one in (79), represented with the tree in (80).

- (79) gɛza miʔinti k-i-gɛziʔ mɛkin-ay ʃɛt'i-ɛ
 in_order_to house for_the_sake_of SBJ-SM.1S-buy.IPFV car sell.GER-SM.1S
 'I sold my car in order to buy a house.'



The tree in (80) shows a subject-control syntax in which the grammatical subject of the sentence is in the matrix clause and controls a PRO subject in the embedded clause. The embedded clause, an adverbial clause, is adjoined to the matrix VP. For simplicity, I label *mɛkinay* ‘my car’ as DP without discussing the possessive marker in detail. *miʔinti* is a preposition merged as a *vP* adjunct, exactly as I proposed for *kɛm* in Chapter 3.

4.3.2 Sentential Subjects

Another type of construction in which $k(i)$ - appears involves sentential subjects, or more specifically, structures in which $k(i)$ -clauses seem to function as the subject of another clause. Instances of these clauses are provided in (81)–(83).²¹

- (81) **k-i**-ʕewit ʔagedasi ʔijj-u
 SBJ-SM.3MS-win.IPFV important BE₂.PRES-SM.3MS
 ‘That he wins is important.’
- (82) ʔit-a kelbi **k-i**-ti-beki ji-ħzin ʔijj-u
 DIST-FS dog SBJ-SM.3FS-cry.IPFV SM.3MS-be_sad.IPFV BE₂.PRES-SM.3MS
 ‘That the (female) dog cries is sad.’
- (83) ʔit-i dimu **k-i**-zarib məṣḥaṣ ʔijj-u
 DIST-MS cat SBJ-SM.3MS-talk.IPFV funny BE₂.PRES-SM.3MS
 ‘That the cat talks is funny.’

The copula *ʔijju* in (81)–(83) carries default 3MS agreement. This default agreement could result from an AGREE relationship between the verb and the $k(i)$ -clause or between the verb and a phonologically null expletive. In the latter scenario, the examples above would be translated as *It is important that he wins*, *It is sad that the (female) dog cries*, and *It is funny that the cat talks*. Further discussion of this point is provided in the analysis section.

If the sentential subject contains an event that has taken place in the past, $k(i)$ - does not appear. This is because, as already noted, this clausal prefix can attach only to verbs in the Imperfective. This is shown in (84).

- (84) a. nisu **k-i**-xɛjjid zihzin ʔijj-u
 he SBJ-SM.3MS-leave.IPFV sad BE₂.PRES-SM.3MS
 ‘That he leaves is sad.’
- b. nisu bi-mi-xed-u zihzin ʔijj-u
 he by-NMZ-leave-3MS sad BE₂.PRES-SM.3MS
 ‘That he left is sad.’

Instead of $k(i)$ -, the verb is prefixed with the nominalizer $m(i)$ - and the preposition *bi*-

²¹These clauses are ungrammatical with the clausal prefix $kemz(i)$ - and the final complementizer *ʔilu*.

‘by’. Why this configuration is triggered when the verb is in the past is left to future research.

Other instances of *k(i)*-clauses as sentential subjects appear in (85)–(87) with the adjectives *kɛbid* ‘difficult’ and *qɛlil* ‘easy’.

- (85) (n-)ʔiz-i q^wanq^wa k-i-tiridiʔ-o qɛlil
 DOM-PROX-MS language.MS SBJ-SM.3MS-understand.IPFV-OM.3MS easy
 ʔijj-u
 BE₂.PRES-SM.3MS
 ‘To understand this language is easy.’
- (86) (n-)ʔiz-a mɛts’ħaf k-i-ti-nbib-a qɛlil
 DOM-PROX-FS book.FS SBJ-SM.3FS-read.IPFV-OM.3FS easy
 ʔijj-u/{*ʔijj-a}
 BE₂.PRES-SM.3MS/3FS
 ‘To read this book is easy.’
- (87) (n-)ʔit-ɛn ʔanisti k-i-tiridiʔ-ɛn kɛbid
 DOM-PROX-FP woman.P SBJ-SM.3MS-understand.IPFV-OM.3FP difficult
 ʔijj-u/{*ʔijj-ɛn}
 BE₂.PRES-SM.3MS/3FP
 ‘To understand these women is difficult.’

As with the first series of examples presented in this section, in (85)–(87) the default 3MS agreement on the copula could result from an AGREE relationship between the copula and the *k(i)*-clause or between the copula and a phonologically null expletive. In this case, the examples above would be translated as *It is easy to understand this language*, *It is easy to read this book*, and *It is difficult to understand these women*. Further discussion of this point is provided in the analysis section.

The ungrammaticality of the examples above when the copulas agree with *q^wanq^wa* ‘language’, *mɛts’ħaf* ‘book’, and *ʔanisti* ‘women’ can be explained by the absence of *tough*-movement constructions in Tigrinya (Wilder 1991, Rezac 2006, Hicks 2009): these three nouns cannot serve as the subject of the matrix sentences.

Moreover, the DOM and accusative marker *n(i)*- is optional. In addition, unlike the first three examples in this section, which in English involved the complementizer *that*, this second series of examples involves the complementizer *to*.

Before turning to the analysis, we can ask the same question posed in each previous section: why would sentential subjects involve *k(i)*- if this were a marker of subjunctive mood? The answer can be found cross-linguistically. In Romance languages - at least in Italian, Spanish, and French, to my knowledge - the subjunctive mood is required in the first series of examples presented in this section, such as example (81). With regard to the second series of examples, such as (85), Romance languages use the infinitive (a dependent mood like the subjunctive (Picallo 1984)). This is illustrated in (88) and (89) for Italian.

- (88) a. È importante che vinc-**a**
is important that win.3MS.SUBJ
- b. *È importante che vinc-e
is important that win.3MS.IND
'That he wins is important.'
- (89) a. È facile capire questa lingua
is easy understand.INF this language
- b. *È facile capisce questa lingua
is easy understand.3MS.PRES this language
'It is easy to understand this language.'

Sentential subjects thus require *k(i)*- in Tigrinya and non-finite dependent verbal forms (such as the subjunctive and the infinitive) in other languages.

Let us now turn to the analysis. For clarity, I have presented two types of sentential subject constructions, which I repeat in (90) and (91).

- (90) **k-i**-ŋɛwɪt ʔagɛdasi ʔijj-u
SBJ-SM.3MS-win.IPFV important BE₂.PRES-SM.3MS
'That he wins is important.'
- (91) (n-)ʔiz-a mɛts'ħaf **kɪ**-ti-nbɪb-a qɛlɪl
DOM-PROX-FS book.FS SBJ-SM.3FS-read.IPFV-OM.3FS easy
ʔijj-u/{*ʔijj-a}
BE₂.PRES-SM.3MS/3FS
'To read this book is easy.'

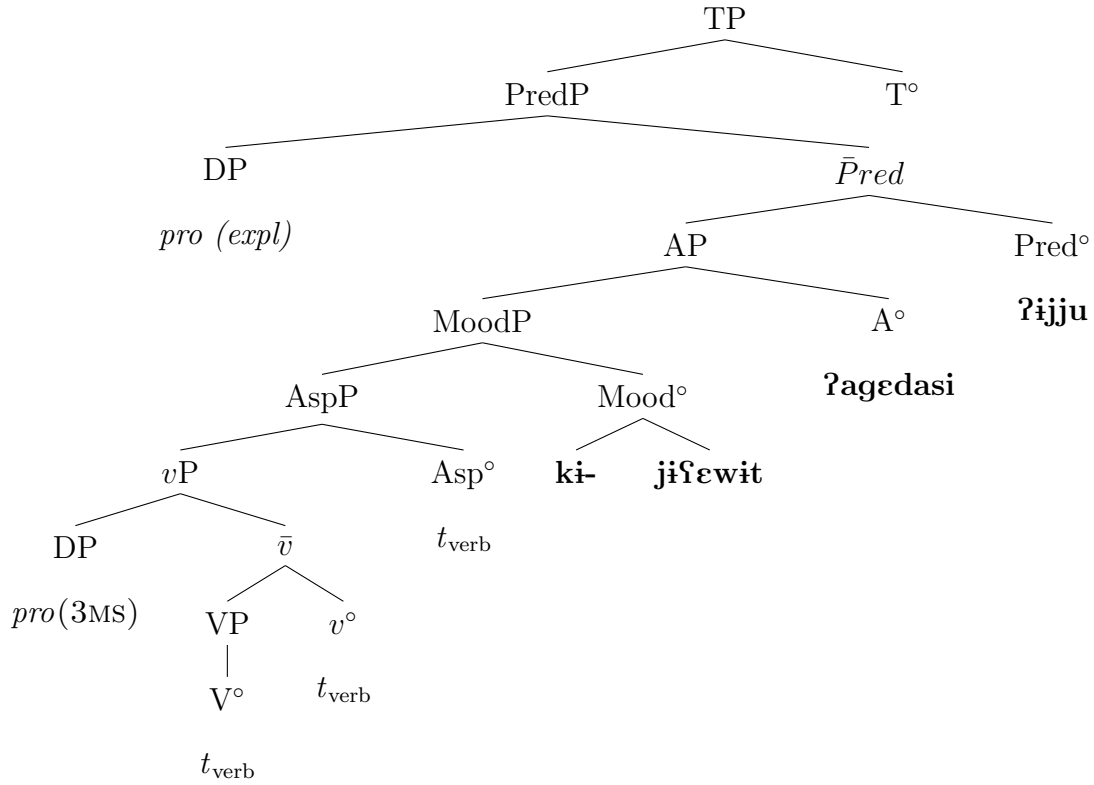
As already noted, the English translation of the first sentence contains *that*, and that of

the second contains *to*. However, both contain *k(i)*- in Tigrinya. I suggest that they also share the same underlying structure.

In the early generative tradition, [Rosenbaum 1967](#) treated the embedded clauses illustrated in the examples above as structural subjects occupying the same position as DP subjects, namely Spec,TP. However, [Ross 1967](#) argued that this cannot be the case, because sentential subjects are far more distributionally restricted than DPs (they cannot appear after sentence-initial topics or preposed auxiliaries ([Alrenga 2005](#))). Subsequently, [Koster 1978](#) proposed that sentential subjects are actually topic phrases base-generated in the left periphery and associated with a null DP that $\bar{A}$ -moves from Spec,IP to Spec,CP. In more recent years, the question of whether sentential subjects exist has not reached consensus, since it is difficult to develop diagnostics that establish whether a clause occupies subject position. The alternative analysis holds that the *that*- and *to*-clauses in the English translations of (90) and (91) are actually complements of the adjectives and that the real subjects are null expletives ([Koster 1978](#)).

If I were to adopt the first approach proposed by [Koster 1978](#), which involves a null DP that $\bar{A}$ -moves, I would expect sentential subjects to be *z(i)*-clauses rather than *k(i)*-clauses, given that in Chapter 2 I claimed that *z(i)*- marks $\bar{A}$ -movement. Therefore, I do not adopt Koster's analysis. Given that I previously suggested that some *k(i)*-clauses - namely those with the modals /gbʔ/ 'should' and /hlw/ 'have to' - involve null expletives, I adopt the alternative analysis proposed by [Koster 1978](#) and suggest that sentential subjects in Tigrinya involve a null expletive that is the matrix subject. I represent the sentence in (90) with the tree in (92).

(92)



The copula *ʔijju* is base-generated in the head of the predication projection PredP (Bowers 1993, Roy 2013), corresponding to the relator RP phrase in Den Dikken 2006. The matrix expletive subject is merged in the Spec,PredP. The *k(i)*-clause is the complement of the adjectival predicate *ʔagedasi* ‘important’.

Before proceeding, note that under the same PredP analysis, one could alternatively posit a derivation without an expletive, in which the *k(i)*-clause itself functions as the subject of the copula. In this case, the clause would occupy Spec,PredP, and the derivation would remain fully consistent, producing the same linear order.

4.3.3 Future Tense

So far in this chapter, I have presented data on *k(i)*-clauses with classes of verbs or constructions claimed to involve subjunctive mood either cross-linguistically or with some variation across languages. In all these cases, at least one language other than Tigrinya is attested to select subjunctive mood. This approach was taken because, at the beginning of this chapter, I showed that the many different contexts in which *k(i)*- appears - although

seemingly unrelated at first sight - can be compared to contexts in which subjunctive mood is used in other languages. The idea is that *k(i)-* could be considered a subjunctive mood marker. I now turn to a construction that involves *k(i)-* in Tigrinya but does not involve subjunctive mood in other languages, at least to my knowledge: future tense.

I have already noted that some future-referring verbs such as ‘want’, ‘wish’ and ‘expect’ select *k(i)-*-marked verbs. Another use of *k(i)-* in Tigrinya is to construct future tense (as briefly mentioned in the Introduction, Section 1.5.2, in the discussion of copulas). This tense is expressed in Tigrinya by a *k(i)-*-prefixed verb followed by the copula *?ijju* (cf. also Nazareth 2011:45). In these constructions the copula is obligatory and must agree with the grammatical subject of the sentence. Consider (93) and (94).

- (93) Rut’ dimu **ki-ti-gezi?** *?ijj-a/*{?ijj-u}*
 Ruth cat SBJ-SM.3FS-buy.IPFV BE₂.PRES-SM.3FS/3MS
 ‘Ruth will buy a cat.’
- (94) bet timihirti ts’bah **k-i-ḡimīr** *?ijj-ε/*{?ijj-u}*
 school tomorrow SBJ-SM.1S-start.IPFV BE₂.PRES-SM.1S/3MS
 ‘I will start school tomorrow.’

How can future tense be linked to subjunctive mood? As previously noted, the subjunctive is semantically associated with irrealis. Because future tense refers to events that have not yet occurred, it can be viewed as an irrealis context and, consequently, as a subjunctive context. Following this reasoning, it is expected that *ki-* plays a role in the construction of future tense in Tigrinya. It is time to ask the question: how can the syntax of these clauses be accounted for?

To my knowledge, two studies discuss potential analyses of future constructions in Tigrinya: Tesfay 2016 and van Urk 2025. According to Tesfay, in future-tense constructions, *k(i)-* signals, together with the copula, “possibility and, hence, mood/modality” (Tesfay 2016:202). The author further proposes analyzing *k(i)-* and the copula as T° heads, but provides no additional details. In contrast, van Urk argues that future clauses in Tigrinya are necessarily biclausal, with the copula embedding an infinitival clause (van Urk 2025:15). This claim is based on the observation that *kemz(i)-* appears only on the

copula and not on the *k(i)*-marked verb, in sentences such as the one in (95).

- (95) kab geza ki-ti-wets'i? kəmzi-xon-ət
 DIR house SBJ-SM.3FS-leave.IPFV COMP-become.PFV-SM.3FS
 ħabir-ε
 report.GER-SM.1S
 'I reported that she will leave the house.' (adapted from [van Urk 2025:15,\(32a\)](#))

Another work discusses future tense in Ethio-Semitic but does not provide a syntactic analysis specifically for Tigrinya: [Girma 2003](#). In this study, the author gives the following Amharic example, which shows that this language also uses the particle *li-* (glossed as INT 'intention' by Demeke) together with an obligatory copula to express immediate or near future.

- (96) l-i-mät'a n-äw
 INT-SM.3MS-come.IPFV AUX.PRES-SM.3MS
 'He is about to come.' *Amharic* (from [Girma 2003:142, \(35\)](#))

In Section 1.5.2 of the Introduction, I explained that the copula *?ijju* has three functions: first, it is the copula used with individual-level predicates; second, it marks Verum, Contrastive, and Exhaustive Focus; and third, it is used in future constructions. The question is therefore whether in future constructions *?ijju* functions as a copula, a focus marker, or something else. This is puzzling: when *?ijju* marks Verum Focus ([Cacchioli 2024](#)), it cannot be negated, as illustrated in (97). However, in future constructions some speakers accept negation on it and some do not, as (98) shows.

- (97) a. Tesfai ?ahmilti ?aj-Ø-beliʕi-n ?ijj-u
 Tesfay vegetables NEG-SM.3MS-eat.IPFV-NEG VER-SM.3MS
 'Indeed, Tesfay doesn't eat vegetables.'
- b. *Tesfai ?ahmilti ji-beliʕ ?aj-kon-ε-n
 Tesfay vegetables SM.3MS-eat.IPFV NEG-VER-SM.3MS-NEG
 'Indeed, Tesfay doesn't eat vegetables.' (from [Cacchioli 2024:37,\(30,31\)](#))
- (98) a. ħanti dmu ?aj-**ki**-geziʕi-n ?ijj-ε
 one.FS cat NEG-SBJ-SM.1S-buy.IPFV-NEG BE₂.PRES-SM.1S
 'I will buy a cat.'

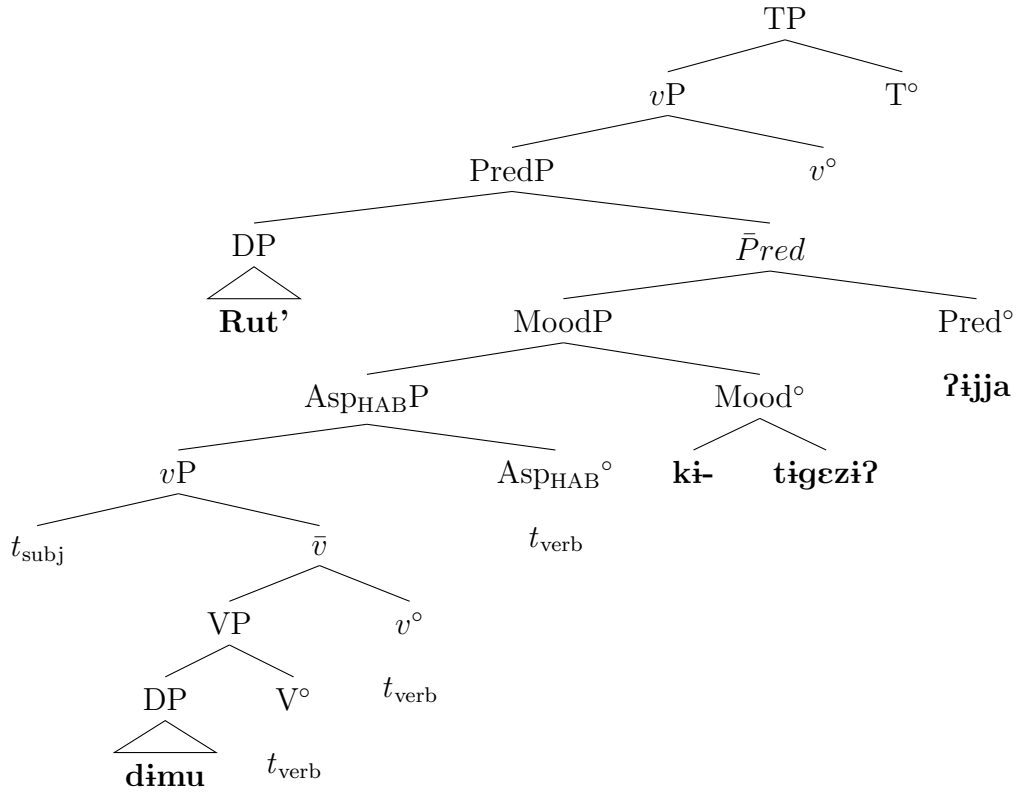
- b. %hanti dmu **ki**-geziʔi ʔaj-kon-ε-n
 one.FS cat SBJ-SM.1S-buy.IPFV NEG-BECOME.PFV-SM.1S-NEG
 ‘I will buy a cat.’

The percentage sign indicates that one of my three consultants does not accept the sentence as grammatical; the same holds for Jason Overfelt’s Ethiopian consultant and Milena Abraham’s (p.c. in [Overfelt & Cacchioli submitted](#)) consultants. It therefore appears that half of the speakers consulted accept the sentence and half do not. This evidence is not strong enough to establish whether *ʔijju* in future constructions is a (Verum) Focus marker or a copula. For simplicity, in order to give a tentative analysis I claim that *ʔijju* functions as a copula in future constructions. This is consistent with the fact that *ʔijju* obligatorily agrees with the subject of the sentence, with further investigation left to Chapter 5.

I therefore claim that the copula *ʔijju* is base-generated in Pred° , and that the $k(i)$ -clause is selected by the predication phrase, as illustrated in the tree in (99). With respect to the subject of the sentence - e.g., ‘Ruth’ in (93) - I propose that it occupies Spec,PredP . The question is whether it has moved out of the $k(i)$ -clause via raising, or is base-generated there and controls a PRO element in the embedded clause.

If ‘Ruth’ were first merged in the matrix clause, the issue would be from where it receives the AGENT theta-role. Because copulas are semantically empty, they cannot introduce arguments or assign theta-roles. Thus, ‘Ruth’ cannot be base-generated in the matrix clause and must instead move there. I therefore suggest that, despite appearances suggesting that $k(i)$ -clauses involve control constructions, future tense constructions in fact involve raising. On this basis, I propose the syntax in (99) for future tense constructions such as the one in (93).

(99)



As noted above, to construct future tense in Tigrinya a *k(i)*-clause is selected by a PredP whose head is occupied by the copula *ʔijju*. This copula agrees with the subject of the sentence, which is merged in the embedded clause and raises into the matrix one.

4.4 Simultaneity

In this last section, I show how *k(i)*- expresses simultaneity when it is prefixed to both the verb and the auxiliary in periphrastic tense forms.

4.4.1 The Data

The last context in which *k(i)*- appears in Tigrinya that does not involve subjunctive mood in other languages (to my knowledge) is in clauses expressing simultaneity. In some constructions with periphrastic tense forms, the appearance of *k(i)*- on both the verb and the auxiliary signals an event simultaneous with the matrix event. Consider (100)–(102).

- (100) **k-i-beliʃ** **k-ell-exu** **nissu riʔij-u-ni**
 SBJ-SM.1S-eat.IPFV SBJ-BE1.PRES-SM.1S he see.GER-SM.3MS
 ‘While I was eating, he saw me.’ (from [Cacchioli 2023:256,\(32\)](#))
- (101) **ʔit-a** **dimu ki-ti-bɛrabɛr-ini** **k-ell-a**
 DIST-FS cat.FS SBJ-SM.3FS-wake_up.IPFV-OM.1S SBJ-BE1.PRES-SM.3FS
 (ʔanɛ) dɛkis-ɛ nɛyr-ɛ
 (I) sleep.GER-SM.1S BE1.GER-SM.1S
 ‘I was asleep while the cat was waking me up.’
- (102) **ʔit-om** **polis nab-t-i** **geza ki-i-ʔati-wu**
 DIST-MP police DIR-DIST-MS house SBJ-SM.3MP-enter.IPFV-SM.3MP
k-ellɛ-wu (ʔanɛ) genzɛb ji-ħabiʔ nɛyr-ɛ
 SBJ-BE1.PRES-SM.3FS (I) money hide.GER-SM.1S BE1.GER-SM.1S
 ‘While the police was entering the house, I hid the money.’

In (100)–(102), the matrix verb forms, *riʔijuni* ‘he saw me’, *dɛkise nɛyrɛ* ‘I was asleep’ and *jiħabiʔ nɛyrɛ* ‘I was hiding’, express a past action. The embedded verb forms to which *k(i)-* is prefixed, *jibeliʃ ʔallɛxu* ‘I was eating’, *tibɛrabɛrini ʔalla* ‘is waking me up’ and *jiʔatiwu ʔallɛwu* ‘they are entering’, are in the present progressive.

However, they do not express present but past: they denote an action simultaneous with that expressed by the matrix verbs.²²

A similar phenomenon is found in Amharic. In this language, simultaneity is conveyed by the prefix *s-* attached to a verb in the imperfective, as in (103). Imperfective verb forms express present tense in Amharic as well, but the addition of this prefix conveys simultaneity with the matrix verb.

- (103) **s-i-bəla** **mətt’a-h^w**
 while-3SM-eat.IPFV arrive.PFV-1SG
 ‘I arrived while he was eating.’ *Amharic* (from [Amberber 2023:440, \(51\)](#))

²²For the sake of completeness, note that it is also possible to convey simultaneity as follows.

- (i) a. **ʔin(d)a bɛliʃɛ-ku** **k-ell-exu** **ħaftei** **geza mɛts’iʔ-et**
 when eat.PFV-SM.1S SBJ-BE1.PRES-SM.1S sister-POSS.1S house come.GER-SM.3FS
 ‘When I was eating, my sister came home.’

The element *ʔinda* (and also *ʔina*, as produced by my consultants) precedes a verb in the Perfective, which in turn is followed by a *k(i)-*marked auxiliary. This element is probably a loanword from Amharic.

The difference between Tigrinya and Amharic is that a periphrastic tense form is used in the former and a simple verb form in the latter; consequently, *k(i)-* appears twice and *s-* once. For completeness, in Amharic it is also possible to convey a temporal meaning with the discontinuous morpheme *kə-...bəhwala*, as in (104). This morpheme corresponds to *after* in English.

- (104) *kə-mətt’attf* *bəh^wala hed-ə*
 after-arrive.PFV.3SF after go.PFV-3SM
 ‘He left after she had arrived.’ *Amharic* (from [Amberber 2023:440](#), (50))

Returning to Tigrinya, if a modal occurs in a *k(i)-* clause that expresses simultaneity, three of these prefixes appear in the structure. This is shown in (105).

- (105) *bun ki-səti* *ki-i-dəli* *k-ell-əxu* *nab*
 coffee ki-SM.1S-drink.IPFV SBJ-SM.1S-want.IPFV SBJ-BE₁.PRES-SM.1S DIR
 kafə ji-xəjjid
 bar SM.1S-go.IPFV
 ‘When I want to drink coffee, I go to a café.’

I used the temporal adverb ‘when’ in the clause above instead of ‘while’ because the appearance of *k(i)-* on both the verb and the auxiliary in periphrastic tense forms can also express a specific point in time and not only a continuous action. The important point is that there is always simultaneity with the matrix verb. However, in root clauses and embedded interrogatives ‘when’ is not expressed by the appearance of *k(i)-* on both the verb and the auxiliary but by the *wh*-element *mɛʃas*, as in (106).

- (106) a. *mɛʃas mɛts’iʔi-ka*
 when come.GER-SM.2MS
 ‘When did you come?’
 b. *mɛʃas kəm-zi-bəts’ih-ə* *ninəbsey hatit-ə-jo*
 when COMP-REL-arrive.PFV-SM.3MS myself ask.GER-SM.1S-OM.3MS
 ‘I wonder when he arrived.’

To link these constructions to the use of subjunctive mood, it is important to note that, in this case as well, the temporal interpretation of *k(i)-* clauses depends on the tense

of the matrix verb: it is simultaneous with it. Thus, as mentioned at the end of the previous section, it would not be surprising to find languages that require subjunctive mood (morphology or particles) when constructing simultaneous clauses, given that this mood is considered dependent on the tense/aspect of the matrix clause.

It is now time to turn to the analysis.

4.4.2 The Analysis

In the previous section, I presented data showing how $k(i)$ -clauses can express simultaneity with matrix clauses when $k(i)$ - is prefixed to both the verb and the auxiliary in periphrastic tense forms. A similar configuration has already been examined in Chapter 2 and Chapter 3: the clausal prefix $z(i)$ - also appears on both the verb and the auxiliary in periphrastic tense forms. However, $z(i)$ - cannot replace $k(i)$ -, hence the ungrammaticality of (107) and (108).

- (107) ***z**-i-beliŋ **z**-ell-εxu nissu riʔij-u-ni
REL-SM.1S-eat.IPFV REL-BE1.PRES-SM.1S he see.GER-SM.3MS
Intended: ‘While I was eating, he saw me.’

- (108) ***z**-i-beliŋ **z**-ell-εxu nissu televiŋin ji-riʔi
REL-SM.1S-eat.IPFV REL-BE1.PRES-SM.1S he television SM.3MS-see.GER
Intended: ‘While I am eating, he watches television.’

The question is therefore how $k(i)$ - can appear twice in the structure while at the same time expressing simultaneity with the matrix clause. Given that the syntactic behavior of $k(i)$ - is similar to that of $z(i)$ -, I suggest that constructions in which $k(i)$ - appears twice involve $\bar{A}$ -movement. However, in contrast with $z(i)$ -clauses, $k(i)$ -clauses do not involve the $\bar{A}$ -movement of nominals or of operators co-indexed with a nominal head but of an operator co-indexed with a temporal specification. In more formal semantic terms, the lambda operator binds a nominal x in $z(i)$ -clauses and an event e in $k(i)$ -clauses. The clausal prefixes $z(i)$ - and $k(i)$ - are thus sensitive to the nature of the relativized head.

In other words, in the presence of periphrastic tense forms, $k(i)$ -clauses involve the relativization of duratives: think of the meaning of *While I was eating*; it can be para-

phrased as *the time during which I was eating* (a relative clause). More concretely, I suggest that there is a silent TIME in the structure, or, more specifically, a temporal operator. This proposal aligns with what we find in other Ethio-Semitic languages: in the Chaha Group, temporal clauses are often supplemented with *gi/gizjə/gam^{wə}* ‘time’, as in (109). In the Chaha Group TIME is phonologically overt, as in (109), whereas in Tigrinya it is phonologically covert.

- (109) k’əja-ta t-i-səra gam^{wə}
 village-POSS.3SM TEMP-SBJ.3SM-arrive.IPFV time
 ‘when he arrives at his village’ (from [Menuta & Völlmin 2023:495,\(60\)](#))

Moreover, it is not surprising that *k(i)-*, which I have claimed is an irrealis mood marker, appears in these contexts. Cognates of *k(i)-* are known to express temporal (and causal) subordination across Central Semitic languages ([Pat-El 2012:62](#)), and in Yucatec Maya ‘when’-clauses involve an irrealis marker ([Guerrero 2021:65](#)). Claiming that these *k(i)-* clauses involve a temporal operator also helps explain the apparent lack of difference in meaning between the use of *kəmz(i)-* in (41) and *k(i)-* in (42) in the Tigrinya sentence *I see that he is eating injera*: as already pointed out, the sentence with *k(i)-* can more precisely be translated as *I see (him) while he is eating injera*.

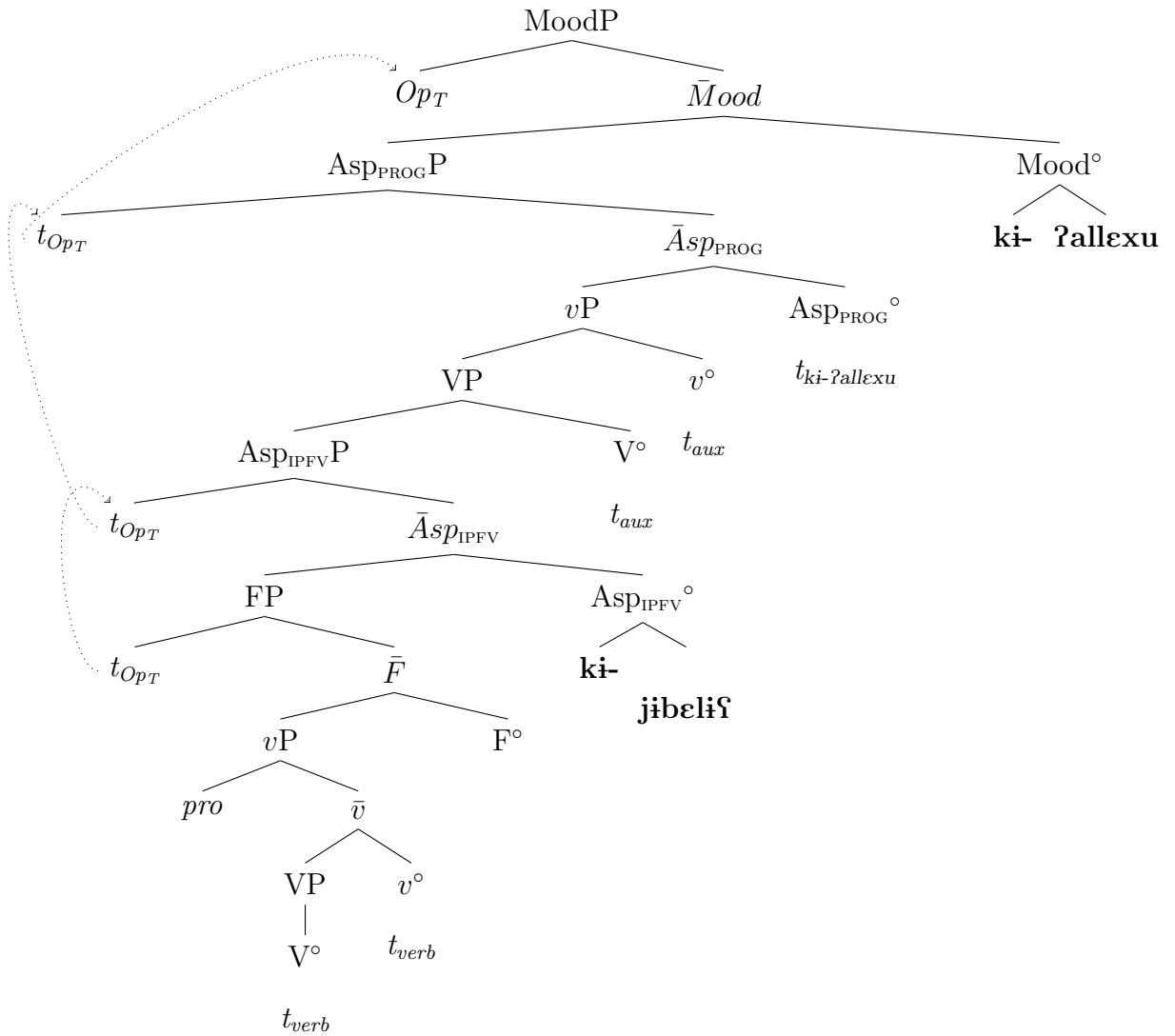
With regard to the internal syntax of these clauses, it has already been claimed in the literature - first by [Geis 1970](#) and subsequently in several implementations - that *while*- and *when*-clauses involve a temporal operator moving to the left periphery of the clause. For specific languages, [Zribi-Hertz & Diagne 2003](#) propose that Wolof adverbial clauses are relative clauses, and [Lipták 2005](#) argues that Hungarian temporal clauses are relative clauses. For a cross-linguistic analysis, in languages in which temporal clauses are introduced by a *when* element, it has been suggested that this element is a temporal operator that moves from the IP domain - the position corresponding to the temporal adverb *then* à la [Cinque 1999](#) - to the CP domain ([Haegeman 2007](#)). In languages in which temporal clauses are introduced by a preposition, the element that moves to the C° domain is a phonologically non-overt temporal operator merged in the AspP domain, and the complement of this preposition is a relative clause with a silent antecedent noun

(Demirdache & Uribe-Etxebarria 2004).

I propose that in clauses expressing simultaneity, $k(i)$ - is not merged in the Mood^o head, but its occurrence is triggered in the same way as the occurrence of $z(i)$:- a temporal operator is merged in the specifier position of a functional projection FP in the aspectual domain and raises via $\bar{A}$ -movement to the edge of the clause (Spec,MoodP). Instead of moving in one step, it stops along the path, landing in the specifier positions of the aspectual phrases of both the verb and the auxiliary. Spec-head agreement between the temporal operator and the verb or auxiliary that has moved into the heads of these projections triggers the appearance of $k(i)$ -.

The derivation of (110) is represented in the tree in (111).

- (110) **k-i-bɛliŋ** **k-ɛll-ɛxu...**
 SBJ-SM.1S-eat.IPFV SBJ-BE1.PRES-SM.1S
 ‘While I am eating,...’



Thus, the syntactic behavior of $z(i)$ - and $k(i)$ - in clauses involving periphrastic tense forms is very similar, but the element that undergoes $\bar{A}$ -movement involved in the derivation differs.

4.5 Interim Summary of Chapter 4

In this chapter, I investigated the clausal prefix $k(i)$ - in Tigrinya. I showed that this prefix appears in several types of clauses, summarized in Table 4.1.

$k(i)$ -	
DIRECTIVES	e.g. ‘order’, ‘suggest’
DESIRE & FUTURE REFERRING	e.g. ‘prefer’, ‘wish’, ‘want’, ‘expect’
ROOT MODALS	e.g. ‘must’, ‘should’, ‘can’
EMOTIVE FACTIVES	e.g. ‘regret’, ‘be happy’
CONTROL	e.g. ‘try’, ‘hope’, ‘choose’, ‘ask’
ECM	e.g. ‘expect’, ‘see’
PURPOSE CLAUSES	
FUTURE TENSE	
SENTENTIAL SUBJECTS	
SIMULTANEITY	

Table 4.1: Classes of verbs and constructions that require $k(i)$ -.

I proposed that $k(i)$ - is a subjunctive (or irrealis) marker that appears in two scenarios. In the first, which involves simple verb forms, $k(i)$ - is merged in the head of a low MoodP projection. $k(i)$ -clauses are reduced clauses whose temporal interpretation depends on the matrix tense/aspect. Because of this, the embedded verb raises to the Mood° head. I provided a syntactic analysis for the four different types of constructions at the bottom of Table 4.1, as well as for the constructions selected by different verb classes discussing first those that involve subject obviation and then those that do not.

In the second scenario, which involves periphrastic tense forms, $k(i)$ - appears on both the verb and the auxiliary, expressing simultaneity with the matrix clause. In this case, I suggested that $k(i)$ - is triggered by the $\bar{A}$ -movement of a temporal operator, as if it were involved in the relativization of duratives. This analysis therefore claims that $k(i)$ - and $z(i)$ - are two sides of the same coin: both are reflexes of $\bar{A}$ -movement but of different elements.

Chapter 5

The Circumfix *ʔaj-...-(i)n*

5.1 Setting the Scene

The fourth and last morpheme investigated in this dissertation is not a clausal prefix but a circumfix: *ʔaj-...-(i)n*, consisting of the prefix *ʔaj-* and the suffix *-(i)n* ([i] is an epenthetic vowel; the morpheme itself is *n*). In the introduction I gave the example in (1) to show the context in which this circumfix occurs in Tigrinya:

- (1) mis dim-ay **ʔaj-Ø-ts'awet-in**
 with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG
 'I do not play with my cat.'

ʔaj-...-(i)n expresses what has been called discontinuous or bipartite sentential negation: two negative elements expressing a single semantic negation. In the Semitic language family, each language possesses a very specific set of negative particles, and no single set is found in all of them (Pat-El 2012:17). Sentential negation in at least two other languages of the Ethio-Semitic branch is also morphologically realized as a circumfix attached to the verb, as shown in (2).

- (2) a. **ji-wæ:sæl-ko-n**
 NEG-continue.PFV-1S-NEG
 'I did not continue...'

Tigre (from Elias 2019:164)

- b. **ʔal-kəffət-ə-mm**
 NEG-open.PFV-3MS-NEG
 ‘He did not open...’

Amharic (from [Endalew 2018:17,\(1a\)](#))

In Tigre, the circumfix is *ji/ʔi...-n(ni)*: *ji-* precedes laryngeals, whereas *ʔi-* precedes all other consonants ([Shlomo 1980](#)). In Amharic, the form of the first element in the circumfix depends on the aspect of the verb: before a verb in the Perfective, as in the example above, the circumfix is *ʔal...-mm*, while before a verb in the Imperfective it is *ʔa...-mm* ([Edzard 2019](#)).¹

All natural languages express negation, making it worth investigating as many languages as possible to see how it varies cross-linguistically. Since the late 1980s, researchers have done exactly that. [Pollock 1989](#), for example, was one of the first to suggest that negation is encoded as a functional projection in the syntax. Today, the literature shows that sentential negation can be expressed at different levels of the clausal spine: it can occur in the C° domain (higher than TP) or in the inflectional domain (as the complement of TP or lower in the aspectual layer). Establishing exactly how negation is encoded in the syntax remains one of the central issues in current syntactic theory ([Cruschina et al. 2017](#)).

When it comes to sentential negation, some languages express it with a single morpheme. This is the case in English, as shown in (3). In this language, *not* is a free morpheme.

- (3) I do **not** like linguistics.

Other languages, such as Bantawa, can express a single semantic negation with five negative morphemes, as in (4). In this language, the morphemes are bound.

¹Note that in Amharic the allomorphy of negation is conditioned by aspect, similarly to the allomorphy of Amharic relative markers, which is also aspect-dependent, as quickly mentioned at the beginning of Chapter 2.

- (4) **i-ciN-nin** set-**nin**-Ø-**nin-ci-n**
 neg1-hang-neg2 kill-neg3-prog-neg4-refl-neg5
 ‘He is not killing himself by hanging.’

Bantawa (from [van der Auwera & Krasnoukhova 2020:94](#), (8))

The goal of this chapter is to fill out the typological picture by giving an account of the phenomenon in Tigrinya, which has also been partly investigated in [Cao & Liotta 2023](#) and [Abraham 2024](#), though these works focus primarily on negative polarity items. I claim that *ʔaj-* heads a low NegP projection that dominates *vP* and carries an [iNeg] feature, whereas *-(i)n* heads a high PolP projection that dominates TP and carries a [uNeg] feature. I show that in root declarative clauses both negative morphemes are required to express sentential negation. However, in *zi-*, (some) *ki-*, and imperative clauses only the presence of *ʔaj-* is grammatical and *-(i)n* does not occur. I provide a syntactic analysis to explain why this is the case for all the clauses mentioned. I also show that in periphrastic tense forms the bipartite negation can appear on either the verb or the auxiliary without an apparent change in meaning.

This chapter is structured as follows. In Section 5.2.1 I focus on and analyze sentential negation with simple verb forms, considering declarative, *z(i)-*, *k(i)-* and imperative clauses. Section (46) is dedicated to sentential negation in the past tense, which again raises the issue of clausal prefixes that cannot be attached to the Gerund. After the analysis, I turn to sentential negation in the present progressive in Section 5.3, concluding with an analysis of negative future clauses.

5.2 Negation in Simple Verb Forms

In this first section I present the behavior of sentential negation in clauses with simple verb forms. I provide data on negative root declarative clauses, negative *z(i)-*-clauses, *k(i)-*-clauses and negative imperative clauses. I then propose an analysis to account for the syntax of these negative sentences.

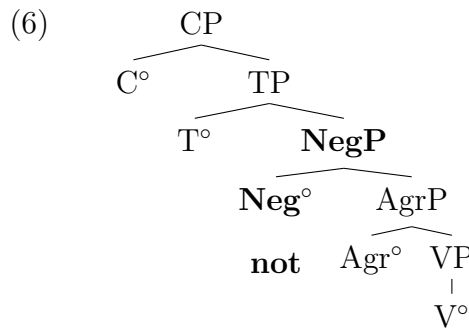
5.2.1 Negative Root Declarative Clauses

This section examines sentential negation in declarative clauses expressing habitual present tense. As noted above, sentential negation is marked by two negative morphemes affixed to the verb: *?aj-* as a prefix and *-(i)n* as a suffix. Consider (5), which presents an affirmative clause and its negative counterpart.

- (5) a. mis dim-ay ji-ts'awet
with cat-POSS.1S SM.1S-play.IPFV
'I play with my cat.'
- b. mis dim-ay **?aj-Ø-ts'aweti-n**
with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG
'I do not play with my cat.'

The data above raise several questions: How can a single semantic negation be expressed by two negative morphemes? What is *?aj-*? What is *-(i)n*? Where are these elements located in the syntax?

Let us consider what has been proposed so far in the literature in order to develop a possible solution. One of the first authors to address negation examined English and argued that *not* is structurally encoded in a functional projection, NegP, positioned between TP and AgrP (Pollock 1989:397,(77)), as shown in (6).

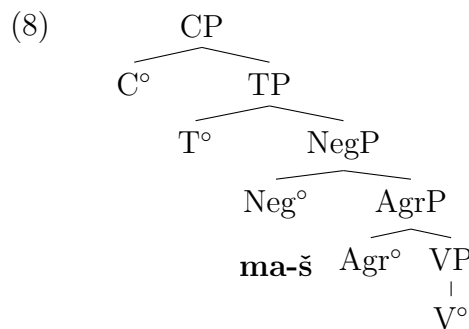


Can this approach account for a language with bipartite negation? It could. For example, in Arabic², where negation is expressed by the two negative morphemes *ma-* and *-š*, Benmamoun et al. 2013 adapt Pollock's approach to account for the following.

²Many modern colloquial dialects of Arabic express sentential negation with two discontinuous morphemes (Shlonsky 1997).

- (7) a. **ma-ža-t-š** Nadia
 neg-come.past.3fs-neg Nadia
 ‘Nadia didn’t come.’ *Moroccan Arabic*
 (from Benmamoun 2000:69,(1))
- b. Nadia **ma-ši** fə-l-madrasa
 Nadia neg-neg in-the-school
 ‘Nadia is not at school.’ *Moroccan Arabic*
 (from Benmamoun 2000:73,(13))

The examples above show that in verbal clauses *ma-* and *-š* attach to the verb, the former as a prefix and the latter as a suffix. In copular clauses, however, *ma-* and *-š* can cliticize onto each other. The authors argue that this pattern indicates that *ma-* and *-š* are two realizations of a single negative morpheme. They therefore propose that both are generated under the Neg° head, as shown in the tree in (8).

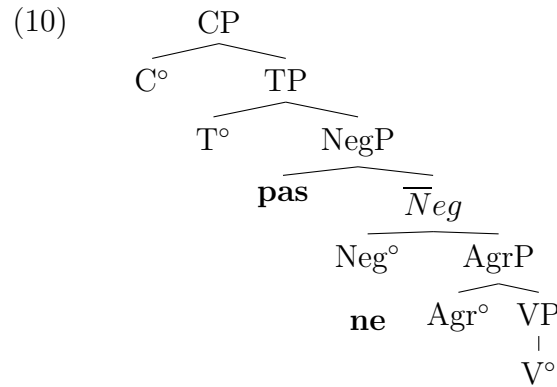


This approach appears to account for bipartite negation, but it is not the only one that has been proposed. Let us return to Pollock 1989. In his work he also examined sentential negation in French. In this language, sentential negation is expressed by the free morphemes *ne*, which precedes the verb, and *pas*, which follows the verb, as in (9). In spoken French, *ne* is typically absent.

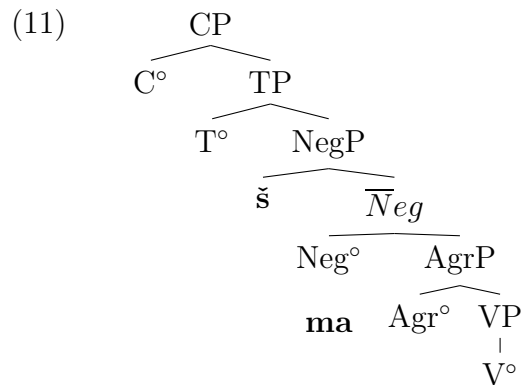
- (9) Je **ne** jou-e **pas** avec mon chat.
 I NEG play.PRES-S.1S NEG with my cat
 ‘I do not play with my cat.’ *French*

Pollock argues that *ne* is the head of the French NegP and *pas* is a phrase in its specifier. Similarly, Rowlett 1998 also places *pas* in Spec,NegP but proposes that it moves there from

a *vP*-adjunct position rather than being base-generated. Pollock’s approach is illustrated in the tree in (10). The correct word order results from the movement of *ne* to a position above *pas* and from the movement of the verb to “(some) Tense (position)” (Pollock 1989:414) between the two negative particles.



Returning to the Arabic case with the particles *ma-* and *-š*, some authors have adopted this analysis to explain their syntax. Shlonsky 1997 for Palestinian Arabic and Benmamoun 2000 for Moroccan Arabic propose that *ma-* heads NegP and *-š* occupies its specifier, as shown in (11).



So far, we have seen two approaches that could, in principle, account for bipartite negation in Tigrinya. In the first, the discontinuous morphemes of a bipartite negation are both merged in the same head (Neg°). In the second, the preverbal morpheme is a head merged in Neg°, and the postverbal morpheme is a phrase in Spec,NegP. The first approach cannot straightforwardly be applied to Tigrinya by claiming that *?aj-* and *-(i)n* both head the NegP projection, since these two morphemes cannot cliticize onto each other.³ The second

³However, Sjörs claims that *-n* can also appear as a prefix to the preverbal negative exponent in the

approach, however, can be applied to Tigrinya by analyzing *?aj-* and *-(i)n* as elements of the NegP projection, one as a head and the other as a phrase. But which is the head and which is the phrase? Moreover, since French is a head-initial language and Pollock 1989 uses a head-initial structure, would the result in Tigrinya - a head-final language - be the opposite, with *-(i)n* as the head and *?aj-* as the phrase? These questions must be resolved before adopting a specific analysis.

At this point, it is important to address the following issue: if Tigrinya exhibits a single semantic negation expressed by two negative morphemes rather than a double negation, which morpheme is the “real” negator? In more formal terms, one of the elements - either the pre- or the post-verbal morpheme - must carry a [iNeg] feature. Which one? To answer this question, it is necessary to identify a context in the language (a negative sentence) in which only one of the two negative morphemes can express negation.

We can already gain some insight by examining how Tigrinya speakers respond negatively to yes/no questions. The form most commonly used by my consultants is *no* ‘no’, which, in my view, is borrowed from either Italian (Eritrea was an Italian colony from 1890 to 1941) or English. Another form used is *?ajfal* ‘no’, which, as the reader will note, contains only the prefixal negative morpheme *?aj-* and not the suffixal *-(i)n*.

This said, in the next sections I present two more robust contexts in which only one of the two morphemes expresses negation: *z(i)-*clauses, (some) *k(i)-* clauses and imperative clauses.

5.2.2 Negative *Z(i)-*clauses

I have already investigated *z(i)-* clauses in Chapter 2; however, apart from one example in Chapter 3 (when arguing that *kəmz(i)-* is bimorphemic), I have not examined these clauses in the presence of negation. Let us therefore consider the relative clauses in (12).

form of *naj-*, but this form is said to be less common than the one discussed here, and I did not encounter such cases in my fieldwork (Sjörs 2018:333).

- (12) a. ʔit-om ʔanɛ z-ɛj-Ø-nbɪb-om mets'hafti
DIST-MP I REL-NEG-SM.1S-read.IPFV-OM.3MP book.MP
 ʔab-t-i ʔarmadyo ʔall-ɛwo
PREP-DIST-MS cabinet BE₁.PRES-SM.3MP
'The books that I do not read are in the cabinet.'
- b. *ʔit-om ʔanɛ z-i-nbɪb-om-n mets'hafti
DIST-MP I REL-SM.1S-read.IPFV-OM.3MP-NEG book.MP
 ʔab-t-i ʔarmadyo ʔall-ɛwo
PREP-DIST-MS cabinet BE₁.PRES-SM.3MP
- c. *ʔit-om ʔanɛ z-ɛj-Ø-nbɪb-om-n mets'hafti
DIST-MP I REL-NEG-SM.1S-read.IPFV-OM.3MP-NEG book.MP
 ʔab-t-i ʔarmadyo ʔall-ɛwo
PREP-DIST-MS cabinet BE₁.PRES-SM.3MP

The sentences above illustrate several points. Example (12a) shows that to negate a relative clause, the (allomorphic version of the) prefixal negative morpheme *ʔaj-* appears between the *z(i)-* prefix and the verb.⁴ It appears alone, without the suffixal negative morpheme *-(i)n*, which cannot negate a relative clause by itself, as shown by the ungrammatical sentence in (12b).⁵ Example (12c) shows that the appearance of both negative morphemes likewise results in ungrammaticality. The impossibility of having *-(i)n* in relative clauses also holds when these clauses are in the past tense (further discussion of sentential negation in past clauses follows later in this chapter), as in (13).

- (13) ʔit-om ʔanɛ z-ɛj-Ø-nbib-ku-wom mets'hafti ʔab-t-i
DIST-MP I REL-NEG-read.PFV-SM.1S-OM.3MP book.MP PREP-DIST-MS
ʔarmadyo ʔall-ɛwo
cabinet BE₁.PRES-SM.3MP
'The books that I did not read are in the cabinet.'

For the sake of completeness and to avoid any doubts, in Chapter 2 I showed that $z(i)$ -also appears in clause types other than relative clauses. It is crucial to examine what happens when we negate also these clauses.

⁴The negative prefix *?aj-* occurs as *ɛj-* because of a phonological process triggered by the presence of the prefix *z(i)-*, which causes the glottal stop to drop with a consequent vowel change from [a] to [ɛ]. This was already mentioned in Chapter 2. Note also that the prefixal *ϕ*-features of the verb are dropped.

⁵This is also the case in six other Ethio-Semitic languages (Amharic, Argobba, Gafat, Harari, Ennemor, and Endegeñ): the preverbal negative marker can negate a clause on its own, but the postverbal negative marker cannot (Girma 2003:217).

clause in (15) and one of a negative $k\epsilon m z(i)$ -clause in (16).⁶

- (14) Kidane meṭs’hafti z-**ɛj**-Ø-nbib ji-mesil
Kidane book.MP REL-NEG-SM.3MS-read.IPFV SM.3MS-seem.IPFV
‘Kidane seems to not read books.’⁷
- (15) nissu ?intε z-**ɛj**-Ø-xejid k-i-beli ?ijj-ε
he if REL-NEG-SM.3MS-leave.IPFV SBJ-SM.1S-cry.IPFV BE₂.PRES-SM.1S
‘If he doesn’t leave, I’ll cry.’
- (16) ?aman ?intay kem-z-**ɛj**-Ø-fettu hatit-u-ni
Aman what COMP-REL-NEG-SM.1S-like.IPFV ask.GER-SM.3MS-OM.1S
‘Aman asked me what I don’t like.’

Before moving forward, I would also like to mention another context in which $z(i)$ - and $\mathcal{I}aj$ - occur without $-(i)n$: lexical negation (see also [Bulakh 2019](#), [Cao & Liotta 2023](#)). Consider (17).

- (17) a. higus , $\text{z-}\mathbf{\text{ej}}\text{-higus}$
happy, REL-NEG-happy
'happy, unhappy'
- b. $\text{zi-}\text{?imen}$, $\text{z-}\mathbf{\text{ej}}\text{-?imen}$
REL-'believe', REL-NEG-'believe'
'believable, unbelievable'

Example (17a) shows that an adjective can be negated by adding both *z(i)-* and *?aj-* to it. Example (17b) illustrates that some adjectives already display the *z(i)-* prefix in their affirmative form (see the section on superlatives in Chapter 2). According to Mason 1996, their literal translation would be ‘(*the one*) that is believ-ed/-able’. Their negative counterparts are negated with *?aj-*.

The data presented so far answer the question posed in the previous section: since $z(i)$ -clauses are negated solely by the negative prefix $?aj$ -, without $-(i)n$, this element must

⁶I am not proposing an example with a negative comparative clause because I have been unable to construct a sentence suitable for testing with my consultants (think of something like *#Tefay is taller than I didn't think...*).

⁷I am aware that the translation might sound bizarre to an English speaker's ear, but I put here the literal translation of the Tigrinya example.

For anyone who might think that this is the result of a phonological process, let me stop you right there. In Tigrinya there is another suffix homophonous with the negative morpheme *-(i)n*: the marker of coordination, already mentioned in Chapter 3, as in (18).⁸

- Example (18) shows that in coordination the $-(i)n$ suffix appears together with $z(i)-$. Therefore, the absence of the suffixal negative marker $-(i)n$ in $z(i)-$ clauses cannot be explained in phonological terms. Moreover, the negative marker $-(i)n$ and the $-(i)n$ used in coordination structures cannot be the same element, contra [Sjörs 2018](#). For the sake of completeness, let us negate (18).

- Because the coordination suffix $-(i)n$ appears in the example above, phonology cannot account for the dropping of the homophonous negative suffix $-(i)n$.

⁸According to [Sjörs 2018](#), the negative suffix and the coordination suffix are actually the same morpheme. This is also claimed in [Dawkins 1969](#) for Amharic.

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5.2.3 Negative $K(i)$ -clauses

I now focus on clauses containing the clausal prefix $k(i)$ -, examined in Chapter 4. Consider (20)–(24) and note where, whether, and how negation occurs.

- (20) $k\text{-}\epsilon j\text{-}\emptyset\text{-}b\epsilon ki$ $f\epsilon ttin\text{-}\epsilon$
 SBJ-NEG-SM.1S-cry.IPFV try.GER-SM.1S
 ‘I tried not to cry.’
- (21) $d\epsilon rho$ $k\text{-}\epsilon j\text{-}\emptyset\text{-}n\epsilon fir$ $?aziz\text{-}u$
 chicken SBJ-NEG-SM.3MS-fly.IPFV order.GER-SM.3MS
 ‘He ordered the chicken not to fly.’
- (22) $?it\text{-}a$ $s\epsilon b\epsilon jiti$ $k\text{-}\epsilon j\text{-}ti\text{-}x\epsilon jjid$ $d\epsilon li\text{-}na$
 DIST-FS woman SBJ-NEG-SM.3FS-go.IPFV want.GER-SM.1P
 ‘We wanted the woman not to leave.’
- (23) $n\text{-}it\text{-}i$ $?anb\epsilon sa$ $k\text{-}\epsilon j\text{-}\emptyset\text{-}b\epsilon r\epsilon b\epsilon r$ $b\epsilon s$ $t\epsilon zawir\text{-}\epsilon$
 DOM-DIST-MS lion SBJ-NEG-SM.3MS-wake_up.IPFV slowly walk.GER-SM.1S
 ‘I walked slowly to not wake up the lion.’
- (24) $T\epsilon w\epsilon ld\epsilon$ $?ihmilti$ $?aj\text{-}k\text{-}i\text{-}b\epsilon l?i\text{-}n$ $?ijj\text{-}u$
 Tewelde vegetables NEG-SBJ-SM.3MS-eat.IPFV-NEG BE₂.PRES-SM.3MS
 ‘Tewelde will not eat vegetables.’

Examples (20)–(24) show that in all types of $k(i)$ -clauses, except for the future tense construction, negation behaves as in $z(i)$ -clauses: only the prefix $?aj\text{-}$ appears and the suffix $-(i)n$ is absent. Moreover, the prefix $?aj\text{-}$ follows $k(i)$ -.¹⁰ In the future tense construction in (24), however, both $?aj\text{-}$ and $-(i)n$ appear and the negative prefix precedes $k(i)$ -. Although ungrammatical examples are not provided here, the configuration of sentential negation shown in the sentences above is the only possible one.

For the moment, let us set aside the negative future construction (whose analysis will be provided later in section 5.2.6) and return to the question posed in the previous section: is it correct to generalize that the negative suffix is absent in subordinate clauses? We still need one more data point to answer this question properly. Although not the focus of this dissertation, in Chapter 3 I showed that Tigrinya possesses a head-final complementizer

¹⁰Sjörs 2018 notes that $kaj\text{-}$ means *so that not* or *in order not to*, while Leslau 1941 instead proposes the meaning *without* for this form.

that also introduces subordinate clauses: *?ilu*. Example (25) illustrates how sentential negation behave in *?ilu* clauses.

- (25) dɛrho **?aj**-Ø-nifiri-*(n) ?il-ε gɛlits'-ε
 chicken NEG-SM.3MS-fly.IPFV-NEG COMP-SM.1S explain.GER-SM.1S
 'I explained that a chicken does not fly.'

As shown in the example above, *?ilu* clauses are negated with both *?aj*- and *-(i)n*, and the absence of *-(i)n* results in ungrammaticality.

Therefore, we cannot generalize that *-(i)n* is present in root clauses and absent in subordinate clauses. What we have seen so far is that *-(i)n* is absent in (some) *k(i)*-clauses and in *z(i)*-clauses. The question thus becomes: what is *-(i)n*, and why is it absent in *z(i)*-clauses and in (some) *k(i)*-clauses? This question can be addressed in two ways. On the one hand, the absence could result from competition between elements that share one (or more) feature and therefore occupy the same syntactic position. Although I have not yet proposed a position for *-(i)n*, I argued in Chapter 2 that *z(i)*- occupies the head of an aspectual projection and in Chapter 4 that *k(i)*- heads a low MoodP. Therefore, I cannot explain the omission of *-(i)n* by claiming that it occupies the same position as *z(i)*- or *k(i)*-, since these two elements occur in different positions.

On the other hand, the incompatibility between the negative suffix and the clausal prefixes could be explained in terms of movement: one element blocks the movement of the other, or one of them blocks the movement of (an)other element(s) in the syntax (an argument, a verb, ...), leading to the omission of *-(i)n* to prevent the derivation from crashing. I argued in Chapter 2 that *z(i)*- does not move and in Chapter 4 that *k(i)*- also does not move. Thus, the incompatibility could only arise from the blocking of the movement of either *-(i)n* or (an)other element(s). Which path to pursue? More data are needed to clarify this issue.

Before turning to the next section, I want to point out for the sake of completeness that the clausal prefix *k(i)*- and the negative prefix *?aj*- are also used together in Tigrinya to express expletive negation.

Expletive negation — or paratactic negation (Zeijlstra 2004) — is a construction in

cases in which Tigrinya displays expletive negation are shown below.¹¹

- French*

7aj-.

We now consider additional data.

producing similar clauses, they write *kej* detached from the verb, as in the following example.

- ‘I cannot sleep before eating.’

This may indicate that *kɛj* is in the process of grammaticalizing into a preposition or an adverb.

¹²Zeijlstra 2004 does not further explain the term “negative prepositions”.

5.2.4 Negative Imperative Clauses

In Tigrinya, imperatives are conjugated for all persons. When not in the second person, these forms have the meaning of the jussive, such as *Let me help!* or *Let's go!* in English (Mason 1996:52). Consider (28) and (29), which present affirmative and negative imperatives.

- (28) a. *ti-sitε*
SM.2MS-drink.IMP
'Drink!'
- b. *?aj-ti-sitε*
NEG-SM.2MS-drink.IMP
'Don't drink!'
- (29) a. *ti-re?ajε-ni*
SM.2MS-look.IMP-OM.1S
'Look at me!'
- b. *?aj-ti-re?ajε-ni*
NEG-SM.2MS-look.IMP-OM.1S
'Don't look at me!'

Examples (29) and (28) show the same phenomenon observed in the previous section with *z(i)-* and (some) *k(i)-* clauses: the suffixal negative morpheme *-(i)n* does not appear together with the prefixal negative morpheme *?aj-* in imperatives.¹³

Digressing briefly, note also that negative imperatives can be expressed with the addition of the *k(i)-* prefix, as in (30).¹⁴ My consultants do not distinguish in meaning a negative imperative with or without *k(i)-*. As in other *k(i)-* clauses (with the exception of negative future tense constructions), *k(i)-* must obligatorily precede the negative marker *?aj-*, and *-(i)n* is absent.

¹³This is also the case in Amharic (Girma 2003, Kramer 2023): the suffix *-(i)mm* is absent in imperatives. Cao & Liotta 2023 argue that a prohibitive suffix *a* appears instead of *-(i)n* in Tigrinya; however, I have not encountered this phenomenon in my fieldwork.

¹⁴First-person jussives ('let me' in English) are expressed in Amharic with *li-*, the counterpart of *k(i)-* (Leung & Girma 2017).

- (30) a. k-**ej**-ti-sitε
 SBJ-NEG-SM.2MS-drink.IMP
 ‘Don’t drink!’
- b. k-**ej**-ti-reʔajε-ni
 SBJ-NEG-SM.2MS-look.IMP-OM.1S
 ‘Don’t look at me!’

One could wonder why *k(i)-* appears in (negative) imperative clauses in Tigrinya. Recall that I claimed that this clausal prefix marks subjunctive mood. In other languages, such as Hungarian, subjunctive and imperative moods share (some of) the same morphology, to the point that it has been argued that certain imperative clauses are special cases of subjunctive - that is, matrix subjunctive clauses (Baunaz & Puskás 2022:157). Remember also that *k(i)-* is involved in the construction of the future tense and appears with future-oriented verbs. Imperatives are likewise future oriented, so it is not surprising to see *k(i)-* playing a role in their formation as well. Note also that in Standard Arabic the subjunctive mood is used when negating a future tense, as illustrated in (31) below.

- (31) a. sa-ʔa-ðhab-u ʔilā ʔal-madrasa yaðan
 FUT-SM.1S-go.IPFV to the-school tomorrow
 ‘I will go to school tomorrow.’
- b. **lan** ʔa-ðhab-a ʔilā ʔal-madrasa yaðan
 NEG SM.1S-go.SBJ to the-school tomorrow
 ‘I won’t go to school tomorrow.’

Example (30) shows that in Standard Arabic the particle *sa-* (or *sawfa*) is used to express future tense together with a verb in the Imperfective. Example (31) shows that the particle *lan* is used to negate the future; in this case the verb appears in the subjunctive.

Now, returning to the issue investigated in this section: why can’t *-(i)n* appear in imperative clauses? Can this observation shed light on why *-(i)n* also cannot appear in *z(i)-* and *k(i)-* clauses?

Let us take a moment to consider imperatives and negation. The literature offers numerous approaches explaining the incompatibility of negation and imperatives, a phenomenon found in many languages from different families. For example, in Romance

languages such as Spanish, Catalan, Romanian, and Italian (Zanuttini 1991), as well as in Modern Greek (Rivero & Terzi 1995), true negative imperatives are ungrammatical, and a suppletive verb form is required to express them (Zeijlstra 2004). Consider (32), in which an infinitive is inserted to express a negative command in Italian.

- (32)
- a. Guard-a-mi
look.IMP-SM.2S-OM.1S
'Look at me!'
 - b. Non guardar-mi
NEG look.INF-OM.1S
'Don't look at me!'
 - c. *Non guard-a-mi
NEG look.IMP-SM.2S-OM.1S
'Don't look at me!'
 - d. Non mi guardare
NEG OM.1S look.INF
'Don't look at me!'
 - e. *Non mi guard-i
NEG OM.1S look.SBJ-SM.2MS
'Don't look at me!'

Italian

In the literature, several analyses attempt to account for the phenomenon described above. One of the first approaches to address negation and imperatives is Kayne 1991, 1992 (republished in Kayne 2000). Examining Italian, Kayne notes that having an infinitive (a non-finite verbal form) in a root clause (such as the one in (32b)) is striking. He therefore proposes that the negative marker *non* licenses an empty modal in imperative clauses, which co-occurs with the infinitive (Kayne 2000:99). Zanuttini 1991 follows Kayne and suggests that, because imperatives are tenseless, they lack a TP projection, which makes the presence of certain negative markers impossible (Zanuttini 1991:73). Looking at Spanish, Laka 1994 claims that negation and imperatives are incompatible because they belong to the same syntactic category, Σ . Belonging to the same category, negative markers and imperatives cannot co-occur in a sentence. Also examining Spanish but comparing it to Modern Greek, Rivero & Terzi 1995 argue that negative exponents block the raising of the verb to C, where it would need to check a Mood feature to express

an imperative. [Zanutini 1997](#) proposes that in imperatives the C° head carries an [Imp] feature that must be checked; since imperative verbs are morphologically defective, the feature cannot be checked. Another approach attributes the incompatibility between negation and imperatives to competition: the verb checks its imperative mood feature in an operator node that is also projected by negative exponents ([Tomić 1999:194](#)). [Han 2001](#) suggests that negative imperatives are unavailable in languages where negation takes syntactic scope over the imperative operator, leading to ungrammaticality. Finally, within a Distributed Morphology framework, [Kramer 2023](#) examines imperatives in Amharic and argues that these involve morphological operations such as haplology that cannot be explained solely by syntax.

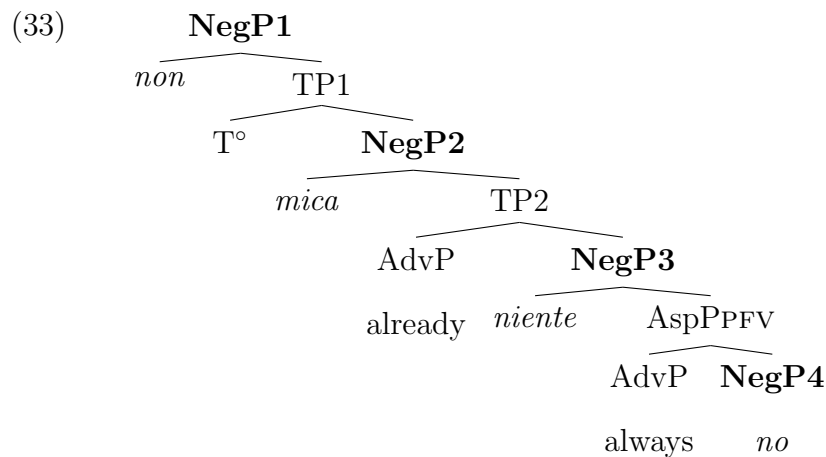
Let us now return to the question raised in the previous section: why can't the $-(i)n$ suffix of bipartite negation in Tigrinya appear in $z(i)-$ and $k(i)-$ clauses and in imperatives? What is the *fil rouge* connecting these three types of clauses? In my view, there is not a single factor that accounts for the disappearance of $-(i)n$ in all three contexts. Rather, I propose that $-(i)n$ is absent in $k(i)-$ clauses for one reason and in $z(i)-$ and imperative clauses for another. I will build toward this analysis in the next section.

5.2.5 Toward an Analysis

At the beginning of this chapter I showed that sentential negation in Tigrinya declarative clauses expressing habitual present tense is formed by affixing two negative morphemes to the verb: the prefix $?aj-$ and the suffix $-(i)n$. Because these two elements express a single semantic negation, the question was which of the two is the “real” negator. In the previous sections I showed that $z(i)-$, $k(i)-$ (except in future constructions), and imperative clauses are negated only by $?aj-$, while $-(i)n$ is absent. This indicates that $?aj-$ is the “real” negator of the two. To propose a syntactic derivation that explains this pattern, we must determine where the two negative exponents are placed in the structure.

I presented two different approaches in the literature addressing bipartite negation in Arabic, composed of the negative exponents *ma-* and *-š*. [Benmamoun et al. 2013](#) claim that these elements are base-generated under the same head, Neg° , whereas [Benmamoun](#)

2000 and Shlonsky 1997 place them in the head and the specifier of NegP. There is, however, yet another approach that can be taken to analyze bipartite negation: the SplitNegP approach. Zanuttini 1997 examined several Northern Italian dialects and argued that the different negative markers used (*non*, *mica*, *niente*, *no*) vary in their distributional properties. By placing these markers within the hierarchy of adverbs proposed by Cinque 1999, she claimed that there must be at least four distinct positions for negative markers in Italian. This is shown in (33).



Zanuttini’s idea was further developed by Poletto 2017 (also for Italian dialects), who argued for a cartographic approach to NegP that she calls BigNegP.¹⁵ According to Poletto, all negative markers of a discontinuous negation “start out as a unit, and that unit is first merged inside the vP” (Poletto 2017:82). She further develops Zanuttini’s proposal by associating the four NegPs with other projections already known to be part of the syntactic map: *no* heads FocP, *mica* heads MinimizerP, *non* heads ScalarP, and *(ni)ente* heads ExistentialP (Poletto 2017:99).

Other authors have adapted Zanuttini’s idea that in bipartite negation the preverbal and postverbal negative exponents are generated in two different projections (either as heads or as phrases). For example, Bell 2004 suggests that in Afrikaans the preverbal *nooit* and the postverbal *nie* head NegP₁ and NegP₂ respectively, with the former immediately dominating the latter. For Modern Arabic dialects, already mentioned in the previous

¹⁵Note that the cartographic enterprise has also suggested the possibility of a NegP projection “on top of every adverb-related functional projection, even simultaneously, up to a certain height” (Cinque 1999:126).

sections, [Soltan 2011](#) proposes that the preverbal *maa* is found in the head of a polarity phrase (PolP) and the postverbal *š* in the head of NegP; whereas [Alqassas 2019](#), discussing Jordanian Arabic specifically, proposes that *maa* heads NegP and *š* (Agr)PolP, similar to [Julien 2002](#) for Moroccan Arabic. In her nanosyntactic account, [De Clercq 2013](#) analyzes French *ne* and *pas* as both part of the specifier of a NegP projection, claiming that they move higher in the structure, landing in PolP and a low FocP respectively. In a more recent publication, De Clercq develops her account and proposes that in French *ne* consists of Mood and Tense features, whereas *pas* consists of Focus, Class(ifier), Q(uantifier), and Neg features ([De Clercq 2020:194](#)).

What about Tigrinya? To my knowledge — except for the already mentioned [Cao & Liotta 2023](#) and [Abraham 2024](#) that focus more on negative polarity items — there are no works that propose a syntactic account of negation in Tigrinya to date. Nevertheless, [Girma 2003](#) investigates the syntax of sentential negation in the Ethio-Semitic branch and adopts a SplitNegP approach. In [Girma 2003](#), seven Ethio-Semitic languages are considered.¹⁶ In line with the direction taken so far in this paper, the author notes that the postverbal suffix in three Ethio-Semitic languages (Amharic, Endegeñ, and Ennemor) is absent in the presence of complementizers and imperatives ([Girma 2003:222–224](#)). Recall, however, that this generalization is not entirely accurate: in Tigrinya, for instance, the suffixal negative exponent $-(i)n$ is absent in the presence of $z(i)-$ and $kemz(i)-$, but neither in all types of clauses involving $k(i)-$ nor $?ilu$ -clauses. In [Girma 2003](#), it is argued that complementizers, imperatives, and postverbal negative markers share a common feature ([Force]) that leads to an incompatibility between the first two and the last. On the basis of Amharic, the author suggests that preverbal negative markers in Ethio-Semitic head a NegP projection because they are able to negate a clause on their own ([Girma 2003:230](#)). Postverbal negative markers, by contrast, head a PolP projection (and carry a [Pol] feature) “given that negation is sensitive to polarity,” but then raise under ForceP because they are specified for a [Force] feature ([Girma 2003:229–230](#)).

In this dissertation I adapt Zanuttini’s approach and claim that the negative mor-

¹⁶The languages are Amharic, Argobba, Gafat, Harari, Tigrinya, Ennemor, and Endegeñ.

phemes of bipartite negation in Tigrinya are found in two different projections. I propose that - because it can negate clauses alone - *?aj-* is the element that carries a [iNeg] feature and therefore heads the negative functional projection NegP (following [Girma 2003](#)), which is lower than TP (see also [Cao & Liotta 2023](#)). I further claim that NegP is merged in the aspectual layer of the structure, dominating *vP* (contra [Poletto 2017](#), who argues that all negative markers are first merged *inside vP*). The idea of a very low placement for NegP is also supported by the fact that, as mentioned in the introductory section of this chapter, in Amharic the form of the first element in the circumfix depends on the aspect of the verb: even though this difference does not exist in Tigrinya, a reflex of it may still be present. The proposed position for *?aj-* also explains why this prefix appears between the verb and the clausal prefix *z(i)-*: the verb first moves to Neg°, where it is prefixed by *?aj-*, and then moves under Asp°.

Now, what about *-(i)n*? In the literature, this suffix has been claimed to derive etymologically from an emphatic particle that “marked the negative predicate as ranking on the extreme end of a pragmatic scale” ([Sjörs 2018:332](#)). Discussing Amharic, [Sjörs 2018](#) argues that the postverbal negative marker in this language, *-(i)mm* - cognate with Tigrinya *-(i)n* - is a scalar additive focus particle ([Sjörs 2018:341](#)) that has also been described as an emphatic and topicalizing particle ([Hetzron 1972](#)). This is also argued in [Girma & Meyer 2008](#), contra [Workneh 2022](#) who claims that the negation particle and the focus particle are distinct elements, suggesting that *-(i)mm* in Amharic is a marker of finiteness. [Sjörs](#) claims that negation with *-(i)mm* in Amharic should originally be understood as a form of emphatic negation, such as *not even* in English, but that through grammaticalization it lost its emphatic force ([Sjörs 2018:343](#)). This would mean that verbs suffixed with *-(i)mm* were more informative than predicates without it. [Sjörs](#) claims that the same holds for Tigrinya, whose speakers would have identified their suffix with the Amharic suffix and consequently extended its use in a similar way ([Sjörs 2018:348](#)), although this final point seems to me somewhat speculative. [Sjörs 2018](#) also claims that the *-(i)n* suffix in negation and the *-(i)n* suffix in coordination in Tigrinya, briefly mentioned in section 5.2.2, are not only homophonous but actually the same element deriving from a focus particle ([Sjörs](#)

2018:336). We have seen that this cannot be the case given their different behavior in the presence of $z(i)$ - (the former is absent, whereas the latter is not). This is also claimed for the Amharic $-(i)mm$ (Leslau 1995).

The fact that $-(i)n$ is an emphatic particle that once reinforced $?aj$ - and is now absent in some cases in Tigrinya may evoke Jespersen’s cycle (Dahl 1979). This cycle, named after Danish linguist Otto Jespersen, is the diachronic tendency in which “pre-verbal negation is phonologically weakened, gradually reinforced and eventually replaced by a postverbal negative marker that was originally a polarity item or a minimizer” (Cruschina et al. 2017:10). This is, for example, the case in French, where the minimizer *pas* reinforced the pre-verbal marker *ne* which is now absent in spoken language. I think that $-(i)n$ could be analyzed as a minimizer (Gonçalves Pinto 2021), similar to *pas* in French (‘step’), *mica* in Italian (‘crumb’), and $-f$ in Moroccan Arabic (from *faj* ‘thing’), namely originating from *neger* ‘thing’. However, in Tigrinya, it is not $?aj$ - that is absent but $-(i)n$.

With all this in mind, I suggest that $-(i)n$ heads a Polarity projection PolP. Finite clauses are said to have a polarity value, either affirmative or negative, the former usually being unmarked (Breithbarth et al. 2013:1). This functional projection is also found in the literature under the label ΣP (Laka 1990) and can be seen as parallel to Ouhalla 1990’s NegP dominating TP. This projection carries an affirmative or a negative feature that must be checked by a lexical item at LF (Zanuttini 1994:430). Because in certain contexts the polarity value of a sentence “may be associated with a particular emphasis” (Breithbarth et al. 2013:2) and because, as seen above, $-(i)n$ is said to be etymologically an emphatic particle, I claim that it is base-generated in the Pol° head. PolP also carry a negative operator in its specifier. Given that finite clauses are said to have a polarity value, I suggest that PolP dominates TP, can only select for TP, and thus can only be present when TP is present.

Moreover, I claim that PolP can host $-(i)n$ only when NegP is present in the structure; that is, PolP cannot occur without NegP in the syntax. I propose this because, unlike NegP, which carries an interpretable [iNEG] feature, PolP carries an uninterpretable [uNEG] feature. This pattern can be understood in terms of Negative Concord (Zei-

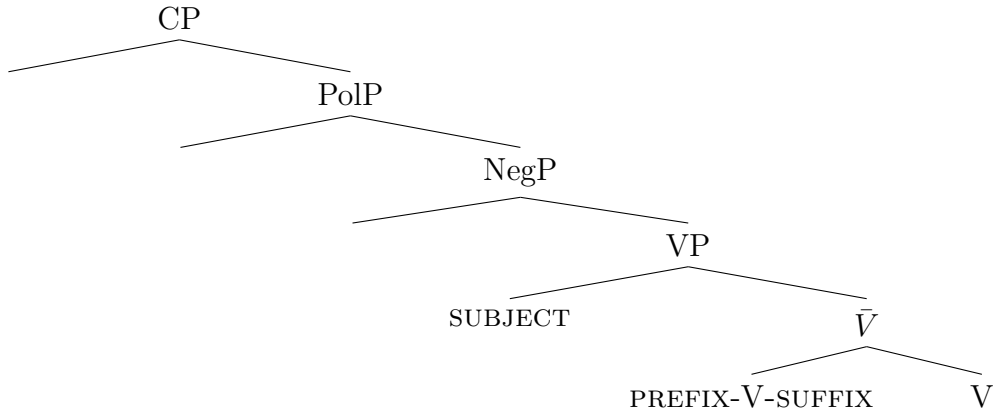
jlstra 2004, Biberauer & Zeijlstra 2012, Zeijlstra 2016, 2022). Negative Concord is a phenomenon whereby two or more negative elements appear in a sentence but yield a single semantic negation. For the syntactic derivation to converge, the two elements must enter into an AGREE relation (see Section 5.3 on negation in progressive clauses). This is similar to Julien 2002’s analysis of bipartite negation in Arabic, where the negative morpheme functions as an operator that must bind a polarity variable (Julien 2002:299).

In the literature on Negative Concord, it is commonly proposed that the higher element bears [iNEG] and the lower element [uNEG]. By contrast, the present proposal reverses this: PolP, which is structurally higher, carries [uNEG], and NegP, which is lower, carries [iNEG]. Under mainstream Agree-based analyses, [uNEG] cannot be valued, because Agree probes downwards. How, then, can we account for the positions of NegP and PolP? Two options are available. One possibility is Upward Agree (Zeijlstra 2012, Bjorkman & Zeijlstra 2019, Bárány & van der Wal 2022), whereby in some languages [uNEG] on a higher element can be valued by [iNEG] on a lower element. Alternatively, [uNEG] can be valued through covert movement of Neg° (or NegP) into a c-commanding position above PolP, or through covert feature percolation, whereby the [iNEG] feature is copied or moved to a head higher than PolP (Zeijlstra 2017). I leave the choice between these explanations to future research, and here tentatively propose that *?aj-* heads NegP (which carries [iNEG]) and *-(i)n* heads PolP (which carries [uNEG]).¹⁷

This is exactly (except for the features) what is also proposed in Girma 2003: he claims that in Amharic (and in Ethio-Semitic more generally) the prefixal negative exponent heads NegP and the suffixal one heads PolP. However, this work does not provide sufficient detail in its analysis. In fact, the author adopts an antisymmetric and LCA approach to head-finality (Kayne 1994) and places PolP higher than NegP, which makes the suffix appear before the prefix. In his lexicalist derivation, the verb (already carrying both the prefix and the suffix) is placed under V°. In (34) I present the tree he provides.

¹⁷That both morphemes are heads is supported by the fact that, according to Breithbarth 2020, free morphemes occupy specifier positions, whereas clitics or affixes occupy head positions.

(34)



(adapted from [Girma 2003:231,\(42\)](#))

I think that Girma’s analysis is not sufficiently explicative to account for negation in Ethio-Semitic. More specifically, it lacks an explanation of how the word order of the elements is derived, especially how the negative exponents appear before and after the verb if they are generated higher and in inverse order. This would presumably involve a considerable amount of movement that is not accounted for by the analysis. I hope that the analysis I propose in the next section will provide a clearer and more detailed solution for the syntax of negation in Tigrinya.

5.2.6 The Analysis of Sentential Negation in Simple Verb Forms

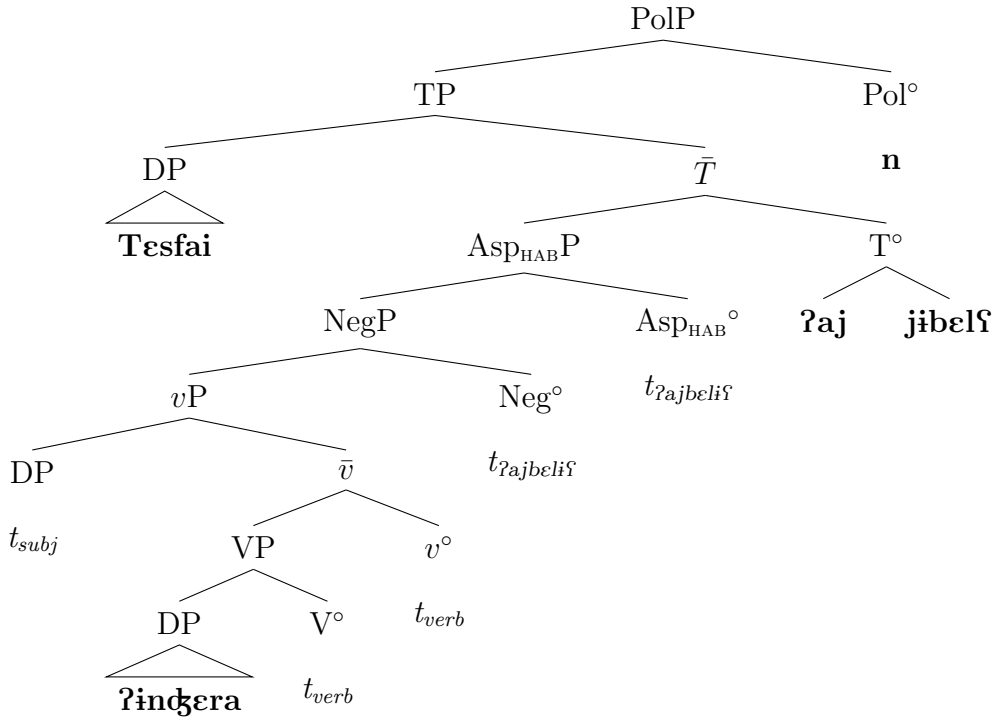
I first provide a syntactic analysis of negative declarative root clauses expressing present tense, and then turn to negative $z(i)$ -, imperative, and $k(i)$ -clauses in this order.

Analysis of Negative Declarative Root Clauses in Present Tense

I suggest that negative root declarative clauses in the habitual present tense in Tigrinya, such as the one in (35), can be represented as in (36) below.

- (35) Tesfaj ʔinɕɛra ʔaj-Ø-bɛliʃ-in
 Tesfaj injera NEG-SM.3MS-eat.IPFV-NEG
 ‘Tesfaj doesn’t eat injera.’

(36)



I propose that *ʔaj-* heads NegP and *-(i)n* heads PolP. The verb raises from its base-generated position and lands in Neg°, creating a complex word to which the negative prefix is attached. The verb and the prefix then raise by head movement to T°. *-(i)n* is suffixed to the verb because it is adjacent to it in the syntactic tree.

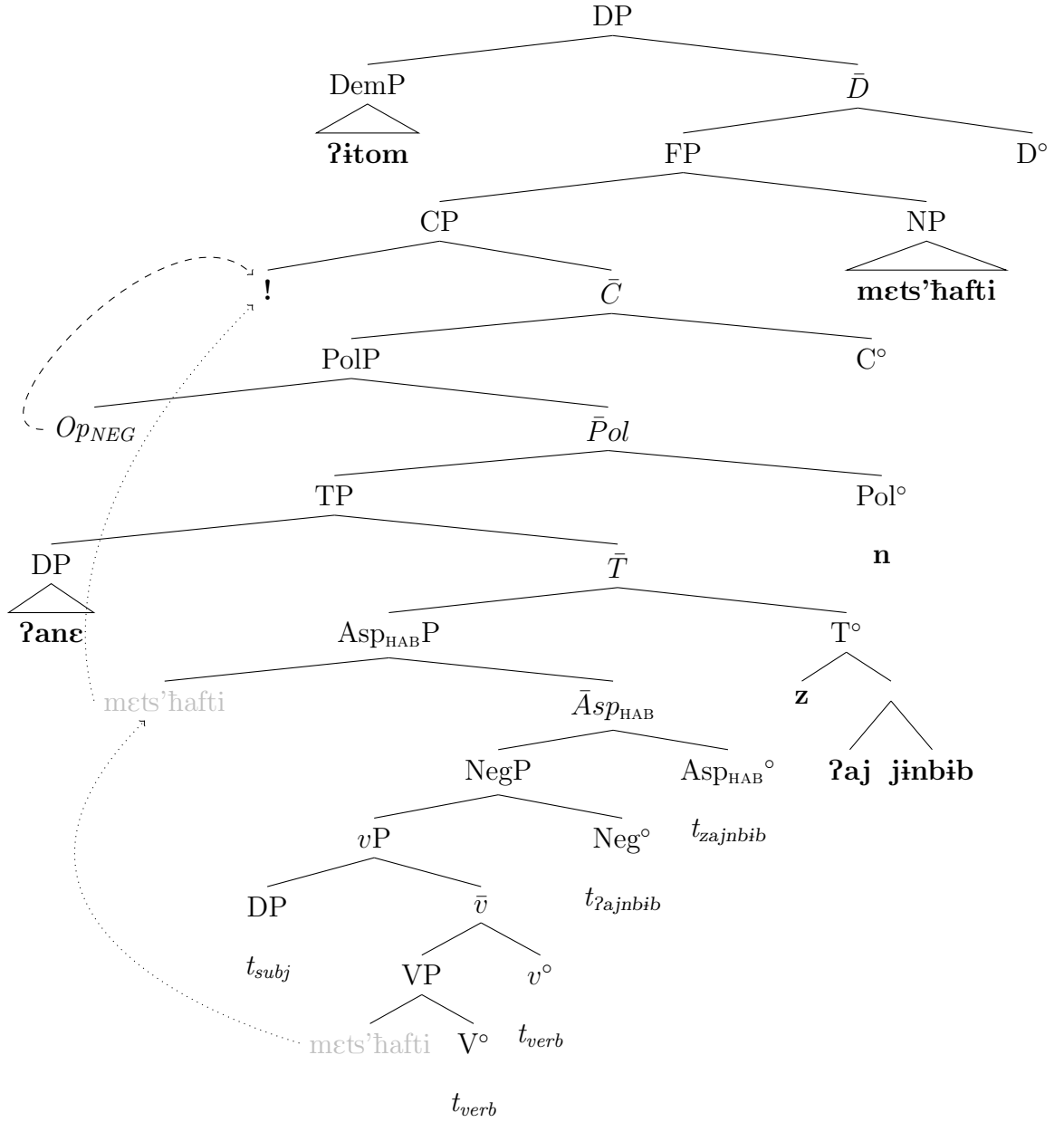
Let us now move to the more complex scenarios of subordinate clauses.

Analysis of Negative *Z(i)*-clauses

I showed that *z(i)*-clauses are negated only by the negative prefix *ʔaj-* and that *-(i)n* is absent. Let us now examine what happens in the derivation of a negative relative clause, such as the one in (37), in the tree in (38).

- (37) ʔit-om ʔanɛ z-ɛj-Ø-nbɪb-om mets'hafti...
 DIST-MP I REL-NEG-SM.1S-read.IPFV-OM.3MP book.MP
 'The books that I do not read...'

(38)



The tree above illustrates the presence of a Relativized Minimality effect (Rizzi 1990) (indicated by the exclamation mark) caused by the intervention of the polarity operator. Relativized Minimality can be stated as in (39).

(39) *Relativized Minimality*:

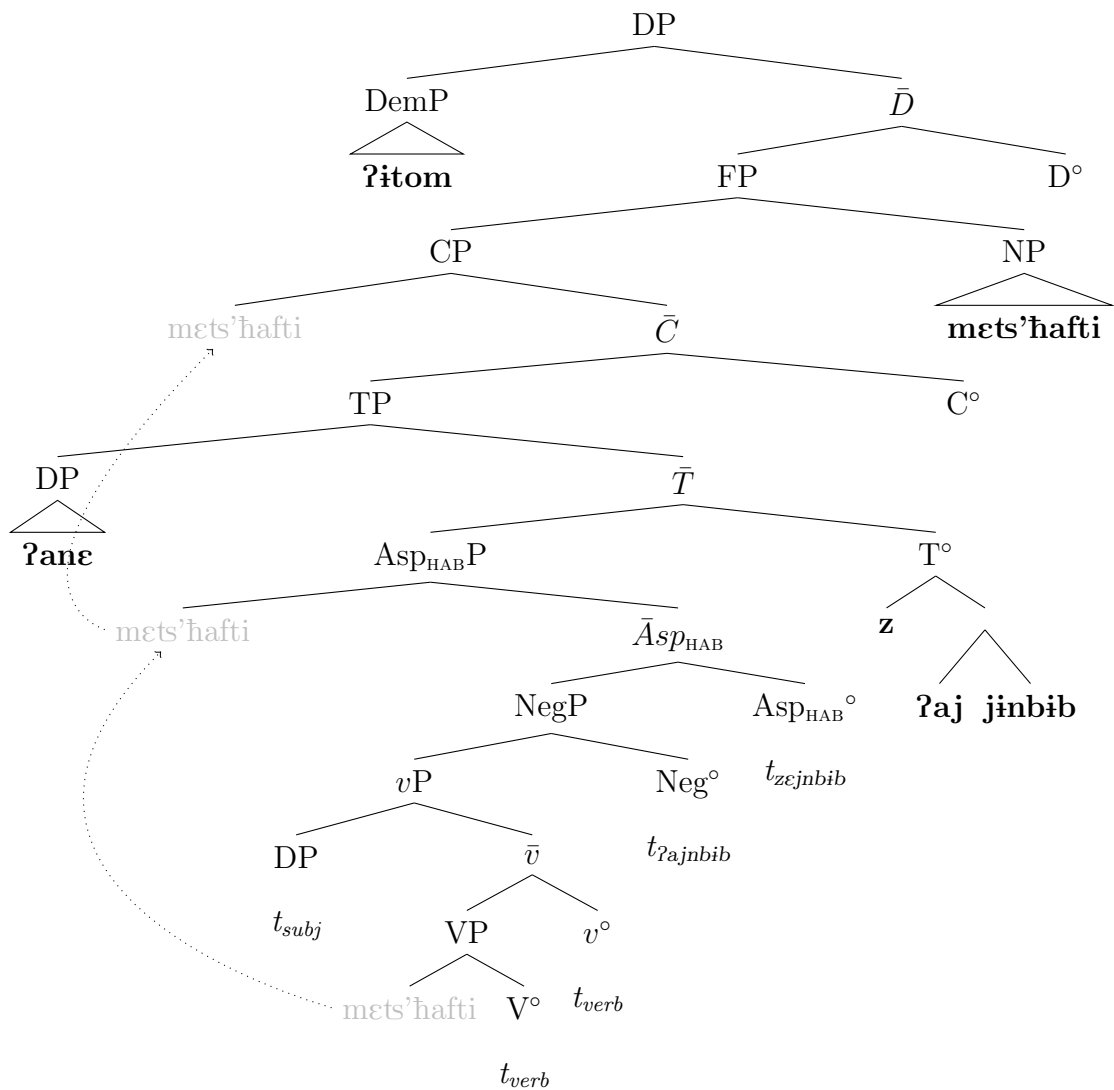
X α -governs Y only if there is no such Z such that:

- (i) Z is a typical potential α -governor for Y;
- (ii) Z c-command Y and does not c-command X.

(from Rizzi 1990:7)

In (38) the negative operator in the specifier of PolP competes with the relativized head *mets'hafti* “books” for the Spec,CP position. Because the negative operator is more local than the relativized head, it lands in Spec,CP. This, however, prevents the relativized head from checking the uninterpretable [Rel] feature on C° (the feature that triggers the raising of the relativized head in the first place, as mentioned in Chapter 2). If the relativized head does not land in Spec,CP, the uninterpretable [Rel] feature cannot be checked and the derivation cannot converge and be sent to Spell-Out. The only way to save the syntactic derivation is not to merge PolP. Consequently, the negative suffix *-(i)n* does not appear. This is shown in (40).

(40)



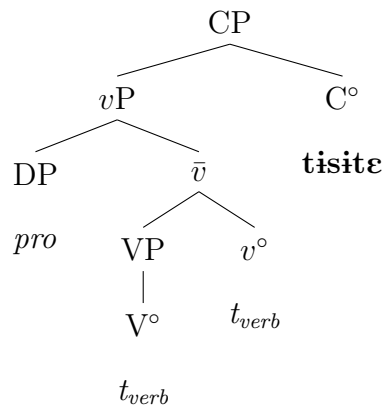
In the tree above PolP is not merged. There are thus no unchecked or uninterpretable features, since NegP in the structure carries [iNeg]. The relativized head can raise to Spec,CP without difficulty, and the sentence is well formed. I claim that this is also similar to what happens in imperative clauses and explains why $-(i)n$ does not appear in the next section.

Analysis of Negative Imperative Clauses

First, let me present the structure of an affirmative imperative clause, such as the one in (41). I follow [Zanuttini 1991](#) and [Kayne 1991](#) in suggesting that, because imperatives are tenseless, they do not project a TP in their structure. I also follow [Zanuttini 1997](#) and propose that in imperatives the verb must raise to C° to check an [IMP] feature. Note that, similarly to what I just mentioned, for Amharic [Kramer 2023](#) assumes that imperatives contain a functional head in the C° domain that carries the imperative meaning ([Kramer 2023:198](#)). For simplicity, I take C° to be this functional head and leave a more in-depth analysis of the syntax and morphology of imperative clauses in Tigrinya to future research.

- (41) ti-site
 SM.2MS-drink.IMP
 ‘Drink!’

(42)

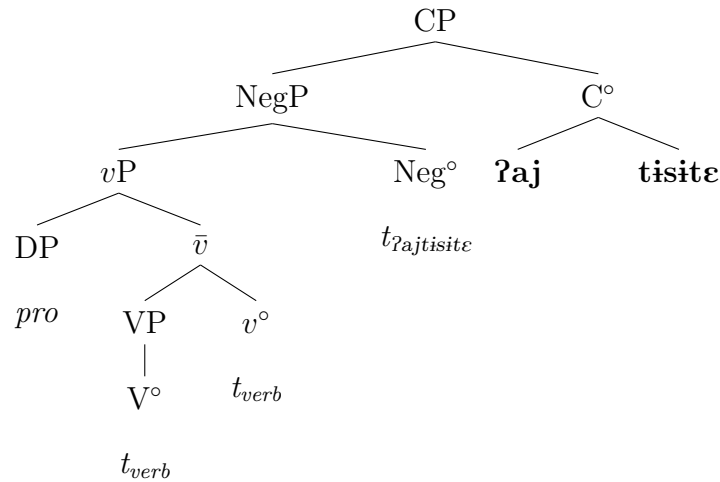


Let us now negate the sentence below, as in (43). Because the polarity phrase PolP selects for a TP and imperative clauses lack this projection, the negative suffix $-(i)n$ does

not appear in the structure and therefore imperatives are negated only by *ʔaj-*. This is shown in (44).

- (43) **ʔaj-ti-siɛ**
 NEG-SM.2MS-drink.IMP
 ‘Don’t drink!’

(44)



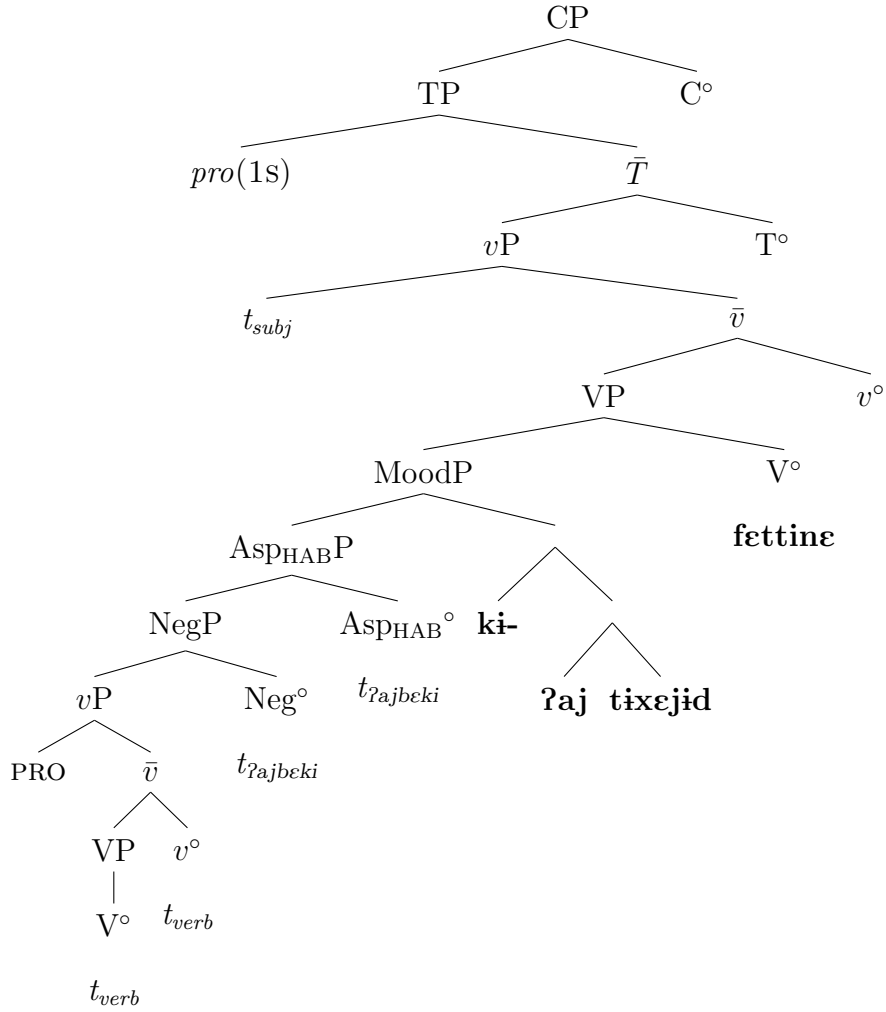
Note that in the tree in (44) there are no unchecked or uninterpretable features, since NegP carries [iNeg].

Analysis of Negative *K(i)*-clauses

I have shown that all *k(i)*-clauses are negated only with the negative prefix *ʔaj-*, except for future constructions, which are negated with both *ʔaj-* and the negative suffix *-(i)n*. I suggest that *k(i)*-clauses are negated only by the negative prefix because they are reduced clauses: lacking a TP projection in their syntax, they consequently cannot project a PolP, and therefore *-(i)n* does not appear. This is similar to what I proposed for imperative clauses in the previous section. Thus, a sentence such as the one in (45) can be derived as illustrated in the tree in (46).

- (45) **k-ɛj-Ø-beki** **fəttin-ɛ**
 SBJ-NEG-SM.1S-cry.IPFV try.GER-SM.1S
 ‘I tried not to cry.’

(46)



As in imperative clauses, *k(i)*-clauses do not involve a PolP because they are reduced clauses and do not project a tense phrase (TP). Because of this, they are negated only by the negative prefix *?aj-*.

I propose an analysis of negative future tense constructions at the end of this chapter, after addressing negation in progressive clauses.

Negation in Past Tense

So far, I have examined negation in clauses expressing habitual present tense. I now turn to negative clauses in the past. These clauses present an interesting puzzle that involves not only negation but prefixation more generally in Tigrinya, specifically with the Gerund verb form. The issue of the Gerund in Tigrinya being unable to carry prefixes has already been raised several times throughout this dissertation. Let us now see what happens with negation.

I have shown that to express past tense in Tigrinya, a verb must be conjugated according to the Gerund paradigm, as in (47).

- (47) mis dim-ay tɛts'iwit-ε
 with cat-POSS.1S play.GER-S.1S
 'I played with my cat.'

In (47) above, the sentence expresses past tense (we could even add 'yesterday' to it) and displays the verb *tɛts'iwitε* '(I) played', which is in the Gerund form. Let us now negate this sentence, as in (48).

- (48) a. *mis dim-ay ?aj-tɛts'iwit-ε-n
 with cat-POSS.1S NEG-play.GER-S.1S-NEG
 b. mis dim-ay ?aj-tɛts'iwɛti-ku-n
 with cat-POSS.1S NEG-play.PFV-S.1S-NEG
 'I did not play with my cat.'

Interestingly, if one negates this sentence, as in (48a), one cannot use the same verb form: the ungrammaticality of this example shows that bipartite negation cannot appear on a Gerund verb form. To negate (47), a suppletive Perfective verb form must be introduced (as also seen for relative clauses in Chapter 2), as in (48b), to obtain a grammatical sentence with the correct meaning. One might ask why, if the Gerund cannot carry prefixes, it cannot at least carry the negative suffix *-(i)n*. However, as stated previously, I claim that the presence of PolP - and especially the presence of the negative suffix *-(i)n* - is dependent on the presence of NegP.

As already argued for relative clauses, I suggest that the Gerund cannot carry the negative prefix *?aj-* because it does not raise to the aspectual layer. Therefore, to express a negative sentence in the past, a suppletive Perfective form must be inserted in the syntax to rescue the derivation. The syntactic derivation of the sentence in (48b) is thus identical to the one presented in the tree in (36); the only difference is the type of aspectual projection involved (Asp_{PFV}P instead of Asp_{HAB}P).

After examining sentential negation in clauses expressing habitual present tense and

past tense, I now turn to the final section, which considers negation in clauses that express progressive aspect.

5.3 Negation in Progressive Aspect

In this second and last part of the chapter I present the behavior of sentential negation in clauses with periphrastic tense forms. I first provide data on these sentences and then propose an analysis to account for them. I also discuss and propose an analysis for negative future tense constructions.

5.3.1 An apparent optionality

As shown throughout the chapters, progressive aspect in Tigrinya is formed by means of a periphrastic tense construction with a verb in the Imperfective and an auxiliary (a present auxiliary in the present progressive and a past auxiliary in the past progressive), as illustrated in (49).

- (49) mis dim-ay ji-ts'awet ?all-εxu
 with cat-POSS.1S SM.1S-play.IPFV BE₁.PRES-SM.1S
 'I am playing with my cat.'

Let us now negate the sentence above.

- (50) a. mis dim-ay ji-ts'awet **j-ell-εxu-n**
 with cat-POSS.1S SM.1S-play.IPFV NEG-BE₁.PRES-SM.1S-NEG
 'I am not playing with my cat.'
- b. mis dim-ay **?aj-Ø-ts'awet-n** ?all-εxu
 with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG BE₁.PRES-SM.1S
 'I am not playing with my cat.'
- c. *mis dim-ay **?aj-Ø-ts'awet-n** **j-ell-εxu-n**
 with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG NEG-BE₁.PRES-SM.1S-NEG
 'I am not playing with my cat.'

The examples above illustrate that to negate the sentence in (49) there are two options: on the one hand, the bipartite negation can appear on the auxiliary, as in (50a); on the

other hand, it can appear on the verb, as in (50b). However, bipartite negation cannot appear on both the verb and the auxiliary, as (50c) shows. For the sake of completeness, note that the same scenario is found in sentences in the past progressive.

- (51) a. mis dim-ay ji-ts'awet neir-ε
with cat-POSS.1S SM.1S-play.IPFV BE₁.GER-SM.1S
'I was playing with my cat.'
- b. mis dim-ay ji-ts'awet **?aj-neber-ku-n**
with cat-POSS.1S SM.1S-play.IPFV NEG-BE₁.PFV-SM.1S-NEG
'I was not playing with my cat.'
- c. mis dim-ay **?aj-Ø-ts'awet-n** nejr-ε
with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG BE₁.GER-SM.1S
'I was not playing with my cat.'
- d. *mis dim-ay **?aj-Ø-ts'awet-n** **?aj-neber-ku-n**
with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG NEG-BE₁.PFV-SM.1S-NEG
Intended: 'I was not playing with my cat.'

Note that the past auxiliary *nejre* '(I) was' is conjugated according to the Gerund paradigm. As seen previously, a Gerund verb form cannot be negated. Hence, the suppletive Perfective form *neberku* '(I) was' is introduced in the structure.

Again, for the sake of completeness, note that the optionality observed in the examples above is also present in negative relative clauses expressing progressive aspect, as in (52).

- (52) a. ?it-i ?anε z-i-nbib-o z-ell-εxu
DIST-MS I REL-SM.1S-read.IPFV-OM.3MS REL-BE₁.PRES-SM.1S
mets'haf ?ab-ti ?armadyo ?all-o
book.MS PREP-DEM.MS cabinet BE₁.PRES.SM.3MS
'The book that I am reading is in the cabinet.'
- b. ?it-i ?anε z-i-nbib-o z-εj-ll-εxu
DIST-MS I REL-SM.1S-read.IPFV-OM.3MS REL-NEG-BE₁.PRES-SM.1S
mets'haf ?ab-ti ?armadyo ?all-o
book.MS PREP-DEM.MS cabinet BE₁.PRES.SM.3MS
'The book that I am not reading is in the cabinet.'
- c. ?it-i ?anε z-εj-Ø-nbib-o z-ell-εxu
DIST-MS I REL-NEG-SM.1S-read.IPFV-OM.3MS REL-BE₁.PRES-SM.1S
mets'haf ?ab-ti ?armadyo ?all-o
book.MS PREP-DEM.MS cabinet BE₁.PRES.SM.3MS
'The book that I am not reading is in the cabinet.'

- d. *ʔit-i ʔanɛ z-ɛj-Ø-nbib-o
DIST-MS I REL-NEG-SM.1S-read.IPFV-OM.3MS
z-ɛj-ll-ɛxu mets'ħaf ʔab-ti ʔarmadyo
REL-NEG-BE₁.PRES-SM.1S book.MS PREP-DEM.MS cabinet
ʔall-o
BE₁.PRES-SM.3MS

Note also, in case one might think that the difference between the position of the negative circumfix on either the verb or the auxiliary in the examples seen so far reflects a difference between sentential negation and constituent negation, that in the latter the negative circumfix can likewise occur on either the verb or the auxiliary. This is illustrated in (53) for sentential negation and in (54) for constituent negation (the constituent negated in (54) is the adverbial phrase ‘long ago’).¹⁸

- (53) a. zinab ʔaj-Ø-zenibi-n ʔall-o
rain NEG-SM.3MS.rain.IPFV-NEG BE₁.PRES-SM.3MS
‘It is not raining.’
b. zinab ji-zenib j-ell-ɛ-n
rain SM.3MS.rain.IPFV NEG-BE₁.PRES-SM.3MS-NEG
‘It is not raining.’
- (54) a. zinab qidmi nɛwiħ ʔiwen ʔaj-Ø-zenibi-n nejr-u
rain before long ago NEG-SM.3MS.rain.IPFV-NEG BE₁.GER-SM.3MS
‘It was raining not long ago.’
b. zinab ipaqldmi nɛwiħ ʔiwen ji-zenib ʔaj-neber-ɛ-n
rain before long ago SM.3MS.rain.IPFV NEG-BE₁.PFV-SM.3MS-NEG
‘It was raining not long ago.’

The data above show that the occurrence of negation on either the verb or the auxiliary in periphrastic tense forms does not depend on whether it involves sentential or constituent negation.

Moreover, in the presence of the element *wala* ‘any’ (which I showed in Chapter 2 is probably not an NPI) it is also possible to place the bipartite negation on either the verb or the auxiliary in periphrastic tense forms, as shown in (55).

¹⁸Note that ‘not long ago’ is usually translated with the insertion of a negative copula: *qidmi nɛwiħʔiwen ʔajkonɛn* ‘long ago is not’.

- (55) a. wala hədε sɛb mis dim-ay ji-ts'awɛt
 any one.MS man with cat-POSS.1S SM.1S-play.IPFV
j-ell-ɛxu-n
 NEG-BE₁.PRES-SM.1S-NEG
 'Nobody is playing with my cat.'
- b. wala hədε sɛb mis dim-ay **?aj-Ø-ts'awɛt-n**
 any one.MS man with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG
 ?all-ɛxu
 BE₁.PRES-SM.1S
 'Nobody is playing with my cat.'

From the examples above, the position of negation in periphrastic tense forms expressing progressive aspect does not affect the meaning of the sentence. According to my (non-linguist) consultants, (50a) and (50b) mean exactly the same thing.¹⁹ Although they seem to show a slight preference for (50a), they cannot explain why. According to four other Eritrean Tigrinya native speakers (who are linguists), (50a) is correct and the common, more standard way to form a negative progressive clause, whereas (50b) is a more marginally accepted form.²⁰ Moreover, according to these four Eritrean linguists, (50b) sounds like a “not properly and fully constructed cleft clause”, such as the one in (56).²¹

- (56) mis dim-ay **?aj-Ø-ts'awɛt-n** ?ijj-ɛ
 with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG BE₂.PRES-SM.1S
 z-ell-ɛxu
 REL-AUX.PRES-SM.1S
 'I am not playing with my cat.'

The sentence in (56) is similar to an English cleft clause, since it contains both a copula and a relativized verbal form. It could therefore be translated more precisely as something like *It is not playing with my cat that I am*. However, note that the copula agrees with the grammatical subject of the sentence ‘I’ and does not display default agreement. Moreover, for Jason Overfelt’s Ethiopian consultant (p.c.), both (50a) and (50b) are ungrammatical, and the only option to convey the meaning of the negative sentences above is (56) but

¹⁹I personally do not believe in optionality in language. I think there must be a very slight difference between the sentences that native speakers are not fully able to perceive. With a more in-depth analysis involving prosody and information structure, such a difference could probably be found.

²⁰I thank Nazareth Amlesom Kifle, who kindly provided her native speaker opinion and gathered acceptability judgments from three other Eritrean linguists.

²¹Again, thanks to Nazareth Amlesom Kifle for pointing this out to me.

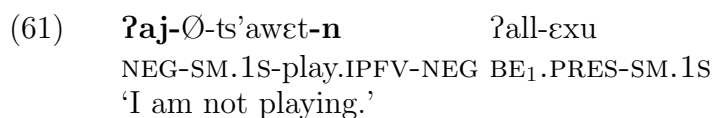
(57) mis dim-ay **?aj-Ø-ts'awet-n** z-ell-exu
 with cat-POSS.1S NEG-SM.1S-play.IPFV-NEG REL-AUX.PRES-SM.1S
 'I am not playing with my cat.'
 (Jason Overfelt, p.c.)

Before turning to the analysis, I would like to point out and remind the reader that progressive aspect is not the only meaning conveyed by periphrastic tense forms. As mentioned in the first two chapters, if a Gerund verb occurs with the present tense auxiliary *?allo* it expresses a resultative state or perfect aspect. In contrast with progressive aspect clauses, when these clauses are negated the circumfix can only appear on the verb in the Perfective and not on the auxiliary, as shown in (58).

- (58) a. kab gɛza ʔaj-wɛts' iʔ-ɛti-n ʔall-a
 from house NEG-leave.PFV-SM.3FS-NEG BE₁.PRES-SM.3FS
 'She has not left the house.'
- b. *kab gɛza wɛts' i-a j-ɛll-a-n
 from house leave.GER-SM.3FS NEG-BE₁.PRES-SM.3FS-NEG

I suggest the syntax in (60) for a negative sentence expressing progressive aspect with the bipartite negation on the auxiliary (the variant preferred by my consultants), as shown in (59).

- (60)



If NegP is merged above *v*P where the auxiliary is merged, the verb would have to cross the auxiliary to be prefixed by *?aj-*, which would violate locality constraints. If NegP is merged above *v*P where the verb is merged, the complex formed by *?aj-* and the verb would again have to cross the auxiliary. One way to avoid these locality violations is to consider an empirical observation from native speakers: when the negative circumfix appears on the verb in periphrastic tense forms, speakers perceive it as resembling a cleft clause.

Cleft clauses in Tigrinya take the following form, as shown in (62).

- (62) a. *ʕimbabatat ?ijj-om ?ab ts'idija z-i-ʕimbib-u*
flower.MP FOC-SM.3MP LOC spring REL-SM.3MP-bloom.IPFV-SM.3MP
'It is flowers that bloom in spring.'
- b. *hanti tixili ?ijj-ε z-i-gezi? z-ell-εxu*
one.F plant.F FOC-SM.3MP REL-SM.1S-buy.IPFV REL-BE₁.PRES-SM.1S
'It is a plant that I am buying.'

In the examples above, the copula functions as a focus marker and is right-adjacent to and agrees with the focalized element. The verb in (62a) and the verb and auxiliary in (62b) are prefixed by *z(i)-*. The structures are therefore similar to the negative sentence in (56): in (56) the object and the negative verb appear to be focalized, just as *ʕimbabatat* 'flowers' and *tixili* 'plant' are in the examples above.

Returning to the syntax of sentences such as that in (61), if NegP were merged above the lower *v*P, one could suggest that *Asp_{IPFV}*, which contains the phonologically null subject, the object *mis dimay* 'my cat,' and the complex formed by the prefix *?aj-* and the verb *jits'awet* '(he) plays,' could cross the auxiliary because it is attracted by a Focus phrase - merged above *PolP* - and lands in *Spec,FocP*. This would avoid locality constraints. However, it would still be difficult not only to explain how the negative suffix *-(i)n* could attach to the verb, but also to justify a further movement of the auxiliary to a position higher than *FocusP* in order to derive the correct surface order of the constituents in the sentence.

Because the sentence in (61) is only marginally accepted and in light of the syntactic

puzzle discussed above, a more detailed analysis of negative clauses with periphrastic tense forms containing the negative circumfix that appears on the verb will be left to future research.

To conclude this chapter, I now turn to the analysis of negation in *k(i)*-clauses expressing future tense.

5.3.3 Analysis of Negative Future Clauses

Recall that unlike other *k(i)*-clauses, future tense constructions are negated by both *?aj-* and *-(i)n*, as in (63) repeated from a previous section.

- (63) *Teweldε ?ihmilti ?aj-k-i-bel?i-n ?ijj-u*
 Tewelde vegetables NEG-SBJ-SM.3MS-eat.IPFV-NEG BE₂.PRES-SM.3MS
 ‘Tewelde will not eat vegetables.’

I have deferred a syntactic analysis of negative future constructions until now, because these clauses obligatorily contain, in addition to a *k(i)*-marked verb, the copula *?ijju*. Examination of (63) shows that the constituent order matches that of a negative clause containing a periphrastic tense form expressing progressive aspect with the negative circumfix on the verb, as in (61): the verb carries the negative circumfix and is followed by a bare auxiliary/copula. The only additional element is *k(i)-*. As discussed in the previous section, this configuration is nontrivial: the verb must not cross the copula if the circumfix is to attach. I proceed step by step.

Note that, unlike negative clauses with progressive aspect, negative future clauses do not exhibit genuine optionality in the placement of the bipartite negation. It can appear on the verb, as in (63), but placement on the copula, as in (64), lacks consistent judgement among native speakers.

- (64) *%Teweldε ?ihmilti k-i-bel? ?aj-kon-ε-n*
 Tewelde vegetables SBJ-SM.3MS-eat.IPFV NEG-become.PFV-SM.3MS
 ‘Tewelde won’t eat vegetables.’

Note that in the context of an individual-level predicate, negating the copula *?ijju* does

not result in ungrammaticality, as shown in (65).

- (65) a. *tɛmɛharai ʔijj-ɛ*
 student BE₂.PRES-SM.1S
 ‘I am a student.’
 b. *tɛmɛharai ʔaj-kon-ku-n*
 student NEGbecome.PFV-SM.1S-NEG
 ‘I am not a student.’

Why is the copula *ʔijju* negatable in the context of individual-level predicates but not in negative future constructions? I propose that this asymmetry arises because *ʔijju* is undergoing grammaticalization from a copula to a (Verum) Focus marker (Cacchioli 2024); see also Section 1.5.2 of the Introduction. When *ʔijju* functions as a Verum Focus marker, it cannot be negated, as shown in (66).

- (66) a. *Tɛwɛɫɛ ʔihmilti ʔaj-Ø-bɛlʔi-n ʔijj-u*
Tewelde vegetables NEG-SM.3MS-eat.IPFV BE₂.PRES-SM.3MS
 ‘Indeed, Tewelde doesn’t eat vegetables.’
 b. **Tɛwɛɫɛ ʔihmilti ji-bɛliʔ ʔaj-kon-ɛ-n*
Tewelde vegetables SM.3MS-eat.IPFV NEG-become.PFV-SM.3MS
 ‘Indeed, Tewelde doesn’t eat vegetables.’

The probable grammaticalization of *ʔijju* as a Verum Focus marker in future tense constructions may inform the syntactic analysis of these sentences. One might argue that, because *ʔijju* is in a transitional stage of grammaticalization — retaining certain copular properties, such as subject agreement with the verb, while also exhibiting focus-marker properties, such as resistance to negation — predication is still syntactically projected, but *ʔijju* no longer functions as a copula. Instead, it occupies the left periphery as a focus head. Consequently, the predication phrase (PredP) remains in the structure, but its head is empty.

This scenario closely parallels a principle frequently invoked in the grammaticalization literature: “an element A probably never ‘becomes’ B without an intermediary stage in which A and B coexist” (Hopper & Traugott 2003:49). I propose that this postulate

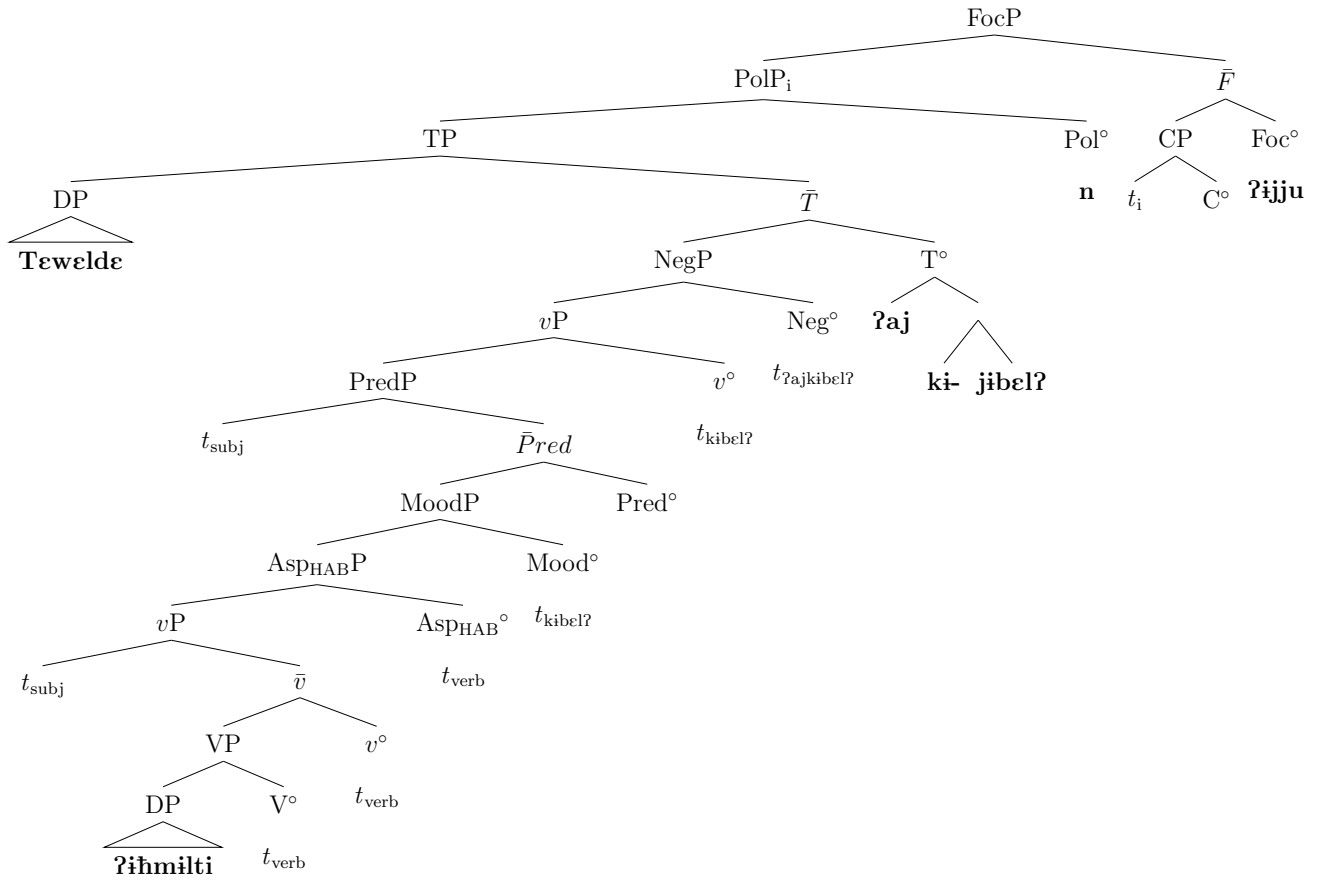
should also be reflected in the syntax. Indeed, such an approach has been suggested, for instance, for the Modern Greek subjunctive particle *na* - discussed in Chapter 4 in connection with *k(i)*- — which grammaticalized from the Classical Greek relative pronoun *hina* following the loss of morphological mood in the language (Philippaki-Warbuton & Spyropoulos 2000). This is illustrated in (67).

- (67) a. $[_{CP} \textit{ina} [_{IP} I_{[Mood/Tense]} [_{VP} V]]]$ > *Classical Greek*
 b. $[_{CP} \textit{ina} [_{MP} M [_{IP} I [_{VP} V]]]]$ > INTERMEDIATE STAGE
 c. $[_{CP} C [_{MP} \textit{ina} [_{IP} I [_{VP} V]]]]$ *Modern Greek*
 (from Roberts & Roussou 2003:83,(15))

The example illustrates several key points. In Classical Greek, Mood and Tense are not distinct projections. In the intermediate stage, Mood (M) emerges as an independent projection with an empty head. In Modern Greek, *ina* has been reanalyzed as a subjunctive marker. Crucially for the analysis of negative future constructions in Tigrinya, the intermediate stage exemplifies the coexistence of projections in the syntax before and after the reanalysis (or grammaticalization) of an element.

Following this line of reasoning, the tree in (68) presents a tentative syntactic analysis of a negative future construction such as that in (63) above.

(68)



In the tree above, *?ijju* does not head a **PredP** projection - whose complement is the *k(i)*- clause - but is instead merged in the **Foc°** head. **NegP** is merged not above the lower *vP* but above the higher one. The verb raises through the structure and is prefixed with *k(i)*-. Crossing the empty **Pred°** (or possibly landing there, although no trace is shown), the *k(i)*-marked verb moves to **Neg°**, where *?aj*- attaches. The suffix *-(i)n* appears on the verb because of its linear adjacency in the syntactic tree. Finally, as a focus marker in **Foc°**, *?ijju* attracts **PolP** to the specifier of **FocusP**.

5.4 Interim Summary of Chapter 5

In this chapter, I investigated the clausal circumfix *?aj-...-(i)n* that expresses sentential negation in Tigrinya. I proposed that *?aj-* heads a low NegP projection dominating *vP* and carrying an [iNeg] feature, and that *-(i)n* heads a high PolP projection dominating TP and carrying a [uNeg] feature. I showed that this circumfix is used to negate declarative root clauses; however, *zi*-clauses, imperatives, and almost all *ki*-clauses are negated only by *?aj-*. I provided syntactic analyses for all of these clause types and explained that *-(i)n* is absent in some of them either because of competition among operators or because PolP cannot be merged in structures lacking TP. I also showed that in clauses with periphrastic tense forms the circumfix can appear on either the verb or the auxiliary, and I provided (tentative) analyses to explain their syntax.

Chapter 6

Conclusions

6.1 Looking Back

This dissertation investigated the syntax of clausal *prefixes* in Tigrinya. I began by showing that Tigrinya, at first glance, appears to be a typological anomaly: it is a head-final language that nevertheless displays prepositions, as well as nominal and verbal *prefixes*. Moreover, clausal *prefixes* are used to introduce subordinate clauses, and sentential negation is bipartite, consisting of both a *prefix* and a suffix. This study addressed these puzzles by proposing syntactic derivations that account for the underlying architecture of sentences containing clausal prefixes and for their resulting word order. I challenged the view that prefixes in head-final languages possess an intrinsic morphological prefixal feature, suggesting instead that they appear to the left of verbs as a consequence of specific syntactic constraints.

The first clausal prefix examined was $z(i)-$. I showed that this element introduces relative clauses, noun clausal complements, comparative clauses, and superlatives. Treating $z(i)-$ as inherently tied to relative clauses would miss the generalization that it also appears in other clause types. I therefore argued that $z(i)-$ is a reflex of successive-cyclic $\bar{A}$ -movement of an element that raises from its argument position to the left periphery of the clause. More specifically, the presence of $z(i)-$ is triggered by a Spec–Head agreement relation between the $\bar{A}$ -moved element and the verb/auxiliary in the aspectual projection of the clause, which I claimed is phasal in Tigrinya. Using binding and case assignment

properties as evidence, I claimed that the relativized head is both internal and external to CP and thus suggested a matching analysis for relative clauses in Tigrinya. I further proposed that what undergoes $\bar{A}$ -movement in noun clausal complements and comparative clauses is the noun itself and, respectively, the nominal DEGREE.

The second clausal prefix examined was *kεmz(i)-*. I argued that this element is bimorphemic, composed of the preposition *kεm* and the clausal prefix studied in the first chapter, *z(i)-*. One argument supporting this analysis is that some adverbial clauses in Tigrinya are also introduced by a preposition-like element combined with *z(i)-*. Because *kεmz(i)-* clauses contain the clausal prefix *z(i)-*, I proposed that they involve $\bar{A}$ -movement of a null nominal (whose meaning is inferred) that is first-merged in the argument structure of the embedded verb. Since I also suggested that *kεm* merges low in the structure, I argued that it pairs with this null nominal.

The third clausal prefix examined was *k(i)-*. By comparing Tigrinya clauses with those of other languages, I suggested that *k(i)-* is a subjunctive marker and that *k(i)-* clauses are reduced clauses whose temporal interpretation depends on that of the matrix clause. I further proposed that *k(i)-* occurs in two distinct configurations. In the first, *k(i)-* is merged in the head of a low MoodP projection. In the second, specifically in clauses expressing simultaneity, *k(i)-* is triggered by the $\bar{A}$ -movement of a temporal operator, as if it were involved in the relativization of duratives. This analysis therefore claims that *k(i)-* and *z(i)-* are both reflexes of $\bar{A}$ -movement but are sensitive to the type of element relativized (a nominal element or a temporal operator).

In Chapters 3 and 4 I presented a wide range of data showing which verb classes select *kεmz(i)-*, *k(i)-*, and *?ilu-* clauses. This is summarized in Table 6.1.

	<i>kɛmz(i)-</i>	<i>k(i)-</i>	<i>?ilu</i>
FACTIVES	✓	✗	✗
COGNITIVE NON-FACTIVES	✓	✗	✓
FICTION VERBS	✓	✓	✓
UTTERANCE VERBS	✓	✗	✓
PERCEPTION VERBS	✓	✓	✗
DIRECTIVE VERBS	✗	✓	✗
DIRECTIVE VERBS	✗	✓	✗
DESIRE VERBS	✗	✓	✗
MODAL VERBS	✗	✓	✗
EMOTIVE FACTIVES	✗	✓	✗
CONTROL VERBS	✗	✓	✗
ECM VERBS	✗	✓	✗

Table 6.1: Classes of verbs that select for *kɛmz(i)*-clauses, *k(i)*-clauses and *?ilu*-clauses.

The fourth and final element investigated was the circumfix *?aj-...-(i)n*, which expresses sentential negation in Tigrinya. I argued that the prefix *?aj-* is merged in the head of a low NegP projection dominating *vP* and carries an [iNeg] feature, whereas the suffix *-(i)n* is merged in the head of a high PolP projection dominating TP and carries a [uNeg] feature. I showed that both affixes negate a declarative root clause, but only the negative prefix *?aj-* appears in *zi*-clauses, imperative clauses, and (most) *ki*-clauses. I suggested that the negative suffix *-(i)n* does not appear in these constructions either because PolP cannot be merged given the size of the clauses (in the case of *ki*- and imperative clauses) or because of operator competition (in the case of *zi*-clauses). I also addressed the issue raised by negative clauses in the past, namely that verbs conjugated in the Gerund paradigm cannot carry prefixes. I proposed that the Gerund does not move in the structure and thus cannot be prefixed by any element. Finally, I presented data on negation in periphrastic tense forms, which show apparent optionality: the negative circumfix can appear on either the verb or the auxiliary. I provided (tentative) analyses to explained this pattern.

This dissertation shows that clausal prefixes in Tigrinya are not typological anomalies, but predictable outcomes of the syntactic structure of the language. By deriving their distribution and order from operations such as $\bar{A}$ -movement, AGREE, head movement, and selection, it reveals that patterns once seen as “morphological exceptions” can illuminate fundamental syntactic mechanisms. This perspective does not undermine morphology or

post-syntactic analyses; rather, it offers an alternative view, demonstrating that these prefixes can be systematically accounted for within syntax. In doing so, this study establishes a framework for understanding morphosyntactic complexity in other languages, showing the explanatory power of syntax in shaping surface morphology.

6.2 Looking Ahead

Looking beyond this dissertation, I would like to pursue two main directions. First, I intend to investigate the clausal prefixes that did not receive full treatment here but were only sporadically mentioned: the nominalizer *m(i)-*, as in (1) (with the option of attaching the preposition *b(i)-* ‘by’, as in (2)), and the DOM, accusative/dative and purposive marker *n(i)-* (on *k(i)-*-marked verbs), as in the third example below. Note that this preposition can also occur on *z(i)-*-marked verbs, as in the fourth example below.

- (1) gizie ʔintɛ z-i-hiliw-ini, ʔɛrɛfti **mi**-ked-ku
time.MS if REL-SM.3MS-BE₁.IPFV-OM.1S vacation NMZ-go.PFV-SM.1S
‘If I had time, I would go on vacation.’
- (2) nisu **bi-mi**-xɛd-u zihzin ʔijj-u
he by-NMZ-leave-3MS sad BE₂.PRES-SM.3MS
‘That he left is sad.’
- (3) ʔab dirar **ni**-k-i-xɛdin-o ɕaket gɛziʔ-ɛ
at dinner PURP-SBJ-SM.1S-wear.IPFV-OM.3MS jacket buy.GER-SM.1S
‘I bought a jacket to wear at dinner.’
- (4) n-it-i habereta **ni**-zi-ħatɛt-ɛ hab-o
DOM-DIST-MS information DAT-REL-ask.PFV-SM.3MS give.IMP-OM.3MS
‘Give the information to whomever asked.’

Second, I would like to examine more closely the clausal prefixes that were fully analyzed in this dissertation and explore how they interact when they co-occur in the same sentence. This includes sentences involving *z(i)-* and *k(i)-*, as in (5).

- (5) ʔit-a ʔanɛ ni-Tɛsfai **k-i-diribi**
DIST-FS I ACC-Tesfay SBJ-SM.3MS-throw_away.IPFV
zi-ʔaziz-ku-wo katena hasot neir-a
REL-order.PFV-SM.1S-OM.3MS necklace false BE₂.GER-SM.3FS
‘The necklace that I ordered Tesfay to throw away was fake.’

Sentences involving *z(i)-* and *kɛmz(i)-*, as in (6).

- (6) ʔit-i ʔanɛ Solomon **kɛm-zi**-geziɛ-o
DEM.MS I Solomon COMP-REL-buy.PFV-SM.3MS-OM.3MS
zi-Ø-hasb-o kɛlbi wididdir ti-ʕawit-u
REL-SM.1S-think.IPFV-OM.3MS dog.MS competition win.GER-SM.3MS
‘The dog that I think Solomon bought won the competition.’

Sentences involving *k(i)-* and *kɛmz(i)-*, as in (7).

- (7) nissu kab geza **k-i-wɛts’** iʔ **kɛm-z-i-kiʔil**
he from house SBJ-SM.3MS-leave.IPFV COMP-REL-SM.3MS-can.IPFV
ji-fɛllit’
SM.1S-know.IPFV
‘I know that he can leave the house.’

Sentences involving *z(i)-*, *k(i)-* and *kɛmz(i)-*, as in (8).

- (8) n-it-i Tɛsfai **kɛm-z-i-mɛsl-o**
DOM-DIST-MS Tesfay COMP-REL-SM.3MS-seem.IPFV-OM.3MS
k-i-kits’ir-o **z-i-dɛli** **z-ɛll-o**
SBJ-SM.3MS-hire.IPFV-OM.3MS REL-SM.3MS-want.IPFV REL-BE₁.PRES-SM.3MS
wɛdi ɥawɛb-oi ʔijj-u
cousin-poss.1s BE₂.PRES-SM.3MS
‘The man that Tesfay thinks that he wants to hire is my cousin.’

And finally, sentences involving *z(i)-*, *k(i)-* (that expresses simultaneity) and *ʔaj-...-(i)n-*, as in (9).

- (9) sirah **z-ɛj-k-i-sɛrriɥ** **k-ɛll-ɛxu,** n-ad-ay
work REL-NEG-SBJ-SM.1S-work.IPFV SBJ-BE₁-SM.1S DOM-mother-POSS.1S
ji-ɥiggiz-a
SM.1S-help.IPFV-OM.3FS
‘When I am not working, I help my mother.’

To conclude, I think that particularly promising avenues for future research lie in descriptive, comparative, and cross-linguistic investigations of the syntax of the cognates of these clausal prefixes in other understudied Ethio-semitic languages.

I hope that this dissertation will be of interest both to scholars working in linguistic theory and to those engaged in the description of Semitic languages. The observations made in this thesis highlight the considerable value of Tigrinya as a source of empirical evidence for typological comparison and theoretical refinement, and they point to the broader potential of Ethio-semitic languages for deepening our understanding of syntax.

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